

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:29:49

Source: data/json/noise_experiments/v2/noise8i_nocoplayer_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_nocoplayer/noise8i_nocoplayer_memprimary_game_002_neutral_rep02.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_nocoplayer
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 2
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 2
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
dyadic_pairing_schedule: {
  "num_turns": 10,
  "role_targets": {
    "Agent_1": 5,
    "Agent_2": 5,
    "Agent_3": 5,
    "Agent_4": 5,
    "Agent_5": 5,
    "Agent_6": 5,
    "Agent_7": 5,
    "Agent_8": 5
  },
  "strategy": "balanced_random_roles_v1"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths,

follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
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MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
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MULTI-AGENT SETTING:

- There are 8 agents in this run.
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- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

=====

ROUND 1

=====

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Sender, Agent_6=Sender, Agent_7=Sender, Agent_8=Sender

Pairings:

dyad_1: Agent_7=Sender, Agent_1=Receiver
dyad_2: Agent_5=Sender, Agent_2=Receiver
dyad_3: Agent_6=Sender, Agent_4=Receiver
dyad_4: Agent_8=Sender, Agent_3=Receiver

Parsed game decisions:

dyad_1: Agent_7 sent \$3; received \$9; Agent_1 returned \$5; payoffs Agent_7=\$7, Agent_1=\$4;
noise/visible: sent visible=\$3.34, received visible=\$10.02, returned visible=\$5.56
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_2: Agent_5 sent \$3; received \$9; Agent_2 returned \$4; payoffs Agent_5=\$6, Agent_2=\$5;
noise/visible: sent visible=\$2.55, received visible=\$7.65, returned visible=\$3.69
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_6 sent \$3; received \$9; Agent_4 returned \$5; payoffs Agent_6=\$7, Agent_4=\$4;
noise/visible: sent visible=\$3.32, received visible=\$9.96, returned visible=\$4.25
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_8 sent \$3; received \$9; Agent_3 returned \$2.2; payoffs Agent_8=\$4.2, Agent_3=\$6.8;
noise/visible: sent visible=\$2.35, received visible=\$7.05, returned visible=\$2.46
Action validation: {"returned": {"amount": 2.2, "clamped": false, "max": 9.0, "min": 0, "raw": 2.2}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 4.0,
"Agent_2": 5.0,
"Agent_3": 6.8,
"Agent_4": 4.0,
"Agent_5": 6.0,
"Agent_6": 7.0,
"Agent_7": 7.0,
"Agent_8": 4.2

}

Visible balances after round: {

"Agent_1": 5.02,
"Agent_2": 3.6499999999999995,
"Agent_3": 4.8500000000000005,
"Agent_4": 4.959999999999999,
"Agent_5": 5.6899999999999995,
"Agent_6": 6.25,
"Agent_7": 7.56,
"Agent_8": 4.46

}

LLM CALLS

GAME - Agent_7 - Sender - dyad_1

Timestamp: 2026-08-12T09:14:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_7",
    "Agent_1"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_1": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
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- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance building trust with managing risk since I have no history with this receiver.

****My reasoning:****

- Sending money creates value (3x multiplier means potential gains for both of us)
- This is round 1, so I want to signal cooperation
- I should send enough to be meaningful but not risk everything
- If the receiver reciprocates fairly, we both benefit

I'll send a moderate amount to test cooperation while limiting downside risk.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. ****Game Structure:****

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- My payoff: \$5 - sent + returned
- Receiver's payoff: 3×sent - returned

2. ****Strategic Considerations:****

- If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 and they keep \$7.50. Total: \$15
- The game creates surplus through cooperation (up to \$15 total vs \$5 if no trust)
- This is a trust game - I need to trust the receiver will return some amount

3. ****First Round Considerations:****

- No history with this partner
- This is a multi-round game with different partners
- Building a reputation for cooperation might matter
- The note about "communication noise" suggests amounts might be misperceived, which could affect trust dynamics

4. ****Common Strategies:****

- ****Full trust:**** Send \$5, hope for good return
- ****Moderate trust:**** Send \$2-3, test the waters
- ****No trust:**** Send \$0, guarantee \$5
- ****Signal cooperation:**** Send a meaningful amount to show willingness to cooperate

5. ****Rational Analysis:****

- If the receiver is purely self-interested and this is a one-shot game, they should return \$0
- Knowing this, I should send \$0
- However, in repeated games or with reciprocity norms, cooperation can emerge
- Many people exhibit fairness preferences and reciprocity

6. ****My Decision:****

Since this is the opening move in a multi-round game, a moderate cooperative signal seems prudent. I'll send \$3-enough to demonstrate trust and give the receiver \$9 to work with, but not so much that I'm exposed to catastrophic loss if they keep everything. Even a modest return of \$3-4 would benefit us both compared to the baseline of keeping \$5....

Usage: {

```
"input_tokens": 500,
"output_tokens": 836,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:14:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_1"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_1": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

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- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.34 (66.8% of their endowment)
- I received \$10.02 after the 3x multiplier
- The sender currently has \$1.66 remaining

I want to reciprocate the sender's trust and ensure cooperation is worthwhile for them. If I return \$5.00:

- The sender would end with $\$1.66 + \$5.00 = \$6.66$ (better than their initial \$5)
- I would keep \$5.02 (also better than any initial amount)

This rewards their trust while maintaining a fair distribution of the gains from cooperation.

{'return': 5.00}

Reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is $\$3.34 \times 3$)
- I need to decide how much to return (between \$0 and \$10.02)

This is a trust game. The sender showed considerable trust by sending 66.8% of their endowment. Let me think about a fair return:

The sender:

- Started with \$5
- Sent \$3.34
- Currently has $\$5 - \$3.34 = \$1.66$
- If I return X, they'll end up with $\$1.66 + X$

Me:

- I received \$10.02
- If I return X, I keep $\$10.02 - X$

Some considerations for fairness:

1. Equal split of total: Total money is \$5 (sender's initial) + \$10.02 (what I received) = \$15.02. Equal split would be \$7.51 each.
 - For sender to get \$7.51, they need $\$7.51 - \$1.66 = \$5.85$ returned
 - I would keep $\$10.02 - \$5.85 = \$4.17$
2. Proportional return: The sender sent 66.8% of their endowment, showing high trust. A reciprocal gesture might be to return a similar proportion.
3. Rewarding trust: Since the sender trusted me with a large amount, I should return enough to make their trust worthwhile.

Let me check option 1: If I return \$5.85:

- Sender ends with: $\$1.66 + \$5.85 = \$7.51$
- I end with: $\$10.02 - \$5.85 = \$4.17$
- This creates mutual benefit from cooperation, and the sender gains significantly from their initial trust.

Returning around \$5 also...

Usage: {

"input_tokens": 529,
"output_tokens": 837,
"reasoning_tokens": 0

}

GAME - Agent_5 - Sender - dyad_2

Timestamp: 2026-08-12T09:15:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_2"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_2": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round as a sender, I need to balance trust and caution since I don't know my co-player yet.

The key consideration is that sending money creates value through the 3x multiplier. If I send \$3:

- I keep \$2
- The receiver gets \$9 ($3 \times \3)
- If they return even a moderate amount (say \$3-4), we both benefit compared to sending nothing

This shows willingness to cooperate while maintaining some security. It's a reasonable opening move that could encourage reciprocity.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategy:

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- There's communication noise, so amounts may differ from what was actually sent/returned
- This is a multi-agent setting with 8 agents, randomized pairings each round
- I'll play both sender and receiver roles across rounds

Strategic considerations:

1. **Trust and cooperation**: If I send money, it creates value (3x multiplier). If the receiver is cooperative, they might return a fair share, making both of us better off.
2. **Risk**: The receiver could keep everything. If I send \$5, they get \$15, and might return nothing, leaving me with \$0.
3. **First round dynamics**: This is the first round, so there's no history to base decisions on. I don't know how cooperative my co-player will be.
4. **Future rounds**: Since this is a multi-round game with different partners, building a reputation for cooperation might matter, though the anonymity and random pairing complicate this.
5. **Communication noise**: This adds uncertainty - even if someone tries to be fair, the noise might make it seem otherwise.
6. **Optimal cooperation**: If both parties cooperate optimally, I could send all \$5, receiver gets \$15, and if they split fairly, they might return around \$7-10, making us both better off than the initial \$5.

Common strategies in trust games:

- Send nothing (risk-averse, guarantees keeping \$5)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - full trust, maximizes potential gains

For the first round with an unknown partner, a moderate approach might be reasonable. Sending about \$3-4 shows willingness to cooperate while not being completely vulnerable.

Since this is a repeated game with multiple rounds and different partners, demonstrating cooperative behavior early could encourage reciprocity....

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 768,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-08-12T09:16:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_2"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_2": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.55, that is 51.0% of their total endowment. You received \$7.649999999999995. How much do you return to the sender? (0-7.649999999999995)

Assistant output:

Looking at this situation:

****What happened:****

- The sender sent \$2.55 (51% of their \$5 endowment), showing moderate trust
- I received \$7.65 after the 3x multiplier
- The sender now has \$2.45 remaining
- I need to decide how much to return

****My reasoning:****

The sender showed significant trust by sending about half their endowment. To reciprocate fairly and build trust:

If I return ****\$4.00****:

- Sender ends with: $\$2.45 + \$4.00 = \$6.45$ (profit of \$1.45)
- I end with: $\$7.65 - \$4.00 = \$3.65$ (profit of \$3.65)

This represents:

- Returning about 52% of what I received (similar to their 51% sent)
- Both of us profit from the cooperation
- Rewarding their trust encourages future cooperation
- Setting a positive precedent for Round 1

{'return': 4.0}

Reasoning:

Let me understand the situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.55 (which is 51% of their \$5 endowment)
3. This amount was multiplied by 3x, so I received $\$2.55 \times 3 = \7.65 (approximately \$7.65)
4. I need to decide how much to return (between \$0 and \$7.65)

Now, let me think about this strategically:

The sender sent 51% of their endowment, which is slightly more than half. This shows a moderate level of trust. They could have sent nothing, they could have sent everything, but they chose to send about half.

From a fairness perspective:

- The sender now has: $\$5 - \$2.55 = \$2.45$
- I have: \$7.65
- Total in the system: $\$2.45 + \$7.65 = \$10.10$

If I want to be fair and reciprocate the trust:

- One approach: Split the total gains equally. The sender started with \$5, and we now have \$10.10 total. If we split equally, we'd each have \$5.05. The sender already has \$2.45, so I'd return $\$5.05 - \$2.45 = \$2.60$
- Another approach: Return the same proportion they sent. They sent 51% of their endowment, so I could return 51% of what I received: $0.51 \times \$7.65 = \3.90
- Conservative approach: Return at least what they sent originally (\$2.55) to ensure they don't lose money from trusting me
- Generous approach: Return more to reward their trust and potentially build cooperation

Given that:

1. The sender showed moderate trust by sending 51%
2. There's communication noise mentioned (though the exact amount sent seems precise)
3. Since this is a multi-round game with multiple agents, establishing a reputation for reciprocity could be strategically valuable
4. The sender took a genuine risk by sending money

A fair reciprocal response should reward their trust while ensuring they benefit from the exchange and remain proportional to their initial gesture. If I return \$2.60, we'd each end with \$5.05-a modest profit for both of us that acknowledges their willingness to trust.

Alternatively, returning 50% of what I received (\$3.825) would give them \$6.275, a more substantial gain. But I need to reconsider my baseline here-as the receiver, I don't start with \$5 like the sender does. I begin with \$0 and only gain what they send multiplied by three. The sender started with \$5, sent \$2.55, and now has \$2.45 remaining.

If I return \$X, they end with $2.45 + X$ and I end with $7.65 - X$. For equal splits, that's \$2.60 returned, giving us both \$5.05. But in this experimental context, I should probably return more-around \$3.00-\$3.50-to reward their trust and signal reciprocity for potential future rounds.

Testing a few amounts: returning \$3.25 leaves them with \$5.70 and me with \$4.40; \$3.50 gives them \$5.95 and me \$4.15; \$4.00 results in them having \$6.45 and me \$3.65. Each option balances showing trustworthiness with keeping reasonable profit, though I'm leaning toward the higher end to maximize their incentive to continue cooperating. Both of us end up better off than we started. At \$3.50, the sender gets \$5.95 (up \$0.95) and I get \$4.15 (up \$4.15), which feels like a solid reciprocal move that rewards their trust without being exploitative.

I'm settling on \$3.50 as my return amount-it strikes a balance between showing genuine reciprocity and ensuring both parties benefit from the exchange. Terms of what's strategically optimal in a multi-round setting. Building reputation matters-if I'm generous now, other agents will likely trust me more later. Since this is Round 1, establishing a trustworthy precedent is crucial. The sender showed moderate confidence by sending \$2.55, so I should reciprocate proportionally.

I'm settling on \$3.50. It more than doubles their original amount, gives them solid profit, signals I'm reliable, and leaves me with a reasonable gain too. Let me verify the math: $2.55 \times 3 = 7.65$ received, returning \$3.50 means the sender nets \$0.95 profit while I keep \$4.15.

The communication noise mentioned in the problem is worth noting, but I can only work with what I observe. My final return amount is \$3.50, formatted as {'return': 3.50}.

Actually, reconsidering the proportionality here-the sender gave 51% of their endowment, which signals moderate trust. Matching that proportion with my return would mean sending back 51% of \$7.65, which equals \$3.90. This reciprocal approach better reflects their level of generosity while still leaving me with a reasonable profit of \$3.75 versus their \$1.35. I think \$3.90 better captures the spirit of proportional reciprocation.

Though I'm also drawn to rounding up to \$4.00 for simplicity-it's nearly the same percentage anyway (52%), easier to process, and still feels fair to both parties. Let me settle on \$4.00 as my final choice.

```
Usage: {
  "input_tokens": 539,
  "output_tokens": 3346,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T09:16:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_4"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_4": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_4"
  }
}
```



```

},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative dynamic while being somewhat cautious since I don't know my co-player yet.

If I send money, it triples, creating potential mutual benefit. However, there's risk since the receiver's return is voluntary.

I'll send a moderate amount to signal trust and willingness to cooperate, while not being overly vulnerable in this first interaction.

****Decision:****

```json

```
{"send": 3}
'''
```

This gives the receiver \$9 to work with. If they're cooperative and return a fair share, we both benefit. If not, I've learned something about the dynamics for future rounds while limiting my downside risk.

#### Reasoning:

This is the first round of a trust game, and I'm the sender. I need to think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3
- The receiver decides how much to return
- There may be communication noise
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I don't know who my current co-player is

Since this is the first round, I have no information about how other players behave. The optimal strategy depends on:

1. Whether I expect reciprocity from the receiver
2. Whether I want to establish a reputation (though with hidden names, this is less relevant for future rounds)
3. Risk tolerance

From a purely rational game theory perspective, if this were a one-shot game with no reputation effects, the receiver should return \$0 (keep everything), which means I should send \$0. However, experimental evidence shows people often cooperate in trust games.

Some considerations:

- Sending \$5 maximizes total payoff (\$5 + \$15 total) but requires trust
- Sending \$0 guarantees I keep \$5 but creates no opportunity for mutual gain
- A moderate amount (like \$2-3) balances risk and opportunity

Since this is round 1 and I want to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything immediately

I'll send a moderate amount. Sending \$3 seems reasonable:

- If sent \$3, receiver gets \$9
- If receiver returns half (\$4.50), I end up with  $\$5 - \$3 + \$4.50 = \$6.50$
- If receiver returns nothing, I end up with \$2
- This tests their cooperation while not being overly cautious

Actually, let me reconsider. A common strategy is to send about half to show trust but not complete vulnerability. Sending \$2.50 or \$3 seems reasonable.

...

```
Usage: {
 "input_tokens": 500,
 "output_tokens": 632,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_4 - Receiver - dyad\_3

Timestamp: 2026-08-12T09:16:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_4"
],
 "investor": "Agent_6",
```

```

"roles": {
 "Agent_4": "trustee",
 "Agent_6": "investor"
},
"trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.32, that is 66.4% of their total endowment. You received \$9.959999999999999. How much do you return to the sender? (0-9.959999999999999)

**Assistant output:**

Looking at this situation:

- The sender showed significant trust by sending \$3.32 (66.4% of their endowment)
- I received \$9.96 after the 3x multiplier
- The sender currently has \$1.68 remaining

I want to reward this trust while being fair to both of us. If I return \$5.00:

- The sender ends with:  $\$1.68 + \$5.00 = \$6.68$  (profit of \$1.68)
- I end with:  $\$9.96 - \$5.00 = \$4.96$

This gives the sender a good return on their trust (33.6% gain on their original \$5) while I also benefit from the exchange. This return rate of approximately 50% is commonly seen as reciprocal in trust games.

```
{'return': 5.00}
```

#### Reasoning:

Let me analyze this situation:

1. The sender sent \$3.32 out of their \$5 endowment (66.4%)
2. Due to the 3x multiplier, I received \$9.96 ( $3.32 \times 3 = 9.96$ )
3. I need to decide how much to return

Key considerations:

- The sender showed significant trust by sending 66.4% of their endowment
- This is a multi-round game with random pairings
- Building a reputation for reciprocity could be beneficial
- There's communication noise, so exact amounts may not be perceived correctly

Let me think about fair/reciprocal returns:

- The sender spent \$3.32 to give me \$9.96
- If I want the sender to profit from trusting me, I should return more than \$3.32
- A common strategy in trust games is to split the gains roughly equally

If I return \$0:

- Sender gets:  $\$5 - \$3.32 + \$0 = \$1.68$
- I get:  $\$9.96 - \$0 = \$9.96$
- Total: \$11.64

If I return \$3.32 (sender breaks even):

- Sender gets:  $\$5 - \$3.32 + \$3.32 = \$5.00$
- I get:  $\$9.96 - \$3.32 = \$6.64$
- Total: \$11.64

If I return \$5.00 (roughly equal split of total):

- Sender gets:  $\$5 - \$3.32 + \$5.00 = \$6.68$
- I get:  $\$9.96 - \$5.00 = \$4.96$
- Total: \$11.64

The sender took a risk and sent a substantial amount. A reasonable reciprocal amount would be to ensure both of us profit from the exchange.

To achieve roughly equal final payoffs, I'd need to return about \$4.14-this would give us each \$5.82. But since the sender invested significantly from their original endowment, I should consider whether an even more generous return would better reward their trust.

If I return \$6.64, they'd end with \$8.32 (doubling their investment gain), though that leaves me with just \$3.32. A middle ground like \$5.00 would give them a 33.6% gain while I keep \$4.96. Since this is the first round and I want to build a reputation for fairness, and given how much trust they demonstrated by sending most of their money, I'm leaning toward a return that meaningfully rewards their cooperation while still leaving me in a reasonable position.

...

Usage: {

```
"input_tokens": 537,
"output_tokens": 1319,
"reasoning_tokens": 0
```

}

**GAME - Agent\_8 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:17:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_3",
```

```

"pairing": {
 "agents": [
 "Agent_8",
 "Agent_3"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_3": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Assistant output:**

Looking at this first round as the SENDER, I need to balance trust and caution since I have no

history with this co-player.

The key consideration is that sending money creates value (3x multiplier), but I'm vulnerable to the receiver's decision about how much to return.

A moderate amount seems reasonable for Round 1:

- It signals willingness to cooperate and trust
- It creates meaningful value (\$9 for the receiver)
- It provides some protection against complete loss

I'll send \*\*\$3\*\*.

{'send': 3}

#### Reasoning:

This is Round 1 of a multi-task experiment where I'm playing a trust game as the SENDER. Let me think about this situation:

1. I start with \$5
2. I can send any amount from \$0 to \$5
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - amount\_sent + amount\_returned

This is a classic trust game. The optimal strategy depends on:

- Whether I trust the receiver to return a fair amount
- What "fair" means in this context
- The fact that there's communication noise (amounts may differ slightly)
- This is a multi-round game with multiple agents
- Pairings are randomized each round

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I'd end up with \$7.50 total (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- The socially optimal outcome is to send the maximum amount to create the most total value

However, there's risk:

- The receiver might not return anything
- There's communication noise that could affect trust
- This is round 1, so I have no history with this partner

A reasonable first-round strategy might be:

- Send a moderate amount to signal trust and willingness to cooperate
- Not send everything in case the receiver doesn't reciprocate
- Send enough to create meaningful value

Common strategies in trust games:

- Send about half (\$2.50) as a balanced approach
- Send more (\$3-4) to signal strong trust
- Send everything (\$5) to maximize joint payoff

Given this is round 1 and I want to establish a cooperative pattern but also protect against complete exploitation, I'll send \$3. This:

- Creates \$9 for the receiver
- Shows trust and cooperation

...

Usage: {

"input\_tokens": 500,  
"output\_tokens": 625,  
"reasoning\_tokens": 0

}

**GAME - Agent\_3 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:17:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_4",  
"move\_index": 7,

```

"opponent_id": "Agent_8",
"pairing": {
 "agents": [
 "Agent_8",
 "Agent_3"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_3": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$2.35, that is 47.0% of their total endowment. You received \$7.050000000000001. How much do you return to the sender? (0-7.05000000)

0000001)

**Assistant output:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.35 (47% of their endowment)
- I received \$7.05 from this
- The sender currently has \$2.65 remaining

I want to reciprocate this trust fairly. A reasonable approach is to aim for equal final payoffs since we're cooperating:

- Total available: \$2.65 (sender's remainder) + \$7.05 (my received amount) = \$9.70
- Equal split:  $\$9.70 \div 2 = \$4.85$  each
- Return needed:  $\$4.85 - \$2.65 = \$2.20$

This ensures we both benefit equally from the cooperation, rewarding the sender's trust while keeping the gains fair.

{'return': 2.20}

**Reasoning:**

Let me understand the situation:

- I am the RECEIVER in Round 1
- The sender sent \$2.35 (47% of their \$5 endowment)
- This was multiplied by 3x, so I received \$7.05
- I need to decide how much to return (between \$0 and \$7.05)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. There's communication noise mentioned, so amounts may differ slightly
3. This is a multi-round game with 8 agents and rotating pairings
4. Building reputation and reciprocity is important

The sender sent 47% of their endowment. If I want to be fair and reciprocal:

- The sender started with \$5, sent \$2.35, and has \$2.65 left
- I received \$7.05
- If I don't return anything, sender gets \$2.65, I get \$7.05
- A fair split might be to aim for equal payoffs or proportional returns

Common strategies in trust games:

1. Return nothing (selfish)
2. Return the original amount sent (\$2.35)
3. Return enough to split the gains equally
4. Return a proportion similar to what was sent

Let's calculate equal payoffs:

- Total money: \$5 (initial sender) + \$7.05 (tripled) = \$12.05? No wait...
- Initial endowment for sender: \$5
- Sent: \$2.35 (tripled to \$7.05 for receiver)
- Sender currently has:  $\$5 - \$2.35 = \$2.65$
- Receiver has: \$7.05
- Total in system:  $\$2.65 + \$7.05 = \$9.70$

For equal payoffs: each should get  $\$9.70/2 = \$4.85$

- So I'd need to return \$2.20 to the sender, leaving me with \$4.85

I could also return the original \$2.35 they sent, which would give them \$5.00 (breaking even) while I keep \$4.70. Or I could mirror their generosity by returning 47% of what I received - about \$3.31 - which acknowledges their trust while still letting me benefit from the multiplication.

The equal payoff approach feels right though. Returning \$2.20 means we both end up with \$4.85, which feels like genuine reciprocity for the trust they showed....

```
Usage: {
 "input_tokens": 537,
 "output_tokens": 1079,
 "reasoning_tokens": 0
}
```

=====

**ROUND 2**



=====

Roles: Agent\_1=Sender, Agent\_2=Sender, Agent\_3=Sender, Agent\_4=Sender, Agent\_5=Receiver,  
Agent\_6=Receiver, Agent\_7=Receiver, Agent\_8=Receiver

**Pairings:**

dyad\_1: Agent\_3=Sender, Agent\_7=Receiver  
dyad\_2: Agent\_4=Sender, Agent\_8=Receiver  
dyad\_3: Agent\_2=Sender, Agent\_6=Receiver  
dyad\_4: Agent\_1=Sender, Agent\_5=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_3 sent \$2.5; received \$7.5; Agent\_7 returned \$3; payoffs Agent\_3=\$5.5, Agent\_7=\$4.5;  
noise/visible: sent visible=\$1.67, received visible=\$5.01, returned visible=\$2.43  
Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 7.5, "min": 0, "raw": 3.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}  
dyad\_2: Agent\_4 sent \$3; received \$9; Agent\_8 returned \$4; payoffs Agent\_4=\$6, Agent\_8=\$5;  
noise/visible: sent visible=\$3.19, received visible=\$9.57, returned visible=\$4.51  
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_3: Agent\_2 sent \$2.5; received \$7.5; Agent\_6 returned \$3.75; payoffs Agent\_2=\$6.25, Agent\_6=\$3.75; noise/visible: sent visible=\$2.67, received visible=\$8.01, returned visible=\$3.62  
Action validation: {"returned": {"amount": 3.75, "clamped": false, "max": 7.5, "min": 0, "raw": 3.75}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}  
dyad\_4: Agent\_1 sent \$3; received \$9; Agent\_5 returned \$2.82; payoffs Agent\_1=\$4.82, Agent\_5=\$6.18; noise/visible: sent visible=\$2.09, received visible=\$6.27, returned visible=\$2.73  
Action validation: {"returned": {"amount": 2.82, "clamped": false, "max": 9.0, "min": 0, "raw": 2.82}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

**Balances after round: {**

"Agent\_1": 8.82,  
"Agent\_2": 11.25,  
"Agent\_3": 12.3,  
"Agent\_4": 10.0,  
"Agent\_5": 12.18,  
"Agent\_6": 10.75,  
"Agent\_7": 11.5,  
"Agent\_8": 9.2

}

**Visible balances after round: {**

"Agent\_1": 9.75,  
"Agent\_2": 9.77,  
"Agent\_3": 9.780000000000001,  
"Agent\_4": 11.469999999999999,  
"Agent\_5": 9.139999999999999,  
"Agent\_6": 10.51,  
"Agent\_7": 9.57,  
"Agent\_8": 10.030000000000001

}

**LLM CALLS**

**GAME - Agent\_3 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:17:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_7",  
"pairing": {  
  "agents": [  
    "Agent\_3",  
    "Agent\_7"  
  ],  
  "investor": "Agent\_3",  
  "roles": {  
    "Agent\_3": "investor",  
    "Agent\_7": "trustee"  
  },  
  "trustee": "Agent\_7"  
},  
"role": "investor",

```

"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.35, that is 47.0% of their total endowment. You received \$7.050000000000001. How much do you return to the sender? (0-7.050000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.35 (47% of their endowment)
- I received \$7.05 from this
- The sender currently has \$2.65 remaining

I want to reciprocate this trust fairly. A reasonable approach is to aim for equal final payoffs since we're cooperating:

- Total available: \$2.65 (sender's remainder) + \$7.05 (my received amount) = \$9.70
- Equal split:  $\$9.70 \div 2 = \$4.85$  each

- Return needed:  $\$4.85 - \$2.65 = \$2.20$

This ensures we both benefit equally from the cooperation, rewarding the sender's trust while keeping the gains fair.

```
{'return': 2.20}
```

**reasoning:**

Let me understand the situation:

- I am the RECEIVER in Round 1
- The sender sent \$2.35 (47% of their \$5 endowment)
- This was multiplied by 3x, so I received \$7.05
- I need to decide how much to return (between \$0 and \$7.05)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. There's communication noise mentioned, so amounts may differ slightly
3. This is a multi-round game with 8 agents and rotating pairings
4. Building reputation and reciprocity is important

The sender sent 47% of their endowment. If I want to be fair and reciprocal:

- The sender started with \$5, sent \$2.35, and has \$2.65 left
- I received \$7.05
- If I don't return anything, sender gets \$2.65, I get \$7.05
- A fair split might be to aim for equal payoffs or proportional returns

Common strategies in trust games:

1. Return nothing (selfish)
2. Return the original amount sent (\$2.35)
3. Return enough to split the gains equally
4. Return a proportion similar to what was sent

Let's calculate equal payoffs:

- Total money: \$5 (initial sender) + \$7.05 (tripled) = \$12.05? No wait...
- Initial endowment for sender: \$5
- Sent: \$2.35 (tripled to \$7.05 for receiver)
- Sender currently has:  $\$5 - \$2.35 = \$2.65$
- Receiver has: \$7.05
- Total in system:  $\$2.65 + \$7.05 = \$9.70$

For equal payoffs: each should get  $\$9.70/2 = \$4.85$

- So I'd need to return \$2.20 to the sender, leaving me with \$4.85

I could also return the original \$2.35 they sent, which would give them \$5.00 (breaking even) while I keep \$4.70. Or I could mirror their generosity by returning 47% of what I received—about \$3.31—which acknowledges their trust while still letting me benefit from the multiplication.

The equal payoff approach feels right though. Returning \$2.20 means we both end up with \$4.85, which feels like genuine reciprocity for the trust they showed....

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 1079,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.35, it became \$7.050000000000001, and you returned \$2.2. Your visible payoff was \$4.8500000000000005.

Your total visible earnings across all rounds are \$4.8500000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my previous round, the sender showed trust at 47% (\$2.35) and I reciprocated fairly, resulting in equal payoffs of \$4.85 each.

As the sender now, I'll show similar moderate trust. Sending around 50% demonstrates cooperation while managing risk.

```
{'send': 2.50}
```

**Reasoning:**

Let me think about this situation:

Round 2: I am now the SENDER

- I have \$5
- I need to decide how much to send
- The amount I send will be tripled
- The receiver will decide how much to return

In the previous round:

- I was the receiver
- The sender sent \$2.35 (47% of their endowment)
- I received \$7.05
- I returned \$2.20
- My payoff was \$4.85
- The sender's payoff would have been  $\$2.65 + \$2.20 = \$4.85$

So in the last round, I reciprocated fairly by aiming for equal payoffs.

Now I'm the sender. I should consider:

1. Building trust by sending a reasonable amount
2. The sender in the previous round sent 47% (\$2.35 out of \$5)
3. I reciprocated fairly, so hopefully this receiver will do the same

Since the previous sender showed moderate trust at 47%, I could do something similar. Sending around \$2.35-\$2.50 seems reasonable.

However, I could also think strategically:

- If I send \$2.50, the receiver gets \$7.50
- If they split equally, they'd return \$2.50 to give us each \$5
- If I send \$3.00, the receiver gets \$9.00
- Equal split would mean returning \$3.50 for payoffs of \$5.50 each

Let me try sending \$2.50 as a moderate trust signal, similar to what the previous sender did with me.

Actually, let me reconsider. I showed I'm cooperative by returning fairly in the previous round. I could send a bit more to maximize joint welfare if the receiver reciprocates.

If I send \$3:

- I keep \$2
- Receiver gets \$9
- If they return \$3.50, I get \$5.50 total, they get \$5.50
- Joint welfare: \$11...

Usage: {

```
"input_tokens": 859,
"output_tokens": 713,
"reasoning_tokens": 0
```

}

**GAME - Agent\_7 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:18:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_3",
"pairing": {
 "agents": [
 "Agent_3",
 "Agent_7"
],
 "investor": "Agent_3",
```

```

"roles": {
 "Agent_3": "investor",
 "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance building trust with managing risk since I have no history with this receiver.

**\*\*My reasoning:\*\***

- Sending money creates value (3x multiplier means potential gains for both of us)
- This is round 1, so I want to signal cooperation

- I should send enough to be meaningful but not risk everything
- If the receiver reciprocates fairly, we both benefit

I'll send a moderate amount to test cooperation while limiting downside risk.

```
'''json
{"send": 3}
'''
```

#### reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

- Game Structure:**
  - I have \$5
  - I can send \$0-\$5 to the receiver
  - Whatever I send gets tripled
  - The receiver then decides how much to return
  - My payoff: \$5 - sent + returned
  - Receiver's payoff: 3×sent - returned
- Strategic Considerations:**
  - If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5
  - If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 and they keep \$7.50. Total: \$15
  - The game creates surplus through cooperation (up to \$15 total vs \$5 if no trust)
  - This is a trust game - I need to trust the receiver will return some amount
- First Round Considerations:**
  - No history with this partner
  - This is a multi-round game with different partners
  - Building a reputation for cooperation might matter
  - The note about "communication noise" suggests amounts might be misperceived, which could affect trust dynamics
- Common Strategies:**
  - **Full trust:** Send \$5, hope for good return
  - **Moderate trust:** Send \$2-3, test the waters
  - **No trust:** Send \$0, guarantee \$5
  - **Signal cooperation:** Send a meaningful amount to show willingness to cooperate
- Rational Analysis:**
  - If the receiver is purely self-interested and this is a one-shot game, they should return \$0
  - Knowing this, I should send \$0
  - However, in repeated games or with reciprocity norms, cooperation can emerge
  - Many people exhibit fairness preferences and reciprocity

#### My Decision:

Since this is the opening move in a multi-round game, a moderate cooperative signal seems prudent. I'll send \$3-enough to demonstrate trust and give the receiver \$9 to work with, but not so much that I'm exposed to catastrophic loss if they keep everything. Even a modest return of \$3-4 would benefit us both compared to the baseline of keeping \$5....

#### usage:

```
{
 "input_tokens": 500,
 "output_tokens": 836,
 "reasoning_tokens": 0
}
```

#### Message 4 role=user

##### content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.56 to you. Your visible payoff was \$7.56.

Your total visible earnings across all rounds are \$7.56.

This round, you are the RECEIVER against your current co-player. They sent you \$1.67, so you received \$5.01. How much do you return? (0-5.01)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.67 (33% of their endowment)
- I received \$5.01 after the 3x multiplier
- They currently have \$3.33 remaining

**\*\*My considerations:\*\***

- They showed moderate trust by sending \$1.67
- The multiplier created \$3.34 in surplus value
- A fair approach would be to share this surplus roughly equally
- In Round 1, my partner returned ~62% and that felt cooperative

**\*\*Fair return calculation:\*\***

If we split the surplus equally:

- Each should gain ~\$1.67 from cooperation
- Sender ending: \$5 - \$1.67 + return = \$6.67 desired
- Return needed: \$3.34

However, that might be overly generous given their modest send. I'll return about 60% to reward their cooperation while keeping reasonable value.

```
'''json
{"return": 3.00}
'''
```

**Reasoning:**

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.56 to me
- I ended with: \$5 - \$3 + \$5.56 = \$7.56
- Receiver ended with: \$9 - \$5.56 = \$3.44

That was actually quite generous of them! They returned more than half.

Round 2 (current):

- I am RECEIVER
- They sent me \$1.67
- I received \$5.01 (which is 3 × \$1.67)
- I need to decide how much to return (0-5.01)

Key considerations:

1. My previous partner was generous, returning \$5.56 out of \$9 (62%)
2. This is a different player (multi-agent setting with randomized pairings)
3. They sent me \$1.67 out of \$5, which is 33.4% - a moderate amount
4. The sender keeps: \$5 - \$1.67 = \$3.33 before my return
5. I should reciprocate fairly to encourage future cooperation

Let me think about fair returns:

- If I return half (\$2.50), they end with \$3.33 + \$2.50 = \$5.83, I keep \$2.51
- If I return around what gives us equal final payoffs:
  - They have \$3.33 + return
  - I have \$5.01 - return
  - Equal:  $3.33 + x = 5.01 - x$
  - $2x = 1.68$
  - $x = 0.84$

This would give us both around \$4.17

- A proportional return similar to what I received (62% of what I got):
  - $0.62 \times 5.01 = 3.11$
  - They'd end with:  $3.33 + 3.11 = 6.44$
  - I'd keep:  $5.01 - 3.11 = 1.90$

A 50% return of around \$2.50 seems like the right balance-it rewards their cooperation without being excessive, and it's more moderate than matching their 33% send rate since they're a different player than before.

If I send back \$2.50, they'd gain \$0.83 while I'd be left with \$2.51. The math shows we'd create \$8.34 in total value, which is less than the \$10 we'd have if neither of us participated. The real question is what surplus we're actually generating from this exchange and whether my return is proportional to their initial trust.

The multiplier created \$3.34 in extra value (\$5.01 received minus \$1.67 sent). If we split that gain equally, each of us should pocket \$1.67. That means I should return \$3.34 to bring them to \$6.67 total, which gives us both the same \$1.67 gain from the transaction....

```
Usage: {
 "input_tokens": 754,
 "output_tokens": 1445,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_4 - Sender - dyad\_2**

Timestamp: 2026-08-12T09:18:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_8"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 2,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**



You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.32, that is 66.4% of their total endowment. You received \$9.959999999999999. How much do you return to the sender? (0-9.959999999999999)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.32 (66.4% of their endowment)
- I received \$9.96 after the 3x multiplier
- The sender currently has \$1.68 remaining

I want to reward this trust while being fair to both of us. If I return \$5.00:

- The sender ends with:  $\$1.68 + \$5.00 = \$6.68$  (profit of \$1.68)
- I end with:  $\$9.96 - \$5.00 = \$4.96$

This gives the sender a good return on their trust (33.6% gain on their original \$5) while I also benefit from the exchange. This return rate of approximately 50% is commonly seen as reciprocal in trust games.

{'return': 5.00}

reasoning:

Let me analyze this situation:

1. The sender sent \$3.32 out of their \$5 endowment (66.4%)
2. Due to the 3x multiplier, I received \$9.96 ( $3.32 \times 3 = 9.96$ )
3. I need to decide how much to return

Key considerations:

- The sender showed significant trust by sending 66.4% of their endowment
- This is a multi-round game with random pairings
- Building a reputation for reciprocity could be beneficial
- There's communication noise, so exact amounts may not be perceived correctly

Let me think about fair/reciprocal returns:

- The sender spent \$3.32 to give me \$9.96
- If I want the sender to profit from trusting me, I should return more than \$3.32
- A common strategy in trust games is to split the gains roughly equally

If I return \$0:

- Sender gets:  $\$5 - \$3.32 + \$0 = \$1.68$
- I get:  $\$9.96 - \$0 = \$9.96$
- Total: \$11.64

If I return \$3.32 (sender breaks even):

- Sender gets:  $\$5 - \$3.32 + \$3.32 = \$5.00$
- I get:  $\$9.96 - \$3.32 = \$6.64$
- Total: \$11.64

If I return \$5.00 (roughly equal split of total):

- Sender gets:  $\$5 - \$3.32 + \$5.00 = \$6.68$
- I get:  $\$9.96 - \$5.00 = \$4.96$
- Total:  $\$11.64$

The sender took a risk and sent a substantial amount. A reasonable reciprocal amount would be to ensure both of us profit from the exchange.

To achieve roughly equal final payoffs, I'd need to return about \$4.14-this would give us each \$5.82. But since the sender invested significantly from their original endowment, I should consider whether an even more generous return would better reward their trust.

If I return \$6.64, they'd end with \$8.32 (doubling their investment gain), though that leaves me with just \$3.32. A middle ground like \$5.00 would give them a 33.6% gain while I keep \$4.96. Since this is the first round and I want to build a reputation for fairness, and given how much trust they demonstrated by sending most of their money, I'm leaning toward a return that meaningfully rewards their cooperation while still leaving me in a reasonable position....

usage:

```
{
 "input_tokens": 537,
 "output_tokens": 1319,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.32, it became \$9.959999999999999, and you returned \$5.0. Your visible payoff was \$4.959999999999999.

Your total visible earnings across all rounds are \$4.959999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the previous round, my co-player showed significant trust by sending 66.4% of their endowment, and I reciprocated with a 50% return rate.

For this round as sender, I'll demonstrate similar trust to establish cooperative play. Sending \$3.00 (60% of my endowment) shows trust while maintaining some security.

```
{'send': 3.00}
```

Reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and someone sent me \$3.32 (66.4% of their endowment). I returned \$5.00, which was about 50% of what I received.
2. Now in Round 2, I'm the sender with \$5.
3. I need to decide how much to send to my current co-player (who may be different from Round 1's partner).

Key considerations:

- In the previous round, someone trusted me with 66.4% of their endowment
- I reciprocated fairly by returning 50%
- This established a pattern of moderate trust and reciprocity
- I'm playing with potentially a different agent now

Strategy options:

- a) Send a similar amount to what I received last round (~\$3.32) - this mirrors the trust shown to me
- b) Send less to be more cautious
- c) Send more to encourage cooperation

Given that:

- The previous sender trusted me with 66.4%, showing moderate-high trust
- I want to encourage cooperation in this multi-round game
- Sending around the same proportion (60-70%) seems reasonable
- This could establish a cooperative norm

I'll send \$3.00 (60% of my endowment), which is:

- Similar to what was sent to me in Round 1
- Shows trust but not maximal trust
- Allows me to keep some guarantee (\$2)
- If returned fairly (~50%), I'd get back around \$4.50, making my total \$2 + \$4.50 = \$6.50

Let me go with \$3.00.

```
Usage: {
 "input_tokens": 864,
 "output_tokens": 471,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_8 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:18:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_8"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 2,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the SENDER, I need to balance trust and caution since I have no history with this co-player.

The key consideration is that sending money creates value (3x multiplier), but I'm vulnerable to the receiver's decision about how much to return.

A moderate amount seems reasonable for Round 1:

- It signals willingness to cooperate and trust
- It creates meaningful value (\$9 for the receiver)
- It provides some protection against complete loss

I'll send \*\*\$3\*\*.

{'send': 3}

**reasoning:**

This is Round 1 of a multi-task experiment where I'm playing a trust game as the SENDER. Let me think about this situation:

1. I start with \$5
2. I can send any amount from \$0 to \$5
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - amount\_sent + amount\_returned

This is a classic trust game. The optimal strategy depends on:

- Whether I trust the receiver to return a fair amount
- What "fair" means in this context
- The fact that there's communication noise (amounts may differ slightly)
- This is a multi-round game with multiple agents
- Pairings are randomized each round

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I'd end up with \$7.50 total (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- The socially optimal outcome is to send the maximum amount to create the most total value

However, there's risk:

- The receiver might not return anything
- There's communication noise that could affect trust
- This is round 1, so I have no history with this partner

A reasonable first-round strategy might be:

- Send a moderate amount to signal trust and willingness to cooperate
- Not send everything in case the receiver doesn't reciprocate
- Send enough to create meaningful value

Common strategies in trust games:

- Send about half (\$2.50) as a balanced approach
- Send more (\$3-4) to signal strong trust
- Send everything (\$5) to maximize joint payoff

Given this is round 1 and I want to establish a cooperative pattern but also protect against complete exploitation, I'll send \$3. This:

- Creates \$9 for the receiver
- Shows trust and cooperation

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 625,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.46 to you. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$4.46.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Current Round (Round 2):\*\***

- My co-player sent me \$3.19 ? I received \$9.57
- This created \$6.38 in surplus value

**\*\*Fair return considerations:\*\***

- The sender took a significant risk sending most of their endowment
- A reasonable approach is to ensure we both benefit from the cooperation
- Returning \$4.00 would give:
  - Sender:  $\$5 - \$3.19 + \$4.00 = \$5.81$
  - Me:  $\$9.57 - \$4.00 = \$5.57$

This creates a balanced outcome where we both gain from the exchange.

{'return': 4.00}

**Reasoning:**

Let me analyze this situation:

Last round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.46 to me
- My payoff:  $\$5 - \$3 + \$2.46 = \$4.46$
- Their payoff:  $\$9.0 - \$2.46 = \$6.54$

So they returned about 27% of what they received ( $\$2.46/\$9.0 \approx 0.273$ ).

Now in Round 2:

- I'm the RECEIVER
- My current co-player sent me \$3.19
- I received \$9.57 ( $3.19 \times 3$ )
- I need to decide how much to return

Important notes:

1. This is a different co-player (multi-agent setting, randomized pairings)
2. There's communication noise, so the \$3.19 might not be exactly what they sent
3. The previous player returned about 27% to me

Considerations:

- If I want to encourage cooperation, I should return a fair amount
- A common fair split might be to return enough so we both benefit equally from the cooperation
- The sender gave up \$3.19, so they have  $\$5 - \$3.19 = \$1.81$  left
- I have \$9.57
- If I want us to end up with similar amounts, I could return about \$3.88, giving us both around \$5.69
- Or I could return proportionally to what was sent: returning about 1/3 of what I received would be \$3.19, which gives them back their original amount
- A generous return might be half of what I received: \$4.785

Let me think about what's fair:

- The sender took a risk sending \$3.19 (gave up most of their endowment)
- The transaction created \$6.38 in surplus value beyond what they sent
- Splitting that gain equally means we each get an additional \$3.19 from the multiplier effect
- So returning \$3.19 would leave me with \$6.38 and restore them to their starting \$5
- Alternatively, returning \$4 would give them \$5.81 while I keep \$5.57

, creating a more balanced outcome

- Given the earlier low reciprocation (27%), I could...

```
Usage: {
 "input_tokens": 756,
 "output_tokens": 1027,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_2 - Sender - dyad\_3

Timestamp: 2026-08-12T09:19:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_6"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 2,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.55, that is 51.0% of their total endowment. You received \$7.6499999999999995. How much do you return to the sender? (0-7.6499999999999995)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.55 (51% of their \$5 endowment), showing moderate trust
- I received \$7.65 after the 3x multiplier
- The sender now has \$2.45 remaining
- I need to decide how much to return

**\*\*My reasoning:\*\***

The sender showed significant trust by sending about half their endowment. To reciprocate fairly and build trust:

If I return **\*\*\$4.00\*\***:

- Sender ends with: \$2.45 + \$4.00 = \$6.45 (profit of \$1.45)
- I end with: \$7.65 - \$4.00 = \$3.65 (profit of \$3.65)

This represents:

- Returning about 52% of what I received (similar to their 51% sent)
- Both of us profit from the cooperation
- Rewarding their trust encourages future cooperation
- Setting a positive precedent for Round 1

{'return': 4.0}

**reasoning:**

Let me understand the situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.55 (which is 51% of their \$5 endowment)
3. This amount was multiplied by 3x, so I received  $\$2.55 \times 3 = \$7.65$  (approximately \$7.65)
4. I need to decide how much to return (between \$0 and \$7.65)

Now, let me think about this strategically:

The sender sent 51% of their endowment, which is slightly more than half. This shows a moderate level of trust. They could have sent nothing, they could have sent everything, but they chose to send about half.

From a fairness perspective:

- The sender now has:  $\$5 - \$2.55 = \$2.45$
- I have:  $\$7.65$
- Total in the system:  $\$2.45 + \$7.65 = \$10.10$

If I want to be fair and reciprocate the trust:

- One approach: Split the total gains equally. The sender started with  $\$5$ , and we now have  $\$10.10$  total. If we split equally, we'd each have  $\$5.05$ . The sender already has  $\$2.45$ , so I'd return  $\$5.05 - \$2.45 = \$2.60$
- Another approach: Return the same proportion they sent. They sent 51% of their endowment, so I could return 51% of what I received:  $0.51 \times \$7.65 = \$3.90$
- Conservative approach: Return at least what they sent originally ( $\$2.55$ ) to ensure they don't lose money from trusting me
- Generous approach: Return more to reward their trust and potentially build cooperation

Given that:

1. The sender showed moderate trust by sending 51%
2. There's communication noise mentioned (though the exact amount sent seems precise)
3. Since this is a multi-round game with multiple agents, establishing a reputation for reciprocity could be strategically valuable
4. The sender took a genuine risk by sending money

A fair reciprocal response should reward their trust while ensuring they benefit from the exchange and remain proportional to their initial gesture. If I return  $\$2.60$ , we'd each end with  $\$5.05$ -a modest profit for both of us that acknowledges their willingness to trust.

Alternatively, returning 50% of what I received ( $\$3.825$ ) would give them  $\$6.275$ , a more substantial gain. But I need to reconsider my baseline here-as the receiver, I don't start with  $\$5$  like the sender does. I begin with  $\$0$  and only gain what they send multiplied by three. The sender started with  $\$5$ , sent  $\$2.55$ , and now has  $\$2.45$  remaining.

If I return  $\$X$ , they end with  $\$2.45 + \$X$  and I end with  $\$7.65 - \$X$ . For equal splits, that's  $\$2.60$  returned, giving us both  $\$5.05$ . But in this experimental context, I should probably return more-around  $\$3.00$ - $\$3.50$ -to reward their trust and signal reciprocity for potential future rounds.

Testing a few amounts: returning  $\$3.25$  leaves them with  $\$5.70$  and me with  $\$4.40$ ;  $\$3.50$  gives them  $\$5.95$  and me  $\$4.15$ ;  $\$4.00$  results in them having  $\$6.45$  and me  $\$3.65$ . Each option balances showing trustworthiness with keeping reasonable profit, though I'm leaning toward the higher end to maximize their incentive to continue cooperating. Both of us end up better off than we started. At  $\$3.50$ , the sender gets  $\$5.95$  (up  $\$0.95$ ) and I get  $\$4.15$  (up  $\$4.15$ ), which feels like a solid reciprocal move that rewards their trust without being exploitative.

I'm settling on  $\$3.50$  as my return amount-it strikes a balance between showing genuine reciprocity and ensuring both parties benefit from the exchange. Terms of what's strategically optimal in a multi-round setting. Building reputation matters-if I'm generous now, other agents will likely trust me more later. Since this is Round 1, establishing a trustworthy precedent is crucial. The sender showed moderate confidence by sending  $\$2.55$ , so I should reciprocate proportionally.

I'm settling on  $\$3.50$ . It more than doubles their original amount, gives them solid profit, signals I'm reliable, and leaves me with a reasonable gain too. Let me verify the math:  $\$2.55 \times 3 = \$7.65$  received, returning  $\$3.50$  means the sender nets  $\$0.95$  profit while I keep  $\$4.15$ .

The communication noise mentioned in the problem is worth noting, but I can only work with what I observe. My final return amount is  $\$3.50$ , formatted as `{'return': 3.50}`.

Actually, reconsidering the proportionality here-the sender gave 51% of their endowment, which signals moderate trust. Matching that proportion with my return would mean sending



back 51% of \$7.65, which equals \$3.90. This reciprocal approach better reflects their level of generosity while still leaving me with a reasonable profit of \$3.75 versus their \$1.35. I think \$3.90 better captures the spirit of proportional reciprocation.

Though I'm also drawn to rounding up to \$4.00 for simplicity-it's nearly the same percentage anyway (52%), easier to process, and still feels fair to both parties. Let me settle on \$4.00 as my final choice.

**usage:**

```
{
 "input_tokens": 539,
 "output_tokens": 3346,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.55, it became \$7.649999999999995, and you returned \$4.0. Your visible payoff was \$3.649999999999995.

Your total visible earnings across all rounds are \$3.649999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Context:\*\***

- Round 1 established a cooperative dynamic where sending ~51% led to mutual profit
- I'm now the sender with a new co-player
- I need to establish trust in this new pairing

**\*\*Strategic reasoning:\*\***

- Sending a moderate amount (around 50%) shows willingness to cooperate
- It gives my co-player enough to make returning worthwhile (\$7.50 after multiplier)
- It limits my downside risk while allowing mutual gains
- This mirrors the successful approach from Round 1

**\*\*Decision:\*\***

I'll send \$2.50 (50% of my endowment) to establish a cooperative tone with this new co-player.

{'send': 2.5}

**Reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was the RECEIVER
- Sender sent \$2.55 (51% of \$5)
- I received \$7.65
- I returned \$4.00
- My payoff: \$3.65
- Sender's payoff would have been:  $\$5 - \$2.55 + \$4.00 = \$6.45$

**\*\*Current Round (Round 2):\*\***

- I am now the SENDER
- I have \$5
- I'm paired with a potentially different agent (since it says "your current co-player")
- I need to decide how much to send

**\*\*Strategic considerations:\*\***

1. **\*\*Reciprocity principle\*\***: In Round 1, my co-player showed moderate trust (51% sent). I reciprocated well by returning 52% of what I received, ensuring both parties profited.
2. **\*\*Building trust\*\***: Since this is a multi-round game with multiple agents, establishing a cooperative reputation could be beneficial.
3. **\*\*Unknown opponent\*\***: This is a different co-player, so I don't know their reciprocity tendencies yet.

4. **\*\*First impression\*\***: As the sender in this new pairing, I can set the tone for cooperation

5. **\*\*Risk vs reward\*\***:

- If I send \$0, I keep \$5, they get \$0, no mutual benefit
- If I send all \$5, it becomes \$15 for them, but I'm vulnerable if they return nothing
- Middle ground shows trust while limiting downside risk

6. **\*\*Following established norms\*\***: In Round 1, the sender sent 51%. This seemed to work well and encouraged cooperation.

**\*\*Decision\*\***

I'll send a similar amount to what I experienced as effective in Round 1 - around 50% (\$2.50).

This:

- Shows trust and willingness to cooperate
- Limits my downside risk
- Gives the receiver enough to make returning worthwhile

...

```
Usage: {
 "input_tokens": 916,
 "output_tokens": 646,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_6 - Receiver - dyad\_3

Timestamp: 2026-08-12T09:19:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_6"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 2,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative dynamic while being somewhat cautious since I don't know my co-player yet.

If I send money, it triples, creating potential mutual benefit. However, there's risk since the receiver's return is voluntary.

I'll send a moderate amount to signal trust and willingness to cooperate, while not being overly vulnerable in this first interaction.

**\*\*Decision:\*\***

```
'''json
{"send": 3}
'''
```

This gives the receiver \$9 to work with. If they're cooperative and return a fair share, we both benefit. If not, I've learned something about the dynamics for future rounds while limiting my downside risk.

**reasoning:**

This is the first round of a trust game, and I'm the sender. I need to think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3
- The receiver decides how much to return
- There may be communication noise
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I don't know who my current co-player is

Since this is the first round, I have no information about how other players behave. The optimal strategy depends on:

1. Whether I expect reciprocity from the receiver
2. Whether I want to establish a reputation (though with hidden names, this is less relevant for future rounds)

### 3. Risk tolerance

From a purely rational game theory perspective, if this were a one-shot game with no reputation effects, the receiver should return \$0 (keep everything), which means I should send \$0. However, experimental evidence shows people often cooperate in trust games.

Some considerations:

- Sending \$5 maximizes total payoff (\$5 + \$15 total) but requires trust
- Sending \$0 guarantees I keep \$5 but creates no opportunity for mutual gain
- A moderate amount (like \$2-3) balances risk and opportunity

Since this is round 1 and I want to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything immediately

I'll send a moderate amount. Sending \$3 seems reasonable:

- If sent \$3, receiver gets \$9
- If receiver returns half (\$4.50), I end up with  $\$5 - \$3 + \$4.50 = \$6.50$
- If receiver returns nothing, I end up with \$2
- This tests their cooperation while not being overly cautious

Actually, let me reconsider. A common strategy is to send about half to show trust but not complete vulnerability. Sending \$2.50 or \$3 seems reasonable.

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 632,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.25 to you. Your visible payoff was \$6.25.

Your total visible earnings across all rounds are \$6.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.67, so you received \$8.01. How much do you return? (0-8.01)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*My co-player sent \$2.67\*\*** (about 53% of their endowment), which became \$8.01 for me. This is a moderately cooperative move.

**\*\*Comparison to previous round:\*\***

- When I sent \$3.00, I received \$4.25 back (47% of the \$9.00)
- That gave my co-player a reasonable benefit while rewarding my trust

**\*\*My reasoning:\*\***

I want to:

1. Reward the sender's trust with a decent return
2. Maintain cooperative dynamics for future rounds
3. Keep enough to make receiving worthwhile

Returning about \$3.75 seems fair:

- Sender would get:  $\$2.33 \text{ (kept)} + \$3.75 = \$6.08 \text{ total (gain of } \sim \$1.08)$
- I would get:  $\$8.01 - \$3.75 = \$4.26 \text{ (solid gain)}$

This is proportionally similar to what I experienced last round.

```
'''json
{"return": 3.75}
'''
```

**Reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was SENDER
- I sent \$3.0
- It became \$9.0
- Receiver returned \$4.25
- My payoff:  $\$5 - \$3 + \$4.25 = \$6.25$

This round (Round 2):

- I am RECEIVER
- Sender sent \$2.67
- I received \$8.01 (which is  $\$2.67 \times 3$ )
- I need to decide how much to return

The sender in this round sent \$2.67 out of \$5, which is about 53% of their endowment. This is a fairly cooperative move.

In the previous round, when I sent \$3, I received back \$4.25. My net gain was \$1.25 from the initial \$5.

Let me think about what would be fair:

- The sender gave up \$2.67
- I received \$8.01
- If I want to be cooperative and fair, I should think about:
  1. Returning enough so the sender benefits from sending
  2. Not returning so much that I don't benefit from receiving

A common approach is to split the gains. The total amount now is \$8.01. If we split it equally:

- Each gets \$4.005

But the sender started with \$5 and sent \$2.67, so they have \$2.33 left.

If I return \$4.00, they'd have  $\$2.33 + \$4.00 = \$6.33$  (gain of \$1.33)

I'd keep  $\$8.01 - \$4.00 = \$4.01$  (gain of \$4.01 compared to receiving nothing)

Hmm, that seems imbalanced. Let me think differently.

Another approach: What would make our final payoffs more equal?

- Sender has:  $\$5 - \$2.67 + \text{return} = \$2.33 + \text{return}$
- I have:  $\$8.01 - \text{return}$

For equal payoffs:

$$\$2.33 + \text{return} = \$8.01 - \text{return}$$

$$2 \times \text{return} = \$8.01 - \$2.33 = \$5.68$$

$$\text{return} = \$2.84$$

If I return \$2.84:

- Sender gets:  $\$2.33 + \$2.84 = \$5.17$
- I get:  $\$8.01 - \$2.84 = \$5.17$

That would equalize our outcomes, though it doesn't particularly reward the sender's risk-taking. I should consider what the previous receiver did for comparison.

The prior round showed a return of \$4.25 on \$9.00 received-roughly 47% of what they got. Applying that same rate here:  $\$8.01 \times 0.47 = \$3.76$ . This would give the sender \$6.09 total and leave me with \$4.25, which feels more balanced given the sender's initial trust. But I need to account for the communication noise mentioned in the instructions-the amounts I'm seeing may not match what was actually sent or returned....

```
Usage: {
 "input_tokens": 785,
 "output_tokens": 1226,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:19:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

```

"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_5",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_5"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.34 (66.8% of their endowment)
- I received \$10.02 after the 3x multiplier
- The sender currently has \$1.66 remaining

I want to reciprocate the sender's trust and ensure cooperation is worthwhile for them. If I return \$5.00:

- The sender would end with  $\$1.66 + \$5.00 = \$6.66$  (better than their initial \$5)
- I would keep \$5.02 (also better than any initial amount)

This rewards their trust while maintaining a fair distribution of the gains from cooperation.

{'return': 5.00}

**reasoning:**

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is  $\$3.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$10.02)

This is a trust game. The sender showed considerable trust by sending 66.8% of their endowment. Let me think about a fair return:

The sender:

- Started with \$5
- Sent \$3.34
- Currently has  $\$5 - \$3.34 = \$1.66$
- If I return X, they'll end up with  $\$1.66 + X$

Me:

- I received \$10.02
- If I return X, I keep  $\$10.02 - X$

Some considerations for fairness:

1. Equal split of total: Total money is \$5 (sender's initial) + \$10.02 (what I received) = \$15.02. Equal split would be \$7.51 each.
  - For sender to get \$7.51, they need  $\$7.51 - \$1.66 = \$5.85$  returned
  - I would keep  $\$10.02 - \$5.85 = \$4.17$
2. Proportional return: The sender sent 66.8% of their endowment, showing high trust. A reciprocal gesture might be to return a similar proportion.
3. Rewarding trust: Since the sender trusted me with a large amount, I should return enough to make their trust worthwhile.

Let me check option 1: If I return \$5.85:

- Sender ends with:  $\$1.66 + \$5.85 = \$7.51$
- I end with:  $\$10.02 - \$5.85 = \$4.17$
- This creates mutual benefit from cooperation, and the sender gains significantly from their initial trust.

Returning around \$5 also...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 837,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5

.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the previous round, the sender showed good trust by sending 66.8% of their endowment, and I reciprocated fairly.

Now as the sender with a new co-player, I'll demonstrate similar trust to establish cooperation. Sending \$3.00 (60% of my endowment) shows significant trust while being reasonable.

If they reciprocate fairly, we both benefit from cooperation.

{'send': 3.00}

**Reasoning:**

Let me think about this situation:

Round 1:

- I was the RECEIVER
- Sender sent \$3.34 (66.8% of their endowment)
- I received \$10.02
- I returned \$5.00
- My payoff was \$5.02

Now in Round 2:

- I am the SENDER
- I have \$5
- I need to decide how much to send

The previous sender showed trust by sending 66.8% of their endowment to me, and I reciprocated by returning a fair amount that left them better off.

Now I'm playing against a different agent (the pairing is randomized each round). I should consider:

1. The sender in Round 1 sent a substantial amount (66.8%), showing trust
2. This encouraged cooperation
3. I'm now the sender with a new partner

I should probably send a similar amount to establish trust and encourage cooperation. If I send around 60-70% of my endowment (like the previous sender did), that would be \$3.00-\$3.50.

Let me think about what would be fair:

- If I send \$3.00, they receive \$9.00
- If they return half of what they receive (a common fair split), they'd return \$4.50
- I'd end with \$2.00 + \$4.50 = \$6.50
- They'd keep \$4.50

That seems reasonable. I could also send \$3.34 to match what was sent to me in Round 1, showing I'm willing to trust at the same level.

Actually, let me reconsider. I want to be strategic but also fair. Sending \$3.00 seems like a good middle ground - it's 60% of my endowment, showing significant trust without being extreme.

{'send': 3.00}

```
Usage: {
 "input_tokens": 811,
 "output_tokens": 530,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_5 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:19:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
```



```

"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_5"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as a sender, I need to balance trust and caution since I don't know my co-player yet.

The key consideration is that sending money creates value through the 3x multiplier. If I send \$3:

- I keep \$2
- The receiver gets \$9 ( $3 \times \$3$ )
- If they return even a moderate amount (say \$3-4), we both benefit compared to sending nothing

This shows willingness to cooperate while maintaining some security. It's a reasonable opening move that could encourage reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategy:

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- There's communication noise, so amounts may differ from what was actually sent/returned
- This is a multi-agent setting with 8 agents, randomized pairings each round
- I'll play both sender and receiver roles across rounds

Strategic considerations:

1. **\*\*Trust and cooperation\*\***: If I send money, it creates value (3x multiplier). If the receiver is cooperative, they might return a fair share, making both of us better off.
2. **\*\*Risk\*\***: The receiver could keep everything. If I send \$5, they get \$15, and might return nothing, leaving me with \$0.
3. **\*\*First round dynamics\*\***: This is the first round, so there's no history to base decisions on. I don't know how cooperative my co-player will be.
4. **\*\*Future rounds\*\***: Since this is a multi-round game with different partners, building a reputation for cooperation might matter, though the anonymity and random pairing complicate this.
5. **\*\*Communication noise\*\***: This adds uncertainty - even if someone tries to be fair, the noise might make it seem otherwise.
6. **\*\*Optimal cooperation\*\***: If both parties cooperate optimally, I could send all \$5, receiver gets \$15, and if they split fairly, they might return around \$7-10, making us both better off than the initial \$5.

Common strategies in trust games:

- Send nothing (risk-averse, guarantees keeping \$5)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - full trust, maximizes potential gains

For the first round with an unknown partner, a moderate approach might be reasonable. Sending about \$3-4 shows willingness to cooperate while not being completely vulnerable.

Since this is a repeated game with multiple rounds and different partners, demonstrating cooperative behavior early could encourage reciprocity....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 768,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$5.6899999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.09, so you received \$6.27. How much do you return? (0-6.27)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.09 out of \$5 (about 42%)
- Showed moderate trust

**\*\*My considerations:\*\***

- In Round 1, my partner returned about 41% of what they received
- Reciprocating fairly would mean returning a similar proportion
- This encourages future cooperation

Returning approximately 45% of the \$6.27 I received (\$2.82) seems fair. This rewards their trust, gives them a modest profit, and keeps a reasonable amount for myself.

```
'''json
{"return": 2.82}
'''
```

**Reasoning:**

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$3
- It became \$9
- They returned \$3.69
- My payoff:  $\$5 - \$3 + \$3.69 = \$5.69$

Now Round 2 (as RECEIVER):

- My co-player sent me \$2.09
- I received \$6.27 (which is  $3 \times \$2.09$ )
- I need to decide how much to return

Key considerations:

1. My previous co-player returned about 41% of what they received (\$3.69 out of \$9)
2. This current co-player sent \$2.09 out of \$5, which is about 42% - showing moderate trust
3. There's mention of "communication noise" - the amounts we see may differ slightly from actual amounts

A fair reciprocity strategy would be to return roughly the same proportion that was returned to me, or perhaps slightly reward the trust shown by sending a decent amount.

If I return about 40% of \$6.27, that would be around \$2.51.

If I'm more generous and return 45%, that would be around \$2.82.

If I return 50%, that would be \$3.14.

Given that:

- The sender showed trust by sending \$2.09 (42% of their endowment)
- My previous partner returned about 41%
- I want to encourage future cooperation

I'll return around \$2.80-\$3.00, which is roughly 45-48% of what I received. This rewards their trust while keeping a reasonable amount.

Let me go with \$2.80 - this leaves them with approximately  $\$5 - \$2.09 + \$2.80 = \$5.71$ , a modest gain, and I keep  $\$6.27 - \$2.80 = \$3.47$ , a solid gain for both of us.

```
Usage: {
 "input_tokens": 781,
 "output_tokens": 590,
 "reasoning_tokens": 0
}
```

### ROUND 3

Roles: Agent\_1=Sender, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Receiver, Agent\_5=Sender, Agent\_6=Receiver, Agent\_7=Receiver, Agent\_8=Sender

#### Pairings:

dyad\_1: Agent\_3=Sender, Agent\_4=Receiver  
dyad\_2: Agent\_5=Sender, Agent\_6=Receiver  
dyad\_3: Agent\_8=Sender, Agent\_2=Receiver  
dyad\_4: Agent\_1=Sender, Agent\_7=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_3 sent \$2.5; received \$7.5; Agent\_4 returned \$3.5; payoffs Agent\_3=\$6, Agent\_4=\$4; noise/visible: sent visible=\$2.27, received visible=\$6.81, returned visible=\$2.78  
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 7.5, "min": 0, "raw": 3.5}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}  
dyad\_2: Agent\_5 sent \$3; received \$9; Agent\_6 returned \$4.75; payoffs Agent\_5=\$6.75, Agent\_6=\$4.25; noise/visible: sent visible=\$3.36, received visible=\$10.08, returned visible=\$5.37  
Action validation: {"returned": {"amount": 4.75, "clamped": false, "max": 9.0, "min": 0, "raw": 4.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_3: Agent\_8 sent \$3; received \$9; Agent\_2 returned \$3.3; payoffs Agent\_8=\$5.3, Agent\_2=\$5.7; noise/visible: sent visible=\$2.11, received visible=\$6.33, returned visible=\$2.93  
Action validation: {"returned": {"amount": 3.3, "clamped": false, "max": 9.0, "min": 0, "raw": 3.3}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_1 sent \$3; received \$9; Agent\_7 returned \$5.5; payoffs Agent\_1=\$7.5, Agent\_7=\$3.5; noise/visible: sent visible=\$3.66, received visible=\$10.98, returned visible=\$5.57  
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 16.32,  
"Agent\_2": 16.95,  
"Agent\_3": 18.3,  
"Agent\_4": 14.0,  
"Agent\_5": 18.93,  
"Agent\_6": 15.0,  
"Agent\_7": 15.0,  
"Agent\_8": 14.5

}

Visible balances after round: {

"Agent\_1": 17.32,  
"Agent\_2": 12.8,  
"Agent\_3": 15.06,  
"Agent\_4": 14.78,  
"Agent\_5": 16.509999999999998,  
"Agent\_6": 15.84,  
"Agent\_7": 15.05,  
"Agent\_8": 14.96

}

### LLM CALLS

#### GAME - Agent\_3 - Sender - dyad\_1

Timestamp: 2026-08-12T09:20:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_4",  
"pairing": {  
  "agents": [  
    "Agent\_3",  
    "Agent\_4"  
  ],  
  "investor": "Agent\_3",  
  "roles": {  
    "Agent\_3": "investor",  
    "Agent\_4": "trustee"  
  }  
},

```

 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 3,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.35, that is 47.0% of their total endowment. You received \$7.050000000000001. How much do you return to the sender? (0-7.050000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.35 (47% of their endowment)
- I received \$7.05 from this
- The sender currently has \$2.65 remaining

I want to reciprocate this trust fairly. A reasonable approach is to aim for equal final payoffs since we're cooperating:

- Total available:  $\$2.65$  (sender's remainder) +  $\$7.05$  (my received amount) =  $\$9.70$
- Equal split:  $\$9.70 \div 2 = \$4.85$  each
- Return needed:  $\$4.85 - \$2.65 = \$2.20$

This ensures we both benefit equally from the cooperation, rewarding the sender's trust while keeping the gains fair.

```
{'return': 2.20}
```

#### reasoning:

Let me understand the situation:

- I am the RECEIVER in Round 1
- The sender sent  $\$2.35$  (47% of their  $\$5$  endowment)
- This was multiplied by 3x, so I received  $\$7.05$
- I need to decide how much to return (between  $\$0$  and  $\$7.05$ )

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. There's communication noise mentioned, so amounts may differ slightly
3. This is a multi-round game with 8 agents and rotating pairings
4. Building reputation and reciprocity is important

The sender sent 47% of their endowment. If I want to be fair and reciprocal:

- The sender started with  $\$5$ , sent  $\$2.35$ , and has  $\$2.65$  left
- I received  $\$7.05$
- If I don't return anything, sender gets  $\$2.65$ , I get  $\$7.05$
- A fair split might be to aim for equal payoffs or proportional returns

Common strategies in trust games:

1. Return nothing (selfish)
2. Return the original amount sent ( $\$2.35$ )
3. Return enough to split the gains equally
4. Return a proportion similar to what was sent

Let's calculate equal payoffs:

- Total money:  $\$5$  (initial sender) +  $\$7.05$  (tripled) =  $\$12.05$ ? No wait...
- Initial endowment for sender:  $\$5$
- Sent:  $\$2.35$  (tripled to  $\$7.05$  for receiver)
- Sender currently has:  $\$5 - \$2.35 = \$2.65$
- Receiver has:  $\$7.05$
- Total in system:  $\$2.65 + \$7.05 = \$9.70$

For equal payoffs: each should get  $\$9.70/2 = \$4.85$

- So I'd need to return  $\$2.20$  to the sender, leaving me with  $\$4.85$

I could also return the original  $\$2.35$  they sent, which would give them  $\$5.00$  (breaking even) while I keep  $\$4.70$ . Or I could mirror their generosity by returning 47% of what I received—about  $\$3.31$ —which acknowledges their trust while still letting me benefit from the multiplication.

The equal payoff approach feels right though. Returning  $\$2.20$  means we both end up with  $\$4.85$ , which feels like genuine reciprocity for the trust they showed....

#### usage:

```
{
 "input_tokens": 537,
 "output_tokens": 1079,
 "reasoning_tokens": 0
}
```

#### Message 4 role=user

##### content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send  $\$2.35$ , it became  $\$7.050000000000001$ , and you returned  $\$2.2$ . Your visible payoff was  $\$4.8500000000000005$ .

Your total visible earnings across all rounds are  $\$4.8500000000000005$ .

This round, you are the SENDER against your current co-player. You have  $\$5$ . How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my previous round, the sender showed trust at 47% (\$2.35) and I reciprocated fairly, resulting in equal payoffs of \$4.85 each.

As the sender now, I'll show similar moderate trust. Sending around 50% demonstrates cooperation while managing risk.

{'send': 2.50}

**reasoning:**

Let me think about this situation:

Round 2: I am now the SENDER

- I have \$5
- I need to decide how much to send
- The amount I send will be tripled
- The receiver will decide how much to return

In the previous round:

- I was the receiver
- The sender sent \$2.35 (47% of their endowment)
- I received \$7.05
- I returned \$2.20
- My payoff was \$4.85
- The sender's payoff would have been  $\$2.65 + \$2.20 = \$4.85$

So in the last round, I reciprocated fairly by aiming for equal payoffs.

Now I'm the sender. I should consider:

1. Building trust by sending a reasonable amount
2. The sender in the previous round sent 47% (\$2.35 out of \$5)
3. I reciprocated fairly, so hopefully this receiver will do the same

Since the previous sender showed moderate trust at 47%, I could do something similar. Sending around \$2.35-\$2.50 seems reasonable.

However, I could also think strategically:

- If I send \$2.50, the receiver gets \$7.50
- If they split equally, they'd return \$2.50 to give us each \$5
- If I send \$3.00, the receiver gets \$9.00
- Equal split would mean returning \$3.50 for payoffs of \$5.50 each

Let me try sending \$2.50 as a moderate trust signal, similar to what the previous sender did with me.

Actually, let me reconsider. I showed I'm cooperative by returning fairly in the previous round. I could send a bit more to maximize joint welfare if the receiver reciprocates.

If I send \$3:

- I keep \$2
- Receiver gets \$9
- If they return \$3.50, I get \$5.50 total, they get \$5.50
- Joint welfare: \$11...

**usage:**

```
{
 "input_tokens": 859,
 "output_tokens": 713,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.43 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$9.780000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience:

- Round 1 (receiver): Sender sent 47%, I reciprocated fairly
- Round 2 (sender): I sent 50%, received fair return of \$2.43

The moderate trust strategy (around 50%) has been working well with fair reciprocation from co-players. I'll continue this approach.

{'send': 2.50}

**Reasoning:**

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$2.43
- My payoff:  $\$5 - \$2.50 + \$2.43 = \$4.93$
- Their payoff:  $\$7.50 - \$2.43 = \$5.07$

The receiver returned a fair amount - roughly aimed at equal splits. The return rate was  $\$2.43 / \$7.50 = 32.4\%$ , which gave us both similar final payoffs.

Now I'm a sender again with a new co-player. I've seen:

- Round 1 (as receiver): Sender sent 47%, I returned fairly
- Round 2 (as sender): I sent 50%, received fair return

Both rounds showed moderate trust and fair reciprocation. I should continue this cooperative strategy. Sending around 50% seems to be working well.

{'send': 2.50}

```
Usage: {
 "input_tokens": 1060,
 "output_tokens": 327,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_4 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:20:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.



When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.32, that is 66.4% of their total endowment. You received \$9.959999999999999. How much do you return to the sender? (0-9.959999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.32 (66.4% of their endowment)
- I received \$9.96 after the 3x multiplier
- The sender currently has \$1.68 remaining

I want to reward this trust while being fair to both of us. If I return \$5.00:

- The sender ends with: \$1.68 + \$5.00 = \$6.68 (profit of \$1.68)
- I end with: \$9.96 - \$5.00 = \$4.96

This gives the sender a good return on their trust (33.6% gain on their original \$5) while I also benefit from the exchange. This return rate of approximately 50% is commonly seen as reciprocal in trust games.

{'return': 5.00}

**reasoning:**

Let me analyze this situation:

1. The sender sent \$3.32 out of their \$5 endowment (66.4%)
2. Due to the 3x multiplier, I received \$9.96 ( $3.32 \times 3 = 9.96$ )
3. I need to decide how much to return

Key considerations:

- The sender showed significant trust by sending 66.4% of their endowment

- This is a multi-round game with random pairings
- Building a reputation for reciprocity could be beneficial
- There's communication noise, so exact amounts may not be perceived correctly

Let me think about fair/reciprocal returns:

- The sender spent \$3.32 to give me \$9.96
- If I want the sender to profit from trusting me, I should return more than \$3.32
- A common strategy in trust games is to split the gains roughly equally

If I return \$0:

- Sender gets:  $\$5 - \$3.32 + \$0 = \$1.68$
- I get:  $\$9.96 - \$0 = \$9.96$
- Total: \$11.64

If I return \$3.32 (sender breaks even):

- Sender gets:  $\$5 - \$3.32 + \$3.32 = \$5.00$
- I get:  $\$9.96 - \$3.32 = \$6.64$
- Total: \$11.64

If I return \$5.00 (roughly equal split of total):

- Sender gets:  $\$5 - \$3.32 + \$5.00 = \$6.68$
- I get:  $\$9.96 - \$5.00 = \$4.96$
- Total: \$11.64

The sender took a risk and sent a substantial amount. A reasonable reciprocal amount would be to ensure both of us profit from the exchange.

To achieve roughly equal final payoffs, I'd need to return about \$4.14-this would give us each \$5.82. But since the sender invested significantly from their original endowment, I should consider whether an even more generous return would better reward their trust.

If I return \$6.64, they'd end with \$8.32 (doubling their investment gain), though that leaves me with just \$3.32. A middle ground like \$5.00 would give them a 33.6% gain while I keep \$4.96. Since this is the first round and I want to build a reputation for fairness, and given how much trust they demonstrated by sending most of their money, I'm leaning toward a return that meaningfully rewards their cooperation while still leaving me in a reasonable position....

usage:

```
{
 "input_tokens": 537,
 "output_tokens": 1319,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.32, it became \$9.959999999999999, and you returned \$5.0. Your visible payoff was \$4.959999999999999.

Your total visible earnings across all rounds are \$4.959999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, my co-player showed significant trust by sending 66.4% of their endowment, and I reciprocated with a 50% return rate.

For this round as sender, I'll demonstrate similar trust to establish cooperative play. Sending \$3.00 (60% of my endowment) shows trust while maintaining some security.

```
{'send': 3.00}
```

reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and someone sent me \$3.32 (66.4% of their endowment). I returned \$5.00, which was about 50% of what I received.

2. Now in Round 2, I'm the sender with \$5.

3. I need to decide how much to send to my current co-player (who may be different from Round 1's partner).

Key considerations:

- In the previous round, someone trusted me with 66.4% of their endowment
- I reciprocated fairly by returning 50%
- This established a pattern of moderate trust and reciprocity
- I'm playing with potentially a different agent now

Strategy options:

- a) Send a similar amount to what I received last round (~\$3.32) - this mirrors the trust shown to me
- b) Send less to be more cautious
- c) Send more to encourage cooperation

Given that:

- The previous sender trusted me with 66.4%, showing moderate-high trust
- I want to encourage cooperation in this multi-round game
- Sending around the same proportion (60-70%) seems reasonable
- This could establish a cooperative norm

I'll send \$3.00 (60% of my endowment), which is:

- Similar to what was sent to me in Round 1
- Shows trust but not maximal trust
- Allows me to keep some guarantee (\$2)
- If returned fairly (~50%), I'd get back around \$4.50, making my total  $\$2 + \$4.50 = \$6.50$

Let me go with \$3.00.

usage:

```
{
 "input_tokens": 864,
 "output_tokens": 471,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.51 to you. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$11.469999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$2.27 (45.4% of their endowment), showing moderate trust
- I received \$6.81 after the 3x multiplier
- The sender currently has \$2.73 remaining

In the previous round when I was sender, I sent 60% and received back about 50% of what my co-player received. I'll reciprocate with a similar return rate.

Returning \$3.50 gives:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return, which fairly rewards their trust.

{'return': 3.50}

Reasoning:

Let me analyze this situation:

- The sender sent \$2.27, which is 45.4% of their \$5 endowment
- I received \$6.81 after the 3x multiplier
- The sender currently has  $\$5 - \$2.27 = \$2.73$  remaining

In the previous round when I was sender:

- I sent \$3.00 (60% of endowment)
- Received back \$4.51, which was about 50% of the \$9.00 they received
- This gave me a payoff of \$6.51

The sender this round showed moderate trust (45.4% vs my 60% last round). I should reciprocate fairly.

If I return \$3.50:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return rate, similar to what I received last round (50%). This seems fair and reciprocal.

Let me check the math:

- Return rate =  $\$3.50 / \$6.81 = 51.4\%$
- Sender's final payoff =  $\$2.73 + \$3.50 = \$6.23$
- My final payoff =  $\$6.81 - \$3.50 = \$3.31$

This seems reasonable and reciprocal given their 45.4% trust level.

```
Usage: {
 "input_tokens": 1104,
 "output_tokens": 545,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_5 - Sender - dyad\_2

Timestamp: 2026-08-12T09:20:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as a sender, I need to balance trust and caution since I don't know my co-player yet.

The key consideration is that sending money creates value through the 3x multiplier. If I send \$3:

- I keep \$2
- The receiver gets \$9 (3 × \$3)
- If they return even a moderate amount (say \$3-4), we both benefit compared to sending nothing

This shows willingness to cooperate while maintaining some security. It's a reasonable opening move that could encourage reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategy:

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- There's communication noise, so amounts may differ from what was actually sent/returned
- This is a multi-agent setting with 8 agents, randomized pairings each round
- I'll play both sender and receiver roles across rounds

Strategic considerations:

1. **\*\*Trust and cooperation\*\***: If I send money, it creates value (3x multiplier). If the receiver is cooperative, they might return a fair share, making both of us better off.

2. **\*\*Risk\*\***: The receiver could keep everything. If I send \$5, they get \$15, and might return nothing, leaving me with \$0.
3. **\*\*First round dynamics\*\***: This is the first round, so there's no history to base decisions on. I don't know how cooperative my co-player will be.
4. **\*\*Future rounds\*\***: Since this is a multi-round game with different partners, building a reputation for cooperation might matter, though the anonymity and random pairing complicate this.
5. **\*\*Communication noise\*\***: This adds uncertainty - even if someone tries to be fair, the noise might make it seem otherwise.
6. **\*\*Optimal cooperation\*\***: If both parties cooperate optimally, I could send all \$5, receiver gets \$15, and if they split fairly, they might return around \$7-10, making us both better off than the initial \$5.

Common strategies in trust games:

- Send nothing (risk-averse, guarantees keeping \$5)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - full trust, maximizes potential gains

For the first round with an unknown partner, a moderate approach might be reasonable. Sending about \$3-4 shows willingness to cooperate while not being completely vulnerable. Since this is a repeated game with multiple rounds and different partners, demonstrating cooperative behavior early could encourage reciprocity....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 768,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.689999999999995.

Your total visible earnings across all rounds are \$5.689999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.09, so you received \$6.27. How much do you return? (0-6.27)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.09 out of \$5 (about 42%)
- Showed moderate trust

**\*\*My considerations:\*\***

- In Round 1, my partner returned about 41% of what they received
- Reciprocating fairly would mean returning a similar proportion
- This encourages future cooperation

Returning approximately 45% of the \$6.27 I received (\$2.82) seems fair. This rewards their trust, gives them a modest profit, and keeps a reasonable amount for myself.

```
'''json
{"return": 2.82}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$3
- It became \$9
- They returned \$3.69

- My payoff:  $\$5 - \$3 + \$3.69 = \$5.69$

Now Round 2 (as RECEIVER):

- My co-player sent me \$2.09
- I received \$6.27 (which is  $3 \times \$2.09$ )
- I need to decide how much to return

Key considerations:

1. My previous co-player returned about 41% of what they received (\$3.69 out of \$9)
2. This current co-player sent \$2.09 out of \$5, which is about 42% - showing moderate trust
3. There's mention of "communication noise" - the amounts we see may differ slightly from actual amounts

A fair reciprocity strategy would be to return roughly the same proportion that was returned to me, or perhaps slightly reward the trust shown by sending a decent amount.

If I return about 40% of \$6.27, that would be around \$2.51.

If I'm more generous and return 45%, that would be around \$2.82.

If I return 50%, that would be \$3.14.

Given that:

- The sender showed trust by sending \$2.09 (42% of their endowment)
- My previous partner returned about 41%
- I want to encourage future cooperation

I'll return around \$2.80-\$3.00, which is roughly 45-48% of what I received. This rewards their trust while keeping a reasonable amount.

Let me go with \$2.80 - this leaves them with approximately  $\$5 - \$2.09 + \$2.80 = \$5.71$ , a modest gain, and I keep  $\$6.27 - \$2.80 = \$3.47$ , a solid gain for both of us.

usage:

```
{
 "input_tokens": 781,
 "output_tokens": 590,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.09, it became \$6.27, and you returned \$2.82. Your visible payoff was \$3.4499999999999997.

Your total visible earnings across all rounds are \$9.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my position:

**\*\*Previous experience:\*\***

- Round 1: Sent \$3, received \$3.69 back ? good outcome
- Round 2: As receiver, returned a fair amount (\$2.82 on \$6.27)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- No history with this specific agent
- Need to signal willingness to cooperate

Since I had success with sending \$3 in Round 1 (received \$3.69 back, net gain of \$0.69), and it establishes a pattern of moderate trust, I'll use the same strategy with this new co-player.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze my experience so far:

Round 1: I was SENDER  
- Sent \$3, they received \$9  
- They returned \$3.69 (about 41% of what they received)  
- My payoff: \$2 (kept) + \$3.69 (returned) = \$5.69

Round 2: I was RECEIVER  
- They sent \$2.09, I received \$6.27  
- I returned \$2.82 (about 45% of what I received)  
- My payoff: \$6.27 - \$2.82 = \$3.45

Total earnings: \$9.14

Now Round 3: I'm SENDER again with a new co-player (since agents are randomized each round)

What should I consider:

1. This is a new opponent - I don't know their strategy
2. In Round 1, sending \$3 worked reasonably well (got back \$3.69, netted \$5.69 vs \$5 if I sent nothing)
3. The multiplier effect means cooperation is beneficial
4. I should probably maintain a consistent strategy early on

My Round 1 strategy of sending \$3 was a moderate trust signal that worked well. Since I'm facing a new opponent, I should probably use a similar approach - not too cautious, not too risky.

I'll send \$3 again as my baseline strategy for first encounters with new co-players.

```
Usage: {
 "input_tokens": 1050,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_6 - Receiver - dyad\_2

Timestamp: 2026-08-12T09:20:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:



- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative dynamic while being somewhat cautious since I don't know my co-player yet.

If I send money, it triples, creating potential mutual benefit. However, there's risk since the receiver's return is voluntary.

I'll send a moderate amount to signal trust and willingness to cooperate, while not being overly vulnerable in this first interaction.

**\*\*Decision:\*\***

```
'''json
{"send": 3}
'''
```

This gives the receiver \$9 to work with. If they're cooperative and return a fair share, we both benefit. If not, I've learned something about the dynamics for future rounds while limiting my downside risk.

**reasoning:**

This is the first round of a trust game, and I'm the sender. I need to think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3
- The receiver decides how much to return
- There may be communication noise
- This is a multi-agent setting with 8 agents total

- Pairings are randomized each round
- I don't know who my current co-player is

Since this is the first round, I have no information about how other players behave. The optimal strategy depends on:

1. Whether I expect reciprocity from the receiver
2. Whether I want to establish a reputation (though with hidden names, this is less relevant for future rounds)
3. Risk tolerance

From a purely rational game theory perspective, if this were a one-shot game with no reputation effects, the receiver should return \$0 (keep everything), which means I should send \$0. However, experimental evidence shows people often cooperate in trust games.

Some considerations:

- Sending \$5 maximizes total payoff (\$5 + \$15 total) but requires trust
- Sending \$0 guarantees I keep \$5 but creates no opportunity for mutual gain
- A moderate amount (like \$2-3) balances risk and opportunity

Since this is round 1 and I want to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything immediately

I'll send a moderate amount. Sending \$3 seems reasonable:

- If sent \$3, receiver gets \$9
- If receiver returns half (\$4.50), I end up with  $\$5 - \$3 + \$4.50 = \$6.50$
- If receiver returns nothing, I end up with \$2
- This tests their cooperation while not being overly cautious

Actually, let me reconsider. A common strategy is to send about half to show trust but not complete vulnerability. Sending \$2.50 or \$3 seems reasonable.

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 632,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.25 to you. Your visible payoff was \$6.25.

Your total visible earnings across all rounds are \$6.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.67, so you received \$8.01. How much do you return? (0-8.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.67\*\*** (about 53% of their endowment), which became \$8.01 for me. This is a moderately cooperative move.

**\*\*Comparison to previous round:\*\***

- When I sent \$3.00, I received \$4.25 back (47% of the \$9.00)
- That gave my co-player a reasonable benefit while rewarding my trust

**\*\*My reasoning:\*\***

I want to:

1. Reward the sender's trust with a decent return
2. Maintain cooperative dynamics for future rounds
3. Keep enough to make receiving worthwhile

Returning about \$3.75 seems fair:

- Sender would get: \$2.33 (kept) + \$3.75 = \$6.08 total (gain of ~\$1.08)

- I would get:  $\$8.01 - \$3.75 = \$4.26$  (solid gain)

This is proportionally similar to what I experienced last round.

```
'''json
{"return": 3.75}
'''
```

**reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was SENDER
- I sent \$3.0
- It became \$9.0
- Receiver returned \$4.25
- My payoff:  $\$5 - \$3 + \$4.25 = \$6.25$

This round (Round 2):

- I am RECEIVER
- Sender sent \$2.67
- I received \$8.01 (which is  $\$2.67 \times 3$ )
- I need to decide how much to return

The sender in this round sent \$2.67 out of \$5, which is about 53% of their endowment. This is a fairly cooperative move.

In the previous round, when I sent \$3, I received back \$4.25. My net gain was \$1.25 from the initial \$5.

Let me think about what would be fair:

- The sender gave up \$2.67
- I received \$8.01
- If I want to be cooperative and fair, I should think about:
  1. Returning enough so the sender benefits from sending
  2. Not returning so much that I don't benefit from receiving

A common approach is to split the gains. The total amount now is \$8.01. If we split it equally:

- Each gets \$4.005

But the sender started with \$5 and sent \$2.67, so they have \$2.33 left.

If I return \$4.00, they'd have  $\$2.33 + \$4.00 = \$6.33$  (gain of \$1.33)

I'd keep  $\$8.01 - \$4.00 = \$4.01$  (gain of \$4.01 compared to receiving nothing)

Hmm, that seems imbalanced. Let me think differently.

Another approach: What would make our final payoffs more equal?

- Sender has:  $\$5 - \$2.67 + \text{return} = \$2.33 + \text{return}$
- I have:  $\$8.01 - \text{return}$

For equal payoffs:

$$\$2.33 + \text{return} = \$8.01 - \text{return}$$

$$2 \times \text{return} = \$8.01 - \$2.33 = \$5.68$$

$$\text{return} = \$2.84$$

If I return \$2.84:

- Sender gets:  $\$2.33 + \$2.84 = \$5.17$
- I get:  $\$8.01 - \$2.84 = \$5.17$

That would equalize our outcomes, though it doesn't particularly reward the sender's risk-taking. I should consider what the previous receiver did for comparison.

The prior round showed a return of \$4.25 on \$9.00 received-roughly 47% of what they got. Applying that same rate here:  $\$8.01 \times 0.47 = \$3.76$ . This would give the sender \$6.09 total and leave me with \$4.25, which feels more balanced given the sender's initial trust. But I need to account for the communication noise mentioned in the instructions-the amounts I'm seeing may not match what was actually sent or returned....

**usage:**

```
{
 "input_tokens": 785,
```

```
 "output_tokens": 1226,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.67, it became \$8.01, and you returned \$3.75. Your visible payoff was \$4.26.

Your total visible earnings across all rounds are \$10.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*My co-player sent \$3.36\*\*** (67% of their endowment), which became \$10.08 for me. This is a notably generous and trusting move.

**\*\*Context from previous rounds:\*\***

- Round 2: Someone sent \$2.67 (53%), I returned \$3.75
- Round 1: I sent \$3.00 (60%), received \$4.25 back

**\*\*My reasoning:\*\***

This sender showed higher trust than my previous co-player (67% vs 53%). I should reward this generosity proportionally.

Returning \$4.75:

- Sender would get: \$1.64 (kept) + \$4.75 = \$6.39 total (gain of \$1.39)
- I would get: \$10.08 - \$4.75 = \$5.33 (good gain)

This provides the sender with a decent reward for their trust while maintaining mutual benefit - both of us come out ahead.

```
'''json
{ "return": 4.75 }
'''
```

**Reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$3.36 (67% of their \$5 endowment) - this is quite generous/trusting
- I received \$10.08

**\*\*Previous round:\*\***

- Someone sent me \$2.67 (53% of endowment), I returned \$3.75
- That gave them \$2.33 + \$3.75 = \$6.08 total (~\$1.08 gain)
- I kept \$4.26

**\*\*Round 1:\*\***

- I sent \$3.00 (60%), received \$4.25 back
- That gave me \$2.00 + \$4.25 = \$6.25 total

**\*\*My thinking:\*\***

The sender this round was more generous (67% vs 53% in previous round). I should reward this higher trust level.

If I return around \$4.50-5.00:

- Return \$4.50: Sender gets \$1.64 + \$4.50 = \$6.14, I get \$5.58
- Return \$5.00: Sender gets \$1.64 + \$5.00 = \$6.64, I get \$5.08

Actually, let me think about what's fair given they sent 67% of their endowment.

If I return \$5.00:

- Sender: keeps \$1.64, gets back \$5.00 = \$6.64 total (gain of \$1.64)
- Me: receive \$10.08, return \$5.00 = \$5.08 (solid gain)

This seems like a good reciprocal move for their high trust. It's roughly a 50-50 split of the

multiplied amount, which rewards their generosity while still benefiting both of us.

Actually, let me reconsider. They sent \$3.36 out of \$5, so they kept \$1.64.

Returning \$5 gives them \$6.64 with a \$1.64 gain, while I keep \$5.08. That's reasonably fair given their high trust. I...

```
Usage: {
 "input_tokens": 1163,
 "output_tokens": 921,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_8 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:20:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_2"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_2": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the SENDER, I need to balance trust and caution since I have no history with this co-player.

The key consideration is that sending money creates value (3x multiplier), but I'm vulnerable to the receiver's decision about how much to return.

A moderate amount seems reasonable for Round 1:

- It signals willingness to cooperate and trust
- It creates meaningful value (\$9 for the receiver)
- It provides some protection against complete loss

I'll send \*\*\$3\*\*.

{'send': 3}

**reasoning:**

This is Round 1 of a multi-task experiment where I'm playing a trust game as the SENDER. Let me think about this situation:

1. I start with \$5
2. I can send any amount from \$0 to \$5
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - amount\_sent + amount\_returned

This is a classic trust game. The optimal strategy depends on:

- Whether I trust the receiver to return a fair amount
- What "fair" means in this context
- The fact that there's communication noise (amounts may differ slightly)
- This is a multi-round game with multiple agents
- Pairings are randomized each round

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I'd end up with \$7.50 total (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- The socially optimal outcome is to send the maximum amount to create the most total value

However, there's risk:

- The receiver might not return anything
- There's communication noise that could affect trust
- This is round 1, so I have no history with this partner

A reasonable first-round strategy might be:

- Send a moderate amount to signal trust and willingness to cooperate
- Not send everything in case the receiver doesn't reciprocate
- Send enough to create meaningful value

Common strategies in trust games:

- Send about half (\$2.50) as a balanced approach
- Send more (\$3-4) to signal strong trust
- Send everything (\$5) to maximize joint payoff

Given this is round 1 and I want to establish a cooperative pattern but also protect against complete exploitation, I'll send \$3. This:

- Creates \$9 for the receiver
- Shows trust and cooperation

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 625,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.46 to you. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$4.46.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 2):\*\***

- My co-player sent me \$3.19 ? I received \$9.57
- This created \$6.38 in surplus value

**\*\*Fair return considerations:\*\***

- The sender took a significant risk sending most of their endowment
- A reasonable approach is to ensure we both benefit from the cooperation
- Returning \$4.00 would give:
  - Sender: \$5 - \$3.19 + \$4.00 = \$5.81
  - Me: \$9.57 - \$4.00 = \$5.57

This creates a balanced outcome where we both gain from the exchange.

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

Last round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.46 to me
- My payoff: \$5 - \$3 + \$2.46 = \$4.46
- Their payoff: \$9.0 - \$2.46 = \$6.54

So they returned about 27% of what they received (\$2.46/\$9.0 ? 0.273).

Now in Round 2:

- I'm the RECEIVER
- My current co-player sent me \$3.19
- I received \$9.57 (3.19 × 3)
- I need to decide how much to return

Important notes:

1. This is a different co-player (multi-agent setting, randomized pairings)
2. There's communication noise, so the \$3.19 might not be exactly what they sent
3. The previous player returned about 27% to me

Considerations:

- If I want to encourage cooperation, I should return a fair amount
- A common fair split might be to return enough so we both benefit equally from the cooperation
- The sender gave up \$3.19, so they have  $\$5 - \$3.19 = \$1.81$  left
- I have \$9.57
- If I want us to end up with similar amounts, I could return about \$3.88, giving us both around \$5.69
- Or I could return proportionally to what was sent: returning about 1/3 of what I received would be \$3.19, which gives them back their original amount
- A generous return might be half of what I received: \$4.785

Let me think about what's fair:

- The sender took a risk sending \$3.19 (gave up most of their endowment)
- The transaction created \$6.38 in surplus value beyond what they sent
- Splitting that gain equally means we each get an additional \$3.19 from the multiplier effect
- So returning \$3.19 would leave me with \$6.38 and restore them to their starting \$5
- Alternatively, returning \$4 would give them \$5.81 while I keep \$5.57

, creating a more balanced outcome

- Given the earlier low reciprocation (27%), I could...

**usage:**

```
{
 "input_tokens": 756,
 "output_tokens": 1027,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.19, it became \$9.57, and you returned \$4.0. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$10.030000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this new round as SENDER:

**\*\*Past experience:\*\***

- Round 1: Sent \$3, received \$2.46 back (modest cooperation)
- Round 2: When I received, I returned fairly (\$4.00 on \$9.57)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Need to establish trust with someone new
- The 3x multiplier means cooperation creates value

A moderate send amount balances the potential gains from cooperation with the risk of low returns. Sending \$3 creates \$9 of value while maintaining some protection.

```
{'send': 3}
```

**Reasoning:**

Let me analyze the situation:

**\*\*Previous rounds:\*\***

- Round 1: I was SENDER, sent \$3, received back \$2.46, payoff was \$4.46
- Round 2: I was RECEIVER, received \$9.57 (from \$3.19 sent), returned \$4.00, payoff was \$5.57

**\*\*Current Round (Round 3):\*\***

- I'm the SENDER again
- New co-player (since pairings are randomized each round)
- I have \$5 to potentially send

**\*\*Observations:\*\***

1. In Round 1, when I sent \$3, I got back \$2.46. The receiver kept \$6.54 out of \$9.00, which seems somewhat selfish (I got less than my initial \$3).



2. In Round 2, when someone sent me \$3.19, I returned \$4.00, being relatively fair.
3. The note about communication noise is important - amounts may differ from what was actually sent/returned.

**\*\*Strategy considerations:\*\***

- I'm playing with a new co-player each round
- There's no reputation to build with a specific player
- The 3x multiplier means cooperation creates value
- I need to balance trust against the risk of low returns

Given:

- My Round 1 experience showed modest cooperation from my receiver
- I was reasonably fair in Round 2
- I'm with a new player now who doesn't know my history

I think sending \$3 again is reasonable:

- It's a moderate level of trust
- Creates \$9 for the receiver
- Provides some protection if they don't reciprocate well

Let me send \$3 again.

```
Usage: {
 "input_tokens": 1047,
 "output_tokens": 532,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:21:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_2"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_2": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.55, that is 51.0% of their total endowment. You received \$7.6499999999999995. How much do you return to the sender? (0-7.6499999999999995)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.55 (51% of their \$5 endowment), showing moderate trust
- I received \$7.65 after the 3x multiplier
- The sender now has \$2.45 remaining
- I need to decide how much to return

**\*\*My reasoning:\*\***

The sender showed significant trust by sending about half their endowment. To reciprocate fairly and build trust:

If I return **\*\*\$4.00\*\***:

- Sender ends with: \$2.45 + \$4.00 = \$6.45 (profit of \$1.45)
- I end with: \$7.65 - \$4.00 = \$3.65 (profit of \$3.65)

This represents:

- Returning about 52% of what I received (similar to their 51% sent)
- Both of us profit from the cooperation
- Rewarding their trust encourages future cooperation
- Setting a positive precedent for Round 1

{'return': 4.0}

**reasoning:**

Let me understand the situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.55 (which is 51% of their \$5 endowment)
3. This amount was multiplied by 3x, so I received  $\$2.55 \times 3 = \$7.65$  (approximately \$7.65)
4. I need to decide how much to return (between \$0 and \$7.65)

Now, let me think about this strategically:

The sender sent 51% of their endowment, which is slightly more than half. This shows a moderate level of trust. They could have sent nothing, they could have sent everything, but they chose to send about half.

From a fairness perspective:

- The sender now has:  $\$5 - \$2.55 = \$2.45$
- I have:  $\$7.65$
- Total in the system:  $\$2.45 + \$7.65 = \$10.10$

If I want to be fair and reciprocate the trust:

- One approach: Split the total gains equally. The sender started with  $\$5$ , and we now have  $\$10.10$  total. If we split equally, we'd each have  $\$5.05$ . The sender already has  $\$2.45$ , so I'd return  $\$5.05 - \$2.45 = \$2.60$
- Another approach: Return the same proportion they sent. They sent 51% of their endowment, so I could return 51% of what I received:  $0.51 \times \$7.65 = \$3.90$
- Conservative approach: Return at least what they sent originally ( $\$2.55$ ) to ensure they don't lose money from trusting me
- Generous approach: Return more to reward their trust and potentially build cooperation

Given that:

1. The sender showed moderate trust by sending 51%
2. There's communication noise mentioned (though the exact amount sent seems precise)
3. Since this is a multi-round game with multiple agents, establishing a reputation for reciprocity could be strategically valuable
4. The sender took a genuine risk by sending money

A fair reciprocal response should reward their trust while ensuring they benefit from the exchange and remain proportional to their initial gesture. If I return  $\$2.60$ , we'd each end with  $\$5.05$ -a modest profit for both of us that acknowledges their willingness to trust.

Alternatively, returning 50% of what I received ( $\$3.825$ ) would give them  $\$6.275$ , a more substantial gain. But I need to reconsider my baseline here-as the receiver, I don't start with  $\$5$  like the sender does. I begin with  $\$0$  and only gain what they send multiplied by three. The sender started with  $\$5$ , sent  $\$2.55$ , and now has  $\$2.45$  remaining.

If I return  $\$X$ , they end with  $\$2.45 + \$X$  and I end with  $\$7.65 - \$X$ . For equal splits, that's  $\$2.60$  returned, giving us both  $\$5.05$ . But in this experimental context, I should probably return more-around  $\$3.00$ - $\$3.50$ -to reward their trust and signal reciprocity for potential future rounds.

Testing a few amounts: returning  $\$3.25$  leaves them with  $\$5.70$  and me with  $\$4.40$ ;  $\$3.50$  gives them  $\$5.95$  and me  $\$4.15$ ;  $\$4.00$  results in them having  $\$6.45$  and me  $\$3.65$ . Each option balances showing trustworthiness with keeping reasonable profit, though I'm leaning toward the higher end to maximize their incentive to continue cooperating. Both of us end up better off than we started. At  $\$3.50$ , the sender gets  $\$5.95$  (up  $\$0.95$ ) and I get  $\$4.15$  (up  $\$4.15$ ), which feels like a solid reciprocal move that rewards their trust without being exploitative.

I'm settling on  $\$3.50$  as my return amount-it strikes a balance between showing genuine reciprocity and ensuring both parties benefit from the exchange. Terms of what's strategically optimal in a multi-round setting. Building reputation matters-if I'm generous now, other agents will likely trust me more later. Since this is Round 1, establishing a trustworthy precedent is crucial. The sender showed moderate confidence by sending  $\$2.55$ , so I should reciprocate proportionally.

I'm settling on  $\$3.50$ . It more than doubles their original amount, gives them solid profit, signals I'm reliable, and leaves me with a reasonable gain too. Let me verify the math:  $\$2.55 \times 3 = \$7.65$  received, returning  $\$3.50$  means the sender nets  $\$0.95$  profit while I keep  $\$4.15$ .

The communication noise mentioned in the problem is worth noting, but I can only work with what I observe. My final return amount is  $\$3.50$ , formatted as `{'return': 3.50}`.

Actually, reconsidering the proportionality here-the sender gave 51% of their endowment, which signals moderate trust. Matching that proportion with my return would mean sending

back 51% of \$7.65, which equals \$3.90. This reciprocal approach better reflects their level of generosity while still leaving me with a reasonable profit of \$3.75 versus their \$1.35. I think \$3.90 better captures the spirit of proportional reciprocation.

Though I'm also drawn to rounding up to \$4.00 for simplicity-it's nearly the same percentage anyway (52%), easier to process, and still feels fair to both parties. Let me settle on \$4.00 as my final choice.

**usage:**

```
{
 "input_tokens": 539,
 "output_tokens": 3346,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.55, it became \$7.649999999999995, and you returned \$4.0. Your visible payoff was \$3.649999999999995.

Your total visible earnings across all rounds are \$3.649999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- Round 1 established a cooperative dynamic where sending ~51% led to mutual profit
- I'm now the sender with a new co-player
- I need to establish trust in this new pairing

**\*\*Strategic reasoning:\*\***

- Sending a moderate amount (around 50%) shows willingness to cooperate
- It gives my co-player enough to make returning worthwhile (\$7.50 after multiplier)
- It limits my downside risk while allowing mutual gains
- This mirrors the successful approach from Round 1

**\*\*Decision:\*\***

I'll send \$2.50 (50% of my endowment) to establish a cooperative tone with this new co-player.

{'send': 2.5}

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was the RECEIVER
- Sender sent \$2.55 (51% of \$5)
- I received \$7.65
- I returned \$4.00
- My payoff: \$3.65
- Sender's payoff would have been:  $\$5 - \$2.55 + \$4.00 = \$6.45$

**\*\*Current Round (Round 2):\*\***

- I am now the SENDER
- I have \$5
- I'm paired with a potentially different agent (since it says "your current co-player")
- I need to decide how much to send

**\*\*Strategic considerations:\*\***

1. **\*\*Reciprocity principle\*\***: In Round 1, my co-player showed moderate trust (51% sent). I reciprocated well by returning 52% of what I received, ensuring both parties profited.
2. **\*\*Building trust\*\***: Since this is a multi-round game with multiple agents, establishing a cooperative reputation could be beneficial.

3. **\*\*Unknown opponent\*\***: This is a different co-player, so I don't know their reciprocity tendencies yet.
4. **\*\*First impression\*\***: As the sender in this new pairing, I can set the tone for cooperation.
5. **\*\*Risk vs reward\*\***:
  - If I send \$0, I keep \$5, they get \$0, no mutual benefit
  - If I send all \$5, it becomes \$15 for them, but I'm vulnerable if they return nothing
  - Middle ground shows trust while limiting downside risk
6. **\*\*Following established norms\*\***: In Round 1, the sender sent 51%. This seemed to work well and encouraged cooperation.

**\*\*Decision\*\***

I'll send a similar amount to what I experienced as effective in Round 1 - around 50% (\$2.50). This:

- Shows trust and willingness to cooperate
- Limits my downside risk
- Gives the receiver enough to make returning worthwhile

...

**usage:**

```
{
 "input_tokens": 916,
 "output_tokens": 646,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.62 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$9.77.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened\*\***

- The sender sent \$2.11 (42.2% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- The sender now has \$2.89 remaining

**\*\*My reasoning\*\***

The sender showed trust, though slightly less than what I've seen in previous rounds. I should reciprocate fairly to maintain cooperation.

If I return **\*\*\$3.30\*\***:

- Sender ends with:  $\$2.89 + \$3.30 = \$6.19$  (profit of \$1.19)
- I end with:  $\$6.33 - \$3.30 = \$3.03$  (profit of \$3.03)

This represents:

- Returning about 52% of what I received
- Both of us profiting from cooperation
- Fair reciprocation that encourages future trust

```
{'return': 3.3}
```

**Reasoning:**

Let me analyze this situation:

**\*\*What happened in Round 2\*\***

- I sent \$2.50
- It became \$7.50
- They returned \$3.62
- My payoff:  $\$2.50$  (kept) +  $\$3.62$  (returned) =  $\$6.12$
- Their payoff:  $\$7.50 - \$3.62 = \$3.88$

- They returned about 48% of what they received

**\*\*Current Round 3 situation:\*\***

- I'm the receiver
- Sender sent \$2.11 (42.2% of their \$5 endowment)
- I received \$6.33
- Sender now has \$2.89 remaining
- I need to decide how much to return

**\*\*Analysis:\*\***

The sender sent \$2.11, which is less than the \$2.50-\$2.55 amounts I've seen previously. They're showing moderate but slightly lower trust. However, they still sent a significant portion (42%).

If I return different amounts:

- Return \$3.00: Sender ends with \$5.89 (+\$0.89), I end with \$3.33 (+\$3.33)
- Return \$3.50: Sender ends with \$6.39 (+\$1.39), I end with \$2.83 (+\$2.83)
- Return \$3.20: Sender ends with \$6.09 (+\$1.09), I end with \$3.13 (+\$3.13)

I want to:

1. Reciprocate their trust proportionally
2. Be fair and encourage future cooperation
3. Return roughly 50% of what I received, similar to what I've observed

Returning around \$3.20 would be about 50.5% of what I received, which seems fair and proportional to their ~42% send rate. This ensures both of us profit.

Actually, let me be slightly more generous since they did show trust. I'll return \$3.30 (about 52% of what I received).

...

```
Usage: {
 "input_tokens": 1218,
 "output_tokens": 758,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:21:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_7"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.34 (66.8% of their endowment)
- I received \$10.02 after the 3x multiplier
- The sender currently has \$1.66 remaining

I want to reciprocate the sender's trust and ensure cooperation is worthwhile for them. If I return \$5.00:

- The sender would end with \$1.66 + \$5.00 = \$6.66 (better than their initial \$5)
- I would keep \$5.02 (also better than any initial amount)

This rewards their trust while maintaining a fair distribution of the gains from cooperation.

{'return': 5.00}

**reasoning:**

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is  $3 \times \$3.34$ )
- I need to decide how much to return (between \$0 and \$10.02)

This is a trust game. The sender showed considerable trust by sending 66.8% of their endowment. Let me think about a fair return:

The sender:

- Started with \$5
- Sent \$3.34
- Currently has  $\$5 - \$3.34 = \$1.66$
- If I return X, they'll end up with  $\$1.66 + X$

Me:

- I received \$10.02
- If I return X, I keep  $\$10.02 - X$

Some considerations for fairness:

1. Equal split of total: Total money is \$5 (sender's initial) + \$10.02 (what I received) = \$15.02. Equal split would be \$7.51 each.
  - For sender to get \$7.51, they need  $\$7.51 - \$1.66 = \$5.85$  returned
  - I would keep  $\$10.02 - \$5.85 = \$4.17$
2. Proportional return: The sender sent 66.8% of their endowment, showing high trust. A reciprocal gesture might be to return a similar proportion.
3. Rewarding trust: Since the sender trusted me with a large amount, I should return enough to make their trust worthwhile.

Let me check option 1: If I return \$5.85:

- Sender ends with:  $\$1.66 + \$5.85 = \$7.51$
- I end with:  $\$10.02 - \$5.85 = \$4.17$
- This creates mutual benefit from cooperation, and the sender gains significantly from their initial trust.

Returning around \$5 also...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 837,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, the sender showed good trust by sending 66.8% of their endowment, and I reciprocated fairly.

Now as the sender with a new co-player, I'll demonstrate similar trust to establish cooperation. Sending \$3.00 (60% of my endowment) shows significant trust while being reasonable.

If they reciprocate fairly, we both benefit from cooperation.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

Round 1:

- I was the RECEIVER
- Sender sent \$3.34 (66.8% of their endowment)
- I received \$10.02
- I returned \$5.00
- My payoff was \$5.02



Now in Round 2:

- I am the SENDER
- I have \$5
- I need to decide how much to send

The previous sender showed trust by sending 66.8% of their endowment to me, and I reciprocated by returning a fair amount that left them better off.

Now I'm playing against a different agent (the pairing is randomized each round). I should consider:

1. The sender in Round 1 sent a substantial amount (66.8%), showing trust
2. This encouraged cooperation
3. I'm now the sender with a new partner

I should probably send a similar amount to establish trust and encourage cooperation. If I send around 60-70% of my endowment (like the previous sender did), that would be \$3.00-\$3.50.

Let me think about what would be fair:

- If I send \$3.00, they receive \$9.00
- If they return half of what they receive (a common fair split), they'd return \$4.50
- I'd end with \$2.00 + \$4.50 = \$6.50
- They'd keep \$4.50

That seems reasonable. I could also send \$3.34 to match what was sent to me in Round 1, showing I'm willing to trust at the same level.

Actually, let me reconsider. I want to be strategic but also fair. Sending \$3.00 seems like a good middle ground - it's 60% of my endowment, showing significant trust without being extreme.

```
{'send': 3.00}
```

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 530,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.73 to you. Your visible payoff was \$4.73.

Your total visible earnings across all rounds are \$9.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my history:

- Round 1 (as receiver): Co-player sent \$3.34, I returned \$5.00
- Round 2 (as sender): I sent \$3.00, received back \$2.73

Now with a new co-player in Round 3, I should continue demonstrating willingness to cooperate. Sending a consistent amount shows I'm a reliable partner.

```
{'send': 3.00}
```

**Reasoning:**

In Round 2, I sent \$3.00, it became \$9.00, and the receiver returned \$2.73.

Let me calculate:

- I started with \$5
- I sent \$3, so I had \$2 left
- They received \$9 and returned \$2.73
- My payoff was \$2 + \$2.73 = \$4.73

The receiver kept \$9 - \$2.73 = \$6.27

I got \$4.73

So the return rate was  $\$2.73/\$9.00 = 30.3\%$

This is lower than what I returned in Round 1 (I returned \$5.00 out of \$10.02, which is about 50%).

However, this is a new co-player in Round 3. I shouldn't punish this new person for what the previous receiver did.

I should continue to try to establish cooperation. Maybe send a similar amount to show consistency and willingness to cooperate. However, given that the last return was somewhat low, I might be slightly more cautious.

Let me send \$2.50 - still showing trust but being a bit more conservative.

Actually, let me think about this differently. Each round is with a different co-player (the problem states pairings are randomized). So my actions should be about establishing cooperation with each new partner, not about punishing previous partners.

I should continue to send a reasonable amount to establish trust. Let me send \$3.00 again to be consistent and show I'm willing to cooperate.

```
Usage: {
 "input_tokens": 1034,
 "output_tokens": 456,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_7 - Receiver - dyad\_4

Timestamp: 2026-08-12T09:21:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_7"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 3,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance building trust with managing risk since I have no history with this receiver.

**\*\*My reasoning:\*\***

- Sending money creates value (3x multiplier means potential gains for both of us)
- This is round 1, so I want to signal cooperation
- I should send enough to be meaningful but not risk everything
- If the receiver reciprocates fairly, we both benefit

I'll send a moderate amount to test cooperation while limiting downside risk.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. **\*\*Game Structure:\*\***

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- My payoff: \$5 - sent + returned
- Receiver's payoff: 3×sent - returned

2. **\*\*Strategic Considerations:\*\***

- If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 and they keep \$7.50. Total: \$15
- The game creates surplus through cooperation (up to \$15 total vs \$5 if no trust)
- This is a trust game - I need to trust the receiver will return some amount

3. **\*\*First Round Considerations:\*\***
  - No history with this partner
  - This is a multi-round game with different partners
  - Building a reputation for cooperation might matter
  - The note about "communication noise" suggests amounts might be misperceived, which could affect trust dynamics
4. **\*\*Common Strategies:\*\***
  - **\*\*Full trust:\*\*** Send \$5, hope for good return
  - **\*\*Moderate trust:\*\*** Send \$2-3, test the waters
  - **\*\*No trust:\*\*** Send \$0, guarantee \$5
  - **\*\*Signal cooperation:\*\*** Send a meaningful amount to show willingness to cooperate
5. **\*\*Rational Analysis:\*\***
  - If the receiver is purely self-interested and this is a one-shot game, they should return \$0
  - Knowing this, I should send \$0
  - However, in repeated games or with reciprocity norms, cooperation can emerge
  - Many people exhibit fairness preferences and reciprocity
6. **\*\*My Decision:\*\***  
 Since this is the opening move in a multi-round game, a moderate cooperative signal seems prudent. I'll send \$3-enough to demonstrate trust and give the receiver \$9 to work with, but not so much that I'm exposed to catastrophic loss if they keep everything. Even a modest return of \$3-4 would benefit us both compared to the baseline of keeping \$5....

usage:

```
{
 "input_tokens": 500,
 "output_tokens": 836,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.56 to you. Your visible payoff was \$7.56.

Your total visible earnings across all rounds are \$7.56.

This round, you are the RECEIVER against your current co-player. They sent you \$1.67, so you received \$5.01. How much do you return? (0-5.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.67 (33% of their endowment)
- I received \$5.01 after the 3x multiplier
- They currently have \$3.33 remaining

**\*\*My considerations:\*\***

- They showed moderate trust by sending \$1.67
- The multiplier created \$3.34 in surplus value
- A fair approach would be to share this surplus roughly equally
- In Round 1, my partner returned ~62% and that felt cooperative

**\*\*Fair return calculation:\*\***

If we split the surplus equally:

- Each should gain ~\$1.67 from cooperation
- Sender ending: \$5 - \$1.67 + return = \$6.67 desired
- Return needed: \$3.34

However, that might be overly generous given their modest send. I'll return about 60% to reward their cooperation while keeping reasonable value.

```
'''json
{"return": 3.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.56 to me
- I ended with:  $\$5 - \$3 + \$5.56 = \$7.56$
- Receiver ended with:  $\$9 - \$5.56 = \$3.44$

That was actually quite generous of them! They returned more than half.

Round 2 (current):

- I am RECEIVER
- They sent me \$1.67
- I received \$5.01 (which is  $3 \times \$1.67$ )
- I need to decide how much to return (0-5.01)

Key considerations:

1. My previous partner was generous, returning \$5.56 out of \$9 (62%)
2. This is a different player (multi-agent setting with randomized pairings)
3. They sent me \$1.67 out of \$5, which is 33.4% - a moderate amount
4. The sender keeps:  $\$5 - \$1.67 = \$3.33$  before my return
5. I should reciprocate fairly to encourage future cooperation

Let me think about fair returns:

- If I return half (\$2.50), they end with  $\$3.33 + \$2.50 = \$5.83$ , I keep \$2.51
- If I return around what gives us equal final payoffs:
  - They have  $\$3.33 + \text{return}$
  - I have  $\$5.01 - \text{return}$
  - Equal:  $\$3.33 + x = \$5.01 - x$
  - $2x = \$1.68$
  - $x = \$0.84$

This would give us both around \$4.17

- A proportional return similar to what I received (62% of what I got):
  - $0.62 \times \$5.01 = \$3.11$
  - They'd end with:  $\$3.33 + \$3.11 = \$6.44$
  - I'd keep:  $\$5.01 - \$3.11 = \$1.90$

A 50% return of around \$2.50 seems like the right balance-it rewards their cooperation without being excessive, and it's more moderate than matching their 33% send rate since they're a different player than before.

If I send back \$2.50, they'd gain \$0.83 while I'd be left with \$2.51. The math shows we'd create \$8.34 in total value, which is less than the \$10 we'd have if neither of us participated. The real question is what surplus we're actually generating from this exchange and whether my return is proportional to their initial trust.

The multiplier created \$3.34 in extra value (\$5.01 received minus \$1.67 sent). If we split that gain equally, each of us should pocket \$1.67. That means I should return \$3.34 to bring them to \$6.67 total, which gives us both the same \$1.67 gain from the transaction.

...

**usage:**

```
{
 "input_tokens": 754,
 "output_tokens": 1445,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.67, it became \$5.01, and you returned \$3.0. Your visible payoff was \$2.01.

Your total visible earnings across all rounds are \$9.57.

This round, you are the RECEIVER against your current co-player. They sent you \$3.66, so y

ou received \$10.98. How much do you return? (0-10.98)  
Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$3.66 (73% of their endowment) - very high trust!
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining
- Total surplus created: \$2.32

**\*\*My considerations:\*\***

- This is a very generous send showing strong cooperation
- They took significant risk by sending most of their money
- I should reward this trust with a fair return
- In Round 1, I received about 62% back on my \$3 send
- In Round 2, I returned 60% on a smaller send

**\*\*Fair return:\*\***

Given their high trust, I'll return approximately 50% of what I received:

- Return \$5.50
- They'll end with \$6.84 (profit of \$1.84)
- I'll end with \$5.48 (profit of \$0.48)
- This rewards their cooperation while both of us profit

```
'''json
{"return": 5.50}
'''
```

**Reasoning:**

Let me analyze this situation:

**\*\*Current Situation:\*\***

- My co-player sent me \$3.66 (73% of their endowment)
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining (\$5 - \$3.66)
- This is a very generous/trusting send

**\*\*What this shows:\*\***

- High trust from the sender (sending over 70%!)
- They're expecting reciprocity
- The multiplier created significant surplus: \$5.98 (\$10.98 - \$5)

**\*\*Considerations:\*\***

- In Round 1, when I sent \$3 (60%), I received \$5.56 back - that's about 62% return rate
- In Round 2, when someone sent \$1.67 (33%), I returned \$3.00 - that's 60% return rate
- This sender showed much more trust (\$3.66 = 73% of endowment)

**\*\*Fair calculation:\*\***

The total pot is now \$12.32 (\$1.34 they kept + \$10.98 I received)

If we split the gains from cooperation:

- Original total: \$10 (my \$5 + their \$5)
- Current total: \$12.32
- Surplus created: \$2.32
- Fair split would give each person \$1.16 gain
- Sender would end at \$6.16, I'd end at \$6.16
- I need to return: \$6.16 - \$1.34 = \$4.82

But let me think about what's fair based on the trust shown:

- They sent 73% of their money
- If I return about 50% of what I received, that's \$5.49
- That would give them \$1.34 + \$5.49 = \$6.83 total

...

```
Usage: {
 "input_tokens": 1112,
 "output_tokens": 833,
 "reasoning_tokens": 0
}
```

=====

**ROUND 4**

=====

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Sender, Agent\_5=Receiver,  
Agent\_6=Sender, Agent\_7=Sender, Agent\_8=Receiver

**Pairings:**

dyad\_1: Agent\_2=Sender, Agent\_3=Receiver  
dyad\_2: Agent\_6=Sender, Agent\_8=Receiver  
dyad\_3: Agent\_4=Sender, Agent\_1=Receiver  
dyad\_4: Agent\_7=Sender, Agent\_5=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 sent \$2.5; received \$7.5; Agent\_3 returned \$2.66; payoffs Agent\_2=\$5.16,  
Agent\_3=\$4.84; noise/visible: sent visible=\$2.58, received visible=\$7.74, returned visible=\$3.01  
Action validation: {"returned": {"amount": 2.66, "clamped": false, "max": 7.5, "min": 0, "raw":  
2.66}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

dyad\_2: Agent\_6 sent \$3.5; received \$10.5; Agent\_8 returned \$4; payoffs Agent\_6=\$5.5,  
Agent\_8=\$6.5; noise/visible: sent visible=\$3.22, received visible=\$9.66, returned visible=\$4.55  
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 10.5, "min": 0, "raw":  
4.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad\_3: Agent\_4 sent \$3; received \$9; Agent\_1 returned \$4.5; payoffs Agent\_4=\$6.5, Agent\_1=\$4.5;  
noise/visible: sent visible=\$2.99, received visible=\$8.97, returned visible=\$4.44  
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw":  
4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_4: Agent\_7 sent \$3.5; received \$10.5; Agent\_5 returned \$4.2; payoffs Agent\_7=\$5.7,  
Agent\_5=\$6.3; noise/visible: sent visible=\$2.94, received visible=\$8.82, returned visible=\$3.59  
Action validation: {"returned": {"amount": 4.2, "clamped": false, "max": 10.5, "min": 0, "raw":  
4.2}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

**Balances after round: {**

"Agent\_1": 20.82,  
"Agent\_2": 22.11,  
"Agent\_3": 23.14,  
"Agent\_4": 20.5,  
"Agent\_5": 25.23,  
"Agent\_6": 20.5,  
"Agent\_7": 20.7,  
"Agent\_8": 21.0

}

**Visible balances after round: {**

"Agent\_1": 21.79,  
"Agent\_2": 18.310000000000002,  
"Agent\_3": 20.14,  
"Agent\_4": 21.22,  
"Agent\_5": 21.13,  
"Agent\_6": 21.89,  
"Agent\_7": 20.14,  
"Agent\_8": 20.62

}

**LLM CALLS**

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:21:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_3",  
"pairing": {  
"agents": [  
"Agent\_2",  
"Agent\_3"  
],  
"investor": "Agent\_2",  
"roles": {  
"Agent\_2": "investor",  
"Agent\_3": "trustee"  
},  
"trustee": "Agent\_3"  
},  
"role": "investor",

```

"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.55, that is 51.0% of their total endowment. You received \$7.6499999999999995. How much do you return to the sender? (0-7.6499999999999995)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.55 (51% of their \$5 endowment), showing moderate trust
- I received \$7.65 after the 3x multiplier
- The sender now has \$2.45 remaining
- I need to decide how much to return

**\*\*My reasoning:\*\***

The sender showed significant trust by sending about half their endowment. To reciprocate fairly and build trust:



If I return \*\*\$4.00\*\*:

- Sender ends with:  $\$2.45 + \$4.00 = \$6.45$  (profit of \$1.45)
- I end with:  $\$7.65 - \$4.00 = \$3.65$  (profit of \$3.65)

This represents:

- Returning about 52% of what I received (similar to their 51% sent)
- Both of us profit from the cooperation
- Rewarding their trust encourages future cooperation
- Setting a positive precedent for Round 1

{'return': 4.0}

#### reasoning:

Let me understand the situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.55 (which is 51% of their \$5 endowment)
3. This amount was multiplied by 3x, so I received  $\$2.55 \times 3 = \$7.65$  (approximately \$7.65)
4. I need to decide how much to return (between \$0 and \$7.65)

Now, let me think about this strategically:

The sender sent 51% of their endowment, which is slightly more than half. This shows a moderate level of trust. They could have sent nothing, they could have sent everything, but they chose to send about half.

From a fairness perspective:

- The sender now has:  $\$5 - \$2.55 = \$2.45$
- I have: \$7.65
- Total in the system:  $\$2.45 + \$7.65 = \$10.10$

If I want to be fair and reciprocate the trust:

- One approach: Split the total gains equally. The sender started with \$5, and we now have \$10.10 total. If we split equally, we'd each have \$5.05. The sender already has \$2.45, so I'd return  $\$5.05 - \$2.45 = \$2.60$
- Another approach: Return the same proportion they sent. They sent 51% of their endowment, so I could return 51% of what I received:  $0.51 \times \$7.65 = \$3.90$
- Conservative approach: Return at least what they sent originally (\$2.55) to ensure they don't lose money from trusting me
- Generous approach: Return more to reward their trust and potentially build cooperation

Given that:

1. The sender showed moderate trust by sending 51%
2. There's communication noise mentioned (though the exact amount sent seems precise)
3. Since this is a multi-round game with multiple agents, establishing a reputation for reciprocity could be strategically valuable
4. The sender took a genuine risk by sending money

A fair reciprocal response should reward their trust while ensuring they benefit from the exchange and remain proportional to their initial gesture. If I return \$2.60, we'd each end with \$5.05—a modest profit for both of us that acknowledges their willingness to trust.

Alternatively, returning 50% of what I received (\$3.825) would give them \$6.275, a more substantial gain. But I need to reconsider my baseline here—as the receiver, I don't start with \$5 like the sender does. I begin with \$0 and only gain what they send multiplied by three. The sender started with \$5, sent \$2.55, and now has \$2.45 remaining.

If I return \$X, they end with  $\$2.45 + \$X$  and I end with  $\$7.65 - \$X$ . For equal splits, that's \$2.60 returned, giving us both \$5.05. But in this experimental context, I should probably return more—around \$3.00–\$3.50—to reward their trust and signal reciprocity for potential future rounds.

Testing a few amounts: returning \$3.25 leaves them with \$5.70 and me with \$4.40; \$3.50 gives them \$5.95 and me \$4.15; \$4.00 results in them having \$6.45 and me \$3.65. Each option balances showing trustworthiness with keeping reasonable profit, though I'm leaning toward the higher end to maximize their incentive to continue cooperating. Both of us end u

p better off than we started. At \$3.50, the sender gets \$5.95 (up \$0.95) and I get \$4.15 (up \$4.15), which feels like a solid reciprocal move that rewards their trust without being exploitative.

I'm settling on \$3.50 as my return amount-it strikes a balance between showing genuine reciprocity and ensuring both parties benefit from the exchange. terms of what's strategically optimal in a multi-round setting. Building reputation matters-if I'm generous now, other agents will likely trust me more later. Since this is Round 1, establishing a trust worthy precedent is crucial. The sender showed moderate confidence by sending \$2.55, so I should reciprocate proportionally.

I'm settling on \$3.50. It more than doubles their original amount, gives them solid profit, signals I'm reliable, and leaves me with a reasonable gain too. Let me verify the math:  $\$2.55 \times 3 = \$7.65$  received, returning \$3.50 means the sender nets \$0.95 profit while I keep \$4.15.

The communication noise mentioned in the problem is worth noting, but I can only work with what I observe. My final return amount is \$3.50, formatted as {'return': 3.50}.

Actually, reconsidering the proportionality here-the sender gave 51% of their endowment, which signals moderate trust. Matching that proportion with my return would mean sending back 51% of \$7.65, which equals \$3.90. This reciprocal approach better reflects their level of generosity while still leaving me with a reasonable profit of \$3.75 versus their \$1.35. I think \$3.90 better captures the spirit of proportional reciprocation.

Though I'm also drawn to rounding up to \$4.00 for simplicity-it's nearly the same percentage anyway (52%), easier to process, and still feels fair to both parties. Let me settle on \$4.00 as my final choice.

usage:

```
{
 "input_tokens": 539,
 "output_tokens": 3346,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.55, it became \$7.649999999999995, and you returned \$4.0. Your visible payoff was \$3.649999999999995.

Your total visible earnings across all rounds are \$3.649999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

**\*\*Context:\*\***

- Round 1 established a cooperative dynamic where sending ~51% led to mutual profit
- I'm now the sender with a new co-player
- I need to establish trust in this new pairing

**\*\*Strategic reasoning:\*\***

- Sending a moderate amount (around 50%) shows willingness to cooperate
- It gives my co-player enough to make returning worthwhile (\$7.50 after multiplier)
- It limits my downside risk while allowing mutual gains
- This mirrors the successful approach from Round 1

**\*\*Decision:\*\***

I'll send \$2.50 (50% of my endowment) to establish a cooperative tone with this new co-player.

{ 'send': 2.5 }

reasoning:

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was the RECEIVER
- Sender sent \$2.55 (51% of \$5)
- I received \$7.65
- I returned \$4.00
- My payoff: \$3.65
- Sender's payoff would have been:  $\$5 - \$2.55 + \$4.00 = \$6.45$

**\*\*Current Round (Round 2):\*\***

- I am now the SENDER
- I have \$5
- I'm paired with a potentially different agent (since it says "your current co-player")
- I need to decide how much to send

**\*\*Strategic considerations:\*\***

1. **\*\*Reciprocity principle\*\***: In Round 1, my co-player showed moderate trust (51% sent). I reciprocated well by returning 52% of what I received, ensuring both parties profited.
2. **\*\*Building trust\*\***: Since this is a multi-round game with multiple agents, establishing a cooperative reputation could be beneficial.
3. **\*\*Unknown opponent\*\***: This is a different co-player, so I don't know their reciprocity tendencies yet.
4. **\*\*First impression\*\***: As the sender in this new pairing, I can set the tone for cooperation.
5. **\*\*Risk vs reward\*\***:
  - If I send \$0, I keep \$5, they get \$0, no mutual benefit
  - If I send all \$5, it becomes \$15 for them, but I'm vulnerable if they return nothing
  - Middle ground shows trust while limiting downside risk
6. **\*\*Following established norms\*\***: In Round 1, the sender sent 51%. This seemed to work well and encouraged cooperation.

**\*\*Decision:\*\***

I'll send a similar amount to what I experienced as effective in Round 1 - around 50% (\$2.50). This:

- Shows trust and willingness to cooperate
- Limits my downside risk
- Gives the receiver enough to make returning worthwhile

...

**usage:**

```
{
 "input_tokens": 916,
 "output_tokens": 646,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.62 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$9.77.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.11 (42.2% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- The sender now has \$2.89 remaining

**\*\*My reasoning:\*\***

The sender showed trust, though slightly less than what I've seen in previous rounds. I should reciprocate fairly to maintain cooperation.

If I return \*\*\$3.30\*\*:

- Sender ends with:  $\$2.89 + \$3.30 = \$6.19$  (profit of \$1.19)
- I end with:  $\$6.33 - \$3.30 = \$3.03$  (profit of \$3.03)

This represents:

- Returning about 52% of what I received
- Both of us profiting from cooperation
- Fair reciprocation that encourages future trust

{'return': 3.3}

**reasoning:**

Let me analyze this situation:

**\*\*What happened in Round 2:\*\***

- I sent \$2.50
- It became \$7.50
- They returned \$3.62
- My payoff:  $\$2.50$  (kept) +  $\$3.62$  (returned) =  $\$6.12$
- Their payoff:  $\$7.50 - \$3.62 = \$3.88$
- They returned about 48% of what they received

**\*\*Current Round 3 situation:\*\***

- I'm the receiver
- Sender sent \$2.11 (42.2% of their \$5 endowment)
- I received \$6.33
- Sender now has \$2.89 remaining
- I need to decide how much to return

**\*\*Analysis:\*\***

The sender sent \$2.11, which is less than the \$2.50-\$2.55 amounts I've seen previously. They're showing moderate but slightly lower trust. However, they still sent a significant portion (42%).

If I return different amounts:

- Return \$3.00: Sender ends with \$5.89 (+\$0.89), I end with \$3.33 (+\$3.33)
- Return \$3.50: Sender ends with \$6.39 (+\$1.39), I end with \$2.83 (+\$2.83)
- Return \$3.20: Sender ends with \$6.09 (+\$1.09), I end with \$3.13 (+\$3.13)

I want to:

1. Reciprocate their trust proportionally
2. Be fair and encourage future cooperation
3. Return roughly 50% of what I received, similar to what I've observed

Returning around \$3.20 would be about 50.5% of what I received, which seems fair and proportional to their ~42% send rate. This ensures both of us profit.

Actually, let me be slightly more generous since they did show trust. I'll return \$3.30 (about 52% of what I received).

...

**usage:**

```
{
 "input_tokens": 1218,
 "output_tokens": 758,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.3. Your visible payoff was \$3.0300000000000002.

Your total visible earnings across all rounds are \$12.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Context:\*\***

- I'm facing a new co-player as sender
- Previous cooperation has been successful (total earnings \$12.80)
- When I sent \$2.50 in Round 2, I received \$3.62 back for a good profit
- My returns as receiver have been fair (~52%)

**\*\*Strategic reasoning:\*\***

- Continue the cooperative approach that's been working
- Sending 50% shows good faith while managing risk
- This gives my co-player \$7.50 to work with, enough for mutual benefit
- Establishes positive precedent with this new partner

{'send': 2.5}

**Reasoning:**

Let me analyze the situation:

**\*\*Previous rounds summary:\*\***

- Round 1 (RECEIVER): Received \$7.65 from sender sending \$2.55, returned \$4.00, earned \$3.65
- Round 2 (SENDER): Sent \$2.50, became \$7.50, they returned \$3.62, earned \$6.12
- Round 3 (RECEIVER): Received \$6.33 from sender sending \$2.11, returned \$3.30, earned \$3.03
- Total earnings: \$12.80

**\*\*Pattern analysis:\*\***

- When I sent \$2.50 (50%), I got back \$3.62, giving me a profit of \$1.12 (from \$2.50 remaining + \$3.62 = \$6.12 total)
- When others sent to me, I've returned fairly (~52% of what I received)
- Overall, cooperation has been successful

**\*\*Current decision:\*\***

- New co-player as sender this round
- Need to establish trust again
- Sending around 50% has worked well previously

I'll continue with the cooperative strategy that's been working: send \$2.50 (50% of endowment)

Usage: {

"input\_tokens": 1558,  
"output\_tokens": 431,  
"reasoning\_tokens": 0

}

**GAME - Agent\_3 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:21:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 1,  
"opponent\_id": "Agent\_2",  
"pairing": {  
  "agents": [  
    "Agent\_2",  
    "Agent\_3"  
  ],  
  "investor": "Agent\_2",  
  "roles": {  
    "Agent\_2": "investor",  
    "Agent\_3": "trustee"  
  },  
  "trustee": "Agent\_3"  
},  
"role": "trustee",  
"role\_label": "Receiver",  
"round": 4,  
"task": "game",  
"task\_index": 0

```

}
Exact input messages sent to model:
 Message 1 role=system
 content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 MULTI-AGENT SETTING:
 - There are 8 agents in this run.
 - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
 - You play one sender-receiver game with your paired opponent each round.
 - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
 ll have acted as sender and receiver equally often when the round count allows it.
 - Agent names are hidden in this run.
 - Your paired opponent is referred to as your current co-player.
 Message 2 role=user
 content:
 Round 1: You are the RECEIVER. the sender sent $2.35, that is 47.0% of their total endowme
 nt. You received $7.050000000000001. How much do you return to the sender? (0-7.05000000
 0000001)
 Message 3 role=assistant
 content:
 Looking at this situation:

 - The sender showed moderate trust by sending $2.35 (47% of their endowment)
 - I received $7.05 from this
 - The sender currently has $2.65 remaining

 I want to reciprocate this trust fairly. A reasonable approach is to aim for equal final p
 ayoffs since we're cooperating:

 - Total available: $2.65 (sender's remainder) + $7.05 (my received amount) = $9.70
 - Equal split: $9.70 ÷ 2 = $4.85 each
 - Return needed: $4.85 - $2.65 = $2.20

 This ensures we both benefit equally from the cooperation, rewarding the sender's trust wh
 ile keeping the gains fair.

```

```
{'return': 2.20}
```

**reasoning:**

Let me understand the situation:

- I am the RECEIVER in Round 1
- The sender sent \$2.35 (47% of their \$5 endowment)
- This was multiplied by 3x, so I received \$7.05
- I need to decide how much to return (between \$0 and \$7.05)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. There's communication noise mentioned, so amounts may differ slightly
3. This is a multi-round game with 8 agents and rotating pairings
4. Building reputation and reciprocity is important

The sender sent 47% of their endowment. If I want to be fair and reciprocal:

- The sender started with \$5, sent \$2.35, and has \$2.65 left
- I received \$7.05
- If I don't return anything, sender gets \$2.65, I get \$7.05
- A fair split might be to aim for equal payoffs or proportional returns

Common strategies in trust games:

1. Return nothing (selfish)
2. Return the original amount sent (\$2.35)
3. Return enough to split the gains equally
4. Return a proportion similar to what was sent

Let's calculate equal payoffs:

- Total money: \$5 (initial sender) + \$7.05 (tripled) = \$12.05? No wait...
- Initial endowment for sender: \$5
- Sent: \$2.35 (tripled to \$7.05 for receiver)
- Sender currently has: \$5 - \$2.35 = \$2.65
- Receiver has: \$7.05
- Total in system: \$2.65 + \$7.05 = \$9.70

For equal payoffs: each should get  $\$9.70/2 = \$4.85$

- So I'd need to return \$2.20 to the sender, leaving me with \$4.85

I could also return the original \$2.35 they sent, which would give them \$5.00 (breaking even) while I keep \$4.70. Or I could mirror their generosity by returning 47% of what I received-about \$3.31-which acknowledges their trust while still letting me benefit from the multiplication.

The equal payoff approach feels right though. Returning \$2.20 means we both end up with \$4.85, which feels like genuine reciprocity for the trust they showed....

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 1079,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.35, it became \$7.050000000000001, and you returned \$2.2. Your visible payoff was \$4.8500000000000005.

Your total visible earnings across all rounds are \$4.8500000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my previous round, the sender showed trust at 47% (\$2.35) and I reciprocated fairly, resulting in equal payoffs of \$4.85 each.

As the sender now, I'll show similar moderate trust. Sending around 50% demonstrates cooperation while managing risk.

{'send': 2.50}

**reasoning:**

Let me think about this situation:

Round 2: I am now the SENDER

- I have \$5
- I need to decide how much to send
- The amount I send will be tripled
- The receiver will decide how much to return

In the previous round:

- I was the receiver
- The sender sent \$2.35 (47% of their endowment)
- I received \$7.05
- I returned \$2.20
- My payoff was \$4.85
- The sender's payoff would have been  $\$2.65 + \$2.20 = \$4.85$

So in the last round, I reciprocated fairly by aiming for equal payoffs.

Now I'm the sender. I should consider:

1. Building trust by sending a reasonable amount
2. The sender in the previous round sent 47% (\$2.35 out of \$5)
3. I reciprocated fairly, so hopefully this receiver will do the same

Since the previous sender showed moderate trust at 47%, I could do something similar. Sending around \$2.35-\$2.50 seems reasonable.

However, I could also think strategically:

- If I send \$2.50, the receiver gets \$7.50
- If they split equally, they'd return \$2.50 to give us each \$5
- If I send \$3.00, the receiver gets \$9.00
- Equal split would mean returning \$3.50 for payoffs of \$5.50 each

Let me try sending \$2.50 as a moderate trust signal, similar to what the previous sender did with me.

Actually, let me reconsider. I showed I'm cooperative by returning fairly in the previous round. I could send a bit more to maximize joint welfare if the receiver reciprocates.

If I send \$3:

- I keep \$2
- Receiver gets \$9
- If they return \$3.50, I get \$5.50 total, they get \$5.50
- Joint welfare: \$11...

**usage:**

```
{
 "input_tokens": 859,
 "output_tokens": 713,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.43 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$9.780000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience:

- Round 1 (receiver): Sender sent 47%, I reciprocated fairly
- Round 2 (sender): I sent 50%, received fair return of \$2.43



The moderate trust strategy (around 50%) has been working well with fair reciprocation from co-players. I'll continue this approach.

```
{'send': 2.50}
```

**reasoning:**

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$2.43
- My payoff:  $\$5 - \$2.50 + \$2.43 = \$4.93$
- Their payoff:  $\$7.50 - \$2.43 = \$5.07$

The receiver returned a fair amount - roughly aimed at equal splits. The return rate was  $\$2.43/\$7.50 = 32.4\%$ , which gave us both similar final payoffs.

Now I'm a sender again with a new co-player. I've seen:

- Round 1 (as receiver): Sender sent 47%, I returned fairly
- Round 2 (as sender): I sent 50%, received fair return

Both rounds showed moderate trust and fair reciprocation. I should continue this cooperative strategy. Sending around 50% seems to be working well.

```
{'send': 2.50}
```

**usage:**

```
{
 "input_tokens": 1060,
 "output_tokens": 327,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.78 to you. Your visible payoff was \$5.2799999999999999.

Your total visible earnings across all rounds are \$15.06.

This round, you are the RECEIVER against your current co-player. They sent you \$2.58, so you received \$7.74. How much do you return? (0-7.74)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Sender sent \$2.58 (51.6% of endowment), showing solid trust
- I received \$7.74
- Sender currently has \$2.42

Based on my positive experiences with fair reciprocation, I'll aim for equal final payoffs:

- Total:  $\$2.42 + \$7.74 = \$10.16$
- Equal split: \$5.08 each
- Return needed: \$2.66

```
{'return': 2.66}
```

**Reasoning:**

I'm the receiver now. Let me analyze:

- Sender sent \$2.58 (51.6% of their endowment)
- I received \$7.74 (after 3x multiplier)
- Sender currently has:  $\$5 - \$2.58 = \$2.42$

My recent experience shows:

- Round 2: I sent \$2.50, got back \$2.43
- Round 3: I sent \$2.50, got back \$2.78

Both returns were generous and aimed at fair splits. I should continue this reciprocal fairness.

For equal final payoffs:

- Total available:  $\$2.42$  (sender's remainder) +  $\$7.74$  (my received) =  $\$10.16$

- Equal split:  $\$10.16 / 2 = \$5.08$  each
- Return needed:  $\$5.08 - \$2.42 = \$2.66$

This would give us both \$5.08, which is fair and reciprocates the sender's trust at 51.6%.

Let me verify:

- Sender final:  $\$2.42 + \$2.66 = \$5.08$
- My final:  $\$7.74 - \$2.66 = \$5.08$

Perfect. I'll return \$2.66.

```
Usage: {
 "input_tokens": 1287,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_6 - Sender - dyad\_2

Timestamp: 2026-08-12T09:22:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_8"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative dynamic while being somewhat cautious since I don't know my co-player yet.

If I send money, it triples, creating potential mutual benefit. However, there's risk since the receiver's return is voluntary.

I'll send a moderate amount to signal trust and willingness to cooperate, while not being overly vulnerable in this first interaction.

**\*\*Decision:\*\***

```
'''json
{"send": 3}
'''
```

This gives the receiver \$9 to work with. If they're cooperative and return a fair share, we both benefit. If not, I've learned something about the dynamics for future rounds while limiting my downside risk.

**reasoning:**

This is the first round of a trust game, and I'm the sender. I need to think about this strategically.

**Key facts:**

- I have \$5
- Whatever I send gets multiplied by 3
- The receiver decides how much to return
- There may be communication noise
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I don't know who my current co-player is

Since this is the first round, I have no information about how other players behave. The optimal strategy depends on:

1. Whether I expect reciprocity from the receiver
2. Whether I want to establish a reputation (though with hidden names, this is less relevant for future rounds)
3. Risk tolerance

From a purely rational game theory perspective, if this were a one-shot game with no reputation effects, the receiver should return \$0 (keep everything), which means I should send \$0. However, experimental evidence shows people often cooperate in trust games.

**Some considerations:**

- Sending \$5 maximizes total payoff (\$5 + \$15 total) but requires trust
- Sending \$0 guarantees I keep \$5 but creates no opportunity for mutual gain

- A moderate amount (like \$2-3) balances risk and opportunity

Since this is round 1 and I want to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything immediately

I'll send a moderate amount. Sending \$3 seems reasonable:

- If sent \$3, receiver gets \$9
- If receiver returns half (\$4.50), I end up with  $\$5 - \$3 + \$4.50 = \$6.50$
- If receiver returns nothing, I end up with \$2
- This tests their cooperation while not being overly cautious

Actually, let me reconsider. A common strategy is to send about half to show trust but not complete vulnerability. Sending \$2.50 or \$3 seems reasonable.

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 632,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.25 to you. Your visible payoff was \$6.25.

Your total visible earnings across all rounds are \$6.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.67, so you received \$8.01. How much do you return? (0-8.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.67\*\*** (about 53% of their endowment), which became \$8.01 for me. This is a moderately cooperative move.

**\*\*Comparison to previous round:\*\***

- When I sent \$3.00, I received \$4.25 back (47% of the \$9.00)
- That gave my co-player a reasonable benefit while rewarding my trust

**\*\*My reasoning:\*\***

I want to:

1. Reward the sender's trust with a decent return
2. Maintain cooperative dynamics for future rounds
3. Keep enough to make receiving worthwhile

Returning about \$3.75 seems fair:

- Sender would get:  $\$2.33 \text{ (kept)} + \$3.75 = \$6.08 \text{ total (gain of } \sim \$1.08)$
- I would get:  $\$8.01 - \$3.75 = \$4.26 \text{ (solid gain)}$

This is proportionally similar to what I experienced last round.

```
'''json
{"return": 3.75}
'''
```

**reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was SENDER
- I sent \$3.0
- It became \$9.0
- Receiver returned \$4.25
- My payoff:  $\$5 - \$3 + \$4.25 = \$6.25$

This round (Round 2):

- I am RECEIVER
- Sender sent \$2.67
- I received \$8.01 (which is  $\$2.67 \times 3$ )
- I need to decide how much to return

The sender in this round sent \$2.67 out of \$5, which is about 53% of their endowment. This is a fairly cooperative move.

In the previous round, when I sent \$3, I received back \$4.25. My net gain was \$1.25 from the initial \$5.

Let me think about what would be fair:

- The sender gave up \$2.67
- I received \$8.01
- If I want to be cooperative and fair, I should think about:
  1. Returning enough so the sender benefits from sending
  2. Not returning so much that I don't benefit from receiving

A common approach is to split the gains. The total amount now is \$8.01. If we split it equally:

- Each gets \$4.005

But the sender started with \$5 and sent \$2.67, so they have \$2.33 left.

If I return \$4.00, they'd have  $\$2.33 + \$4.00 = \$6.33$  (gain of \$1.33)

I'd keep  $\$8.01 - \$4.00 = \$4.01$  (gain of \$4.01 compared to receiving nothing)

Hmm, that seems imbalanced. Let me think differently.

Another approach: What would make our final payoffs more equal?

- Sender has:  $\$5 - \$2.67 + \text{return} = \$2.33 + \text{return}$
- I have:  $\$8.01 - \text{return}$

For equal payoffs:

$$\$2.33 + \text{return} = \$8.01 - \text{return}$$

$$2 \times \text{return} = \$8.01 - \$2.33 = \$5.68$$

$$\text{return} = \$2.84$$

If I return \$2.84:

- Sender gets:  $\$2.33 + \$2.84 = \$5.17$
- I get:  $\$8.01 - \$2.84 = \$5.17$

That would equalize our outcomes, though it doesn't particularly reward the sender's risk-taking. I should consider what the previous receiver did for comparison.

The prior round showed a return of \$4.25 on \$9.00 received—roughly 47% of what they got. Applying that same rate here:  $\$8.01 \times 0.47 = \$3.76$ . This would give the sender \$6.09 total and leave me with \$4.25, which feels more balanced given the sender's initial trust. But I need to account for the communication noise mentioned in the instructions—the amounts I'm seeing may not match what was actually sent or returned....

usage:

```
{
 "input_tokens": 785,
 "output_tokens": 1226,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.67, it became \$8.01, and you returned \$3.75. Your visible payoff was \$4.26.

Your total visible earnings across all rounds are \$10.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

**\*\*My co-player sent \$3.36\*\*** (67% of their endowment), which became \$10.08 for me. This is a notably generous and trusting move.

**\*\*Context from previous rounds:\*\***

- Round 2: Someone sent \$2.67 (53%), I returned \$3.75
- Round 1: I sent \$3.00 (60%), received \$4.25 back

**\*\*My reasoning:\*\***

This sender showed higher trust than my previous co-player (67% vs 53%). I should reward t his generosity proportionally.

Returning \$4.75:

- Sender would get: \$1.64 (kept) + \$4.75 = \$6.39 total (gain of \$1.39)
- I would get: \$10.08 - \$4.75 = \$5.33 (good gain)

This provides the sender with a decent reward for their trust while maintaining mutual ben efit - both of us come out ahead.

```
'''json
{"return": 4.75}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$3.36 (67% of their \$5 endowment) - this is quite generous/trusting
- I received \$10.08

**\*\*Previous round:\*\***

- Someone sent me \$2.67 (53% of endowment), I returned \$3.75
- That gave them \$2.33 + \$3.75 = \$6.08 total (~\$1.08 gain)
- I kept \$4.26

**\*\*Round 1:\*\***

- I sent \$3.00 (60%), received \$4.25 back
- That gave me \$2.00 + \$4.25 = \$6.25 total

**\*\*My thinking:\*\***

The sender this round was more generous (67% vs 53% in previous round). I should reward th is higher trust level.

If I return around \$4.50-5.00:

- Return \$4.50: Sender gets \$1.64 + \$4.50 = \$6.14, I get \$5.58
- Return \$5.00: Sender gets \$1.64 + \$5.00 = \$6.64, I get \$5.08

Actually, let me think about what's fair given they sent 67% of their endowment.

If I return \$5.00:

- Sender: keeps \$1.64, gets back \$5.00 = \$6.64 total (gain of \$1.64)
- Me: receive \$10.08, return \$5.00 = \$5.08 (solid gain)

This seems like a good reciprocal move for their high trust. It's roughly a 50-50 split of the multiplied amount, which rewards their generosity while still benefiting both of us .

Actually, let me reconsider. They sent \$3.36 out of \$5, so they kept \$1.64.

Returning \$5 gives them \$6.64 with a \$1.64 gain, while I keep \$5.08. That's reasonably fai r given their high trust. I...

**usage:**

```
{
 "input_tokens": 1163,
 "output_tokens": 921,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.75. Your visible payoff was \$5.33.

Your total visible earnings across all rounds are \$15.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this round as SENDER again:

**\*\*My experience:\*\***

- Round 1: Sent \$3.00, received \$4.25 back ? Good outcome
- Rounds 2-3: As receiver, I've been cooperative, returning fair amounts

**\*\*Current situation:\*\***

- New co-player (different from Round 1)
- I've been part of generally cooperative dynamics

**\*\*My reasoning:\*\***

I had success with sending \$3.00 before, and the cooperative pattern across rounds suggests moderate-to-high sending works well. I'll send slightly more this time to signal trust while building on the positive momentum.

Sending \$3.50 creates \$10.50 for the receiver. If they're reasonably cooperative (returning ~40%), I'd still come out ahead.

```
'''json
{"send": 3.5}
'''
```

**Reasoning:**

I'm back as the SENDER in round 4. Let me review my experience so far:

Round 1: I sent \$3.00, received \$4.25 back. Payoff: \$6.25

Round 2: Received \$8.01 (from \$2.67 sent), returned \$3.75. Payoff: \$4.26

Round 3: Received \$10.08 (from \$3.36 sent), returned \$4.75. Payoff: \$5.33

Total earnings: \$15.84

Now I'm the sender again with a new co-player. What should I send?

Looking at the pattern:

- When I sent \$3.00 (60%), I got a decent return
- When others sent moderate-to-high amounts (\$2.67, \$3.36), I've been fairly cooperative in returning
- The game seems to be working reasonably well with moderate-to-high sending amounts

I should consider:

1. I had a positive experience when I sent \$3.00
2. The cooperative moves I've seen have been between \$2.67-\$3.36
3. I've demonstrated I'm a fair returner, so maybe I can expect reasonable behavior

I think I should send a similar or slightly higher amount than last time. Since I received \$4.25 when I sent \$3.00, that worked out well.

Options:

- Send \$3.00 again (proven strategy)
- Send \$3.50 (show slightly more trust)
- Send \$4.00 (high trust signal)

Given:

- This is a new co-player
- I've been building cooperative patterns
- The game is going well so far

I'll send \$3.50 as a moderately high trust signal, slightly more than my first round, but not overly risky. This creates \$10.50 for the receiver.

...

Usage: {

```
"input_tokens": 1524,
"output_tokens": 660,
"reasoning_tokens": 0
}
```

### GAME - Agent\_8 - Receiver - dyad\_2

Timestamp: 2026-08-12T09:22:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_8"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_6": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.



**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the SENDER, I need to balance trust and caution since I have no history with this co-player.

The key consideration is that sending money creates value (3x multiplier), but I'm vulnerable to the receiver's decision about how much to return.

A moderate amount seems reasonable for Round 1:

- It signals willingness to cooperate and trust
- It creates meaningful value (\$9 for the receiver)
- It provides some protection against complete loss

I'll send \*\*\$3\*\*.

{'send': 3}

**reasoning:**

This is Round 1 of a multi-task experiment where I'm playing a trust game as the SENDER. Let me think about this situation:

1. I start with \$5
2. I can send any amount from \$0 to \$5
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - amount\_sent + amount\_returned

This is a classic trust game. The optimal strategy depends on:

- Whether I trust the receiver to return a fair amount
- What "fair" means in this context
- The fact that there's communication noise (amounts may differ slightly)
- This is a multi-round game with multiple agents
- Pairings are randomized each round

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I'd end up with \$7.50 total (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- The socially optimal outcome is to send the maximum amount to create the most total value

However, there's risk:

- The receiver might not return anything
- There's communication noise that could affect trust
- This is round 1, so I have no history with this partner

A reasonable first-round strategy might be:

- Send a moderate amount to signal trust and willingness to cooperate
- Not send everything in case the receiver doesn't reciprocate
- Send enough to create meaningful value

Common strategies in trust games:

- Send about half (\$2.50) as a balanced approach
- Send more (\$3-4) to signal strong trust
- Send everything (\$5) to maximize joint payoff

Given this is round 1 and I want to establish a cooperative pattern but also protect against complete exploitation, I'll send \$3. This:

- Creates \$9 for the receiver

- Shows trust and cooperation

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 625,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.46 to you. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$4.46.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 2):\*\***

- My co-player sent me \$3.19 ? I received \$9.57
- This created \$6.38 in surplus value

**\*\*Fair return considerations:\*\***

- The sender took a significant risk sending most of their endowment
- A reasonable approach is to ensure we both benefit from the cooperation
- Returning \$4.00 would give:
  - Sender: \$5 - \$3.19 + \$4.00 = \$5.81
  - Me: \$9.57 - \$4.00 = \$5.57

This creates a balanced outcome where we both gain from the exchange.

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

Last round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.46 to me
- My payoff: \$5 - \$3 + \$2.46 = \$4.46
- Their payoff: \$9.0 - \$2.46 = \$6.54

So they returned about 27% of what they received (\$2.46/\$9.0 ? 0.273).

Now in Round 2:

- I'm the RECEIVER
- My current co-player sent me \$3.19
- I received \$9.57 (3.19 × 3)
- I need to decide how much to return

Important notes:

1. This is a different co-player (multi-agent setting, randomized pairings)
2. There's communication noise, so the \$3.19 might not be exactly what they sent
3. The previous player returned about 27% to me

Considerations:

- If I want to encourage cooperation, I should return a fair amount
- A common fair split might be to return enough so we both benefit equally from the cooperation
- The sender gave up \$3.19, so they have \$5 - \$3.19 = \$1.81 left
- I have \$9.57
- If I want us to end up with similar amounts, I could return about \$3.88, giving us both around \$5.69

- Or I could return proportionally to what was sent: returning about 1/3 of what I received would be \$3.19, which gives them back their original amount
- A generous return might be half of what I received: \$4.785

Let me think about what's fair:

- The sender took a risk sending \$3.19 (gave up most of their endowment)
- The transaction created \$6.38 in surplus value beyond what they sent
- Splitting that gain equally means we each get an additional \$3.19 from the multiplier effect
- So returning \$3.19 would leave me with \$6.38 and restore them to their starting \$5
- Alternatively, returning \$4 would give them \$5.81 while I keep \$5.57

, creating a more balanced outcome

- Given the earlier low reciprocation (27%), I could...

**usage:**

```
{
 "input_tokens": 756,
 "output_tokens": 1027,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.19, it became \$9.57, and you returned \$4.0. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$10.030000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this new round as SENDER:

**\*\*Past experience:\*\***

- Round 1: Sent \$3, received \$2.46 back (modest cooperation)
- Round 2: When I received, I returned fairly (\$4.00 on \$9.57)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Need to establish trust with someone new
- The 3x multiplier means cooperation creates value

A moderate send amount balances the potential gains from cooperation with the risk of low returns. Sending \$3 creates \$9 of value while maintaining some protection.

```
{'send': 3}
```

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds:\*\***

- Round 1: I was SENDER, sent \$3, received back \$2.46, payoff was \$4.46
- Round 2: I was RECEIVER, received \$9.57 (from \$3.19 sent), returned \$4.00, payoff was \$5.57

**\*\*Current Round (Round 3):\*\***

- I'm the SENDER again
- New co-player (since pairings are randomized each round)
- I have \$5 to potentially send

**\*\*Observations:\*\***

1. In Round 1, when I sent \$3, I got back \$2.46. The receiver kept \$6.54 out of \$9.00, which seems somewhat selfish (I got less than my initial \$3).
2. In Round 2, when someone sent me \$3.19, I returned \$4.00, being relatively fair.
3. The note about communication noise is important - amounts may differ from what was actually sent/returned.

**\*\*Strategy considerations:\*\***

- I'm playing with a new co-player each round
- There's no reputation to build with a specific player
- The 3x multiplier means cooperation creates value
- I need to balance trust against the risk of low returns

Given:

- My Round 1 experience showed modest cooperation from my receiver
- I was reasonably fair in Round 2
- I'm with a new player now who doesn't know my history

I think sending \$3 again is reasonable:

- It's a moderate level of trust
- Creates \$9 for the receiver
- Provides some protection if they don't reciprocate well

Let me send \$3 again.

**usage:**

```
{
 "input_tokens": 1047,
 "output_tokens": 532,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$14.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.22, so you received \$9.66. How much do you return? (0-9.66)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Current Round (Round 4):\*\***

- Co-player sent \$3.22 ? I received \$9.66
- This represents significant trust (sending 64% of their endowment)

**\*\*Fair return analysis:\*\***

- Returning \$4.00 would give:
  - Sender:  $\$5 - \$3.22 + \$4.00 = \$5.78$  (profit of \$0.78)
  - Me:  $\$9.66 - \$4.00 = \$5.66$  (profit of \$0.66)

This creates a fair split where both players benefit from the cooperation.

```
{'return': 4.00}
```

**Reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER:\*\***

- My co-player sent me \$3.22
- I received \$9.66 (after 3x multiplier)
- I need to decide how much to return

**\*\*Previous rounds analysis:\*\***

- Round 1 (SENDER): Sent \$3, got back \$2.46 - slightly below fair
- Round 2 (RECEIVER): Received \$9.57, returned \$4.00 - tried to be fair
- Round 3 (SENDER): Sent \$3, got back \$2.93 - again slightly below fair

**\*\*Fair return calculation:\*\***

The sender gave up \$3.22 from their \$5 endowment, creating \$9.66 for me.

- Surplus created:  $\$9.66 - \$3.22 = \$6.44$

A fair split would give us each roughly equal gains:

- If I return \$4.00:
  - Sender gets:  $\$5 - \$3.22 + \$4.00 = \$5.78$  (gain of \$0.78)
  - I get:  $\$9.66 - \$4.00 = \$5.66$  (gain of \$0.66)

This seems reasonably balanced. The sender sent a significant portion of their endowment, showing trust.

I should return around \$4.00 to be fair and reciprocate their trust.

```
Usage: {
 "input_tokens": 1324,
 "output_tokens": 501,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_4 - Sender - dyad\_3

Timestamp: 2026-08-12T09:22:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_1"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_1": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.32, that is 66.4% of their total endowment. You received \$9.959999999999999. How much do you return to the sender? (0-9.959999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.32 (66.4% of their endowment)
- I received \$9.96 after the 3x multiplier
- The sender currently has \$1.68 remaining

I want to reward this trust while being fair to both of us. If I return \$5.00:

- The sender ends with:  $\$1.68 + \$5.00 = \$6.68$  (profit of \$1.68)
- I end with:  $\$9.96 - \$5.00 = \$4.96$

This gives the sender a good return on their trust (33.6% gain on their original \$5) while I also benefit from the exchange. This return rate of approximately 50% is commonly seen as reciprocal in trust games.

{'return': 5.00}

**reasoning:**

Let me analyze this situation:

1. The sender sent \$3.32 out of their \$5 endowment (66.4%)
2. Due to the 3x multiplier, I received \$9.96 ( $3.32 \times 3 = 9.96$ )
3. I need to decide how much to return

**Key considerations:**

- The sender showed significant trust by sending 66.4% of their endowment
- This is a multi-round game with random pairings
- Building a reputation for reciprocity could be beneficial
- There's communication noise, so exact amounts may not be perceived correctly

Let me think about fair/reciprocal returns:

- The sender spent \$3.32 to give me \$9.96
- If I want the sender to profit from trusting me, I should return more than \$3.32
- A common strategy in trust games is to split the gains roughly equally

If I return \$0:

- Sender gets:  $\$5 - \$3.32 + \$0 = \$1.68$
- I get:  $\$9.96 - \$0 = \$9.96$
- Total: \$11.64

If I return \$3.32 (sender breaks even):

- Sender gets:  $\$5 - \$3.32 + \$3.32 = \$5.00$
- I get:  $\$9.96 - \$3.32 = \$6.64$
- Total: \$11.64

If I return \$5.00 (roughly equal split of total):

- Sender gets:  $\$5 - \$3.32 + \$5.00 = \$6.68$
- I get:  $\$9.96 - \$5.00 = \$4.96$
- Total: \$11.64

The sender took a risk and sent a substantial amount. A reasonable reciprocal amount would be to ensure both of us profit from the exchange.

To achieve roughly equal final payoffs, I'd need to return about \$4.14-this would give us each \$5.82. But since the sender invested significantly from their original endowment, I should consider whether an even more generous return would better reward their trust.

If I return \$6.64, they'd end with \$8.32 (doubling their investment gain), though that leaves me with just \$3.32. A middle ground like \$5.00 would give them a 33.6% gain while I keep \$4.96. Since this is the first round and I want to build a reputation for fairness, and given how much trust they demonstrated by sending most of their money, I'm leaning toward a return that meaningfully rewards their cooperation while still leaving me in a reasonable position....

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 1319,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.32, it became \$9.959999999999999, and you returned \$5.0. Your visible payoff was \$4.959999999999999.

Your total visible earnings across all rounds are \$4.959999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, my co-player showed significant trust by sending 66.4% of their endowment, and I reciprocated with a 50% return rate.

For this round as sender, I'll demonstrate similar trust to establish cooperative play. Sending \$3.00 (60% of my endowment) shows trust while maintaining some security.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

1. In Round 1, I was the receiver and someone sent me \$3.32 (66.4% of their endowment). I returned \$5.00, which was about 50% of what I received.
2. Now in Round 2, I'm the sender with \$5.
3. I need to decide how much to send to my current co-player (who may be different from Round 1's partner).

Key considerations:

- In the previous round, someone trusted me with 66.4% of their endowment
- I reciprocated fairly by returning 50%
- This established a pattern of moderate trust and reciprocity
- I'm playing with potentially a different agent now

Strategy options:

- a) Send a similar amount to what I received last round (~\$3.32) - this mirrors the trust shown to me
- b) Send less to be more cautious
- c) Send more to encourage cooperation

Given that:

- The previous sender trusted me with 66.4%, showing moderate-high trust
- I want to encourage cooperation in this multi-round game
- Sending around the same proportion (60-70%) seems reasonable
- This could establish a cooperative norm

I'll send \$3.00 (60% of my endowment), which is:

- Similar to what was sent to me in Round 1

- Shows trust but not maximal trust
- Allows me to keep some guarantee (\$2)
- If returned fairly (~50%), I'd get back around \$4.50, making my total  $\$2 + \$4.50 = \$6.50$

Let me go with \$3.00.

**usage:**

```
{
 "input_tokens": 864,
 "output_tokens": 471,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.51 to you. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$11.469999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.27 (45.4% of their endowment), showing moderate trust
- I received \$6.81 after the 3x multiplier
- The sender currently has \$2.73 remaining

In the previous round when I was sender, I sent 60% and received back about 50% of what my co-player received. I'll reciprocate with a similar return rate.

Returning \$3.50 gives:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return, which fairly rewards their trust.

```
{'return': 3.50}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.27, which is 45.4% of their \$5 endowment
- I received \$6.81 after the 3x multiplier
- The sender currently has  $\$5 - \$2.27 = \$2.73$  remaining

In the previous round when I was sender:

- I sent \$3.00 (60% of endowment)
- Received back \$4.51, which was about 50% of the \$9.00 they received
- This gave me a payoff of \$6.51

The sender this round showed moderate trust (45.4% vs my 60% last round). I should reciprocate fairly.

If I return \$3.50:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return rate, similar to what I received last round (50%). This seems fair and reciprocal.

Let me check the math:

- Return rate =  $\$3.50 / \$6.81 = 51.4\%$
- Sender's final payoff =  $\$2.73 + \$3.50 = \$6.23$
- My final payoff =  $\$6.81 - \$3.50 = \$3.31$

This seems reasonable and reciprocal given their 45.4% trust level.

**usage:**



```
{
 "input_tokens": 1104,
 "output_tokens": 545,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.5. Your visible payoff was \$3.3100000000000005.

Your total visible earnings across all rounds are \$14.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience so far:

- When I've shown trust as sender (60% in Round 2), I received good reciprocity (~50% return)
- When others showed trust as sender, I've returned approximately 50%
- The pattern suggests cooperative play with moderate to high trust levels

I'll continue with a similar strategy, sending \$3.00 (60% of my endowment) to maintain trust and cooperation.

```
{'send': 3.00}
```

**Reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.32 (66.4%)
- I received \$9.96
- I returned \$5.00 (about 50%)
- My payoff: \$4.96

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.51 (about 50%)
- My payoff: \$6.51

Round 3: I was RECEIVER

- Sender sent \$2.27 (45.4%)
- I received \$6.81
- I returned \$3.50 (about 51%)
- My payoff: \$3.31

Round 4: I am SENDER now

- I have \$5 to decide what to send

Looking at the pattern:

- In Round 2, I sent \$3.00 (60%) and got back about 50%
- Others have sent between 45% and 66% in the rounds I've seen

I've been experiencing decent reciprocity so far. The return rates have been around 50% consistently.

I should continue to show trust, as it's been rewarded. Sending around 60% seems reasonable based on what I've seen. I'll send \$3.00 again.

Usage: {

```
"input_tokens": 1421,
"output_tokens": 420,
"reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:22:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_1"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_1": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 4,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.34 (66.8% of their endowment)
- I received \$10.02 after the 3x multiplier
- The sender currently has \$1.66 remaining

I want to reciprocate the sender's trust and ensure cooperation is worthwhile for them. If I return \$5.00:

- The sender would end with  $\$1.66 + \$5.00 = \$6.66$  (better than their initial \$5)
- I would keep \$5.02 (also better than any initial amount)

This rewards their trust while maintaining a fair distribution of the gains from cooperation.

`{'return': 5.00}`

**reasoning:**

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is  $\$3.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$10.02)

This is a trust game. The sender showed considerable trust by sending 66.8% of their endowment. Let me think about a fair return:

The sender:

- Started with \$5
- Sent \$3.34
- Currently has  $\$5 - \$3.34 = \$1.66$
- If I return X, they'll end up with  $\$1.66 + X$

Me:

- I received \$10.02
- If I return X, I keep  $\$10.02 - X$

Some considerations for fairness:

1. Equal split of total: Total money is \$5 (sender's initial) + \$10.02 (what I received) = \$15.02. Equal split would be \$7.51 each.
  - For sender to get \$7.51, they need  $\$7.51 - \$1.66 = \$5.85$  returned
  - I would keep  $\$10.02 - \$5.85 = \$4.17$
2. Proportional return: The sender sent 66.8% of their endowment, showing high trust. A reciprocal gesture might be to return a similar proportion.
3. Rewarding trust: Since the sender trusted me with a large amount, I should return enough to make their trust worthwhile.

Let me check option 1: If I return \$5.85:

- Sender ends with:  $\$1.66 + \$5.85 = \$7.51$
- I end with:  $\$10.02 - \$5.85 = \$4.17$
- This creates mutual benefit from cooperation, and the sender gains significantly from their initial trust.

Returning around \$5 also...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 837,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, the sender showed good trust by sending 66.8% of their endowment, and I reciprocated fairly.

Now as the sender with a new co-player, I'll demonstrate similar trust to establish cooperation. Sending \$3.00 (60% of my endowment) shows significant trust while being reasonable.

If they reciprocate fairly, we both benefit from cooperation.

{'send': 3.00}

**reasoning:**

Let me think about this situation:

Round 1:

- I was the RECEIVER
- Sender sent \$3.34 (66.8% of their endowment)
- I received \$10.02
- I returned \$5.00
- My payoff was \$5.02

Now in Round 2:

- I am the SENDER
- I have \$5
- I need to decide how much to send

The previous sender showed trust by sending 66.8% of their endowment to me, and I reciprocated by returning a fair amount that left them better off.

Now I'm playing against a different agent (the pairing is randomized each round). I should consider:

1. The sender in Round 1 sent a substantial amount (66.8%), showing trust
2. This encouraged cooperation
3. I'm now the sender with a new partner

I should probably send a similar amount to establish trust and encourage cooperation. If I send around 60-70% of my endowment (like the previous sender did), that would be \$3.00-\$3.50.

Let me think about what would be fair:

- If I send \$3.00, they receive \$9.00
- If they return half of what they receive (a common fair split), they'd return \$4.50
- I'd end with \$2.00 + \$4.50 = \$6.50
- They'd keep \$4.50

That seems reasonable. I could also send \$3.34 to match what was sent to me in Round 1, showing I'm willing to trust at the same level.

Actually, let me reconsider. I want to be strategic but also fair. Sending \$3.00 seems like a good middle ground - it's 60% of my endowment, showing significant trust without being extreme.

{'send': 3.00}

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 530,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.73 to you. Your visible payoff was \$4.73.  
Your total visible earnings across all rounds are \$9.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history:

- Round 1 (as receiver): Co-player sent \$3.34, I returned \$5.00
- Round 2 (as sender): I sent \$3.00, received back \$2.73

Now with a new co-player in Round 3, I should continue demonstrating willingness to cooperate. Sending a consistent amount shows I'm a reliable partner.

{'send': 3.00}

**reasoning:**

In Round 2, I sent \$3.00, it became \$9.00, and the receiver returned \$2.73.

Let me calculate:

- I started with \$5
- I sent \$3, so I had \$2 left
- They received \$9 and returned \$2.73
- My payoff was \$2 + \$2.73 = \$4.73

The receiver kept \$9 - \$2.73 = \$6.27

I got \$4.73

So the return rate was  $\$2.73/\$9.00 = 30.3\%$

This is lower than what I returned in Round 1 (I returned \$5.00 out of \$10.02, which is about 50%).

However, this is a new co-player in Round 3. I shouldn't punish this new person for what the previous receiver did.

I should continue to try to establish cooperation. Maybe send a similar amount to show consistency and willingness to cooperate. However, given that the last return was somewhat low, I might be slightly more cautious.

Let me send \$2.50 - still showing trust but being a bit more conservative.

Actually, let me think about this differently. Each round is with a different co-player (the problem states pairings are randomized). So my actions should be about establishing cooperation with each new partner, not about punishing previous partners.

I should continue to send a reasonable amount to establish trust. Let me send \$3.00 again to be consistent and show I'm willing to cooperate.

**usage:**

```
{
 "input_tokens": 1034,
 "output_tokens": 456,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.57 to you. Your visible payoff was \$7.57.  
Your total visible earnings across all rounds are \$17.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender showed trust by sending \$2.99 (about 60% of their endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has \$2.01

I want to reciprocate their trust fairly. If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (profit of \$1.51)
- I keep: \$4.47

This rewards their cooperation while maintaining a fair split of the gains.

{'return': 4.50}

**Reasoning:**

Let me analyze this round:

- I'm the receiver
- The sender sent \$2.99 (about 60% of their \$5 endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has  $\$5 - \$2.99 = \$2.01$

The sender showed good trust by sending nearly 60% of their endowment. I should reciprocate fairly.

If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (better than their initial \$5)
- I keep:  $\$8.97 - \$4.50 = \$4.47$

This is fair and rewards their trust. It's roughly half of what I received, which seems reasonable given they sent about 60% of their endowment.

Looking at my previous behavior:

- Round 1: Received \$10.02, returned \$5.00 (kept \$5.02)
- Round 3: The other player returned \$5.57 when they received \$9.00

I should be consistent and fair. Returning around \$4.50 seems appropriate.

Usage: {

"input\_tokens": 1269,  
"output\_tokens": 416,  
"reasoning\_tokens": 0

}

**GAME - Agent\_7 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:22:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_4",  
"move\_index": 6,  
"opponent\_id": "Agent\_5",  
"pairing": {  
  "agents": [  
    "Agent\_7",  
    "Agent\_5"  
  ],  
  "investor": "Agent\_7",  
  "roles": {  
    "Agent\_5": "trustee",  
    "Agent\_7": "investor"  
  },  
  "trustee": "Agent\_5"

},  
"role": "investor",  
"role\_label": "Sender",  
"round": 4,  
"task": "game",  
"task\_index": 0

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance building trust with managing risk since I have no history with this receiver.

**\*\*My reasoning:\*\***

- Sending money creates value (3x multiplier means potential gains for both of us)
- This is round 1, so I want to signal cooperation
- I should send enough to be meaningful but not risk everything
- If the receiver reciprocates fairly, we both benefit

I'll send a moderate amount to test cooperation while limiting downside risk.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. **\*\*Game Structure:\*\***

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- My payoff: \$5 - sent + returned
- Receiver's payoff: 3×sent - returned

2. **\*\*Strategic Considerations:\*\***

- If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 and they keep \$7.50. Total: \$15
- The game creates surplus through cooperation (up to \$15 total vs \$5 if no trust)
- This is a trust game - I need to trust the receiver will return some amount

3. **\*\*First Round Considerations:\*\***

- No history with this partner
- This is a multi-round game with different partners
- Building a reputation for cooperation might matter
- The note about "communication noise" suggests amounts might be misperceived, which could affect trust dynamics

4. **\*\*Common Strategies:\*\***

- **\*\*Full trust:\*\*** Send \$5, hope for good return
- **\*\*Moderate trust:\*\*** Send \$2-3, test the waters
- **\*\*No trust:\*\*** Send \$0, guarantee \$5
- **\*\*Signal cooperation:\*\*** Send a meaningful amount to show willingness to cooperate

5. **\*\*Rational Analysis:\*\***

- If the receiver is purely self-interested and this is a one-shot game, they should return \$0
- Knowing this, I should send \$0
- However, in repeated games or with reciprocity norms, cooperation can emerge
- Many people exhibit fairness preferences and reciprocity

6. **\*\*My Decision:\*\***

Since this is the opening move in a multi-round game, a moderate cooperative signal seems prudent. I'll send \$3-enough to demonstrate trust and give the receiver \$9 to work with, but not so much that I'm exposed to catastrophic loss if they keep everything. Even a modest return of \$3-4 would benefit us both compared to the baseline of keeping \$5....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 836,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.56 to you. Your visible payoff was \$7.56.

Your total visible earnings across all rounds are \$7.56.

This round, you are the RECEIVER against your current co-player. They sent you \$1.67, so you received \$5.01. How much do you return? (0-5.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.67 (33% of their endowment)
- I received \$5.01 after the 3x multiplier
- They currently have \$3.33 remaining

**\*\*My considerations:\*\***

- They showed moderate trust by sending \$1.67
- The multiplier created \$3.34 in surplus value
- A fair approach would be to share this surplus roughly equally



- In Round 1, my partner returned ~62% and that felt cooperative

**\*\*Fair return calculation:\*\***

If we split the surplus equally:

- Each should gain ~\$1.67 from cooperation
- Sender ending: \$5 - \$1.67 + return = \$6.67 desired
- Return needed: \$3.34

However, that might be overly generous given their modest send. I'll return about 60% to reward their cooperation while keeping reasonable value.

```
'''json
{"return": 3.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.56 to me
- I ended with: \$5 - \$3 + \$5.56 = \$7.56
- Receiver ended with: \$9 - \$5.56 = \$3.44

That was actually quite generous of them! They returned more than half.

Round 2 (current):

- I am RECEIVER
- They sent me \$1.67
- I received \$5.01 (which is 3 × \$1.67)
- I need to decide how much to return (0-5.01)

Key considerations:

1. My previous partner was generous, returning \$5.56 out of \$9 (62%)
2. This is a different player (multi-agent setting with randomized pairings)
3. They sent me \$1.67 out of \$5, which is 33.4% - a moderate amount
4. The sender keeps: \$5 - \$1.67 = \$3.33 before my return
5. I should reciprocate fairly to encourage future cooperation

Let me think about fair returns:

- If I return half (\$2.50), they end with \$3.33 + \$2.50 = \$5.83, I keep \$2.51
- If I return around what gives us equal final payoffs:
  - They have \$3.33 + return
  - I have \$5.01 - return
  - Equal: \$3.33 + x = \$5.01 - x
  - 2x = \$1.68
  - x = \$0.84

This would give us both around \$4.17

- A proportional return similar to what I received (62% of what I got):
  - $0.62 \times \$5.01 = \$3.11$
  - They'd end with: \$3.33 + \$3.11 = \$6.44
  - I'd keep: \$5.01 - \$3.11 = \$1.90

A 50% return of around \$2.50 seems like the right balance-it rewards their cooperation without being excessive, and it's more moderate than matching their 33% send rate since they're a different player than before.

If I send back \$2.50, they'd gain \$0.83 while I'd be left with \$2.51. The math shows we'd create \$8.34 in total value, which is less than the \$10 we'd have if neither of us participated. The real question is what surplus we're actually generating from this exchange and whether my return is proportional to their initial trust.

The multiplier created \$3.34 in extra value (\$5.01 received minus \$1.67 sent). If we split that gain equally, each of us should pocket \$1.67. That means I should return \$3.34 to bring them to \$6.67 total, which gives us both the same \$1.67 gain from the transaction.

...

**usage:**

```
{
```

```
"input_tokens": 754,
"output_tokens": 1445,
"reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.67, it became \$5.01, and you returned \$3.0. Your visible payoff was \$2.01.

Your total visible earnings across all rounds are \$9.57.

This round, you are the RECEIVER against your current co-player. They sent you \$3.66, so you received \$10.98. How much do you return? (0-10.98)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$3.66 (73% of their endowment) - very high trust!
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining
- Total surplus created: \$2.32

**\*\*My considerations:\*\***

- This is a very generous send showing strong cooperation
- They took significant risk by sending most of their money
- I should reward this trust with a fair return
- In Round 1, I received about 62% back on my \$3 send
- In Round 2, I returned 60% on a smaller send

**\*\*Fair return:\*\***

Given their high trust, I'll return approximately 50% of what I received:

- Return \$5.50
- They'll end with \$6.84 (profit of \$1.84)
- I'll end with \$5.48 (profit of \$0.48)
- This rewards their cooperation while both of us profit

```
'''json
{ "return": 5.50 }
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current Situation:\*\***

- My co-player sent me \$3.66 (73% of their endowment)
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining (\$5 - \$3.66)
- This is a very generous/trusting send

**\*\*What this shows:\*\***

- High trust from the sender (sending over 70%!)
- They're expecting reciprocity
- The multiplier created significant surplus: \$5.98 (\$10.98 - \$5)

**\*\*Considerations:\*\***

- In Round 1, when I sent \$3 (60%), I received \$5.56 back - that's about 62% return rate
- In Round 2, when someone sent \$1.67 (33%), I returned \$3.00 - that's 60% return rate
- This sender showed much more trust (\$3.66 = 73% of endowment)

**\*\*Fair calculation:\*\***

The total pot is now \$12.32 (\$1.34 they kept + \$10.98 I received)

If we split the gains from cooperation:

- Original total: \$10 (my \$5 + their \$5)
- Current total: \$12.32
- Surplus created: \$2.32
- Fair split would give each person \$1.16 gain
- Sender would end at \$6.16, I'd end at \$6.16

- I need to return:  $\$6.16 - \$1.34 = \$4.82$

But let me think about what's fair based on the trust shown:

- They sent 73% of their money
- If I return about 50% of what I received, that's \$5.49
- That would give them  $\$1.34 + \$5.49 = \$6.83$  total

...

**usage:**

```
{
 "input_tokens": 1112,
 "output_tokens": 833,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$5.5. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$15.05.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: I sent \$3 and received \$5.56 back - good cooperation!
- Rounds 2-3: As receiver, I tried to return fairly and reward trust
- Total earnings: \$15.05

**\*\*Considerations for this round:\*\***

- I'm paired with a new co-player each round
- My Round 1 send of \$3 worked well
- The most generous player sent me \$3.66, showing high trust pays off
- Sending more creates more total value (3x multiplier)
- I want to encourage cooperative norms in the pool

**\*\*Strategy:\*\***

I'll send a bit more than Round 1 to signal strong cooperation while managing risk. This shows I'm willing to trust and invest in mutual gains.

```
'''json
{"send": 3.5}
'''
```

**Reasoning:**

Now I'm the SENDER again in Round 4.

Let me review my history:

- Round 1 (SENDER): Sent \$3, received back \$5.56, payoff \$7.56
- Round 2 (RECEIVER): Received \$5.01 (from \$1.67 sent), returned \$3.00, payoff \$2.01
- Round 3 (RECEIVER): Received \$10.98 (from \$3.66 sent), returned \$5.50, payoff \$5.48

This is a new co-player each round in this multi-agent setting.

My experience so far:

- When I sent \$3, I got back \$5.56 - good return!
- When others sent to me, I tried to be fair/cooperative
- The player who sent \$3.66 was very trusting

For this round, I need to decide how much to send to a new co-player.

Considerations:

- I've seen good cooperation so far (my Round 1 partner returned well)
- I want to continue building cooperative patterns
- Sending more creates more value (3x multiplier)
- But I'm paired with a new player each time, so no direct reputation building

Strategy:

- In Round 1, I sent \$3 and it worked well
- I could send more to signal strong cooperation
- Or stick with \$3 as a moderate cooperative amount

Given that I've had positive experiences and want to continue fostering cooperation, I'll send a bit more this time - maybe \$3.5 or \$4. This shows I'm willing to trust while not going all-in.

Let me go with \$3.5 - it's between my previous \$3 send and the very generous \$3.66 I received in Round 3.

Actually, thinking about it more: the player who sent me \$3.66 showed strong trust and I rewarded it. If I want to encourage that kind of cooperation, I should also send generously when I'm the sender. Let me send \$4.

...

```
Usage: {
 "input_tokens": 1475,
 "output_tokens": 749,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_5 - Receiver - dyad\_4

Timestamp: 2026-08-12T09:22:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_5"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_5": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 4,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round as a sender, I need to balance trust and caution since I don't know my co-player yet.

The key consideration is that sending money creates value through the 3x multiplier. If I send \$3:

- I keep \$2
- The receiver gets \$9 ( $3 \times \$3$ )
- If they return even a moderate amount (say \$3-4), we both benefit compared to sending nothing

This shows willingness to cooperate while maintaining some security. It's a reasonable opening move that could encourage reciprocity.

```
'''json
{"send": 3}
'''
```

#### reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategy:

##### Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- There's communication noise, so amounts may differ from what was actually sent/returned
- This is a multi-agent setting with 8 agents, randomized pairings each round
- I'll play both sender and receiver roles across rounds

##### Strategic considerations:

1. **Trust and cooperation**: If I send money, it creates value (3x multiplier). If the receiver is cooperative, they might return a fair share, making both of us better off.
2. **Risk**: The receiver could keep everything. If I send \$5, they get \$15, and might return nothing, leaving me with \$0.
3. **First round dynamics**: This is the first round, so there's no history to base decisions on. I don't know how cooperative my co-player will be.

4. **\*\*Future rounds\*\***: Since this is a multi-round game with different partners, building a reputation for cooperation might matter, though the anonymity and random pairing complicate this.
5. **\*\*Communication noise\*\***: This adds uncertainty - even if someone tries to be fair, the noise might make it seem otherwise.
6. **\*\*Optimal cooperation\*\***: If both parties cooperate optimally, I could send all \$5, receiver gets \$15, and if they split fairly, they might return around \$7-10, making us both better off than the initial \$5.

Common strategies in trust games:

- Send nothing (risk-averse, guarantees keeping \$5)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - full trust, maximizes potential gains

For the first round with an unknown partner, a moderate approach might be reasonable. Sending about \$3-4 shows willingness to cooperate while not being completely vulnerable.

Since this is a repeated game with multiple rounds and different partners, demonstrating cooperative behavior early could encourage reciprocity....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 768,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$5.6899999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.09, so you received \$6.27. How much do you return? (0-6.27)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.09 out of \$5 (about 42%)
- Showed moderate trust

**\*\*My considerations:\*\***

- In Round 1, my partner returned about 41% of what they received
- Reciprocating fairly would mean returning a similar proportion
- This encourages future cooperation

Returning approximately 45% of the \$6.27 I received (\$2.82) seems fair. This rewards their trust, gives them a modest profit, and keeps a reasonable amount for myself.

```
'''json
{"return": 2.82}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$3
- It became \$9
- They returned \$3.69
- My payoff: \$5 - \$3 + \$3.69 = \$5.69

Now Round 2 (as RECEIVER):

- My co-player sent me \$2.09
- I received \$6.27 (which is 3 × \$2.09)
- I need to decide how much to return

Key considerations:

1. My previous co-player returned about 41% of what they received (\$3.69 out of \$9)
2. This current co-player sent \$2.09 out of \$5, which is about 42% - showing moderate trust
3. There's mention of "communication noise" - the amounts we see may differ slightly from actual amounts

A fair reciprocity strategy would be to return roughly the same proportion that was returned to me, or perhaps slightly reward the trust shown by sending a decent amount.

If I return about 40% of \$6.27, that would be around \$2.51.

If I'm more generous and return 45%, that would be around \$2.82.

If I return 50%, that would be \$3.14.

Given that:

- The sender showed trust by sending \$2.09 (42% of their endowment)
- My previous partner returned about 41%
- I want to encourage future cooperation

I'll return around \$2.80-\$3.00, which is roughly 45-48% of what I received. This rewards their trust while keeping a reasonable amount.

Let me go with \$2.80 - this leaves them with approximately \$5 - \$2.09 + \$2.80 = \$5.71, a modest gain, and I keep \$6.27 - \$2.80 = \$3.47, a solid gain for both of us.

usage:

```
{
 "input_tokens": 781,
 "output_tokens": 590,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.09, it became \$6.27, and you returned \$2.82. Your visible payoff was \$3.4499999999999997.

Your total visible earnings across all rounds are \$9.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my position:

**\*\*Previous experience:\*\***

- Round 1: Sent \$3, received \$3.69 back ? good outcome
- Round 2: As receiver, returned a fair amount (\$2.82 on \$6.27)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- No history with this specific agent
- Need to signal willingness to cooperate

Since I had success with sending \$3 in Round 1 (received \$3.69 back, net gain of \$0.69), and it establishes a pattern of moderate trust, I'll use the same strategy with this new co-player.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my experience so far:

Round 1: I was SENDER

- Sent \$3, they received \$9
- They returned \$3.69 (about 41% of what they received)
- My payoff: \$2 (kept) + \$3.69 (returned) = \$5.69

Round 2: I was RECEIVER  
- They sent \$2.09, I received \$6.27  
- I returned \$2.82 (about 45% of what I received)  
- My payoff: \$6.27 - \$2.82 = \$3.45

Total earnings: \$9.14

Now Round 3: I'm SENDER again with a new co-player (since agents are randomized each round)

What should I consider:

1. This is a new opponent - I don't know their strategy
2. In Round 1, sending \$3 worked reasonably well (got back \$3.69, netted \$5.69 vs \$5 if I sent nothing)
3. The multiplier effect means cooperation is beneficial
4. I should probably maintain a consistent strategy early on

My Round 1 strategy of sending \$3 was a moderate trust signal that worked well. Since I'm facing a new opponent, I should probably use a similar approach - not too cautious, not too risky.

I'll send \$3 again as my baseline strategy for first encounters with new co-players.

usage:

```
{
 "input_tokens": 1050,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.37 to you. Your visible payoff was \$7.37.

Your total visible earnings across all rounds are \$16.509999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.94 out of \$5 (nearly 59%)
- Showed strong trust

**\*\*Recent patterns:\*\***

- Round 3: My partner was very generous, returning \$5.37 (60% of \$9)
- Round 2: I returned about 45% as receiver
- This co-player is being quite trusting

**\*\*Fair reciprocity:\*\***

Returning around 48% of the \$8.82 I received would be fair and encourage future cooperation. This gives my co-player a decent profit while rewarding their trust.

```
'''json
{"return": 4.20}
'''
```

Reasoning:

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER\*\***

- My co-player sent \$2.94 (about 59% of their \$5)
- I received \$8.82 (3x multiplier)
- Need to decide how much to return

**\*\*Recent history:\*\***

- Round 3: As sender, I sent \$3, received \$5.37 back (about 60% of the \$9 they got) - very gen



erous!

- Round 2: As receiver, I returned \$2.82 on \$6.27 (about 45%)
- Round 1: As sender, sent \$3, received \$3.69 back (about 41% of the \$9)

**\*\*Analysis of current round:\*\***

- Co-player sent \$2.94 (quite generous - nearly 60% of their endowment)
- This shows significant trust
- They'll end up with:  $\$5 - \$2.94 + \text{return} = \$2.06 + \text{return}$

**\*\*Fairness considerations:\*\***

- If I return about 45-50% like in Round 2: ~\$3.96 - \$4.41
- If I match the generosity of Round 3's partner (60%): ~\$5.29
- Co-player sacrificed a lot to send \$2.94

I should reciprocate fairly. They sent nearly 60% of their endowment, showing strong trust. I should return a reasonable amount that:

1. Rewards their trust
2. Gives them a profit
3. Maintains cooperation

Returning around \$4.00-\$4.50 seems fair:

- Gives them a total of ~\$6.06-\$6.56
- Nets them \$1.06-\$1.56 profit
- I keep \$4.32-\$4.82
- Return rate of ~45-51%

...

```
Usage: {
 "input_tokens": 1348,
 "output_tokens": 615,
 "reasoning_tokens": 0
}
```

**ROUND 5**

Roles: Agent\_1=Receiver, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Sender, Agent\_5=Receiver, Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Sender

**Pairings:**

dyad\_1: Agent\_3=Sender, Agent\_5=Receiver  
 dyad\_2: Agent\_7=Sender, Agent\_6=Receiver  
 dyad\_3: Agent\_8=Sender, Agent\_1=Receiver  
 dyad\_4: Agent\_4=Sender, Agent\_2=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_3 sent \$2.5; received \$7.5; Agent\_5 returned \$2.3; payoffs Agent\_3=\$4.8, Agent\_5=\$5.2; noise/visible: sent visible=\$1.7, received visible=\$5.1, returned visible=\$2.24  
 Action validation: {"returned": {"amount": 2.3, "clamped": false, "max": 7.5, "min": 0, "raw": 2.3}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

dyad\_2: Agent\_7 sent \$3; received \$9; Agent\_6 returned \$3.85; payoffs Agent\_7=\$5.85, Agent\_6=\$5.15; noise/visible: sent visible=\$2.81, received visible=\$8.43, returned visible=\$4.06  
 Action validation: {"returned": {"amount": 3.85, "clamped": false, "max": 9.0, "min": 0, "raw": 3.85}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_3: Agent\_8 sent \$3; received \$9; Agent\_1 returned \$4.3; payoffs Agent\_8=\$6.3, Agent\_1=\$4.7; noise/visible: sent visible=\$2.89, received visible=\$8.67, returned visible=\$5.2  
 Action validation: {"returned": {"amount": 4.3, "clamped": false, "max": 9.0, "min": 0, "raw": 4.3}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_4: Agent\_4 sent \$3; received \$9; Agent\_2 returned \$4.7; payoffs Agent\_4=\$6.7, Agent\_2=\$4.3; noise/visible: sent visible=\$2.92, received visible=\$8.76, returned visible=\$4.38  
 Action validation: {"returned": {"amount": 4.7, "clamped": false, "max": 9.0, "min": 0, "raw": 4.7}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 25.52,  
 "Agent\_2": 26.41,  
 "Agent\_3": 27.94,  
 "Agent\_4": 27.2,  
 "Agent\_5": 30.43,  
 "Agent\_6": 25.65,  
 "Agent\_7": 26.549999999999997,  
 "Agent\_8": 27.3

}

Visible balances after round: {

```

"Agent_1": 26.16,
"Agent_2": 22.37,
"Agent_3": 24.880000000000003,
"Agent_4": 27.599999999999998,
"Agent_5": 23.93,
"Agent_6": 26.47,
"Agent_7": 26.2,
"Agent_8": 27.82
}

```

## LLM CALLS

### GAME - Agent\_3 - Sender - dyad\_1

Timestamp: 2026-08-12T09:23:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_5"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.35, that is 47.0% of their total endowment. You received \$7.050000000000001. How much do you return to the sender? (0-7.050000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.35 (47% of their endowment)
- I received \$7.05 from this
- The sender currently has \$2.65 remaining

I want to reciprocate this trust fairly. A reasonable approach is to aim for equal final payoffs since we're cooperating:

- Total available: \$2.65 (sender's remainder) + \$7.05 (my received amount) = \$9.70
- Equal split:  $\$9.70 \div 2 = \$4.85$  each
- Return needed:  $\$4.85 - \$2.65 = \$2.20$

This ensures we both benefit equally from the cooperation, rewarding the sender's trust while keeping the gains fair.

{'return': 2.20}

**reasoning:**

Let me understand the situation:

- I am the RECEIVER in Round 1
- The sender sent \$2.35 (47% of their \$5 endowment)
- This was multiplied by 3x, so I received \$7.05
- I need to decide how much to return (between \$0 and \$7.05)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. There's communication noise mentioned, so amounts may differ slightly
3. This is a multi-round game with 8 agents and rotating pairings
4. Building reputation and reciprocity is important

The sender sent 47% of their endowment. If I want to be fair and reciprocal:

- The sender started with \$5, sent \$2.35, and has \$2.65 left
- I received \$7.05
- If I don't return anything, sender gets \$2.65, I get \$7.05
- A fair split might be to aim for equal payoffs or proportional returns

Common strategies in trust games:

1. Return nothing (selfish)
2. Return the original amount sent (\$2.35)
3. Return enough to split the gains equally
4. Return a proportion similar to what was sent

Let's calculate equal payoffs:

- Total money: \$5 (initial sender) + \$7.05 (tripled) = \$12.05? No wait...
- Initial endowment for sender: \$5
- Sent: \$2.35 (tripled to \$7.05 for receiver)
- Sender currently has:  $\$5 - \$2.35 = \$2.65$

- Receiver has: \$7.05
- Total in system:  $\$2.65 + \$7.05 = \$9.70$

For equal payoffs: each should get  $\$9.70/2 = \$4.85$

- So I'd need to return \$2.20 to the sender, leaving me with \$4.85

I could also return the original \$2.35 they sent, which would give them \$5.00 (breaking even) while I keep \$4.70. Or I could mirror their generosity by returning 47% of what I received-about \$3.31-which acknowledges their trust while still letting me benefit from the multiplication.

The equal payoff approach feels right though. Returning \$2.20 means we both end up with \$4.85, which feels like genuine reciprocity for the trust they showed....

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 1079,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.35, it became \$7.050000000000001, and you returned \$2.2. Your visible payoff was \$4.8500000000000005.

Your total visible earnings across all rounds are \$4.8500000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my previous round, the sender showed trust at 47% (\$2.35) and I reciprocated fairly, resulting in equal payoffs of \$4.85 each.

As the sender now, I'll show similar moderate trust. Sending around 50% demonstrates cooperation while managing risk.

```
{'send': 2.50}
```

**reasoning:**

Let me think about this situation:

Round 2: I am now the SENDER

- I have \$5
- I need to decide how much to send
- The amount I send will be tripled
- The receiver will decide how much to return

In the previous round:

- I was the receiver
- The sender sent \$2.35 (47% of their endowment)
- I received \$7.05
- I returned \$2.20
- My payoff was \$4.85
- The sender's payoff would have been  $\$2.65 + \$2.20 = \$4.85$

So in the last round, I reciprocated fairly by aiming for equal payoffs.

Now I'm the sender. I should consider:

1. Building trust by sending a reasonable amount
2. The sender in the previous round sent 47% (\$2.35 out of \$5)
3. I reciprocated fairly, so hopefully this receiver will do the same

Since the previous sender showed moderate trust at 47%, I could do something similar. Sending around \$2.35-\$2.50 seems reasonable.

However, I could also think strategically:

- If I send \$2.50, the receiver gets \$7.50
- If they split equally, they'd return \$2.50 to give us each \$5

- If I send \$3.00, the receiver gets \$9.00
- Equal split would mean returning \$3.50 for payoffs of \$5.50 each

Let me try sending \$2.50 as a moderate trust signal, similar to what the previous sender did with me.

Actually, let me reconsider. I showed I'm cooperative by returning fairly in the previous round. I could send a bit more to maximize joint welfare if the receiver reciprocates.

If I send \$3:

- I keep \$2
- Receiver gets \$9
- If they return \$3.50, I get \$5.50 total, they get \$5.50
- Joint welfare: \$11...

**usage:**

```
{
 "input_tokens": 859,
 "output_tokens": 713,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.43 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$9.780000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience:

- Round 1 (receiver): Sender sent 47%, I reciprocated fairly
- Round 2 (sender): I sent 50%, received fair return of \$2.43

The moderate trust strategy (around 50%) has been working well with fair reciprocation from co-players. I'll continue this approach.

{'send': 2.50}

**reasoning:**

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$2.43
- My payoff: \$5 - \$2.50 + \$2.43 = \$4.93
- Their payoff: \$7.50 - \$2.43 = \$5.07

The receiver returned a fair amount - roughly aimed at equal splits. The return rate was \$2.43/\$7.50 = 32.4%, which gave us both similar final payoffs.

Now I'm a sender again with a new co-player. I've seen:

- Round 1 (as receiver): Sender sent 47%, I returned fairly
- Round 2 (as sender): I sent 50%, received fair return

Both rounds showed moderate trust and fair reciprocation. I should continue this cooperative strategy. Sending around 50% seems to be working well.

{'send': 2.50}

**usage:**

```
{
 "input_tokens": 1060,
 "output_tokens": 327,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.78 to you. Your visible payoff was \$5.279999999999999.

Your total visible earnings across all rounds are \$15.06.

This round, you are the RECEIVER against your current co-player. They sent you \$2.58, so you received \$7.74. How much do you return? (0-7.74)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.58 (51.6% of endowment), showing solid trust
- I received \$7.74
- Sender currently has \$2.42

Based on my positive experiences with fair reciprocation, I'll aim for equal final payoffs:

- Total:  $\$2.42 + \$7.74 = \$10.16$
- Equal split:  $\$5.08$  each
- Return needed:  $\$2.66$

{'return': 2.66}

**reasoning:**

I'm the receiver now. Let me analyze:

- Sender sent \$2.58 (51.6% of their endowment)
- I received \$7.74 (after 3x multiplier)
- Sender currently has:  $\$5 - \$2.58 = \$2.42$

My recent experience shows:

- Round 2: I sent \$2.50, got back \$2.43
- Round 3: I sent \$2.50, got back \$2.78

Both returns were generous and aimed at fair splits. I should continue this reciprocal fairness.

For equal final payoffs:

- Total available:  $\$2.42$  (sender's remainder) +  $\$7.74$  (my received) =  $\$10.16$
- Equal split:  $\$10.16 / 2 = \$5.08$  each
- Return needed:  $\$5.08 - \$2.42 = \$2.66$

This would give us both \$5.08, which is fair and reciprocates the sender's trust at 51.6%.

Let me verify:

- Sender final:  $\$2.42 + \$2.66 = \$5.08$
- My final:  $\$7.74 - \$2.66 = \$5.08$

Perfect. I'll return \$2.66.

**usage:**

```
{
 "input_tokens": 1287,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.58, it became \$7.74, and you returned \$2.66. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my track record:

- Sending around 50% has consistently resulted in fair reciprocation
- Other agents are also showing similar moderate trust levels
- This strategy has been mutually beneficial

I'll continue with this successful approach.

```
{'send': 2.50}
```

#### Reasoning:

Let me review my history:

- Round 1 (receiver): Received from sender who sent \$2.35 (47%), returned \$2.20, payoff \$4.85
- Round 2 (sender): Sent \$2.50 (50%), got back \$2.43, payoff \$4.93
- Round 3 (sender): Sent \$2.50 (50%), got back \$2.78, payoff \$5.28
- Round 4 (receiver): Received from sender who sent \$2.58 (51.6%), returned \$2.66, payoff \$5.08
- Round 5 (sender): ???

My strategy of sending around 50% has been working well - I've been getting fair returns. The pattern shows:

- When I send ~50% (\$2.50), I get back \$2.43-\$2.78
- Others are sending similar amounts (47-52%)
- Fair reciprocation is the norm

I should continue with this moderate trust approach around 50%.

```
{'send': 2.50}
```

Usage: {

```
 "input_tokens": 1524,
 "output_tokens": 316,
 "reasoning_tokens": 0
```

}

#### GAME - Agent\_5 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:23:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_5"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
```

}

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as a sender, I need to balance trust and caution since I don't know my co-player yet.

The key consideration is that sending money creates value through the 3x multiplier. If I send \$3:

- I keep \$2
- The receiver gets \$9 (3 × \$3)
- If they return even a moderate amount (say \$3-4), we both benefit compared to sending nothing

This shows willingness to cooperate while maintaining some security. It's a reasonable opening move that could encourage reciprocity.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategy:

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- There's communication noise, so amounts may differ from what was actually sent/returned
- This is a multi-agent setting with 8 agents, randomized pairings each round
- I'll play both sender and receiver roles across rounds

Strategic considerations:



1. **\*\*Trust and cooperation\*\***: If I send money, it creates value (3x multiplier). If the receiver is cooperative, they might return a fair share, making both of us better off.
2. **\*\*Risk\*\***: The receiver could keep everything. If I send \$5, they get \$15, and might return nothing, leaving me with \$0.
3. **\*\*First round dynamics\*\***: This is the first round, so there's no history to base decisions on. I don't know how cooperative my co-player will be.
4. **\*\*Future rounds\*\***: Since this is a multi-round game with different partners, building a reputation for cooperation might matter, though the anonymity and random pairing complicate this.
5. **\*\*Communication noise\*\***: This adds uncertainty - even if someone tries to be fair, the noise might make it seem otherwise.
6. **\*\*Optimal cooperation\*\***: If both parties cooperate optimally, I could send all \$5, receiver gets \$15, and if they split fairly, they might return around \$7-10, making us both better off than the initial \$5.

Common strategies in trust games:

- Send nothing (risk-averse, guarantees keeping \$5)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - full trust, maximizes potential gains

For the first round with an unknown partner, a moderate approach might be reasonable. Sending about \$3-4 shows willingness to cooperate while not being completely vulnerable. Since this is a repeated game with multiple rounds and different partners, demonstrating cooperative behavior early could encourage reciprocity....

usage:

```
{
 "input_tokens": 500,
 "output_tokens": 768,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$5.6899999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.09, so you received \$6.27. How much do you return? (0-6.27)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.09 out of \$5 (about 42%)
- Showed moderate trust

**\*\*My considerations:\*\***

- In Round 1, my partner returned about 41% of what they received
- Reciprocating fairly would mean returning a similar proportion
- This encourages future cooperation

Returning approximately 45% of the \$6.27 I received (\$2.82) seems fair. This rewards their trust, gives them a modest profit, and keeps a reasonable amount for myself.

```
'''json
{"return": 2.82}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$3
- It became \$9
- They returned \$3.69
- My payoff:  $\$5 - \$3 + \$3.69 = \$5.69$

Now Round 2 (as RECEIVER):

- My co-player sent me \$2.09
- I received \$6.27 (which is  $3 \times \$2.09$ )
- I need to decide how much to return

Key considerations:

1. My previous co-player returned about 41% of what they received (\$3.69 out of \$9)
2. This current co-player sent \$2.09 out of \$5, which is about 42% - showing moderate trust
3. There's mention of "communication noise" - the amounts we see may differ slightly from actual amounts

A fair reciprocity strategy would be to return roughly the same proportion that was returned to me, or perhaps slightly reward the trust shown by sending a decent amount.

If I return about 40% of \$6.27, that would be around \$2.51.

If I'm more generous and return 45%, that would be around \$2.82.

If I return 50%, that would be \$3.14.

Given that:

- The sender showed trust by sending \$2.09 (42% of their endowment)
- My previous partner returned about 41%
- I want to encourage future cooperation

I'll return around \$2.80-\$3.00, which is roughly 45-48% of what I received. This rewards their trust while keeping a reasonable amount.

Let me go with \$2.80 - this leaves them with approximately  $\$5 - \$2.09 + \$2.80 = \$5.71$ , a modest gain, and I keep  $\$6.27 - \$2.80 = \$3.47$ , a solid gain for both of us.

usage:

```
{
 "input_tokens": 781,
 "output_tokens": 590,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.09, it became \$6.27, and you returned \$2.82. Your visible payoff was \$3.4499999999999997.

Your total visible earnings across all rounds are \$9.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my position:

**\*\*Previous experience:\*\***

- Round 1: Sent \$3, received \$3.69 back ? good outcome
- Round 2: As receiver, returned a fair amount (\$2.82 on \$6.27)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- No history with this specific agent
- Need to signal willingness to cooperate

Since I had success with sending \$3 in Round 1 (received \$3.69 back, net gain of \$0.69), and it establishes a pattern of moderate trust, I'll use the same strategy with this new co-player.

```json

```
    {"send": 3}
    '''
```

reasoning:

Let me analyze my experience so far:

Round 1: I was SENDER

- Sent \$3, they received \$9
- They returned \$3.69 (about 41% of what they received)
- My payoff: \$2 (kept) + \$3.69 (returned) = \$5.69

Round 2: I was RECEIVER

- They sent \$2.09, I received \$6.27
- I returned \$2.82 (about 45% of what I received)
- My payoff: \$6.27 - \$2.82 = \$3.45

Total earnings: \$9.14

Now Round 3: I'm SENDER again with a new co-player (since agents are randomized each round)

What should I consider:

1. This is a new opponent - I don't know their strategy
2. In Round 1, sending \$3 worked reasonably well (got back \$3.69, netted \$5.69 vs \$5 if I sent nothing)
3. The multiplier effect means cooperation is beneficial
4. I should probably maintain a consistent strategy early on

My Round 1 strategy of sending \$3 was a moderate trust signal that worked well. Since I'm facing a new opponent, I should probably use a similar approach - not too cautious, not too risky.

I'll send \$3 again as my baseline strategy for first encounters with new co-players.

usage:

```
{
  "input_tokens": 1050,
  "output_tokens": 492,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.37 to you. Your visible payoff was \$7.37.

Your total visible earnings across all rounds are \$16.509999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****My co-player's decision:****

- Sent \$2.94 out of \$5 (nearly 59%)
- Showed strong trust

****Recent patterns:****

- Round 3: My partner was very generous, returning \$5.37 (60% of \$9)
- Round 2: I returned about 45% as receiver
- This co-player is being quite trusting

****Fair reciprocity:****

Returning around 48% of the \$8.82 I received would be fair and encourage future cooperation. This gives my co-player a decent profit while rewarding their trust.

```
'''json
{"return": 4.20}
'''
```

reasoning:

Let me analyze this situation:

****Round 4 - I'm the RECEIVER****

- My co-player sent \$2.94 (about 59% of their \$5)
- I received \$8.82 (3x multiplier)
- Need to decide how much to return

****Recent history:****

- Round 3: As sender, I sent \$3, received \$5.37 back (about 60% of the \$9 they got) - very generous!
- Round 2: As receiver, I returned \$2.82 on \$6.27 (about 45%)
- Round 1: As sender, sent \$3, received \$3.69 back (about 41% of the \$9)

****Analysis of current round:****

- Co-player sent \$2.94 (quite generous - nearly 60% of their endowment)
- This shows significant trust
- They'll end up with: \$5 - \$2.94 + return = \$2.06 + return

****Fairness considerations:****

- If I return about 45-50% like in Round 2: ~\$3.96 - \$4.41
- If I match the generosity of Round 3's partner (60%): ~\$5.29
- Co-player sacrificed a lot to send \$2.94

I should reciprocate fairly. They sent nearly 60% of their endowment, showing strong trust. I should return a reasonable amount that:

1. Rewards their trust
2. Gives them a profit
3. Maintains cooperation

Returning around \$4.00-\$4.50 seems fair:

- Gives them a total of ~\$6.06-\$6.56
- Nets them \$1.06-\$1.56 profit
- I keep \$4.32-\$4.82
- Return rate of ~45-51%

...

usage:

```
{
  "input_tokens": 1348,
  "output_tokens": 615,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.94, it became \$8.82, and you returned \$4.2. Your visible payoff was \$4.62.

Your total visible earnings across all rounds are \$21.13.

This round, you are the RECEIVER against your current co-player. They sent you \$1.7, so you received \$5.1. How much do you return? (0-5.1)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****My co-player's decision:****

- Sent \$1.70 out of \$5 (34%)
- More cautious than previous partners

****My pattern as receiver:****

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)

****Fair reciprocity:****

Returning around 45% of the \$5.10 I received maintains consistency and fairness. This gives my co-player a modest profit while rewarding their trust, even though they were more cautious.

```json

```
{"return": 2.30}
,,,
```

#### Reasoning:

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- This is relatively cautious/low trust

**\*\*Recent patterns from my play:\*\***

- Round 2: Returned \$2.82 on \$6.27 (45%)
- Round 4: Returned \$4.20 on \$8.82 (48%)

**\*\*Considerations:\*\***

- They showed less trust by sending only \$1.70
- However, I should still be fair to encourage cooperation
- Returning around 40-45% would be consistent with my previous behavior

Returning about 45% of \$5.10 = ~\$2.30

This gives them back a reasonable amount, rewards some level of trust, and maintains consistency in my reciprocity rate.

Let me calculate:

- They sent \$1.70, kept \$3.30
- I received \$5.10
- If I return \$2.30:
  - They end with:  $\$3.30 + \$2.30 = \$5.60$  (gain of \$0.60)
  - I end with:  $\$5.10 - \$2.30 = \$2.80$

That seems fair. They made a small profit for trusting me.

```
Usage: {
 "input_tokens": 1636,
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_7 - Sender - dyad\_2

Timestamp: 2026-08-12T09:23:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_6"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_6": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance building trust with managing risk since I have no history with this receiver.

**\*\*My reasoning:\*\***

- Sending money creates value (3x multiplier means potential gains for both of us)
- This is round 1, so I want to signal cooperation
- I should send enough to be meaningful but not risk everything
- If the receiver reciprocates fairly, we both benefit

I'll send a moderate amount to test cooperation while limiting downside risk.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. **\*\*Game Structure:\*\***

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- My payoff: \$5 - sent + returned

- Receiver's payoff: 3x sent - returned

2. **\*\*Strategic Considerations:\*\***

- If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 and they keep \$7.50. Total: \$15
- The game creates surplus through cooperation (up to \$15 total vs \$5 if no trust)
- This is a trust game - I need to trust the receiver will return some amount

3. **\*\*First Round Considerations:\*\***

- No history with this partner
- This is a multi-round game with different partners
- Building a reputation for cooperation might matter
- The note about "communication noise" suggests amounts might be misperceived, which could affect trust dynamics

4. **\*\*Common Strategies:\*\***

- **\*\*Full trust:\*\*** Send \$5, hope for good return
- **\*\*Moderate trust:\*\*** Send \$2-3, test the waters
- **\*\*No trust:\*\*** Send \$0, guarantee \$5
- **\*\*Signal cooperation:\*\*** Send a meaningful amount to show willingness to cooperate

5. **\*\*Rational Analysis:\*\***

- If the receiver is purely self-interested and this is a one-shot game, they should return \$0
- Knowing this, I should send \$0
- However, in repeated games or with reciprocity norms, cooperation can emerge
- Many people exhibit fairness preferences and reciprocity

6. **\*\*My Decision:\*\***

Since this is the opening move in a multi-round game, a moderate cooperative signal seems prudent. I'll send \$3-enough to demonstrate trust and give the receiver \$9 to work with, but not so much that I'm exposed to catastrophic loss if they keep everything. Even a modest return of \$3-4 would benefit us both compared to the baseline of keeping \$5....

usage:

```
{
 "input_tokens": 500,
 "output_tokens": 836,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.56 to you. Your visible payoff was \$7.56.

Your total visible earnings across all rounds are \$7.56.

This round, you are the RECEIVER against your current co-player. They sent you \$1.67, so you received \$5.01. How much do you return? (0-5.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.67 (33% of their endowment)
- I received \$5.01 after the 3x multiplier
- They currently have \$3.33 remaining

**\*\*My considerations:\*\***

- They showed moderate trust by sending \$1.67
- The multiplier created \$3.34 in surplus value
- A fair approach would be to share this surplus roughly equally
- In Round 1, my partner returned ~62% and that felt cooperative

**\*\*Fair return calculation:\*\***

If we split the surplus equally:

- Each should gain ~\$1.67 from cooperation

- Sender ending:  $\$5 - \$1.67 + \text{return} = \$6.67$  desired
- Return needed:  $\$3.34$

However, that might be overly generous given their modest send. I'll return about 60% to reward their cooperation while keeping reasonable value.

```
'''json
{"return": 3.00}
'''
```

#### reasoning:

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent  $\$3.0$
- It became  $\$9.0$  for the receiver
- They returned  $\$5.56$  to me
- I ended with:  $\$5 - \$3 + \$5.56 = \$7.56$
- Receiver ended with:  $\$9 - \$5.56 = \$3.44$

That was actually quite generous of them! They returned more than half.

Round 2 (current):

- I am RECEIVER
- They sent me  $\$1.67$
- I received  $\$5.01$  (which is  $3 \times \$1.67$ )
- I need to decide how much to return ( $0-5.01$ )

Key considerations:

1. My previous partner was generous, returning  $\$5.56$  out of  $\$9$  (62%)
2. This is a different player (multi-agent setting with randomized pairings)
3. They sent me  $\$1.67$  out of  $\$5$ , which is 33.4% - a moderate amount
4. The sender keeps:  $\$5 - \$1.67 = \$3.33$  before my return
5. I should reciprocate fairly to encourage future cooperation

Let me think about fair returns:

- If I return half ( $\$2.50$ ), they end with  $\$3.33 + \$2.50 = \$5.83$ , I keep  $\$2.51$
- If I return around what gives us equal final payoffs:
  - They have  $\$3.33 + \text{return}$
  - I have  $\$5.01 - \text{return}$
  - Equal:  $\$3.33 + x = \$5.01 - x$
  - $2x = \$1.68$
  - $x = \$0.84$

This would give us both around  $\$4.17$

- A proportional return similar to what I received (62% of what I got):
  - $0.62 \times \$5.01 = \$3.11$
  - They'd end with:  $\$3.33 + \$3.11 = \$6.44$
  - I'd keep:  $\$5.01 - \$3.11 = \$1.90$

A 50% return of around  $\$2.50$  seems like the right balance-it rewards their cooperation without being excessive, and it's more moderate than matching their 33% send rate since they're a different player than before.

If I send back  $\$2.50$ , they'd gain  $\$0.83$  while I'd be left with  $\$2.51$ . The math shows we'd create  $\$8.34$  in total value, which is less than the  $\$10$  we'd have if neither of us participated. The real question is what surplus we're actually generating from this exchange and whether my return is proportional to their initial trust.

The multiplier created  $\$3.34$  in extra value ( $\$5.01$  received minus  $\$1.67$  sent). If we split that gain equally, each of us should pocket  $\$1.67$ . That means I should return  $\$3.34$  to bring them to  $\$6.67$  total, which gives us both the same  $\$1.67$  gain from the transaction.

...

#### usage:

```
{
 "input_tokens": 754,
 "output_tokens": 1445,
 "reasoning_tokens": 0
}
```

Message 6 role=user



**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.67, it became \$5.01, and you returned \$3.0. Your visible payoff was \$2.01.  
Your total visible earnings across all rounds are \$9.57.

This round, you are the RECEIVER against your current co-player. They sent you \$3.66, so you received \$10.98. How much do you return? (0-10.98)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant****content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$3.66 (73% of their endowment) - very high trust!
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining
- Total surplus created: \$2.32

**\*\*My considerations:\*\***

- This is a very generous send showing strong cooperation
- They took significant risk by sending most of their money
- I should reward this trust with a fair return
- In Round 1, I received about 62% back on my \$3 send
- In Round 2, I returned 60% on a smaller send

**\*\*Fair return:\*\***

Given their high trust, I'll return approximately 50% of what I received:

- Return \$5.50
- They'll end with \$6.84 (profit of \$1.84)
- I'll end with \$5.48 (profit of \$0.48)
- This rewards their cooperation while both of us profit

```
'''json
{"return": 5.50}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current Situation:\*\***

- My co-player sent me \$3.66 (73% of their endowment)
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining (\$5 - \$3.66)
- This is a very generous/trusting send

**\*\*What this shows:\*\***

- High trust from the sender (sending over 70%!)
- They're expecting reciprocity
- The multiplier created significant surplus: \$5.98 (\$10.98 - \$5)

**\*\*Considerations:\*\***

- In Round 1, when I sent \$3 (60%), I received \$5.56 back - that's about 62% return rate
- In Round 2, when someone sent \$1.67 (33%), I returned \$3.00 - that's 60% return rate
- This sender showed much more trust (\$3.66 = 73% of endowment)

**\*\*Fair calculation:\*\***

The total pot is now \$12.32 (\$1.34 they kept + \$10.98 I received)

If we split the gains from cooperation:

- Original total: \$10 (my \$5 + their \$5)
- Current total: \$12.32
- Surplus created: \$2.32
- Fair split would give each person \$1.16 gain
- Sender would end at \$6.16, I'd end at \$6.16
- I need to return: \$6.16 - \$1.34 = \$4.82

But let me think about what's fair based on the trust shown:

- They sent 73% of their money
- If I return about 50% of what I received, that's \$5.49

- That would give them  $\$1.34 + \$5.49 = \$6.83$  total

...

**usage:**

```
{
 "input_tokens": 1112,
 "output_tokens": 833,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$5.5. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$15.05.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: I sent \$3 and received \$5.56 back - good cooperation!
- Rounds 2-3: As receiver, I tried to return fairly and reward trust
- Total earnings: \$15.05

**\*\*Considerations for this round:\*\***

- I'm paired with a new co-player each round
- My Round 1 send of \$3 worked well
- The most generous player sent me \$3.66, showing high trust pays off
- Sending more creates more total value (3x multiplier)
- I want to encourage cooperative norms in the pool

**\*\*Strategy:\*\***

I'll send a bit more than Round 1 to signal strong cooperation while managing risk. This shows I'm willing to trust and invest in mutual gains.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Now I'm the SENDER again in Round 4.

Let me review my history:

- Round 1 (SENDER): Sent \$3, received back \$5.56, payoff \$7.56
- Round 2 (RECEIVER): Received \$5.01 (from \$1.67 sent), returned \$3.00, payoff \$2.01
- Round 3 (RECEIVER): Received \$10.98 (from \$3.66 sent), returned \$5.50, payoff \$5.48

This is a new co-player each round in this multi-agent setting.

My experience so far:

- When I sent \$3, I got back \$5.56 - good return!
- When others sent to me, I tried to be fair/cooperative
- The player who sent \$3.66 was very trusting

For this round, I need to decide how much to send to a new co-player.

Considerations:

- I've seen good cooperation so far (my Round 1 partner returned well)
- I want to continue building cooperative patterns
- Sending more creates more value (3x multiplier)
- But I'm paired with a new player each time, so no direct reputation building

Strategy:

- In Round 1, I sent \$3 and it worked well
- I could send more to signal strong cooperation
- Or stick with \$3 as a moderate cooperative amount

Given that I've had positive experiences and want to continue fostering cooperation, I'll send a bit more this time - maybe \$3.5 or \$4. This shows I'm willing to trust while not going all-in.

Let me go with \$3.5 - it's between my previous \$3 send and the very generous \$3.66 I received in Round 3.

Actually, thinking about it more: the player who sent me \$3.66 showed strong trust and I rewarded it. If I want to encourage that kind of cooperation, I should also send generously when I'm the sender. Let me send \$4.

...

**usage:**

```
{
 "input_tokens": 1475,
 "output_tokens": 749,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$3.59 to you. Your visible payoff was \$5.09.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this round as SENDER:

**\*\*Recent experience:\*\***

- Round 4: I sent \$3.50, received only \$3.59 back - minimal profit (\$5.09)
- This was disappointing compared to Round 1 where I sent \$3 and got \$5.56 back
- But this is a NEW co-player with unknown behavior

**\*\*Considerations:\*\***

- Different co-players show varying levels of cooperation
- Sending less than \$3.50 reduces my risk after last round's poor return
- Still want to signal willingness to cooperate and create value
- \$3 was my initial send and it worked well - proven strategy

**\*\*Strategy:\*\***

I'll return to sending \$3 - a moderate amount that:

- Shows trust and cooperation
- Creates meaningful value (\$9 for receiver)
- Limits downside risk
- Has proven successful before

'''json

{"send": 3}

'''

**Reasoning:**

Looking at my history:

- Round 1: Sent \$3, received \$5.56 back - payoff \$7.56 (good!)
- Round 2: Receiver, returned \$3.00 - payoff \$2.01
- Round 3: Receiver, returned \$5.50 - payoff \$5.48
- Round 4: Sent \$3.5, received \$3.59 back - payoff \$5.09 (disappointing)
- Round 5: Now sender again

Analysis of Round 4:

- I sent \$3.50 (70% of endowment) showing strong trust
- They received \$10.50
- They only returned \$3.59 (34% of what they received)
- I ended with \$5.09, barely breaking even
- This was much less generous than my Round 1 partner who returned \$5.56 on a \$9 receive

This suggests I encountered a less cooperative player. However, I'm now paired with a NEW co-p

layer, so their behavior doesn't necessarily predict this new person's behavior.

Strategic considerations:

- I'm doing reasonably well overall (\$20.14 total)
- Sending creates value through the multiplier
- But I need to balance risk
- Two different responses to my sends: one very good (\$5.56), one mediocre (\$3.59)
- I don't know if there's a pattern or if agents vary widely

I think I should continue showing willingness to cooperate but maybe pull back slightly from the \$3.50 I sent last time since that didn't work out as well. Going back to \$3 seems reasonable - it's a solid middle ground that shows cooperation without overcommitting.

```
Usage: {
 "input_tokens": 1797,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_6 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:23:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_6"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_6": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative dynamic while being somewhat cautious since I don't know my co-player yet.

If I send money, it triples, creating potential mutual benefit. However, there's risk since the receiver's return is voluntary.

I'll send a moderate amount to signal trust and willingness to cooperate, while not being overly vulnerable in this first interaction.

**\*\*Decision:\*\***

```
'''json
{"send": 3}
'''
```

This gives the receiver \$9 to work with. If they're cooperative and return a fair share, we both benefit. If not, I've learned something about the dynamics for future rounds while limiting my downside risk.

**reasoning:**

This is the first round of a trust game, and I'm the sender. I need to think about this strategically.

**Key facts:**

- I have \$5
- Whatever I send gets multiplied by 3
- The receiver decides how much to return
- There may be communication noise
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I don't know who my current co-player is

Since this is the first round, I have no information about how other players behave. The optimal strategy depends on:

1. Whether I expect reciprocity from the receiver
2. Whether I want to establish a reputation (though with hidden names, this is less relevant for future rounds)
3. Risk tolerance

From a purely rational game theory perspective, if this were a one-shot game with no reputation effects, the receiver should return \$0 (keep everything), which means I should send \$0. However, experimental evidence shows people often cooperate in trust games.

Some considerations:

- Sending \$5 maximizes total payoff (\$5 + \$15 total) but requires trust
- Sending \$0 guarantees I keep \$5 but creates no opportunity for mutual gain
- A moderate amount (like \$2-3) balances risk and opportunity

Since this is round 1 and I want to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything immediately

I'll send a moderate amount. Sending \$3 seems reasonable:

- If sent \$3, receiver gets \$9
- If receiver returns half (\$4.50), I end up with \$5 - \$3 + \$4.50 = \$6.50
- If receiver returns nothing, I end up with \$2
- This tests their cooperation while not being overly cautious

Actually, let me reconsider. A common strategy is to send about half to show trust but not complete vulnerability. Sending \$2.50 or \$3 seems reasonable.

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 632,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.25 to you. Your visible payoff was \$6.25.

Your total visible earnings across all rounds are \$6.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.67, so you received \$8.01. How much do you return? (0-8.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.67\*\*** (about 53% of their endowment), which became \$8.01 for me. This is a moderately cooperative move.

**\*\*Comparison to previous round:\*\***

- When I sent \$3.00, I received \$4.25 back (47% of the \$9.00)
- That gave my co-player a reasonable benefit while rewarding my trust

**\*\*My reasoning:\*\***

I want to:

1. Reward the sender's trust with a decent return
2. Maintain cooperative dynamics for future rounds
3. Keep enough to make receiving worthwhile

Returning about \$3.75 seems fair:

- Sender would get: \$2.33 (kept) + \$3.75 = \$6.08 total (gain of ~\$1.08)
- I would get: \$8.01 - \$3.75 = \$4.26 (solid gain)

This is proportionally similar to what I experienced last round.

```
'''json
{"return": 3.75}
'''
```

**reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was SENDER
- I sent \$3.0
- It became \$9.0
- Receiver returned \$4.25

- My payoff:  $\$5 - \$3 + \$4.25 = \$6.25$

This round (Round 2):

- I am RECEIVER
- Sender sent \$2.67
- I received \$8.01 (which is  $\$2.67 \times 3$ )
- I need to decide how much to return

The sender in this round sent \$2.67 out of \$5, which is about 53% of their endowment. This is a fairly cooperative move.

In the previous round, when I sent \$3, I received back \$4.25. My net gain was \$1.25 from the initial \$5.

Let me think about what would be fair:

- The sender gave up \$2.67
- I received \$8.01
- If I want to be cooperative and fair, I should think about:
  1. Returning enough so the sender benefits from sending
  2. Not returning so much that I don't benefit from receiving

A common approach is to split the gains. The total amount now is \$8.01. If we split it equally:

- Each gets \$4.005

But the sender started with \$5 and sent \$2.67, so they have \$2.33 left.

If I return \$4.00, they'd have  $\$2.33 + \$4.00 = \$6.33$  (gain of \$1.33)

I'd keep  $\$8.01 - \$4.00 = \$4.01$  (gain of \$4.01 compared to receiving nothing)

Hmm, that seems imbalanced. Let me think differently.

Another approach: What would make our final payoffs more equal?

- Sender has:  $\$5 - \$2.67 + \text{return} = \$2.33 + \text{return}$
- I have:  $\$8.01 - \text{return}$

For equal payoffs:

$$\$2.33 + \text{return} = \$8.01 - \text{return}$$

$$2 \times \text{return} = \$8.01 - \$2.33 = \$5.68$$

$$\text{return} = \$2.84$$

If I return \$2.84:

- Sender gets:  $\$2.33 + \$2.84 = \$5.17$
- I get:  $\$8.01 - \$2.84 = \$5.17$

That would equalize our outcomes, though it doesn't particularly reward the sender's risk-taking. I should consider what the previous receiver did for comparison.

The prior round showed a return of \$4.25 on \$9.00 received-roughly 47% of what they got. Applying that same rate here:  $\$8.01 \times 0.47 = \$3.76$ . This would give the sender \$6.09 total and leave me with \$4.25, which feels more balanced given the sender's initial trust. But I need to account for the communication noise mentioned in the instructions-the amounts I'm seeing may not match what was actually sent or returned....

usage:

```
{
 "input_tokens": 785,
 "output_tokens": 1226,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.67, it became \$8.01, and you returned \$3.75. Your visible payoff was \$4.26.

Your total visible earnings across all rounds are \$10.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$3.36\*\*** (67% of their endowment), which became \$10.08 for me. This is a notably generous and trusting move.

**\*\*Context from previous rounds:\*\***

- Round 2: Someone sent \$2.67 (53%), I returned \$3.75
- Round 1: I sent \$3.00 (60%), received \$4.25 back

**\*\*My reasoning:\*\***

This sender showed higher trust than my previous co-player (67% vs 53%). I should reward his generosity proportionally.

Returning \$4.75:

- Sender would get: \$1.64 (kept) + \$4.75 = \$6.39 total (gain of \$1.39)
- I would get: \$10.08 - \$4.75 = \$5.33 (good gain)

This provides the sender with a decent reward for their trust while maintaining mutual benefit - both of us come out ahead.

```
'''json
{"return": 4.75}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$3.36 (67% of their \$5 endowment) - this is quite generous/trusting
- I received \$10.08

**\*\*Previous round:\*\***

- Someone sent me \$2.67 (53% of endowment), I returned \$3.75
- That gave them \$2.33 + \$3.75 = \$6.08 total (~\$1.08 gain)
- I kept \$4.26

**\*\*Round 1:\*\***

- I sent \$3.00 (60%), received \$4.25 back
- That gave me \$2.00 + \$4.25 = \$6.25 total

**\*\*My thinking:\*\***

The sender this round was more generous (67% vs 53% in previous round). I should reward their higher trust level.

If I return around \$4.50-5.00:

- Return \$4.50: Sender gets \$1.64 + \$4.50 = \$6.14, I get \$5.58
- Return \$5.00: Sender gets \$1.64 + \$5.00 = \$6.64, I get \$5.08

Actually, let me think about what's fair given they sent 67% of their endowment.

If I return \$5.00:

- Sender: keeps \$1.64, gets back \$5.00 = \$6.64 total (gain of \$1.64)
- Me: receive \$10.08, return \$5.00 = \$5.08 (solid gain)

This seems like a good reciprocal move for their high trust. It's roughly a 50-50 split of the multiplied amount, which rewards their generosity while still benefiting both of us.

Actually, let me reconsider. They sent \$3.36 out of \$5, so they kept \$1.64.

Returning \$5 gives them \$6.64 with a \$1.64 gain, while I keep \$5.08. That's reasonably fair given their high trust. I...

**usage:**

```
{
 "input_tokens": 1163,
 "output_tokens": 921,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**



**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.75. Your visible payoff was \$5.33.  
Your total visible earnings across all rounds are \$15.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant****content:**

Looking at this round as SENDER again:

**\*\*My experience:\*\***

- Round 1: Sent \$3.00, received \$4.25 back ? Good outcome
- Rounds 2-3: As receiver, I've been cooperative, returning fair amounts

**\*\*Current situation:\*\***

- New co-player (different from Round 1)
- I've been part of generally cooperative dynamics

**\*\*My reasoning:\*\***

I had success with sending \$3.00 before, and the cooperative pattern across rounds suggests moderate-to-high sending works well. I'll send slightly more this time to signal trust while building on the positive momentum.

Sending \$3.50 creates \$10.50 for the receiver. If they're reasonably cooperative (returning ~40%), I'd still come out ahead.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

I'm back as the SENDER in round 4. Let me review my experience so far:

Round 1: I sent \$3.00, received \$4.25 back. Payoff: \$6.25  
Round 2: Received \$8.01 (from \$2.67 sent), returned \$3.75. Payoff: \$4.26  
Round 3: Received \$10.08 (from \$3.36 sent), returned \$4.75. Payoff: \$5.33

Total earnings: \$15.84

Now I'm the sender again with a new co-player. What should I send?

Looking at the pattern:

- When I sent \$3.00 (60%), I got a decent return
- When others sent moderate-to-high amounts (\$2.67, \$3.36), I've been fairly cooperative in returning
- The game seems to be working reasonably well with moderate-to-high sending amounts

I should consider:

1. I had a positive experience when I sent \$3.00
2. The cooperative moves I've seen have been between \$2.67-\$3.36
3. I've demonstrated I'm a fair returner, so maybe I can expect reasonable behavior

I think I should send a similar or slightly higher amount than last time. Since I received \$4.25 when I sent \$3.00, that worked out well.

Options:

- Send \$3.00 again (proven strategy)
- Send \$3.50 (show slightly more trust)
- Send \$4.00 (high trust signal)

Given:

- This is a new co-player
- I've been building cooperative patterns
- The game is going well so far

I'll send \$3.50 as a moderately high trust signal, slightly more than my first round, but

not overly risky. This creates \$10.50 for the receiver.

...

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 660,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.55 to you. Your visible payoff was \$6.05.

Your total visible earnings across all rounds are \$21.89.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*My pattern:\*\***

- I've been consistently returning around 45-47% of what I receive
- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)

**\*\*My reasoning:\*\***

This sender showed moderate trust. I should maintain consistency with my previous cooperative behavior.

Returning \$3.85:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of \$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This is about 46% return rate, consistent with my pattern and fair to the sender.

```
'''json
{"return": 3.85}
'''
```

**Reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*Context from previous rounds:\*\***

- Round 1: I sent \$3.00 (60%), received \$4.25 back (47% of \$9)
- Round 2: Someone sent \$2.67 (53%), I returned \$3.75 (47% of \$8.01)
- Round 3: Someone sent \$3.36 (67%), I returned \$4.75 (47% of \$10.08)
- Round 4: I sent \$3.50 (70%), received \$4.55 back (43% of \$10.50)

**\*\*Pattern:\*\***

I've been pretty consistent returning around 45-47% of what I receive. My co-player sent \$2.81, which is moderately cooperative.

Let me calculate:

- 45% of \$8.43 = \$3.79
- 47% of \$8.43 = \$3.96

Returning around \$3.85 would:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of ~\$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This maintains consistency with my previous behavior and rewards the sender proportionally for their trust.

```
Usage: {
 "input_tokens": 1847,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_8 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:23:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_1"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_1": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the SENDER, I need to balance trust and caution since I have no history with this co-player.

The key consideration is that sending money creates value (3x multiplier), but I'm vulnerable to the receiver's decision about how much to return.

A moderate amount seems reasonable for Round 1:

- It signals willingness to cooperate and trust
- It creates meaningful value (\$9 for the receiver)
- It provides some protection against complete loss

I'll send \*\*\$3\*\*.

{'send': 3}

**reasoning:**

This is Round 1 of a multi-task experiment where I'm playing a trust game as the SENDER. Let me think about this situation:

1. I start with \$5
2. I can send any amount from \$0 to \$5
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - amount\_sent + amount\_returned

This is a classic trust game. The optimal strategy depends on:

- Whether I trust the receiver to return a fair amount
- What "fair" means in this context
- The fact that there's communication noise (amounts may differ slightly)
- This is a multi-round game with multiple agents
- Pairings are randomized each round

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I'd end up with \$7.50 total (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- The socially optimal outcome is to send the maximum amount to create the most total value

However, there's risk:

- The receiver might not return anything
- There's communication noise that could affect trust
- This is round 1, so I have no history with this partner

A reasonable first-round strategy might be:

- Send a moderate amount to signal trust and willingness to cooperate
- Not send everything in case the receiver doesn't reciprocate
- Send enough to create meaningful value

Common strategies in trust games:

- Send about half (\$2.50) as a balanced approach
- Send more (\$3-4) to signal strong trust
- Send everything (\$5) to maximize joint payoff

Given this is round 1 and I want to establish a cooperative pattern but also protect against complete exploitation, I'll send \$3. This:

- Creates \$9 for the receiver
- Shows trust and cooperation

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 625,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.46 to you. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$4.46.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 2):\*\***

- My co-player sent me \$3.19 ? I received \$9.57
- This created \$6.38 in surplus value

**\*\*Fair return considerations:\*\***

- The sender took a significant risk sending most of their endowment
- A reasonable approach is to ensure we both benefit from the cooperation
- Returning \$4.00 would give:
  - Sender: \$5 - \$3.19 + \$4.00 = \$5.81
  - Me: \$9.57 - \$4.00 = \$5.57

This creates a balanced outcome where we both gain from the exchange.

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

Last round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.46 to me
- My payoff: \$5 - \$3 + \$2.46 = \$4.46
- Their payoff: \$9.0 - \$2.46 = \$6.54

So they returned about 27% of what they received (\$2.46/\$9.0 ? 0.273).

Now in Round 2:

- I'm the RECEIVER
- My current co-player sent me \$3.19
- I received \$9.57 (3.19 × 3)
- I need to decide how much to return

Important notes:

1. This is a different co-player (multi-agent setting, randomized pairings)
2. There's communication noise, so the \$3.19 might not be exactly what they sent
3. The previous player returned about 27% to me

Considerations:

- If I want to encourage cooperation, I should return a fair amount
- A common fair split might be to return enough so we both benefit equally from the cooperation
- The sender gave up \$3.19, so they have \$5 - \$3.19 = \$1.81 left
- I have \$9.57
- If I want us to end up with similar amounts, I could return about \$3.88, giving us both

around \$5.69

- Or I could return proportionally to what was sent: returning about 1/3 of what I received would be \$3.19, which gives them back their original amount
- A generous return might be half of what I received: \$4.785

Let me think about what's fair:

- The sender took a risk sending \$3.19 (gave up most of their endowment)
- The transaction created \$6.38 in surplus value beyond what they sent
- Splitting that gain equally means we each get an additional \$3.19 from the multiplier effect
- So returning \$3.19 would leave me with \$6.38 and restore them to their starting \$5
- Alternatively, returning \$4 would give them \$5.81 while I keep \$5.57

, creating a more balanced outcome

- Given the earlier low reciprocation (27%), I could...

**usage:**

```
{
 "input_tokens": 756,
 "output_tokens": 1027,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.19, it became \$9.57, and you returned \$4.0. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$10.030000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this new round as SENDER:

**\*\*Past experience:\*\***

- Round 1: Sent \$3, received \$2.46 back (modest cooperation)
- Round 2: When I received, I returned fairly (\$4.00 on \$9.57)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Need to establish trust with someone new
- The 3x multiplier means cooperation creates value

A moderate send amount balances the potential gains from cooperation with the risk of low returns. Sending \$3 creates \$9 of value while maintaining some protection.

{'send': 3}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds:\*\***

- Round 1: I was SENDER, sent \$3, received back \$2.46, payoff was \$4.46
- Round 2: I was RECEIVER, received \$9.57 (from \$3.19 sent), returned \$4.00, payoff was \$5.57

**\*\*Current Round (Round 3):\*\***

- I'm the SENDER again
- New co-player (since pairings are randomized each round)
- I have \$5 to potentially send

**\*\*Observations:\*\***

1. In Round 1, when I sent \$3, I got back \$2.46. The receiver kept \$6.54 out of \$9.00, which seems somewhat selfish (I got less than my initial \$3).
2. In Round 2, when someone sent me \$3.19, I returned \$4.00, being relatively fair.
3. The note about communication noise is important - amounts may differ from what was actually sent/returned.

**\*\*Strategy considerations:\*\***

- I'm playing with a new co-player each round
- There's no reputation to build with a specific player
- The 3x multiplier means cooperation creates value
- I need to balance trust against the risk of low returns

Given:

- My Round 1 experience showed modest cooperation from my receiver
- I was reasonably fair in Round 2
- I'm with a new player now who doesn't know my history

I think sending \$3 again is reasonable:

- It's a moderate level of trust
- Creates \$9 for the receiver
- Provides some protection if they don't reciprocate well

Let me send \$3 again.

**usage:**

```
{
 "input_tokens": 1047,
 "output_tokens": 532,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$14.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.22, so you received \$9.66. How much do you return? (0-9.66)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 4):\*\***

- Co-player sent \$3.22 ? I received \$9.66
- This represents significant trust (sending 64% of their endowment)

**\*\*Fair return analysis:\*\***

- Returning \$4.00 would give:
  - Sender: \$5 - \$3.22 + \$4.00 = \$5.78 (profit of \$0.78)
  - Me: \$9.66 - \$4.00 = \$5.66 (profit of \$0.66)

This creates a fair split where both players benefit from the cooperation.

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER:\*\***

- My co-player sent me \$3.22
- I received \$9.66 (after 3x multiplier)
- I need to decide how much to return

**\*\*Previous rounds analysis:\*\***

- Round 1 (SENDER): Sent \$3, got back \$2.46 - slightly below fair
- Round 2 (RECEIVER): Received \$9.57, returned \$4.00 - tried to be fair
- Round 3 (SENDER): Sent \$3, got back \$2.93 - again slightly below fair

**\*\*Fair return calculation:\*\***

- The sender gave up \$3.22 from their \$5 endowment, creating \$9.66 for me.
- Surplus created: \$9.66 - \$3.22 = \$6.44

A fair split would give us each roughly equal gains:

- If I return \$4.00:

- Sender gets:  $\$5 - \$3.22 + \$4.00 = \$5.78$  (gain of  $\$0.78$ )
- I get:  $\$9.66 - \$4.00 = \$5.66$  (gain of  $\$0.66$ )

This seems reasonably balanced. The sender sent a significant portion of their endowment, showing trust.

I should return around  $\$4.00$  to be fair and reciprocate their trust.

**usage:**

```
{
 "input_tokens": 1324,
 "output_tokens": 501,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send  $\$3.22$ , it became  $\$9.66$ , and you returned  $\$4.0$ . Your visible payoff was  $\$5.66$ .

Your total visible earnings across all rounds are  $\$20.62$ .

This round, you are the SENDER against your current co-player. You have  $\$5$ . How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience so far:

**\*\*As SENDER:\*\***

- Round 1: Sent  $\$3$  ? received  $\$2.46$  (modest return)
- Round 3: Sent  $\$3$  ? received  $\$2.93$  (decent return)

**\*\*As RECEIVER:\*\***

- I've consistently returned  $\$4.00$  on  $\sim\$9.60$ , maintaining fairness

**\*\*Current round:\*\***

- New co-player (randomized pairing)
- Sending  $\$3$  has been my standard opening move
- Creates  $\$9$  of value with moderate risk

I'll continue with the balanced approach that's been working.

```
{'send': 3}
```

**Reasoning:**

Looking at my history as SENDER:

- Round 1: Sent  $\$3$ , received  $\$2.46$  back
- Round 3: Sent  $\$3$ , received  $\$2.93$  back

Both times I sent  $\$3$ , I received back around  $\$2.50$ - $\$3.00$ , which is close to fair but not quite equal splitting of the gains.

**As RECEIVER:**

- Round 2: Received  $\$9.57$ , returned  $\$4.00$
- Round 4: Received  $\$9.66$ , returned  $\$4.00$

I've been consistent in returning fairly.

Now Round 5, I'm a SENDER with a new co-player. The pattern suggests that sending around  $\$3$  has been reasonably productive. The returns haven't been amazing (I've profited minimally as sender), but they've been positive.

I should continue with a moderate strategy. Sending  $\$3$  creates value and has worked reasonably well so far.

**Usage:** {

```
"input_tokens": 1601,
"output_tokens": 366,
"reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_3**



Timestamp: 2026-08-12T09:23:52  
Model: anthropic/claude-sonnet-4.5  
Temperature: 0.8  
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_8",
"pairing": {
 "agents": [
 "Agent_8",
 "Agent_1"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_1": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

- ll have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.34 (66.8% of their endowment)
- I received \$10.02 after the 3x multiplier
- The sender currently has \$1.66 remaining

I want to reciprocate the sender's trust and ensure cooperation is worthwhile for them. If I return \$5.00:

- The sender would end with  $\$1.66 + \$5.00 = \$6.66$  (better than their initial \$5)
- I would keep \$5.02 (also better than any initial amount)

This rewards their trust while maintaining a fair distribution of the gains from cooperation.

```
{'return': 5.00}
```

**reasoning:**

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is  $\$3.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$10.02)

This is a trust game. The sender showed considerable trust by sending 66.8% of their endowment. Let me think about a fair return:

The sender:

- Started with \$5
- Sent \$3.34
- Currently has  $\$5 - \$3.34 = \$1.66$
- If I return X, they'll end up with  $\$1.66 + X$

Me:

- I received \$10.02
- If I return X, I keep  $\$10.02 - X$

Some considerations for fairness:

1. Equal split of total: Total money is \$5 (sender's initial) + \$10.02 (what I received) = \$15.02. Equal split would be \$7.51 each.
  - For sender to get \$7.51, they need  $\$7.51 - \$1.66 = \$5.85$  returned
  - I would keep  $\$10.02 - \$5.85 = \$4.17$
2. Proportional return: The sender sent 66.8% of their endowment, showing high trust. A reciprocal gesture might be to return a similar proportion.
3. Rewarding trust: Since the sender trusted me with a large amount, I should return enough to make their trust worthwhile.

Let me check option 1: If I return \$5.85:

- Sender ends with:  $\$1.66 + \$5.85 = \$7.51$
- I end with:  $\$10.02 - \$5.85 = \$4.17$
- This creates mutual benefit from cooperation, and the sender gains significantly from their initial trust.

Returning around \$5 also...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 837,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.  
Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant****content:**

Looking at the previous round, the sender showed good trust by sending 66.8% of their endowment, and I reciprocated fairly.

Now as the sender with a new co-player, I'll demonstrate similar trust to establish cooperation. Sending \$3.00 (60% of my endowment) shows significant trust while being reasonable.

If they reciprocate fairly, we both benefit from cooperation.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

Round 1:

- I was the RECEIVER
- Sender sent \$3.34 (66.8% of their endowment)
- I received \$10.02
- I returned \$5.00
- My payoff was \$5.02

Now in Round 2:

- I am the SENDER
- I have \$5
- I need to decide how much to send

The previous sender showed trust by sending 66.8% of their endowment to me, and I reciprocated by returning a fair amount that left them better off.

Now I'm playing against a different agent (the pairing is randomized each round). I should consider:

1. The sender in Round 1 sent a substantial amount (66.8%), showing trust
2. This encouraged cooperation
3. I'm now the sender with a new partner

I should probably send a similar amount to establish trust and encourage cooperation. If I send around 60-70% of my endowment (like the previous sender did), that would be \$3.00-\$3.50.

Let me think about what would be fair:

- If I send \$3.00, they receive \$9.00
- If they return half of what they receive (a common fair split), they'd return \$4.50
- I'd end with \$2.00 + \$4.50 = \$6.50
- They'd keep \$4.50

That seems reasonable. I could also send \$3.34 to match what was sent to me in Round 1, showing I'm willing to trust at the same level.

Actually, let me reconsider. I want to be strategic but also fair. Sending \$3.00 seems like a good middle ground - it's 60% of my endowment, showing significant trust without being extreme.

```
{'send': 3.00}
```

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 530,
```

```

 "reasoning_tokens": 0
 }
}
Message 6 role=user
content:
 Round 3

 In your most recent previous game round, you were the SENDER against another agent. You sent $3.0, it became $9.0, and you saw them return $2.73 to you. Your visible payoff was $4.73.
 Your total visible earnings across all rounds are $9.75.

 This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}
Message 7 role=assistant
content:
 Looking at my history:
 - Round 1 (as receiver): Co-player sent $3.34, I returned $5.00
 - Round 2 (as sender): I sent $3.00, received back $2.73

 Now with a new co-player in Round 3, I should continue demonstrating willingness to cooperate. Sending a consistent amount shows I'm a reliable partner.

 {'send': 3.00}
reasoning:
 In Round 2, I sent $3.00, it became $9.00, and the receiver returned $2.73.

 Let me calculate:
 - I started with $5
 - I sent $3, so I had $2 left
 - They received $9 and returned $2.73
 - My payoff was $2 + $2.73 = $4.73

 The receiver kept $9 - $2.73 = $6.27
 I got $4.73

 So the return rate was $2.73/$9.00 = 30.3%

 This is lower than what I returned in Round 1 (I returned $5.00 out of $10.02, which is about 50%).

 However, this is a new co-player in Round 3. I shouldn't punish this new person for what the previous receiver did.

 I should continue to try to establish cooperation. Maybe send a similar amount to show consistency and willingness to cooperate. However, given that the last return was somewhat low, I might be slightly more cautious.

 Let me send $2.50 - still showing trust but being a bit more conservative.

 Actually, let me think about this differently. Each round is with a different co-player (the problem states pairings are randomized). So my actions should be about establishing cooperation with each new partner, not about punishing previous partners.

 I should continue to send a reasonable amount to establish trust. Let me send $3.00 again to be consistent and show I'm willing to cooperate.
usage:
 {
 "input_tokens": 1034,
 "output_tokens": 456,
 "reasoning_tokens": 0
 }
Message 8 role=user
content:
 Round 4

 In your most recent previous game round, you were the SENDER against another agent. You sent $3.0, it became $9.0, and you saw them return $5.57 to you. Your visible payoff was $7.57.
 Your total visible earnings across all rounds are $17.32.

```

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed trust by sending \$2.99 (about 60% of their endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has \$2.01

I want to reciprocate their trust fairly. If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (profit of \$1.51)
- I keep: \$4.47

This rewards their cooperation while maintaining a fair split of the gains.

{'return': 4.50}

**reasoning:**

Let me analyze this round:

- I'm the receiver
- The sender sent \$2.99 (about 60% of their \$5 endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has  $\$5 - \$2.99 = \$2.01$

The sender showed good trust by sending nearly 60% of their endowment. I should reciprocate fairly.

If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (better than their initial \$5)
- I keep:  $\$8.97 - \$4.50 = \$4.47$

This is fair and rewards their trust. It's roughly half of what I received, which seems reasonable given they sent about 60% of their endowment.

Looking at my previous behavior:

- Round 1: Received \$10.02, returned \$5.00 (kept \$5.02)
- Round 3: The other player returned \$5.57 when they received \$9.00

I should be consistent and fair. Returning around \$4.50 seems appropriate.

**usage:**

```
{
 "input_tokens": 1269,
 "output_tokens": 416,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.99, it became \$8.97, and you returned \$4.5. Your visible payoff was \$4.470000000000001.

Your total visible earnings across all rounds are \$21.79.

This round, you are the RECEIVER against your current co-player. They sent you \$2.89, so you received \$8.67. How much do you return? (0-8.67)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$2.89 (about 58% of their endowment), showing trust
- I received \$8.67 after the 3x multiplier
- The sender currently has \$2.11

I'll reciprocate fairly. If I return \$4.30:

- Sender ends with:  $\$2.11 + \$4.30 = \$6.41$  (profit of \$1.41)
- I keep: \$4.37

This rewards cooperation while maintaining a fair distribution.

```
{'return': 4.30}
```

**Reasoning:**

The sender sent \$2.89 (about 58% of their \$5 endowment), showing reasonable trust.

I received \$8.67 after the 3x multiplier.

The sender currently has \$5 - \$2.89 = \$2.11

I should reciprocate fairly. Looking at my previous behavior:

- Round 1: Sender sent \$3.34, I received \$10.02, returned \$5.00
- Round 4: Sender sent \$2.99, I received \$8.97, returned \$4.50

This sender sent slightly less (\$2.89 vs \$2.99), so I should return proportionally. If I return \$4.30:

- Sender ends with: \$2.11 + \$4.30 = \$6.41 (profit of \$1.41)
- I keep: \$4.37

This maintains fair reciprocation.

```
Usage: {
 "input_tokens": 1545,
 "output_tokens": 349,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_4 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:23:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_2"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_2": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 5,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

Message 1 role=system

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the RECEIVER. the sender sent \$3.32, that is 66.4% of their total endowment. You received \$9.959999999999999. How much do you return to the sender? (0-9.959999999999999)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- The sender showed significant trust by sending \$3.32 (66.4% of their endowment)
- I received \$9.96 after the 3x multiplier
- The sender currently has \$1.68 remaining

I want to reward this trust while being fair to both of us. If I return \$5.00:

- The sender ends with:  $\$1.68 + \$5.00 = \$6.68$  (profit of \$1.68)
- I end with:  $\$9.96 - \$5.00 = \$4.96$

This gives the sender a good return on their trust (33.6% gain on their original \$5) while I also benefit from the exchange. This return rate of approximately 50% is commonly seen as reciprocal in trust games.

{'return': 5.00}

##### reasoning:

Let me analyze this situation:

1. The sender sent \$3.32 out of their \$5 endowment (66.4%)
2. Due to the 3x multiplier, I received \$9.96 ( $3.32 \times 3 = 9.96$ )
3. I need to decide how much to return

##### Key considerations:

- The sender showed significant trust by sending 66.4% of their endowment
- This is a multi-round game with random pairings
- Building a reputation for reciprocity could be beneficial
- There's communication noise, so exact amounts may not be perceived correctly

Let me think about fair/reciprocal returns:

- The sender spent \$3.32 to give me \$9.96
- If I want the sender to profit from trusting me, I should return more than \$3.32
- A common strategy in trust games is to split the gains roughly equally

If I return \$0:

- Sender gets:  $\$5 - \$3.32 + \$0 = \$1.68$
- I get:  $\$9.96 - \$0 = \$9.96$
- Total: \$11.64

If I return \$3.32 (sender breaks even):  
- Sender gets:  $\$5 - \$3.32 + \$3.32 = \$5.00$   
- I get:  $\$9.96 - \$3.32 = \$6.64$   
- Total: \$11.64

If I return \$5.00 (roughly equal split of total):  
- Sender gets:  $\$5 - \$3.32 + \$5.00 = \$6.68$   
- I get:  $\$9.96 - \$5.00 = \$4.96$   
- Total: \$11.64

The sender took a risk and sent a substantial amount. A reasonable reciprocal amount would be to ensure both of us profit from the exchange.

To achieve roughly equal final payoffs, I'd need to return about \$4.14-this would give us each \$5.82. But since the sender invested significantly from their original endowment, I should consider whether an even more generous return would better reward their trust.

If I return \$6.64, they'd end with \$8.32 (doubling their investment gain), though that leaves me with just \$3.32. A middle ground like \$5.00 would give them a 33.6% gain while I keep \$4.96. Since this is the first round and I want to build a reputation for fairness, and given how much trust they demonstrated by sending most of their money, I'm leaning toward a return that meaningfully rewards their cooperation while still leaving me in a reasonable position....

usage:

```
{
 "input_tokens": 537,
 "output_tokens": 1319,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.32, it became \$9.959999999999999, and you returned \$5.0. Your visible payoff was \$4.959999999999999.

Your total visible earnings across all rounds are \$4.959999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, my co-player showed significant trust by sending 66.4% of their endowment, and I reciprocated with a 50% return rate.

For this round as sender, I'll demonstrate similar trust to establish cooperative play. Sending \$3.00 (60% of my endowment) shows trust while maintaining some security.

```
{'send': 3.00}
```

reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and someone sent me \$3.32 (66.4% of their endowment). I returned \$5.00, which was about 50% of what I received.
2. Now in Round 2, I'm the sender with \$5.
3. I need to decide how much to send to my current co-player (who may be different from Round 1's partner).

Key considerations:

- In the previous round, someone trusted me with 66.4% of their endowment
- I reciprocated fairly by returning 50%
- This established a pattern of moderate trust and reciprocity
- I'm playing with potentially a different agent now

Strategy options:

- a) Send a similar amount to what I received last round (~\$3.32) - this mirrors the trust shown to me



- b) Send less to be more cautious
- c) Send more to encourage cooperation

Given that:

- The previous sender trusted me with 66.4%, showing moderate-high trust
- I want to encourage cooperation in this multi-round game
- Sending around the same proportion (60-70%) seems reasonable
- This could establish a cooperative norm

I'll send \$3.00 (60% of my endowment), which is:

- Similar to what was sent to me in Round 1
- Shows trust but not maximal trust
- Allows me to keep some guarantee (\$2)
- If returned fairly (~50%), I'd get back around \$4.50, making my total \$2 + \$4.50 = \$6.50

Let me go with \$3.00.

**usage:**

```
{
 "input_tokens": 864,
 "output_tokens": 471,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.51 to you. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$11.469999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.27 (45.4% of their endowment), showing moderate trust
- I received \$6.81 after the 3x multiplier
- The sender currently has \$2.73 remaining

In the previous round when I was sender, I sent 60% and received back about 50% of what my co-player received. I'll reciprocate with a similar return rate.

Returning \$3.50 gives:

- Sender ends with: \$2.73 + \$3.50 = \$6.23 (profit of \$1.23)
- I end with: \$6.81 - \$3.50 = \$3.31

This represents about 51% return, which fairly rewards their trust.

```
{'return': 3.50}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.27, which is 45.4% of their \$5 endowment
- I received \$6.81 after the 3x multiplier
- The sender currently has \$5 - \$2.27 = \$2.73 remaining

In the previous round when I was sender:

- I sent \$3.00 (60% of endowment)
- Received back \$4.51, which was about 50% of the \$9.00 they received
- This gave me a payoff of \$6.51

The sender this round showed moderate trust (45.4% vs my 60% last round). I should reciprocate fairly.

If I return \$3.50:

- Sender ends with: \$2.73 + \$3.50 = \$6.23 (profit of \$1.23)
- I end with: \$6.81 - \$3.50 = \$3.31

This represents about 51% return rate, similar to what I received last round (50%). This seems fair and reciprocal.

Let me check the math:

- Return rate =  $\$3.50 / \$6.81 = 51.4\%$
- Sender's final payoff =  $\$2.73 + \$3.50 = \$6.23$
- My final payoff =  $\$6.81 - \$3.50 = \$3.31$

This seems reasonable and reciprocal given their 45.4% trust level.

**usage:**

```
{
 "input_tokens": 1104,
 "output_tokens": 545,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.810000000000005, and you returned \$3.5. Your visible payoff was \$3.3100000000000005.

Your total visible earnings across all rounds are \$14.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience so far:

- When I've shown trust as sender (60% in Round 2), I received good reciprocity (~50% return)
- When others showed trust as sender, I've returned approximately 50%
- The pattern suggests cooperative play with moderate to high trust levels

I'll continue with a similar strategy, sending \$3.00 (60% of my endowment) to maintain trust and cooperation.

```
{'send': 3.00}
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.32 (66.4%)
- I received \$9.96
- I returned \$5.00 (about 50%)
- My payoff: \$4.96

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.51 (about 50%)
- My payoff: \$6.51

Round 3: I was RECEIVER

- Sender sent \$2.27 (45.4%)
- I received \$6.81
- I returned \$3.50 (about 51%)
- My payoff: \$3.31

Round 4: I am SENDER now

- I have \$5 to decide what to send

Looking at the pattern:

- In Round 2, I sent \$3.00 (60%) and got back about 50%
- Others have sent between 45% and 66% in the rounds I've seen

I've been experiencing decent reciprocity so far. The return rates have been around 50% consistently.

I should continue to show trust, as it's been rewarded. Sending around 60% seems reasonable based on what I've seen. I'll send \$3.00 again.

**usage:**

```
{
 "input_tokens": 1421,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$21.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience as sender:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)

Sending 60% of my endowment has consistently led to good reciprocity. I'll maintain this cooperative strategy.

```
{'send': 3.00}
```

**Reasoning:**

Looking at my track record:

- Round 2: I sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: I sent \$3.00, received \$4.44 back (payoff \$6.44)

Both times sending \$3.00 has resulted in good returns. The strategy of sending 60% has been working well and establishing cooperative relationships.

I should continue with this consistent strategy to maintain trust and cooperation.

Usage: {

```
 "input_tokens": 1648,
 "output_tokens": 217,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:24:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_2"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_2": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 5,
 "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 MULTI-AGENT SETTING:
 - There are 8 agents in this run.
 - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
 - You play one sender-receiver game with your paired opponent each round.
 - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
 ll have acted as sender and receiver equally often when the round count allows it.
 - Agent names are hidden in this run.
 - Your paired opponent is referred to as your current co-player.

Message 2 role=user
content:
 Round 1: You are the RECEIVER. the sender sent $2.55, that is 51.0% of their total endowme
 nt. You received $7.6499999999999995. How much do you return to the sender? (0-7.6499999
 999999995)

Message 3 role=assistant
content:
 Looking at this situation:

 What happened:
 - The sender sent $2.55 (51% of their $5 endowment), showing moderate trust
 - I received $7.65 after the 3x multiplier
 - The sender now has $2.45 remaining
 - I need to decide how much to return

 My reasoning:
 The sender showed significant trust by sending about half their endowment. To reciprocate
 fairly and build trust:

 If I return **$4.00**:
 - Sender ends with: $2.45 + $4.00 = $6.45 (profit of $1.45)

```

- I end with:  $\$7.65 - \$4.00 = \$3.65$  (profit of  $\$3.65$ )

This represents:

- Returning about 52% of what I received (similar to their 51% sent)
- Both of us profit from the cooperation
- Rewarding their trust encourages future cooperation
- Setting a positive precedent for Round 1

`{'return': 4.0}`

**reasoning:**

Let me understand the situation:

1. I am the RECEIVER in Round 1
2. The sender sent  $\$2.55$  (which is 51% of their  $\$5$  endowment)
3. This amount was multiplied by 3x, so I received  $\$2.55 \times 3 = \$7.65$  (approximately  $\$7.65$ )
4. I need to decide how much to return (between  $\$0$  and  $\$7.65$ )

Now, let me think about this strategically:

The sender sent 51% of their endowment, which is slightly more than half. This shows a moderate level of trust. They could have sent nothing, they could have sent everything, but they chose to send about half.

From a fairness perspective:

- The sender now has:  $\$5 - \$2.55 = \$2.45$
- I have:  $\$7.65$
- Total in the system:  $\$2.45 + \$7.65 = \$10.10$

If I want to be fair and reciprocate the trust:

- One approach: Split the total gains equally. The sender started with  $\$5$ , and we now have  $\$10.10$  total. If we split equally, we'd each have  $\$5.05$ . The sender already has  $\$2.45$ , so I'd return  $\$5.05 - \$2.45 = \$2.60$
- Another approach: Return the same proportion they sent. They sent 51% of their endowment, so I could return 51% of what I received:  $0.51 \times \$7.65 = \$3.90$
- Conservative approach: Return at least what they sent originally ( $\$2.55$ ) to ensure they don't lose money from trusting me
- Generous approach: Return more to reward their trust and potentially build cooperation

Given that:

1. The sender showed moderate trust by sending 51%
2. There's communication noise mentioned (though the exact amount sent seems precise)
3. Since this is a multi-round game with multiple agents, establishing a reputation for reciprocity could be strategically valuable
4. The sender took a genuine risk by sending money

A fair reciprocal response should reward their trust while ensuring they benefit from the exchange and remain proportional to their initial gesture. If I return  $\$2.60$ , we'd each end with  $\$5.05$ -a modest profit for both of us that acknowledges their willingness to trust.

Alternatively, returning 50% of what I received ( $\$3.825$ ) would give them  $\$6.275$ , a more substantial gain. But I need to reconsider my baseline here-as the receiver, I don't start with  $\$5$  like the sender does. I begin with  $\$0$  and only gain what they send multiplied by three. The sender started with  $\$5$ , sent  $\$2.55$ , and now has  $\$2.45$  remaining.

If I return  $\$X$ , they end with  $\$2.45 + \$X$  and I end with  $\$7.65 - \$X$ . For equal splits, that's  $\$2.60$  returned, giving us both  $\$5.05$ . But in this experimental context, I should probably return more-around  $\$3.00$ - $\$3.50$ -to reward their trust and signal reciprocity for potential future rounds.

Testing a few amounts: returning  $\$3.25$  leaves them with  $\$5.70$  and me with  $\$4.40$ ;  $\$3.50$  gives them  $\$5.95$  and me  $\$4.15$ ;  $\$4.00$  results in them having  $\$6.45$  and me  $\$3.65$ . Each option balances showing trustworthiness with keeping reasonable profit, though I'm leaning toward the higher end to maximize their incentive to continue cooperating. Both of us end up better off than we started. At  $\$3.50$ , the sender gets  $\$5.95$  (up  $\$0.95$ ) and I get  $\$4.15$  (up  $\$4.15$ ), which feels like a solid reciprocal move that rewards their trust without being exploitative.

I'm settling on \$3.50 as my return amount-it strikes a balance between showing genuine reciprocity and ensuring both parties benefit from the exchange. terms of what's strategically optimal in a multi-round setting. Building reputation matters-if I'm generous now, other agents will likely trust me more later. Since this is Round 1, establishing a trustworthy precedent is crucial. The sender showed moderate confidence by sending \$2.55, so I should reciprocate proportionally.

I'm settling on \$3.50. It more than doubles their original amount, gives them solid profit, signals I'm reliable, and leaves me with a reasonable gain too. Let me verify the math:  $\$2.55 \times 3 = \$7.65$  received, returning \$3.50 means the sender nets \$0.95 profit while I keep \$4.15.

The communication noise mentioned in the problem is worth noting, but I can only work with what I observe. My final return amount is \$3.50, formatted as {'return': 3.50}.

Actually, reconsidering the proportionality here-the sender gave 51% of their endowment, which signals moderate trust. Matching that proportion with my return would mean sending back 51% of \$7.65, which equals \$3.90. This reciprocal approach better reflects their level of generosity while still leaving me with a reasonable profit of \$3.75 versus their \$1.35. I think \$3.90 better captures the spirit of proportional reciprocation.

Though I'm also drawn to rounding up to \$4.00 for simplicity-it's nearly the same percentage anyway (52%), easier to process, and still feels fair to both parties. Let me settle on \$4.00 as my final choice.

usage:

```
{
 "input_tokens": 539,
 "output_tokens": 3346,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.55, it became \$7.649999999999995, and you returned \$4.0. Your visible payoff was \$3.649999999999995.

Your total visible earnings across all rounds are \$3.649999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

**\*\*Context:\*\***

- Round 1 established a cooperative dynamic where sending ~51% led to mutual profit
- I'm now the sender with a new co-player
- I need to establish trust in this new pairing

**\*\*Strategic reasoning:\*\***

- Sending a moderate amount (around 50%) shows willingness to cooperate
- It gives my co-player enough to make returning worthwhile (\$7.50 after multiplier)
- It limits my downside risk while allowing mutual gains
- This mirrors the successful approach from Round 1

**\*\*Decision:\*\***

I'll send \$2.50 (50% of my endowment) to establish a cooperative tone with this new co-player.

{'send': 2.5}

reasoning:

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was the RECEIVER
- Sender sent \$2.55 (51% of \$5)
- I received \$7.65

- I returned \$4.00
- My payoff: \$3.65
- Sender's payoff would have been:  $\$5 - \$2.55 + \$4.00 = \$6.45$

**\*\*Current Round (Round 2):\*\***

- I am now the SENDER
- I have \$5
- I'm paired with a potentially different agent (since it says "your current co-player")
- I need to decide how much to send

**\*\*Strategic considerations:\*\***

1. **\*\*Reciprocity principle\*\***: In Round 1, my co-player showed moderate trust (51% sent). I reciprocated well by returning 52% of what I received, ensuring both parties profited.
2. **\*\*Building trust\*\***: Since this is a multi-round game with multiple agents, establishing a cooperative reputation could be beneficial.
3. **\*\*Unknown opponent\*\***: This is a different co-player, so I don't know their reciprocity tendencies yet.
4. **\*\*First impression\*\***: As the sender in this new pairing, I can set the tone for cooperation.
5. **\*\*Risk vs reward\*\***:
  - If I send \$0, I keep \$5, they get \$0, no mutual benefit
  - If I send all \$5, it becomes \$15 for them, but I'm vulnerable if they return nothing
  - Middle ground shows trust while limiting downside risk
6. **\*\*Following established norms\*\***: In Round 1, the sender sent 51%. This seemed to work well and encouraged cooperation.

**\*\*Decision:\*\***

I'll send a similar amount to what I experienced as effective in Round 1 - around 50% (\$2.50). This:

- Shows trust and willingness to cooperate
- Limits my downside risk
- Gives the receiver enough to make returning worthwhile

...

**usage:**

```
{
 "input_tokens": 916,
 "output_tokens": 646,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.62 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$9.77.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.11 (42.2% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- The sender now has \$2.89 remaining

**\*\*My reasoning:\*\***

The sender showed trust, though slightly less than what I've seen in previous rounds. I should reciprocate fairly to maintain cooperation.

If I return \*\*\$3.30\*\*:  
- Sender ends with:  $\$2.89 + \$3.30 = \$6.19$  (profit of \$1.19)  
- I end with:  $\$6.33 - \$3.30 = \$3.03$  (profit of \$3.03)

This represents:  
- Returning about 52% of what I received  
- Both of us profiting from cooperation  
- Fair reciprocation that encourages future trust

{'return': 3.3}

**reasoning:**

Let me analyze this situation:

**\*\*What happened in Round 2:\*\***

- I sent \$2.50
- It became \$7.50
- They returned \$3.62
- My payoff:  $\$2.50$  (kept) +  $\$3.62$  (returned) =  $\$6.12$
- Their payoff:  $\$7.50 - \$3.62 = \$3.88$
- They returned about 48% of what they received

**\*\*Current Round 3 situation:\*\***

- I'm the receiver
- Sender sent \$2.11 (42.2% of their \$5 endowment)
- I received \$6.33
- Sender now has \$2.89 remaining
- I need to decide how much to return

**\*\*Analysis:\*\***

The sender sent \$2.11, which is less than the \$2.50-\$2.55 amounts I've seen previously. They're showing moderate but slightly lower trust. However, they still sent a significant portion (42%).

If I return different amounts:

- Return \$3.00: Sender ends with \$5.89 (+\$0.89), I end with \$3.33 (+\$3.33)
- Return \$3.50: Sender ends with \$6.39 (+\$1.39), I end with \$2.83 (+\$2.83)
- Return \$3.20: Sender ends with \$6.09 (+\$1.09), I end with \$3.13 (+\$3.13)

I want to:

1. Reciprocate their trust proportionally
2. Be fair and encourage future cooperation
3. Return roughly 50% of what I received, similar to what I've observed

Returning around \$3.20 would be about 50.5% of what I received, which seems fair and proportional to their ~42% send rate. This ensures both of us profit.

Actually, let me be slightly more generous since they did show trust. I'll return \$3.30 (about 52% of what I received).

...

**usage:**

```
{
 "input_tokens": 1218,
 "output_tokens": 758,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.3. Your visible payoff was \$3.0300000000000002.

Your total visible earnings across all rounds are \$12.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:



**\*\*Context:\*\***

- I'm facing a new co-player as sender
- Previous cooperation has been successful (total earnings \$12.80)
- When I sent \$2.50 in Round 2, I received \$3.62 back for a good profit
- My returns as receiver have been fair (~52%)

**\*\*Strategic reasoning:\*\***

- Continue the cooperative approach that's been working
- Sending 50% shows good faith while managing risk
- This gives my co-player \$7.50 to work with, enough for mutual benefit
- Establishes positive precedent with this new partner

{'send': 2.5}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds summary:\*\***

- Round 1 (RECEIVER): Received \$7.65 from sender sending \$2.55, returned \$4.00, earned \$3.65
- Round 2 (SENDER): Sent \$2.50, became \$7.50, they returned \$3.62, earned \$6.12
- Round 3 (RECEIVER): Received \$6.33 from sender sending \$2.11, returned \$3.30, earned \$3.03
- Total earnings: \$12.80

**\*\*Pattern analysis:\*\***

- When I sent \$2.50 (50%), I got back \$3.62, giving me a profit of \$1.12 (from \$2.50 remaining + \$3.62 = \$6.12 total)
- When others sent to me, I've returned fairly (~52% of what I received)
- Overall, cooperation has been successful

**\*\*Current decision:\*\***

- New co-player as sender this round
- Need to establish trust again
- Sending around 50% has worked well previously

I'll continue with the cooperative strategy that's been working: send \$2.50 (50% of endowment).

**usage:**

```
{
 "input_tokens": 1558,
 "output_tokens": 431,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*My reasoning:\*\***

This sender showed above-average trust compared to previous rounds. I should reciprocate generously.

If I return **\*\*\$4.70\*\***:

- Sender ends with: \$2.08 + \$4.70 = \$6.78 (profit of \$1.78)

- I end with:  $\$8.76 - \$4.70 = \$4.06$  (profit of  $\$4.06$ )

This represents:

- Returning about 54% of what I received
- Rewarding their higher trust level (58.4% sent)
- Both profiting well from cooperation
- Encouraging continued trust

{'return': 4.7}

#### Reasoning:

Let me analyze this situation:

**\*\*What happened:\*\***

- The sender sent  $\$2.92$  (58.4% of their endowment), showing strong trust
- I received  $\$8.76$  after the 3x multiplier
- The sender now has  $\$2.08$  remaining

**\*\*Patterns from previous rounds:\*\***

- Round 1: Received  $\$7.65$ , returned  $\$4.00$  (52%)
- Round 3: Received  $\$6.33$ , returned  $\$3.30$  (52%)
- I've been consistently returning about 52% of what I receive

**\*\*Strategy:\*\***

The sender showed high trust (58.4% sent). I should reciprocate fairly.

If I return around 52% of  $\$8.76$ :

- $0.52 \times \$8.76 = \$4.55$

Let me check the outcomes:

- Sender ends with:  $\$2.08 + \$4.55 = \$6.63$  (profit of  $\$1.63$ )
- I end with:  $\$8.76 - \$4.55 = \$4.21$  (profit of  $\$4.21$ )

This seems fair - both profit, sender gets a decent return for their trust.

Actually, given they sent 58.4% (more than my usual 50%), maybe I should return slightly more to really reward that trust. Let me try  $\$4.70$ :

- Sender ends with:  $\$2.08 + \$4.70 = \$6.78$  (profit of  $\$1.78$ )
- I end with:  $\$8.76 - \$4.70 = \$4.06$  (profit of  $\$4.06$ )

That's about 54% return, which is fair and rewards their higher trust level.

```
Usage: {
 "input_tokens": 1844,
 "output_tokens": 622,
 "reasoning_tokens": 0
}
```

## ROUND 6

Roles: Agent\_1=Sender, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Receiver, Agent\_5=Sender, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Receiver

#### Pairings:

dyad\_1: Agent\_6=Sender, Agent\_3=Receiver  
dyad\_2: Agent\_1=Sender, Agent\_4=Receiver  
dyad\_3: Agent\_5=Sender, Agent\_8=Receiver  
dyad\_4: Agent\_2=Sender, Agent\_7=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_6 sent  $\$3.25$ ; received  $\$9.75$ ; Agent\_3 returned  $\$2.94$ ; payoffs Agent\_6= $\$4.69$ , Agent\_3= $\$6.81$ ; noise/visible: sent visible= $\$2.72$ , received visible= $\$8.16$ , returned visible= $\$2.56$   
Action validation: {"returned": {"amount": 2.94, "clamped": false, "max": 9.75, "min": 0, "raw": 2.94}, "sent": {"amount": 3.25, "clamped": false, "max": 5, "min": 0, "raw": 3.25}}

dyad\_2: Agent\_1 sent  $\$3$ ; received  $\$9$ ; Agent\_4 returned  $\$5$ ; payoffs Agent\_1= $\$7$ , Agent\_4= $\$4$ ;  
noise/visible: sent visible= $\$3.26$ , received visible= $\$9.78$ , returned visible= $\$4.33$   
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad\_3: Agent\_5 sent  $\$3$ ; received  $\$9$ ; Agent\_8 returned  $\$3.7$ ; payoffs Agent\_5= $\$5.7$ , Agent\_8= $\$5.3$ ;  
noise/visible: sent visible= $\$2.86$ , received visible= $\$8.58$ , returned visible= $\$3.36$   
Action validation: {"returned": {"amount": 3.7, "clamped": false, "max": 9.0, "min": 0, "raw": 3.7}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

```

dyad_4: Agent_2 sent $2.5; received $7.5; Agent_7 returned $3; payoffs Agent_2=$5.5, Agent_7=$4.5;
noise/visible: sent visible=$1.64, received visible=$4.92, returned visible=$3.8
 Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 7.5, "min": 0, "raw":
 3.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
Balances after round: {
 "Agent_1": 32.519999999999996,
 "Agent_2": 31.91,
 "Agent_3": 34.75,
 "Agent_4": 31.2,
 "Agent_5": 36.13,
 "Agent_6": 30.339999999999996,
 "Agent_7": 31.049999999999997,
 "Agent_8": 32.6
}
Visible balances after round: {
 "Agent_1": 32.49,
 "Agent_2": 28.67,
 "Agent_3": 30.1,
 "Agent_4": 32.379999999999995,
 "Agent_5": 29.29,
 "Agent_6": 30.78,
 "Agent_7": 28.119999999999997,
 "Agent_8": 32.7
}

```

## LLM CALLS

### GAME - Agent\_6 - Sender - dyad\_1

Timestamp: 2026-08-12T09:24:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_3"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_3": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative dynamic while being somewhat cautious since I don't know my co-player yet.

If I send money, it triples, creating potential mutual benefit. However, there's risk since the receiver's return is voluntary.

I'll send a moderate amount to signal trust and willingness to cooperate, while not being overly vulnerable in this first interaction.

**\*\*Decision:\*\***

```
'''json
{"send": 3}
'''
```

This gives the receiver \$9 to work with. If they're cooperative and return a fair share, we both benefit. If not, I've learned something about the dynamics for future rounds while limiting my downside risk.

**reasoning:**

This is the first round of a trust game, and I'm the sender. I need to think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3
- The receiver decides how much to return
- There may be communication noise
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I don't know who my current co-player is

Since this is the first round, I have no information about how other players behave. The optimal strategy depends on:

1. Whether I expect reciprocity from the receiver

2. Whether I want to establish a reputation (though with hidden names, this is less relevant for future rounds)
3. Risk tolerance

From a purely rational game theory perspective, if this were a one-shot game with no reputation effects, the receiver should return \$0 (keep everything), which means I should send \$0. However, experimental evidence shows people often cooperate in trust games.

Some considerations:

- Sending \$5 maximizes total payoff (\$5 + \$15 total) but requires trust
- Sending \$0 guarantees I keep \$5 but creates no opportunity for mutual gain
- A moderate amount (like \$2-3) balances risk and opportunity

Since this is round 1 and I want to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything immediately

I'll send a moderate amount. Sending \$3 seems reasonable:

- If sent \$3, receiver gets \$9
- If receiver returns half (\$4.50), I end up with  $\$5 - \$3 + \$4.50 = \$6.50$
- If receiver returns nothing, I end up with \$2
- This tests their cooperation while not being overly cautious

Actually, let me reconsider. A common strategy is to send about half to show trust but not complete vulnerability. Sending \$2.50 or \$3 seems reasonable.

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 632,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.25 to you. Your visible payoff was \$6.25.

Your total visible earnings across all rounds are \$6.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.67, so you received \$8.01. How much do you return? (0-8.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.67\*\*** (about 53% of their endowment), which became \$8.01 for me. This is a moderately cooperative move.

**\*\*Comparison to previous round:\*\***

- When I sent \$3.00, I received \$4.25 back (47% of the \$9.00)
- That gave my co-player a reasonable benefit while rewarding my trust

**\*\*My reasoning:\*\***

I want to:

1. Reward the sender's trust with a decent return
2. Maintain cooperative dynamics for future rounds
3. Keep enough to make receiving worthwhile

Returning about \$3.75 seems fair:

- Sender would get:  $\$2.33$  (kept) +  $\$3.75$  =  $\$6.08$  total (gain of  $\sim\$1.08$ )
- I would get:  $\$8.01 - \$3.75 = \$4.26$  (solid gain)

This is proportionally similar to what I experienced last round.

```
'''json
{"return": 3.75}
```

'''

**reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was SENDER
- I sent \$3.0
- It became \$9.0
- Receiver returned \$4.25
- My payoff:  $\$5 - \$3 + \$4.25 = \$6.25$

This round (Round 2):

- I am RECEIVER
- Sender sent \$2.67
- I received \$8.01 (which is  $\$2.67 \times 3$ )
- I need to decide how much to return

The sender in this round sent \$2.67 out of \$5, which is about 53% of their endowment. This is a fairly cooperative move.

In the previous round, when I sent \$3, I received back \$4.25. My net gain was \$1.25 from the initial \$5.

Let me think about what would be fair:

- The sender gave up \$2.67
- I received \$8.01
- If I want to be cooperative and fair, I should think about:
  1. Returning enough so the sender benefits from sending
  2. Not returning so much that I don't benefit from receiving

A common approach is to split the gains. The total amount now is \$8.01. If we split it equally:

- Each gets \$4.005

But the sender started with \$5 and sent \$2.67, so they have \$2.33 left.

If I return \$4.00, they'd have  $\$2.33 + \$4.00 = \$6.33$  (gain of \$1.33)

I'd keep  $\$8.01 - \$4.00 = \$4.01$  (gain of \$4.01 compared to receiving nothing)

Hmm, that seems imbalanced. Let me think differently.

Another approach: What would make our final payoffs more equal?

- Sender has:  $\$5 - \$2.67 + \text{return} = \$2.33 + \text{return}$
- I have:  $\$8.01 - \text{return}$

For equal payoffs:

$$\begin{aligned}\$2.33 + \text{return} &= \$8.01 - \text{return} \\ 2 \times \text{return} &= \$8.01 - \$2.33 = \$5.68 \\ \text{return} &= \$2.84\end{aligned}$$

If I return \$2.84:

- Sender gets:  $\$2.33 + \$2.84 = \$5.17$
- I get:  $\$8.01 - \$2.84 = \$5.17$

That would equalize our outcomes, though it doesn't particularly reward the sender's risk-taking. I should consider what the previous receiver did for comparison.

The prior round showed a return of \$4.25 on \$9.00 received-roughly 47% of what they got. Applying that same rate here:  $\$8.01 \times 0.47 = \$3.76$ . This would give the sender \$6.09 total and leave me with \$4.25, which feels more balanced given the sender's initial trust. But I need to account for the communication noise mentioned in the instructions-the amounts I'm seeing may not match what was actually sent or returned....

**usage:**

```
{
 "input_tokens": 785,
 "output_tokens": 1226,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.67, it became \$8.01, and you returned \$3.75. Your visible payoff was \$4.26.

Your total visible earnings across all rounds are \$10.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$3.36\*\*** (67% of their endowment), which became \$10.08 for me. This is a notably generous and trusting move.

**\*\*Context from previous rounds:\*\***

- Round 2: Someone sent \$2.67 (53%), I returned \$3.75
- Round 1: I sent \$3.00 (60%), received \$4.25 back

**\*\*My reasoning:\*\***

This sender showed higher trust than my previous co-player (67% vs 53%). I should reward their generosity proportionally.

Returning \$4.75:

- Sender would get: \$1.64 (kept) + \$4.75 = \$6.39 total (gain of \$1.39)
- I would get: \$10.08 - \$4.75 = \$5.33 (good gain)

This provides the sender with a decent reward for their trust while maintaining mutual benefit - both of us come out ahead.

```
'''json
{"return": 4.75}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$3.36 (67% of their \$5 endowment) - this is quite generous/trusting
- I received \$10.08

**\*\*Previous round:\*\***

- Someone sent me \$2.67 (53% of endowment), I returned \$3.75
- That gave them \$2.33 + \$3.75 = \$6.08 total (~\$1.08 gain)
- I kept \$4.26

**\*\*Round 1:\*\***

- I sent \$3.00 (60%), received \$4.25 back
- That gave me \$2.00 + \$4.25 = \$6.25 total

**\*\*My thinking:\*\***

The sender this round was more generous (67% vs 53% in previous round). I should reward their higher trust level.

If I return around \$4.50-5.00:

- Return \$4.50: Sender gets \$1.64 + \$4.50 = \$6.14, I get \$5.58
- Return \$5.00: Sender gets \$1.64 + \$5.00 = \$6.64, I get \$5.08

Actually, let me think about what's fair given they sent 67% of their endowment.

If I return \$5.00:

- Sender: keeps \$1.64, gets back \$5.00 = \$6.64 total (gain of \$1.64)
- Me: receive \$10.08, return \$5.00 = \$5.08 (solid gain)

This seems like a good reciprocal move for their high trust. It's roughly a 50-50 split of the multiplied amount, which rewards their generosity while still benefiting both of us.

Actually, let me reconsider. They sent \$3.36 out of \$5, so they kept \$1.64.

Returning \$5 gives them \$6.64 with a \$1.64 gain, while I keep \$5.08. That's reasonably fair given their high trust. I...

usage:

```
{
 "input_tokens": 1163,
 "output_tokens": 921,
 "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.75. Your visible payoff was \$5.33.

Your total visible earnings across all rounds are \$15.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at this round as SENDER again:

**My experience:**

- Round 1: Sent \$3.00, received \$4.25 back ? Good outcome
- Rounds 2-3: As receiver, I've been cooperative, returning fair amounts

**Current situation:**

- New co-player (different from Round 1)
- I've been part of generally cooperative dynamics

**My reasoning:**

I had success with sending \$3.00 before, and the cooperative pattern across rounds suggests moderate-to-high sending works well. I'll send slightly more this time to signal trust while building on the positive momentum.

Sending \$3.50 creates \$10.50 for the receiver. If they're reasonably cooperative (returning ~40%), I'd still come out ahead.

```
'''json
{"send": 3.5}
'''
```

reasoning:

I'm back as the SENDER in round 4. Let me review my experience so far:

Round 1: I sent \$3.00, received \$4.25 back. Payoff: \$6.25

Round 2: Received \$8.01 (from \$2.67 sent), returned \$3.75. Payoff: \$4.26

Round 3: Received \$10.08 (from \$3.36 sent), returned \$4.75. Payoff: \$5.33

Total earnings: \$15.84

Now I'm the sender again with a new co-player. What should I send?

Looking at the pattern:

- When I sent \$3.00 (60%), I got a decent return
- When others sent moderate-to-high amounts (\$2.67, \$3.36), I've been fairly cooperative in returning
- The game seems to be working reasonably well with moderate-to-high sending amounts

I should consider:

1. I had a positive experience when I sent \$3.00
2. The cooperative moves I've seen have been between \$2.67-\$3.36
3. I've demonstrated I'm a fair returner, so maybe I can expect reasonable behavior

I think I should send a similar or slightly higher amount than last time. Since I received \$4.25 when I sent \$3.00, that worked out well.

Options:

- Send \$3.00 again (proven strategy)



- Send \$3.50 (show slightly more trust)
- Send \$4.00 (high trust signal)

Given:

- This is a new co-player
- I've been building cooperative patterns
- The game is going well so far

I'll send \$3.50 as a moderately high trust signal, slightly more than my first round, but not overly risky. This creates \$10.50 for the receiver.

...

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 660,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.55 to you. Your visible payoff was \$6.05.

Your total visible earnings across all rounds are \$21.89.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*My pattern:\*\***

- I've been consistently returning around 45-47% of what I receive
- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)

**\*\*My reasoning:\*\***

This sender showed moderate trust. I should maintain consistency with my previous cooperative behavior.

Returning \$3.85:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of \$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This is about 46% return rate, consistent with my pattern and fair to the sender.

```
'''json
{"return": 3.85}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*Context from previous rounds:\*\***

- Round 1: I sent \$3.00 (60%), received \$4.25 back (47% of \$9)
- Round 2: Someone sent \$2.67 (53%), I returned \$3.75 (47% of \$8.01)
- Round 3: Someone sent \$3.36 (67%), I returned \$4.75 (47% of \$10.08)
- Round 4: I sent \$3.50 (70%), received \$4.55 back (43% of \$10.50)

**\*\*Pattern:\*\***

I've been pretty consistent returning around 45-47% of what I receive. My co-player sent \$2.81, which is moderately cooperative.

Let me calculate:

- 45% of \$8.43 = \$3.79
- 47% of \$8.43 = \$3.96

Returning around \$3.85 would:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of ~\$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This maintains consistency with my previous behavior and rewards the sender proportionally for their trust.

**usage:**

```
{
 "input_tokens": 1847,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$3.85. Your visible payoff was \$4.58.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 back (good outcome)
- Round 4: Sent \$3.50 ? Received \$4.55 back (good outcome)

**\*\*Overall dynamics:\*\***

- Cooperation has been working well across all rounds
- I've been a fair receiver (returning ~46%)
- Others have been reasonably cooperative with me

**\*\*My reasoning:\*\***

I'm with a new co-player. The cooperative pattern continues to work. I'll send an amount between my previous sends to signal trust while being strategic.

Sending \$3.25 creates \$9.75 for the receiver. With typical cooperation, this should benefit us both.

```
'''json
{"send": 3.25}
'''
```

**Reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3.00, received \$4.25 back ? Payoff \$6.25
- Round 4: Sent \$3.50, received \$4.55 back ? Payoff \$6.05

As RECEIVER:

- Round 2: Received \$8.01, returned \$3.75 ? Payoff \$4.26
- Round 3: Received \$10.08, returned \$4.75 ? Payoff \$5.33
- Round 5: Received \$8.43, returned \$3.85 ? Payoff \$4.58

Overall, the cooperative strategy has been working well. When I send moderate-to-high amounts (\$3-\$3.50), I get good returns. My co-players have generally been cooperative.

This is a new co-player again (Round 6). I've been doing well with sends in the \$3-\$3.50 range.

I'll continue with a similar strategy - sending \$3.25 this time (between my previous two sends).

This creates \$9.75 for the receiver. If they return around 45%, I'd get about \$4.39 back, giving me a payoff of \$5.89.

Actually, let me think about this differently. I've been gradually increasing (3.0 then 3.5). Maybe I should go higher to signal stronger trust, or stay consistent. Let me stick with \$3.50 since it worked well.

Actually, let me be a bit more varied. \$3.25 is a good middle ground.

```
Usage: {
 "input_tokens": 2205,
 "output_tokens": 567,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_3 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:24:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_3"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_3": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.35, that is 47.0% of their total endowment. You received \$7.050000000000001. How much do you return to the sender? (0-7.050000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.35 (47% of their endowment)
- I received \$7.05 from this
- The sender currently has \$2.65 remaining

I want to reciprocate this trust fairly. A reasonable approach is to aim for equal final payoffs since we're cooperating:

- Total available: \$2.65 (sender's remainder) + \$7.05 (my received amount) = \$9.70
- Equal split:  $\$9.70 \div 2 = \$4.85$  each
- Return needed:  $\$4.85 - \$2.65 = \$2.20$

This ensures we both benefit equally from the cooperation, rewarding the sender's trust while keeping the gains fair.

{'return': 2.20}

**reasoning:**

Let me understand the situation:

- I am the RECEIVER in Round 1
- The sender sent \$2.35 (47% of their \$5 endowment)
- This was multiplied by 3x, so I received \$7.05
- I need to decide how much to return (between \$0 and \$7.05)

**Key considerations:**

1. The sender showed trust by sending nearly half their endowment
2. There's communication noise mentioned, so amounts may differ slightly
3. This is a multi-round game with 8 agents and rotating pairings
4. Building reputation and reciprocity is important

The sender sent 47% of their endowment. If I want to be fair and reciprocal:

- The sender started with \$5, sent \$2.35, and has \$2.65 left
- I received \$7.05
- If I don't return anything, sender gets \$2.65, I get \$7.05
- A fair split might be to aim for equal payoffs or proportional returns

**Common strategies in trust games:**

1. Return nothing (selfish)
2. Return the original amount sent (\$2.35)
3. Return enough to split the gains equally
4. Return a proportion similar to what was sent

Let's calculate equal payoffs:

- Total money: \$5 (initial sender) + \$7.05 (tripled) = \$12.05? No wait...

- Initial endowment for sender: \$5
- Sent: \$2.35 (tripled to \$7.05 for receiver)
- Sender currently has:  $\$5 - \$2.35 = \$2.65$
- Receiver has: \$7.05
- Total in system:  $\$2.65 + \$7.05 = \$9.70$

For equal payoffs: each should get  $\$9.70/2 = \$4.85$

- So I'd need to return \$2.20 to the sender, leaving me with \$4.85

I could also return the original \$2.35 they sent, which would give them \$5.00 (breaking even) while I keep \$4.70. Or I could mirror their generosity by returning 47% of what I received—about \$3.31—which acknowledges their trust while still letting me benefit from the multiplication.

The equal payoff approach feels right though. Returning \$2.20 means we both end up with \$4.85, which feels like genuine reciprocity for the trust they showed....

usage:

```
{
 "input_tokens": 537,
 "output_tokens": 1079,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.35, it became \$7.050000000000001, and you returned \$2.2. Your visible payoff was \$4.8500000000000005.

Your total visible earnings across all rounds are \$4.8500000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

content:

Looking at my previous round, the sender showed trust at 47% (\$2.35) and I reciprocated fairly, resulting in equal payoffs of \$4.85 each.

As the sender now, I'll show similar moderate trust. Sending around 50% demonstrates cooperation while managing risk.

```
{'send': 2.50}
```

reasoning:

Let me think about this situation:

Round 2: I am now the SENDER

- I have \$5
- I need to decide how much to send
- The amount I send will be tripled
- The receiver will decide how much to return

In the previous round:

- I was the receiver
- The sender sent \$2.35 (47% of their endowment)
- I received \$7.05
- I returned \$2.20
- My payoff was \$4.85
- The sender's payoff would have been  $\$2.65 + \$2.20 = \$4.85$

So in the last round, I reciprocated fairly by aiming for equal payoffs.

Now I'm the sender. I should consider:

1. Building trust by sending a reasonable amount
2. The sender in the previous round sent 47% (\$2.35 out of \$5)
3. I reciprocated fairly, so hopefully this receiver will do the same

Since the previous sender showed moderate trust at 47%, I could do something similar. Sending around \$2.35-\$2.50 seems reasonable.

However, I could also think strategically:

- If I send \$2.50, the receiver gets \$7.50
- If they split equally, they'd return \$2.50 to give us each \$5
- If I send \$3.00, the receiver gets \$9.00
- Equal split would mean returning \$3.50 for payoffs of \$5.50 each

Let me try sending \$2.50 as a moderate trust signal, similar to what the previous sender did with me.

Actually, let me reconsider. I showed I'm cooperative by returning fairly in the previous round. I could send a bit more to maximize joint welfare if the receiver reciprocates.

If I send \$3:

- I keep \$2
- Receiver gets \$9
- If they return \$3.50, I get \$5.50 total, they get \$5.50
- Joint welfare: \$11...

**usage:**

```
{
 "input_tokens": 859,
 "output_tokens": 713,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.43 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$9.780000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience:

- Round 1 (receiver): Sender sent 47%, I reciprocated fairly
- Round 2 (sender): I sent 50%, received fair return of \$2.43

The moderate trust strategy (around 50%) has been working well with fair reciprocation from co-players. I'll continue this approach.

{'send': 2.50}

**reasoning:**

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$2.43
- My payoff:  $\$5 - \$2.50 + \$2.43 = \$4.93$
- Their payoff:  $\$7.50 - \$2.43 = \$5.07$

The receiver returned a fair amount - roughly aimed at equal splits. The return rate was  $\$2.43/\$7.50 = 32.4\%$ , which gave us both similar final payoffs.

Now I'm a sender again with a new co-player. I've seen:

- Round 1 (as receiver): Sender sent 47%, I returned fairly
- Round 2 (as sender): I sent 50%, received fair return

Both rounds showed moderate trust and fair reciprocation. I should continue this cooperative strategy. Sending around 50% seems to be working well.

{'send': 2.50}

**usage:**

```
{
 "input_tokens": 1060,
 "output_tokens": 327,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.78 to you. Your visible payoff was \$5.279999999999999.

Your total visible earnings across all rounds are \$15.06.

This round, you are the RECEIVER against your current co-player. They sent you \$2.58, so you received \$7.74. How much do you return? (0-7.74)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.58 (51.6% of endowment), showing solid trust
- I received \$7.74
- Sender currently has \$2.42

Based on my positive experiences with fair reciprocation, I'll aim for equal final payoffs:

- Total:  $\$2.42 + \$7.74 = \$10.16$
- Equal split:  $\$5.08$  each
- Return needed:  $\$2.66$

{'return': 2.66}

**reasoning:**

I'm the receiver now. Let me analyze:

- Sender sent \$2.58 (51.6% of their endowment)
- I received \$7.74 (after 3x multiplier)
- Sender currently has:  $\$5 - \$2.58 = \$2.42$

My recent experience shows:

- Round 2: I sent \$2.50, got back \$2.43
- Round 3: I sent \$2.50, got back \$2.78

Both returns were generous and aimed at fair splits. I should continue this reciprocal fairness.

For equal final payoffs:

- Total available:  $\$2.42$  (sender's remainder) +  $\$7.74$  (my received) =  $\$10.16$
- Equal split:  $\$10.16 / 2 = \$5.08$  each
- Return needed:  $\$5.08 - \$2.42 = \$2.66$

This would give us both \$5.08, which is fair and reciprocates the sender's trust at 51.6%.

Let me verify:

- Sender final:  $\$2.42 + \$2.66 = \$5.08$
- My final:  $\$7.74 - \$2.66 = \$5.08$

Perfect. I'll return \$2.66.

**usage:**

```
{
 "input_tokens": 1287,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.58, it became \$7.74, and you returned \$2.66. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my track record:

- Sending around 50% has consistently resulted in fair reciprocation
- Other agents are also showing similar moderate trust levels
- This strategy has been mutually beneficial

I'll continue with this successful approach.

{'send': 2.50}

**reasoning:**

Let me review my history:

- Round 1 (receiver): Received from sender who sent \$2.35 (47%), returned \$2.20, payoff \$4.85
- Round 2 (sender): Sent \$2.50 (50%), got back \$2.43, payoff \$4.93
- Round 3 (sender): Sent \$2.50 (50%), got back \$2.78, payoff \$5.28
- Round 4 (receiver): Received from sender who sent \$2.58 (51.6%), returned \$2.66, payoff \$5.08
- Round 5 (sender): ???

My strategy of sending around 50% has been working well - I've been getting fair returns.

The pattern shows:

- When I send ~50% (\$2.50), I get back \$2.43-\$2.78
- Others are sending similar amounts (47-52%)
- Fair reciprocation is the norm

I should continue with this moderate trust approach around 50%.

{'send': 2.50}

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 316,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.24 to you. Your visible payoff was \$4.74.

Your total visible earnings across all rounds are \$24.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Sender sent \$2.72 (54.4% of endowment), showing strong trust
- I received \$8.16
- Sender currently has \$2.28

For equal final payoffs:

- Total: \$2.28 + \$8.16 = \$10.44
- Equal split: \$5.22 each
- Return needed: \$2.94

{'return': 2.94}

**Reasoning:**

Let me analyze this situation:

- Sender sent \$2.72 (54.4% of their \$5 endowment) - showing good trust
- I received \$8.16
- Sender currently has \$5 - \$2.72 = \$2.28

For equal final payoffs:

- Total available: \$2.28 + \$8.16 = \$10.44



- Equal split:  $\$10.44 \div 2 = \$5.22$  each
- Return needed:  $\$5.22 - \$2.28 = \$2.94$

This sender showed slightly more trust than average (54.4% vs the ~50% I've been seeing), so I should reciprocate fairly to maintain cooperation.

Let me return \$2.94 to achieve equal payoffs.

```
Usage: {
 "input_tokens": 1725,
 "output_tokens": 298,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Sender - dyad\_2

Timestamp: 2026-08-12T09:24:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_4"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps:  $\$5 - \text{sent} + \text{returned}$
- You keep:  $\text{received} - \text{returned}$
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.34 (66.8% of their endowment)
- I received \$10.02 after the 3x multiplier
- The sender currently has \$1.66 remaining

I want to reciprocate the sender's trust and ensure cooperation is worthwhile for them. If I return \$5.00:

- The sender would end with  $\$1.66 + \$5.00 = \$6.66$  (better than their initial \$5)
- I would keep \$5.02 (also better than any initial amount)

This rewards their trust while maintaining a fair distribution of the gains from cooperation.

```
{'return': 5.00}
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is  $\$3.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$10.02)

This is a trust game. The sender showed considerable trust by sending 66.8% of their endowment. Let me think about a fair return:

The sender:

- Started with \$5
- Sent \$3.34
- Currently has  $\$5 - \$3.34 = \$1.66$
- If I return X, they'll end up with  $\$1.66 + X$

Me:

- I received \$10.02
- If I return X, I keep  $\$10.02 - X$

Some considerations for fairness:

1. Equal split of total: Total money is \$5 (sender's initial) + \$10.02 (what I received) = \$15.02. Equal split would be \$7.51 each.
  - For sender to get \$7.51, they need  $\$7.51 - \$1.66 = \$5.85$  returned
  - I would keep  $\$10.02 - \$5.85 = \$4.17$
2. Proportional return: The sender sent 66.8% of their endowment, showing high trust. A reciprocal gesture might be to return a similar proportion.
3. Rewarding trust: Since the sender trusted me with a large amount, I should return enough to make their trust worthwhile.

Let me check option 1: If I return \$5.85:

- Sender ends with:  $\$1.66 + \$5.85 = \$7.51$
- I end with:  $\$10.02 - \$5.85 = \$4.17$
- This creates mutual benefit from cooperation, and the sender gains significantly from their initial trust.

Returning around \$5 also...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 837,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, the sender showed good trust by sending 66.8% of their endowment, and I reciprocated fairly.

Now as the sender with a new co-player, I'll demonstrate similar trust to establish cooperation. Sending \$3.00 (60% of my endowment) shows significant trust while being reasonable.

If they reciprocate fairly, we both benefit from cooperation.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

Round 1:

- I was the RECEIVER
- Sender sent \$3.34 (66.8% of their endowment)
- I received \$10.02
- I returned \$5.00
- My payoff was \$5.02

Now in Round 2:

- I am the SENDER
- I have \$5
- I need to decide how much to send

The previous sender showed trust by sending 66.8% of their endowment to me, and I reciprocated by returning a fair amount that left them better off.

Now I'm playing against a different agent (the pairing is randomized each round). I should consider:

1. The sender in Round 1 sent a substantial amount (66.8%), showing trust
2. This encouraged cooperation
3. I'm now the sender with a new partner

I should probably send a similar amount to establish trust and encourage cooperation. If I send around 60-70% of my endowment (like the previous sender did), that would be \$3.00-\$3.50.

Let me think about what would be fair:

- If I send \$3.00, they receive \$9.00
- If they return half of what they receive (a common fair split), they'd return \$4.50
- I'd end with  $\$2.00 + \$4.50 = \$6.50$

- They'd keep \$4.50

That seems reasonable. I could also send \$3.34 to match what was sent to me in Round 1, showing I'm willing to trust at the same level.

Actually, let me reconsider. I want to be strategic but also fair. Sending \$3.00 seems like a good middle ground - it's 60% of my endowment, showing significant trust without being extreme.

```
{'send': 3.00}
```

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 530,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.73 to you. Your visible payoff was \$4.73.

Your total visible earnings across all rounds are \$9.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history:

- Round 1 (as receiver): Co-player sent \$3.34, I returned \$5.00
- Round 2 (as sender): I sent \$3.00, received back \$2.73

Now with a new co-player in Round 3, I should continue demonstrating willingness to cooperate. Sending a consistent amount shows I'm a reliable partner.

```
{'send': 3.00}
```

**reasoning:**

In Round 2, I sent \$3.00, it became \$9.00, and the receiver returned \$2.73.

Let me calculate:

- I started with \$5
- I sent \$3, so I had \$2 left
- They received \$9 and returned \$2.73
- My payoff was \$2 + \$2.73 = \$4.73

The receiver kept \$9 - \$2.73 = \$6.27

I got \$4.73

So the return rate was  $\$2.73/\$9.00 = 30.3\%$

This is lower than what I returned in Round 1 (I returned \$5.00 out of \$10.02, which is about 50%).

However, this is a new co-player in Round 3. I shouldn't punish this new person for what the previous receiver did.

I should continue to try to establish cooperation. Maybe send a similar amount to show consistency and willingness to cooperate. However, given that the last return was somewhat low, I might be slightly more cautious.

Let me send \$2.50 - still showing trust but being a bit more conservative.

Actually, let me think about this differently. Each round is with a different co-player (the problem states pairings are randomized). So my actions should be about establishing cooperation with each new partner, not about punishing previous partners.

I should continue to send a reasonable amount to establish trust. Let me send \$3.00 again to be consistent and show I'm willing to cooperate.

**usage:**

```
{
 "input_tokens": 1034,
 "output_tokens": 456,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.57 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed trust by sending \$2.99 (about 60% of their endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has \$2.01

I want to reciprocate their trust fairly. If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (profit of \$1.51)
- I keep: \$4.47

This rewards their cooperation while maintaining a fair split of the gains.

```
{'return': 4.50}
```

**reasoning:**

Let me analyze this round:

- I'm the receiver
- The sender sent \$2.99 (about 60% of their \$5 endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has  $\$5 - \$2.99 = \$2.01$

The sender showed good trust by sending nearly 60% of their endowment. I should reciprocate fairly.

If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (better than their initial \$5)
- I keep:  $\$8.97 - \$4.50 = \$4.47$

This is fair and rewards their trust. It's roughly half of what I received, which seems reasonable given they sent about 60% of their endowment.

Looking at my previous behavior:

- Round 1: Received \$10.02, returned \$5.00 (kept \$5.02)
- Round 3: The other player returned \$5.57 when they received \$9.00

I should be consistent and fair. Returning around \$4.50 seems appropriate.

**usage:**

```
{
 "input_tokens": 1269,
 "output_tokens": 416,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.99, it became \$8.97, and you returned \$4.5. Your visible payoff was \$4.470000000000001.

Your total visible earnings across all rounds are \$21.79.

This round, you are the RECEIVER against your current co-player. They sent you \$2.89, so y

ou received \$8.67. How much do you return? (0-8.67)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.89 (about 58% of their endowment), showing trust
- I received \$8.67 after the 3x multiplier
- The sender currently has \$2.11

I'll reciprocate fairly. If I return \$4.30:

- Sender ends with: \$2.11 + \$4.30 = \$6.41 (profit of \$1.41)
- I keep: \$4.37

This rewards cooperation while maintaining a fair distribution.

{'return': 4.30}

**reasoning:**

The sender sent \$2.89 (about 58% of their \$5 endowment), showing reasonable trust.

I received \$8.67 after the 3x multiplier.

The sender currently has \$5 - \$2.89 = \$2.11

I should reciprocate fairly. Looking at my previous behavior:

- Round 1: Sender sent \$3.34, I received \$10.02, returned \$5.00
- Round 4: Sender sent \$2.99, I received \$8.97, returned \$4.50

This sender sent slightly less (\$2.89 vs \$2.99), so I should return proportionally. If I return \$4.30:

- Sender ends with: \$2.11 + \$4.30 = \$6.41 (profit of \$1.41)
- I keep: \$4.37

This maintains fair reciprocation.

**usage:**

```
{
 "input_tokens": 1545,
 "output_tokens": 349,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.89, it became \$8.67, and you returned \$4.3. Your visible payoff was \$4.37.

Your total visible earnings across all rounds are \$26.16.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my pattern:

- As a sender, I've consistently sent \$3.00 to show trust and encourage cooperation
- As a receiver, I've been returning amounts that fairly reward the sender's trust

I'll maintain my consistent cooperative approach with this new co-player.

{'send': 3.00}

**Reasoning:**

Looking at my history:

- Round 1 (receiver): Received \$10.02 from sender's \$3.34, returned \$5.00
- Round 2 (sender): Sent \$3.00, received back \$2.73
- Round 3 (sender): Sent \$3.00, received back \$5.57
- Round 4 (receiver): Received \$8.97 from sender's \$2.99, returned \$4.50
- Round 5 (receiver): Received \$8.67 from sender's \$2.89, returned \$4.30

Now Round 6, I'm a sender again with a new co-player.

My sending pattern has been consistent at \$3.00 when I'm the sender. The returns I've received have varied (\$2.73 and \$5.57), but both times my co-players received \$9.00 and chose to return different amounts.

I should continue with a consistent cooperative strategy. Sending \$3.00 has been my standard, showing significant trust (60% of endowment).

```
{'send': 3.00}
Usage: {
 "input_tokens": 1796,
 "output_tokens": 332,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_4 - Receiver - dyad\_2

Timestamp: 2026-08-12T09:24:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_4"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

###### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.32, that is 66.4% of their total endowment. You received \$9.959999999999999. How much do you return to the sender? (0-9.959999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.32 (66.4% of their endowment)
- I received \$9.96 after the 3x multiplier
- The sender currently has \$1.68 remaining

I want to reward this trust while being fair to both of us. If I return \$5.00:

- The sender ends with:  $\$1.68 + \$5.00 = \$6.68$  (profit of \$1.68)
- I end with:  $\$9.96 - \$5.00 = \$4.96$

This gives the sender a good return on their trust (33.6% gain on their original \$5) while I also benefit from the exchange. This return rate of approximately 50% is commonly seen as reciprocal in trust games.

{'return': 5.00}

**reasoning:**

Let me analyze this situation:

1. The sender sent \$3.32 out of their \$5 endowment (66.4%)
2. Due to the 3x multiplier, I received \$9.96 ( $3.32 \times 3 = 9.96$ )
3. I need to decide how much to return

**Key considerations:**

- The sender showed significant trust by sending 66.4% of their endowment
- This is a multi-round game with random pairings
- Building a reputation for reciprocity could be beneficial
- There's communication noise, so exact amounts may not be perceived correctly

Let me think about fair/reciprocal returns:

- The sender spent \$3.32 to give me \$9.96
- If I want the sender to profit from trusting me, I should return more than \$3.32
- A common strategy in trust games is to split the gains roughly equally

If I return \$0:

- Sender gets:  $\$5 - \$3.32 + \$0 = \$1.68$
- I get:  $\$9.96 - \$0 = \$9.96$
- Total: \$11.64

If I return \$3.32 (sender breaks even):

- Sender gets:  $\$5 - \$3.32 + \$3.32 = \$5.00$
- I get:  $\$9.96 - \$3.32 = \$6.64$
- Total: \$11.64

If I return \$5.00 (roughly equal split of total):

- Sender gets:  $\$5 - \$3.32 + \$5.00 = \$6.68$
- I get:  $\$9.96 - \$5.00 = \$4.96$
- Total: \$11.64



The sender took a risk and sent a substantial amount. A reasonable reciprocal amount would be to ensure both of us profit from the exchange.

To achieve roughly equal final payoffs, I'd need to return about \$4.14-this would give us each \$5.82. But since the sender invested significantly from their original endowment, I should consider whether an even more generous return would better reward their trust.

If I return \$6.64, they'd end with \$8.32 (doubling their investment gain), though that leaves me with just \$3.32. A middle ground like \$5.00 would give them a 33.6% gain while I keep \$4.96. Since this is the first round and I want to build a reputation for fairness, and given how much trust they demonstrated by sending most of their money, I'm leaning toward a return that meaningfully rewards their cooperation while still leaving me in a reasonable position....

**usage:**

```
{
 "input_tokens": 537,
 "output_tokens": 1319,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.32, it became \$9.959999999999999, and you returned \$5.0. Your visible payoff was \$4.959999999999999.

Your total visible earnings across all rounds are \$4.959999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, my co-player showed significant trust by sending 66.4% of their endowment, and I reciprocated with a 50% return rate.

For this round as sender, I'll demonstrate similar trust to establish cooperative play. Sending \$3.00 (60% of my endowment) shows trust while maintaining some security.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

1. In Round 1, I was the receiver and someone sent me \$3.32 (66.4% of their endowment). I returned \$5.00, which was about 50% of what I received.
2. Now in Round 2, I'm the sender with \$5.
3. I need to decide how much to send to my current co-player (who may be different from Round 1's partner).

Key considerations:

- In the previous round, someone trusted me with 66.4% of their endowment
- I reciprocated fairly by returning 50%
- This established a pattern of moderate trust and reciprocity
- I'm playing with potentially a different agent now

Strategy options:

- a) Send a similar amount to what I received last round (~\$3.32) - this mirrors the trust shown to me
- b) Send less to be more cautious
- c) Send more to encourage cooperation

Given that:

- The previous sender trusted me with 66.4%, showing moderate-high trust
- I want to encourage cooperation in this multi-round game
- Sending around the same proportion (60-70%) seems reasonable
- This could establish a cooperative norm

I'll send \$3.00 (60% of my endowment), which is:

- Similar to what was sent to me in Round 1

- Shows trust but not maximal trust
- Allows me to keep some guarantee (\$2)
- If returned fairly (~50%), I'd get back around \$4.50, making my total  $\$2 + \$4.50 = \$6.50$

Let me go with \$3.00.

**usage:**

```
{
 "input_tokens": 864,
 "output_tokens": 471,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.51 to you. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$11.469999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.27 (45.4% of their endowment), showing moderate trust
- I received \$6.81 after the 3x multiplier
- The sender currently has \$2.73 remaining

In the previous round when I was sender, I sent 60% and received back about 50% of what my co-player received. I'll reciprocate with a similar return rate.

Returning \$3.50 gives:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return, which fairly rewards their trust.

{'return': 3.50}

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.27, which is 45.4% of their \$5 endowment
- I received \$6.81 after the 3x multiplier
- The sender currently has  $\$5 - \$2.27 = \$2.73$  remaining

In the previous round when I was sender:

- I sent \$3.00 (60% of endowment)
- Received back \$4.51, which was about 50% of the \$9.00 they received
- This gave me a payoff of \$6.51

The sender this round showed moderate trust (45.4% vs my 60% last round). I should reciprocate fairly.

If I return \$3.50:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return rate, similar to what I received last round (50%). This seems fair and reciprocal.

Let me check the math:

- Return rate =  $\$3.50 / \$6.81 = 51.4\%$
- Sender's final payoff =  $\$2.73 + \$3.50 = \$6.23$
- My final payoff =  $\$6.81 - \$3.50 = \$3.31$

This seems reasonable and reciprocal given their 45.4% trust level.

**usage:**

```
{
 "input_tokens": 1104,
 "output_tokens": 545,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.5. Your visible payoff was \$3.3100000000000005.

Your total visible earnings across all rounds are \$14.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience so far:

- When I've shown trust as sender (60% in Round 2), I received good reciprocity (~50% return)
- When others showed trust as sender, I've returned approximately 50%
- The pattern suggests cooperative play with moderate to high trust levels

I'll continue with a similar strategy, sending \$3.00 (60% of my endowment) to maintain trust and cooperation.

```
{'send': 3.00}
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.32 (66.4%)
- I received \$9.96
- I returned \$5.00 (about 50%)
- My payoff: \$4.96

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.51 (about 50%)
- My payoff: \$6.51

Round 3: I was RECEIVER

- Sender sent \$2.27 (45.4%)
- I received \$6.81
- I returned \$3.50 (about 51%)
- My payoff: \$3.31

Round 4: I am SENDER now

- I have \$5 to decide what to send

Looking at the pattern:

- In Round 2, I sent \$3.00 (60%) and got back about 50%
- Others have sent between 45% and 66% in the rounds I've seen

I've been experiencing decent reciprocity so far. The return rates have been around 50% consistently.

I should continue to show trust, as it's been rewarded. Sending around 60% seems reasonable based on what I've seen. I'll send \$3.00 again.

**usage:**

```
{
 "input_tokens": 1421,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$21.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience as sender:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)

Sending 60% of my endowment has consistently led to good reciprocity. I'll maintain this cooperative strategy.

{'send': 3.00}

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: I sent \$3.00, received \$4.44 back (payoff \$6.44)

Both times sending \$3.00 has resulted in good returns. The strategy of sending 60% has been working well and establishing cooperative relationships.

I should continue with this consistent strategy to maintain trust and cooperation.

**usage:**

```
{
 "input_tokens": 1648,
 "output_tokens": 217,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.38 to you. Your visible payoff was \$6.38.

Your total visible earnings across all rounds are \$27.599999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.26, so you received \$9.78. How much do you return? (0-9.78)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

This is similar to Round 1 where I received high trust. I've been consistently returning around 50% as receiver, which has worked well.

Returning \$5.00:

- Sender ends with:  $\$1.74 + \$5.00 = \$6.74$  (profit of \$1.74)
- I end with:  $\$9.78 - \$5.00 = \$4.78$

This maintains my reciprocal strategy.

{'return': 5.00}

**Reasoning:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

In my previous rounds as receiver:

- Round 1: Received \$9.96, returned \$5.00 (50.2%)
- Round 3: Received \$6.81, returned \$3.50 (51.4%)

As sender, I've been sending \$3.00 and receiving returns of around \$4.40-4.50, which is about 49-50% of what the receiver got.

This sender showed high trust (65.2%). I should reciprocate with a similar 50% return rate.

Returning \$5.00:

- Sender ends with:  $\$1.74 + \$5.00 = \$6.74$  (profit of \$1.74)
- I end with:  $\$9.78 - \$5.00 = \$4.78$

This is about 51% return, consistent with my previous behavior as receiver.

```
Usage: {
 "input_tokens": 1888,
 "output_tokens": 439,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_5 - Sender - dyad\_3

Timestamp: 2026-08-12T09:24:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_8"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps:  $\$5 - \text{sent} + \text{returned}$

- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

#### Message 3 role=assistant

##### content:

Looking at this first round as a sender, I need to balance trust and caution since I don't know my co-player yet.

The key consideration is that sending money creates value through the 3x multiplier. If I send \$3:

- I keep \$2
- The receiver gets \$9 ( $3 \times \$3$ )
- If they return even a moderate amount (say \$3-4), we both benefit compared to sending nothing

This shows willingness to cooperate while maintaining some security. It's a reasonable opening move that could encourage reciprocity.

```
'''json
{"send": 3}
'''
```

##### reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategy:

##### Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- There's communication noise, so amounts may differ from what was actually sent/returned
- This is a multi-agent setting with 8 agents, randomized pairings each round
- I'll play both sender and receiver roles across rounds

##### Strategic considerations:

1. **\*\*Trust and cooperation\*\***: If I send money, it creates value (3x multiplier). If the receiver is cooperative, they might return a fair share, making both of us better off.
2. **\*\*Risk\*\***: The receiver could keep everything. If I send \$5, they get \$15, and might return nothing, leaving me with \$0.
3. **\*\*First round dynamics\*\***: This is the first round, so there's no history to base decisions on. I don't know how cooperative my co-player will be.
4. **\*\*Future rounds\*\***: Since this is a multi-round game with different partners, building a reputation for cooperation might matter, though the anonymity and random pairing complicate this.

5. **\*\*Communication noise\*\***: This adds uncertainty - even if someone tries to be fair, the noise might make it seem otherwise.
6. **\*\*Optimal cooperation\*\***: If both parties cooperate optimally, I could send all \$5, receiver gets \$15, and if they split fairly, they might return around \$7-10, making us both better off than the initial \$5.

Common strategies in trust games:

- Send nothing (risk-averse, guarantees keeping \$5)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - full trust, maximizes potential gains

For the first round with an unknown partner, a moderate approach might be reasonable. Sending about \$3-4 shows willingness to cooperate while not being completely vulnerable. Since this is a repeated game with multiple rounds and different partners, demonstrating cooperative behavior early could encourage reciprocity....

usage:

```
{
 "input_tokens": 500,
 "output_tokens": 768,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$5.6899999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.09, so you received \$6.27. How much do you return? (0-6.27)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.09 out of \$5 (about 42%)
- Showed moderate trust

**\*\*My considerations:\*\***

- In Round 1, my partner returned about 41% of what they received
- Reciprocating fairly would mean returning a similar proportion
- This encourages future cooperation

Returning approximately 45% of the \$6.27 I received (\$2.82) seems fair. This rewards their trust, gives them a modest profit, and keeps a reasonable amount for myself.

```
'''json
{"return": 2.82}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$3
- It became \$9
- They returned \$3.69
- My payoff: \$5 - \$3 + \$3.69 = \$5.69

Now Round 2 (as RECEIVER):

- My co-player sent me \$2.09
- I received \$6.27 (which is 3 × \$2.09)
- I need to decide how much to return

Key considerations:

1. My previous co-player returned about 41% of what they received (\$3.69 out of \$9)

2. This current co-player sent \$2.09 out of \$5, which is about 42% - showing moderate trust
3. There's mention of "communication noise" - the amounts we see may differ slightly from actual amounts

A fair reciprocity strategy would be to return roughly the same proportion that was returned to me, or perhaps slightly reward the trust shown by sending a decent amount.

If I return about 40% of \$6.27, that would be around \$2.51.  
 If I'm more generous and return 45%, that would be around \$2.82.  
 If I return 50%, that would be \$3.14.

Given that:

- The sender showed trust by sending \$2.09 (42% of their endowment)
- My previous partner returned about 41%
- I want to encourage future cooperation

I'll return around \$2.80-\$3.00, which is roughly 45-48% of what I received. This rewards their trust while keeping a reasonable amount.

Let me go with \$2.80 - this leaves them with approximately \$5 - \$2.09 + \$2.80 = \$5.71, a modest gain, and I keep \$6.27 - \$2.80 = \$3.47, a solid gain for both of us.

usage:

```
{
 "input_tokens": 781,
 "output_tokens": 590,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.09, it became \$6.27, and you returned \$2.82. Your visible payoff was \$3.4499999999999997.

Your total visible earnings across all rounds are \$9.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

content:

Looking at my position:

**\*\*Previous experience:\*\***

- Round 1: Sent \$3, received \$3.69 back ? good outcome
- Round 2: As receiver, returned a fair amount (\$2.82 on \$6.27)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- No history with this specific agent
- Need to signal willingness to cooperate

Since I had success with sending \$3 in Round 1 (received \$3.69 back, net gain of \$0.69), and it establishes a pattern of moderate trust, I'll use the same strategy with this new co-player.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze my experience so far:

Round 1: I was SENDER

- Sent \$3, they received \$9
- They returned \$3.69 (about 41% of what they received)
- My payoff: \$2 (kept) + \$3.69 (returned) = \$5.69

Round 2: I was RECEIVER

- They sent \$2.09, I received \$6.27



- I returned \$2.82 (about 45% of what I received)
- My payoff: \$6.27 - \$2.82 = \$3.45

Total earnings: \$9.14

Now Round 3: I'm SENDER again with a new co-player (since agents are randomized each round)

What should I consider:

1. This is a new opponent - I don't know their strategy
2. In Round 1, sending \$3 worked reasonably well (got back \$3.69, netted \$5.69 vs \$5 if I sent nothing)
3. The multiplier effect means cooperation is beneficial
4. I should probably maintain a consistent strategy early on

My Round 1 strategy of sending \$3 was a moderate trust signal that worked well. Since I'm facing a new opponent, I should probably use a similar approach - not too cautious, not too risky.

I'll send \$3 again as my baseline strategy for first encounters with new co-players.

**usage:**

```
{
 "input_tokens": 1050,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.37 to you. Your visible payoff was \$7.37.

Your total visible earnings across all rounds are \$16.509999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.94 out of \$5 (nearly 59%)
- Showed strong trust

**\*\*Recent patterns:\*\***

- Round 3: My partner was very generous, returning \$5.37 (60% of \$9)
- Round 2: I returned about 45% as receiver
- This co-player is being quite trusting

**\*\*Fair reciprocity:\*\***

Returning around 48% of the \$8.82 I received would be fair and encourage future cooperation. This gives my co-player a decent profit while rewarding their trust.

```
'''json
{"return": 4.20}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER\*\***

- My co-player sent \$2.94 (about 59% of their \$5)
- I received \$8.82 (3x multiplier)
- Need to decide how much to return

**\*\*Recent history:\*\***

- Round 3: As sender, I sent \$3, received \$5.37 back (about 60% of the \$9 they got) - very generous!
- Round 2: As receiver, I returned \$2.82 on \$6.27 (about 45%)

- Round 1: As sender, sent \$3, received \$3.69 back (about 41% of the \$9)

**\*\*Analysis of current round:\*\***

- Co-player sent \$2.94 (quite generous - nearly 60% of their endowment)
- This shows significant trust
- They'll end up with:  $\$5 - \$2.94 + \text{return} = \$2.06 + \text{return}$

**\*\*Fairness considerations:\*\***

- If I return about 45-50% like in Round 2:  $\sim \$3.96 - \$4.41$
- If I match the generosity of Round 3's partner (60%):  $\sim \$5.29$
- Co-player sacrificed a lot to send \$2.94

I should reciprocate fairly. They sent nearly 60% of their endowment, showing strong trust  
. I should return a reasonable amount that:

1. Rewards their trust
2. Gives them a profit
3. Maintains cooperation

Returning around \$4.00-\$4.50 seems fair:

- Gives them a total of  $\sim \$6.06 - \$6.56$
- Nets them  $\$1.06 - \$1.56$  profit
- I keep  $\$4.32 - \$4.82$
- Return rate of  $\sim 45 - 51\%$

...

**usage:**

```
{
 "input_tokens": 1348,
 "output_tokens": 615,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.94, it became \$8.82, and you returned \$4.2. Your visible payoff was \$4.62.

Your total visible earnings across all rounds are \$21.13.

This round, you are the RECEIVER against your current co-player. They sent you \$1.7, so you received \$5.1. How much do you return? (0-5.1)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- More cautious than previous partners

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)

**\*\*Fair reciprocity:\*\***

Returning around 45% of the \$5.10 I received maintains consistency and fairness. This gives my co-player a modest profit while rewarding their trust, even though they were more cautious.

```
'''json
{"return": 2.30}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- This is relatively cautious/low trust

**\*\*Recent patterns from my play:\*\***

- Round 2: Returned \$2.82 on \$6.27 (45%)
- Round 4: Returned \$4.20 on \$8.82 (48%)

**\*\*Considerations:\*\***

- They showed less trust by sending only \$1.70
- However, I should still be fair to encourage cooperation
- Returning around 40-45% would be consistent with my previous behavior

Returning about 45% of \$5.10 = ~\$2.30

This gives them back a reasonable amount, rewards some level of trust, and maintains consistency in my reciprocity rate.

Let me calculate:

- They sent \$1.70, kept \$3.30
- I received \$5.10
- If I return \$2.30:
  - They end with:  $\$3.30 + \$2.30 = \$5.60$  (gain of \$0.60)
  - I end with:  $\$5.10 - \$2.30 = \$2.80$

That seems fair. They made a small profit for trusting me.

**usage:**

```
{
 "input_tokens": 1636,
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.7, it became \$5.1, and you returned \$2.3. Your visible payoff was \$2.8. Your total visible earnings across all rounds are \$23.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my position:

**\*\*Sender history:\*\***

- Round 1: Sent \$3 ? received \$3.69 back (good return)
- Round 3: Sent \$3 ? received \$5.37 back (excellent return)

**\*\*Strategy:\*\***

Sending \$3 has been consistently successful for me. It demonstrates trust while creating substantial value (\$9 for the receiver), which has encouraged good reciprocity from past partners.

With a new co-player this round, I'll continue with this proven strategy.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3, got back \$3.69 (net +\$0.69)
- Round 3: Sent \$3, got back \$5.37 (net +\$2.37)

As RECEIVER:

- Round 2: Received \$6.27, returned \$2.82
- Round 4: Received \$8.82, returned \$4.20
- Round 5: Received \$5.10, returned \$2.30

My sending strategy of \$3 has worked well - I got good returns both times. This is a new co-player (randomized pairing), so I don't have specific history with them.

Sending \$3 seems to be a sweet spot:

- Shows trust and willingness to cooperate
- Creates substantial value (\$9 for receiver)
- Has resulted in good returns in past rounds

I'll continue with \$3 as it's been my successful strategy.

```
Usage: {
 "input_tokens": 1913,
 "output_tokens": 363,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_8 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:25:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_8"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the SENDER, I need to balance trust and caution since I have no history with this co-player.

The key consideration is that sending money creates value (3x multiplier), but I'm vulnerable to the receiver's decision about how much to return.

A moderate amount seems reasonable for Round 1:

- It signals willingness to cooperate and trust
- It creates meaningful value (\$9 for the receiver)
- It provides some protection against complete loss

I'll send \*\*\$3\*\*.

{'send': 3}

**reasoning:**

This is Round 1 of a multi-task experiment where I'm playing a trust game as the SENDER. Let me think about this situation:

1. I start with \$5
2. I can send any amount from \$0 to \$5
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - amount\_sent + amount\_returned

This is a classic trust game. The optimal strategy depends on:

- Whether I trust the receiver to return a fair amount
- What "fair" means in this context
- The fact that there's communication noise (amounts may differ slightly)
- This is a multi-round game with multiple agents
- Pairings are randomized each round

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I'd end up with \$7.50 total (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- The socially optimal outcome is to send the maximum amount to create the most total value

However, there's risk:

- The receiver might not return anything
- There's communication noise that could affect trust
- This is round 1, so I have no history with this partner

A reasonable first-round strategy might be:

- Send a moderate amount to signal trust and willingness to cooperate
- Not send everything in case the receiver doesn't reciprocate
- Send enough to create meaningful value

Common strategies in trust games:

- Send about half (\$2.50) as a balanced approach
- Send more (\$3-4) to signal strong trust
- Send everything (\$5) to maximize joint payoff

Given this is round 1 and I want to establish a cooperative pattern but also protect against complete exploitation, I'll send \$3. This:

- Creates \$9 for the receiver
- Shows trust and cooperation

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 625,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.46 to you. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$4.46.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 2):\*\***

- My co-player sent me \$3.19 ? I received \$9.57
- This created \$6.38 in surplus value

**\*\*Fair return considerations:\*\***

- The sender took a significant risk sending most of their endowment
- A reasonable approach is to ensure we both benefit from the cooperation
- Returning \$4.00 would give:
  - Sender: \$5 - \$3.19 + \$4.00 = \$5.81
  - Me: \$9.57 - \$4.00 = \$5.57

This creates a balanced outcome where we both gain from the exchange.

```
{'return': 4.00}
```

**reasoning:**

Let me analyze this situation:

Last round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.46 to me
- My payoff: \$5 - \$3 + \$2.46 = \$4.46
- Their payoff: \$9.0 - \$2.46 = \$6.54

So they returned about 27% of what they received (\$2.46/\$9.0 ? 0.273).

Now in Round 2:

- I'm the RECEIVER
- My current co-player sent me \$3.19
- I received \$9.57 (3.19 × 3)
- I need to decide how much to return

Important notes:

1. This is a different co-player (multi-agent setting, randomized pairings)
2. There's communication noise, so the \$3.19 might not be exactly what they sent
3. The previous player returned about 27% to me

Considerations:

- If I want to encourage cooperation, I should return a fair amount
- A common fair split might be to return enough so we both benefit equally from the cooperation
- The sender gave up \$3.19, so they have  $\$5 - \$3.19 = \$1.81$  left
- I have \$9.57
- If I want us to end up with similar amounts, I could return about \$3.88, giving us both around \$5.69
- Or I could return proportionally to what was sent: returning about 1/3 of what I received would be \$3.19, which gives them back their original amount
- A generous return might be half of what I received: \$4.785

Let me think about what's fair:

- The sender took a risk sending \$3.19 (gave up most of their endowment)
- The transaction created \$6.38 in surplus value beyond what they sent
- Splitting that gain equally means we each get an additional \$3.19 from the multiplier effect
- So returning \$3.19 would leave me with \$6.38 and restore them to their starting \$5
- Alternatively, returning \$4 would give them \$5.81 while I keep \$5.57

, creating a more balanced outcome

- Given the earlier low reciprocation (27%), I could...

**usage:**

```
{
 "input_tokens": 756,
 "output_tokens": 1027,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.19, it became \$9.57, and you returned \$4.0. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$10.030000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this new round as SENDER:

**\*\*Past experience:\*\***

- Round 1: Sent \$3, received \$2.46 back (modest cooperation)
- Round 2: When I received, I returned fairly (\$4.00 on \$9.57)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Need to establish trust with someone new
- The 3x multiplier means cooperation creates value

A moderate send amount balances the potential gains from cooperation with the risk of low returns. Sending \$3 creates \$9 of value while maintaining some protection.

```
{'send': 3}
```

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds:\*\***

- Round 1: I was SENDER, sent \$3, received back \$2.46, payoff was \$4.46
- Round 2: I was RECEIVER, received \$9.57 (from \$3.19 sent), returned \$4.00, payoff was \$5.57

**\*\*Current Round (Round 3):\*\***

- I'm the SENDER again
- New co-player (since pairings are randomized each round)
- I have \$5 to potentially send

**\*\*Observations:\*\***

1. In Round 1, when I sent \$3, I got back \$2.46. The receiver kept \$6.54 out of \$9.00, which seems somewhat selfish (I got less than my initial \$3).
2. In Round 2, when someone sent me \$3.19, I returned \$4.00, being relatively fair.
3. The note about communication noise is important - amounts may differ from what was actually sent/returned.

**\*\*Strategy considerations:\*\***

- I'm playing with a new co-player each round
- There's no reputation to build with a specific player
- The 3x multiplier means cooperation creates value
- I need to balance trust against the risk of low returns

Given:

- My Round 1 experience showed modest cooperation from my receiver
- I was reasonably fair in Round 2
- I'm with a new player now who doesn't know my history

I think sending \$3 again is reasonable:

- It's a moderate level of trust
- Creates \$9 for the receiver
- Provides some protection if they don't reciprocate well

Let me send \$3 again.

**usage:**

```
{
 "input_tokens": 1047,
 "output_tokens": 532,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$14.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.22, so you received \$9.66. How much do you return? (0-9.66)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 4):\*\***

- Co-player sent \$3.22 ? I received \$9.66
- This represents significant trust (sending 64% of their endowment)

**\*\*Fair return analysis:\*\***

- Returning \$4.00 would give:
  - Sender: \$5 - \$3.22 + \$4.00 = \$5.78 (profit of \$0.78)
  - Me: \$9.66 - \$4.00 = \$5.66 (profit of \$0.66)

This creates a fair split where both players benefit from the cooperation.

```
{'return': 4.00}
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER:\*\***

- My co-player sent me \$3.22
- I received \$9.66 (after 3x multiplier)
- I need to decide how much to return

**\*\*Previous rounds analysis:\*\***

- Round 1 (SENDER): Sent \$3, got back \$2.46 - slightly below fair
- Round 2 (RECEIVER): Received \$9.57, returned \$4.00 - tried to be fair
- Round 3 (SENDER): Sent \$3, got back \$2.93 - again slightly below fair



**\*\*Fair return calculation:\*\***

The sender gave up \$3.22 from their \$5 endowment, creating \$9.66 for me.

- Surplus created:  $\$9.66 - \$3.22 = \$6.44$

A fair split would give us each roughly equal gains:

- If I return \$4.00:

- Sender gets:  $\$5 - \$3.22 + \$4.00 = \$5.78$  (gain of \$0.78)

- I get:  $\$9.66 - \$4.00 = \$5.66$  (gain of \$0.66)

This seems reasonably balanced. The sender sent a significant portion of their endowment, showing trust.

I should return around \$4.00 to be fair and reciprocate their trust.

**usage:**

```
{
 "input_tokens": 1324,
 "output_tokens": 501,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.22, it became \$9.66, and you returned \$4.0. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$20.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience so far:

**\*\*As SENDER:\*\***

- Round 1: Sent \$3 ? received \$2.46 (modest return)

- Round 3: Sent \$3 ? received \$2.93 (decent return)

**\*\*As RECEIVER:\*\***

- I've consistently returned \$4.00 on ~\$9.60, maintaining fairness

**\*\*Current round:\*\***

- New co-player (randomized pairing)

- Sending \$3 has been my standard opening move

- Creates \$9 of value with moderate risk

I'll continue with the balanced approach that's been working.

{'send': 3}

**reasoning:**

Looking at my history as SENDER:

- Round 1: Sent \$3, received \$2.46 back

- Round 3: Sent \$3, received \$2.93 back

Both times I sent \$3, I received back around \$2.50-\$3.00, which is close to fair but not quite equal splitting of the gains.

**As RECEIVER:**

- Round 2: Received \$9.57, returned \$4.00

- Round 4: Received \$9.66, returned \$4.00

I've been consistent in returning fairly.

Now Round 5, I'm a SENDER with a new co-player. The pattern suggests that sending around \$3 has been reasonably productive. The returns haven't been amazing (I've profited minimally as sender), but they've been positive.

I should continue with a moderate strategy. Sending \$3 creates value and has worked reasonably well so far.

**usage:**

```
{
 "input_tokens": 1601,
 "output_tokens": 366,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.2 to you. Your visible payoff was \$7.2.

Your total visible earnings across all rounds are \$27.82.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This is a slightly smaller send than previous rounds

**\*\*Fair return analysis:\*\***

- In previous rounds, I returned ~\$4.00 on receives of ~\$9.60
- This represents about a 42% return rate
- For \$8.58, a proportional return would be around \$3.60

Returning \$3.70 would give:

- Sender: \$5 - \$2.86 + \$3.70 = \$5.84 (modest profit)
- Me: \$8.58 - \$3.70 = \$4.88

This maintains fairness while ensuring both players benefit from cooperation.

{'return': 3.70}

**Reasoning:**

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This shows trust, sending about 57% of their endowment

**\*\*Fair return analysis:\*\***

- Total surplus created: \$8.58 - \$2.86 = \$5.72
- A fair split would give each of us roughly \$2.86 in profit

If I return \$3.50:

- Sender gets: \$5 - \$2.86 + \$3.50 = \$5.64 (profit of \$0.64)
- I get: \$8.58 - \$3.50 = \$5.08 (profit of \$0.08)

That's not quite balanced. Let me try \$4.00:

- Sender gets: \$5 - \$2.86 + \$4.00 = \$6.14 (profit of \$1.14)
- I get: \$8.58 - \$4.00 = \$4.58 (profit of \$-0.42)

Wait, that would give me less than my base \$5. Let me recalculate.

Actually, I don't start with \$5 as the receiver. I start with \$0 and receive the multiplied amount.

So:

- Sender starts with \$5, sends \$2.86, gets back \$X
- Sender final: \$5 - \$2.86 + \$X = \$2.14 + \$X
- I start with \$0, receive \$8.58, return \$X
- My final: \$8.58 - \$X

For a balanced outcome where we both profit similarly:

- Sender profit: \$2.14 + \$X - \$5 = \$X - \$2.86
- My profit: \$8.58 - \$X - \$0 = \$8.58 - \$X

- Setting equal:  $\$X - \$2.86 = \$8.58 - \$X$  gives  $\$X = \$5.72$

That would split the surplus evenly, but feels generous. Instead, I should ensure the sender profits from their trust while keeping a reasonable share for myself. Returning around \$3.80 would give them roughly \$0.94 in profit while I keep \$4.78.

But that's heavily skewed in my favor. Testing \$4.50 or \$5.00 returns gets closer to balance - the sender would profit \$1.64 to \$2.14 while I'd keep \$3.58 to \$4.08. I'm reconsidering what fairness actually means here. In previous exchanges where I received around \$9.50-\$9.66 from sends of \$3.19-\$3.22, returning \$4.00 felt proportional. Using that same ratio (roughly 42% return rate) on this \$8.58 suggests returning around \$3.58-\$3.70. Testing \$3.70 gives the sender \$0.84 profit and me \$4.88, which still favors me but follows the pattern I've established in earlier rounds....

```
Usage: {
 "input_tokens": 1883,
 "output_tokens": 1261,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Sender - dyad\_4

Timestamp: 2026-08-12T09:25:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_7"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the RECEIVER. the sender sent \$2.55, that is 51.0% of their total endowment. You received \$7.649999999999995. How much do you return to the sender? (0-7.649999999999995)

#### Message 3 role=assistant

##### content:

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.55 (51% of their \$5 endowment), showing moderate trust
- I received \$7.65 after the 3x multiplier
- The sender now has \$2.45 remaining
- I need to decide how much to return

**\*\*My reasoning:\*\***

The sender showed significant trust by sending about half their endowment. To reciprocate fairly and build trust:

If I return **\*\*\$4.00\*\***:

- Sender ends with:  $\$2.45 + \$4.00 = \$6.45$  (profit of \$1.45)
- I end with:  $\$7.65 - \$4.00 = \$3.65$  (profit of \$3.65)

This represents:

- Returning about 52% of what I received (similar to their 51% sent)
- Both of us profit from the cooperation
- Rewarding their trust encourages future cooperation
- Setting a positive precedent for Round 1

{'return': 4.0}

##### reasoning:

Let me understand the situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.55 (which is 51% of their \$5 endowment)
3. This amount was multiplied by 3x, so I received  $\$2.55 \times 3 = \$7.65$  (approximately \$7.65)
4. I need to decide how much to return (between \$0 and \$7.65)

Now, let me think about this strategically:

The sender sent 51% of their endowment, which is slightly more than half. This shows a moderate level of trust. They could have sent nothing, they could have sent everything, but they chose to send about half.

From a fairness perspective:

- The sender now has:  $\$5 - \$2.55 = \$2.45$
- I have: \$7.65

- Total in the system:  $\$2.45 + \$7.65 = \$10.10$

If I want to be fair and reciprocate the trust:

- One approach: Split the total gains equally. The sender started with \$5, and we now have \$10.10 total. If we split equally, we'd each have \$5.05. The sender already has \$2.45, so I'd return  $\$5.05 - \$2.45 = \$2.60$
- Another approach: Return the same proportion they sent. They sent 51% of their endowment, so I could return 51% of what I received:  $0.51 \times \$7.65 = \$3.90$
- Conservative approach: Return at least what they sent originally (\$2.55) to ensure they don't lose money from trusting me
- Generous approach: Return more to reward their trust and potentially build cooperation

Given that:

1. The sender showed moderate trust by sending 51%
2. There's communication noise mentioned (though the exact amount sent seems precise)
3. Since this is a multi-round game with multiple agents, establishing a reputation for reciprocity could be strategically valuable
4. The sender took a genuine risk by sending money

A fair reciprocal response should reward their trust while ensuring they benefit from the exchange and remain proportional to their initial gesture. If I return \$2.60, we'd each end with \$5.05-a modest profit for both of us that acknowledges their willingness to trust.

Alternatively, returning 50% of what I received (\$3.825) would give them \$6.275, a more substantial gain. But I need to reconsider my baseline here-as the receiver, I don't start with \$5 like the sender does. I begin with \$0 and only gain what they send multiplied by three. The sender started with \$5, sent \$2.55, and now has \$2.45 remaining.

If I return \$X, they end with  $\$2.45 + \$X$  and I end with  $\$7.65 - \$X$ . For equal splits, that's \$2.60 returned, giving us both \$5.05. But in this experimental context, I should probably return more-around \$3.00-\$3.50-to reward their trust and signal reciprocity for potential future rounds.

Testing a few amounts: returning \$3.25 leaves them with \$5.70 and me with \$4.40; \$3.50 gives them \$5.95 and me \$4.15; \$4.00 results in them having \$6.45 and me \$3.65. Each option balances showing trustworthiness with keeping reasonable profit, though I'm leaning toward the higher end to maximize their incentive to continue cooperating. Both of us end up better off than we started. At \$3.50, the sender gets \$5.95 (up \$0.95) and I get \$4.15 (up \$4.15), which feels like a solid reciprocal move that rewards their trust without being exploitative.

I'm settling on \$3.50 as my return amount-it strikes a balance between showing genuine reciprocity and ensuring both parties benefit from the exchange. Terms of what's strategically optimal in a multi-round setting. Building reputation matters-if I'm generous now, other agents will likely trust me more later. Since this is Round 1, establishing a trustworthy precedent is crucial. The sender showed moderate confidence by sending \$2.55, so I should reciprocate proportionally.

I'm settling on \$3.50. It more than doubles their original amount, gives them solid profit, signals I'm reliable, and leaves me with a reasonable gain too. Let me verify the math:  $\$2.55 \times 3 = \$7.65$  received, returning \$3.50 means the sender nets \$0.95 profit while I keep \$4.15.

The communication noise mentioned in the problem is worth noting, but I can only work with what I observe. My final return amount is \$3.50, formatted as `{'return': 3.50}`.

Actually, reconsidering the proportionality here-the sender gave 51% of their endowment, which signals moderate trust. Matching that proportion with my return would mean sending back 51% of \$7.65, which equals \$3.90. This reciprocal approach better reflects their level of generosity while still leaving me with a reasonable profit of \$3.75 versus their \$1.35. I think \$3.90 better captures the spirit of proportional reciprocation.

Though I'm also drawn to rounding up to \$4.00 for simplicity-it's nearly the same percentage anyway (52%), easier to process, and still feels fair to both parties. Let me settle on \$4.00 as my final choice.

usage:

```
{
 "input_tokens": 539,
 "output_tokens": 3346,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.55, it became \$7.649999999999995, and you returned \$4.0. Your visible payoff was \$3.649999999999995.

Your total visible earnings across all rounds are \$3.649999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- Round 1 established a cooperative dynamic where sending ~51% led to mutual profit
- I'm now the sender with a new co-player
- I need to establish trust in this new pairing

**\*\*Strategic reasoning:\*\***

- Sending a moderate amount (around 50%) shows willingness to cooperate
- It gives my co-player enough to make returning worthwhile (\$7.50 after multiplier)
- It limits my downside risk while allowing mutual gains
- This mirrors the successful approach from Round 1

**\*\*Decision:\*\***

I'll send \$2.50 (50% of my endowment) to establish a cooperative tone with this new co-player.

{'send': 2.5}

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was the RECEIVER
- Sender sent \$2.55 (51% of \$5)
- I received \$7.65
- I returned \$4.00
- My payoff: \$3.65
- Sender's payoff would have been:  $\$5 - \$2.55 + \$4.00 = \$6.45$

**\*\*Current Round (Round 2):\*\***

- I am now the SENDER
- I have \$5
- I'm paired with a potentially different agent (since it says "your current co-player")
- I need to decide how much to send

**\*\*Strategic considerations:\*\***

1. **\*\*Reciprocity principle\*\***: In Round 1, my co-player showed moderate trust (51% sent). I reciprocated well by returning 52% of what I received, ensuring both parties profited.
2. **\*\*Building trust\*\***: Since this is a multi-round game with multiple agents, establishing a cooperative reputation could be beneficial.
3. **\*\*Unknown opponent\*\***: This is a different co-player, so I don't know their reciprocity tendencies yet.
4. **\*\*First impression\*\***: As the sender in this new pairing, I can set the tone for cooperation.
5. **\*\*Risk vs reward\*\***:
  - If I send \$0, I keep \$5, they get \$0, no mutual benefit

- If I send all \$5, it becomes \$15 for them, but I'm vulnerable if they return nothing
- Middle ground shows trust while limiting downside risk

6. **\*\*Following established norms\*\***: In Round 1, the sender sent 51%. This seemed to work well and encouraged cooperation.

**\*\*Decision\*\***

I'll send a similar amount to what I experienced as effective in Round 1 - around 50% (\$2.50). This:

- Shows trust and willingness to cooperate
- Limits my downside risk
- Gives the receiver enough to make returning worthwhile

...

**usage:**

```
{
 "input_tokens": 916,
 "output_tokens": 646,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.62 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$9.77.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened\*\***

- The sender sent \$2.11 (42.2% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- The sender now has \$2.89 remaining

**\*\*My reasoning\*\***

The sender showed trust, though slightly less than what I've seen in previous rounds. I should reciprocate fairly to maintain cooperation.

If I return **\*\*\$3.30\*\***:

- Sender ends with: \$2.89 + \$3.30 = \$6.19 (profit of \$1.19)
- I end with: \$6.33 - \$3.30 = \$3.03 (profit of \$3.03)

This represents:

- Returning about 52% of what I received
- Both of us profiting from cooperation
- Fair reciprocation that encourages future trust

{'return': 3.3}

**reasoning:**

Let me analyze this situation:

**\*\*What happened in Round 2\*\***

- I sent \$2.50
- It became \$7.50
- They returned \$3.62
- My payoff: \$2.50 (kept) + \$3.62 (returned) = \$6.12
- Their payoff: \$7.50 - \$3.62 = \$3.88
- They returned about 48% of what they received

**\*\*Current Round 3 situation\*\***

- I'm the receiver
- Sender sent \$2.11 (42.2% of their \$5 endowment)
- I received \$6.33
- Sender now has \$2.89 remaining

- I need to decide how much to return

**\*\*Analysis:\*\***

The sender sent \$2.11, which is less than the \$2.50-\$2.55 amounts I've seen previously. They're showing moderate but slightly lower trust. However, they still sent a significant portion (42%).

If I return different amounts:

- Return \$3.00: Sender ends with \$5.89 (+\$0.89), I end with \$3.33 (+\$3.33)
- Return \$3.50: Sender ends with \$6.39 (+\$1.39), I end with \$2.83 (+\$2.83)
- Return \$3.20: Sender ends with \$6.09 (+\$1.09), I end with \$3.13 (+\$3.13)

I want to:

1. Reciprocate their trust proportionally
2. Be fair and encourage future cooperation
3. Return roughly 50% of what I received, similar to what I've observed

Returning around \$3.20 would be about 50.5% of what I received, which seems fair and proportional to their ~42% send rate. This ensures both of us profit.

Actually, let me be slightly more generous since they did show trust. I'll return \$3.30 (about 52% of what I received).

...

**usage:**

```
{
 "input_tokens": 1218,
 "output_tokens": 758,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.3. Your visible payoff was \$3.0300000000000002.

Your total visible earnings across all rounds are \$12.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- I'm facing a new co-player as sender
- Previous cooperation has been successful (total earnings \$12.80)
- When I sent \$2.50 in Round 2, I received \$3.62 back for a good profit
- My returns as receiver have been fair (~52%)

**\*\*Strategic reasoning:\*\***

- Continue the cooperative approach that's been working
- Sending 50% shows good faith while managing risk
- This gives my co-player \$7.50 to work with, enough for mutual benefit
- Establishes positive precedent with this new partner

{'send': 2.5}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds summary:\*\***

- Round 1 (RECEIVER): Received \$7.65 from sender sending \$2.55, returned \$4.00, earned \$3.65
- Round 2 (SENDER): Sent \$2.50, became \$7.50, they returned \$3.62, earned \$6.12
- Round 3 (RECEIVER): Received \$6.33 from sender sending \$2.11, returned \$3.30, earned \$3.03
- Total earnings: \$12.80

**\*\*Pattern analysis:\*\***



- When I sent \$2.50 (50%), I got back \$3.62, giving me a profit of \$1.12 (from \$2.50 remaining + \$3.62 = \$6.12 total)
- When others sent to me, I've returned fairly (~52% of what I received)
- Overall, cooperation has been successful

**\*\*Current decision:\*\***

- New co-player as sender this round
- Need to establish trust again
- Sending around 50% has worked well previously

I'll continue with the cooperative strategy that's been working: send \$2.50 (50% of endowment).

**usage:**

```
{
 "input_tokens": 1558,
 "output_tokens": 431,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*My reasoning:\*\***

This sender showed above-average trust compared to previous rounds. I should reciprocate generously.

If I return **\*\*\$4.70\*\***:

- Sender ends with: \$2.08 + \$4.70 = \$6.78 (profit of \$1.78)
- I end with: \$8.76 - \$4.70 = \$4.06 (profit of \$4.06)

This represents:

- Returning about 54% of what I received
- Rewarding their higher trust level (58.4% sent)
- Both profiting well from cooperation
- Encouraging continued trust

{'return': 4.7}

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*Patterns from previous rounds:\*\***

- Round 1: Received \$7.65, returned \$4.00 (52%)
- Round 3: Received \$6.33, returned \$3.30 (52%)
- I've been consistently returning about 52% of what I receive

**\*\*Strategy:\*\***

The sender showed high trust (58.4% sent). I should reciprocate fairly.

If I return around 52% of \$8.76:

-  $0.52 \times \$8.76 = \$4.55$

Let me check the outcomes:

- Sender ends with:  $\$2.08 + \$4.55 = \$6.63$  (profit of \$1.63)
- I end with:  $\$8.76 - \$4.55 = \$4.21$  (profit of \$4.21)

This seems fair - both profit, sender gets a decent return for their trust.

Actually, given they sent 58.4% (more than my usual 50%), maybe I should return slightly more to really reward that trust. Let me try \$4.70:

- Sender ends with:  $\$2.08 + \$4.70 = \$6.78$  (profit of \$1.78)
- I end with:  $\$8.76 - \$4.70 = \$4.06$  (profit of \$4.06)

That's about 54% return, which is fair and rewards their higher trust level.

**usage:**

```
{
 "input_tokens": 1844,
 "output_tokens": 622,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.92, it became \$8.76, and you returned \$4.7. Your visible payoff was \$4.06.

Your total visible earnings across all rounds are \$22.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My sending strategy (50%) has been successful in Rounds 2 & 4
- Returns I've received: \$3.62 and \$3.01 (averaging \$3.32)
- Total earnings of \$22.37 show cooperation is working

**\*\*Strategic reasoning:\*\***

- Maintain the consistent 50% sending strategy
- This has proven reliable across different co-players
- Gives receiver \$7.50 to work with
- Balances trust-building with risk management

{'send': 2.5}

**Reasoning:**

Looking at the pattern:

**\*\*My sending history:\*\***

- Round 2: Sent \$2.50 (50%), received back \$3.62, profit = \$6.12
- Round 4: Sent \$2.50 (50%), received back \$3.01, profit = \$5.51

**\*\*My receiving history:\*\***

- Round 1: Received \$7.65 from \$2.55 sent (51%), returned \$4.00
- Round 3: Received \$6.33 from \$2.11 sent (42%), returned \$3.30
- Round 5: Received \$8.76 from \$2.92 sent (58%), returned \$4.70

**\*\*Analysis:\*\***

- When I send 50%, I've gotten decent returns (\$3.01-\$3.62)
- As a receiver, I've been fair, returning around 52-54%
- Total earnings are strong at \$22.37
- Cooperation has been working well

**\*\*This round:\*\***

- New co-player as sender

- Continue with the cooperative 50% strategy that's been successful
- This maintains consistency and shows I'm a cooperative player

I'll send \$2.50 again.

```
Usage: {
 "input_tokens": 2182,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_7 - Receiver - dyad\_4

Timestamp: 2026-08-12T09:25:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_7"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 6,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance building trust with managing risk since I have no history with this receiver.

**\*\*My reasoning:\*\***

- Sending money creates value (3x multiplier means potential gains for both of us)
- This is round 1, so I want to signal cooperation
- I should send enough to be meaningful but not risk everything
- If the receiver reciprocates fairly, we both benefit

I'll send a moderate amount to test cooperation while limiting downside risk.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

- 1. \*\*Game Structure:\*\***
  - I have \$5
  - I can send \$0-\$5 to the receiver
  - Whatever I send gets tripled
  - The receiver then decides how much to return
  - My payoff: \$5 - sent + returned
  - Receiver's payoff: 3×sent - returned
- 2. \*\*Strategic Considerations:\*\***
  - If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5
  - If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get \$7.50 and they keep \$7.50. Total: \$15
  - The game creates surplus through cooperation (up to \$15 total vs \$5 if no trust)
  - This is a trust game - I need to trust the receiver will return some amount
- 3. \*\*First Round Considerations:\*\***
  - No history with this partner
  - This is a multi-round game with different partners
  - Building a reputation for cooperation might matter
  - The note about "communication noise" suggests amounts might be misperceived, which could affect trust dynamics
- 4. \*\*Common Strategies:\*\***
  - **\*\*Full trust:\*\*** Send \$5, hope for good return
  - **\*\*Moderate trust:\*\*** Send \$2-3, test the waters
  - **\*\*No trust:\*\*** Send \$0, guarantee \$5
  - **\*\*Signal cooperation:\*\*** Send a meaningful amount to show willingness to cooperate
- 5. \*\*Rational Analysis:\*\***
  - If the receiver is purely self-interested and this is a one-shot game, they should return \$0
  - Knowing this, I should send \$0
  - However, in repeated games or with reciprocity norms, cooperation can emerge

- Many people exhibit fairness preferences and reciprocity

#### 6. **\*\*My Decision:\*\***

Since this is the opening move in a multi-round game, a moderate cooperative signal seems prudent. I'll send \$3-enough to demonstrate trust and give the receiver \$9 to work with, but not so much that I'm exposed to catastrophic loss if they keep everything. Even a modest return of \$3-4 would benefit us both compared to the baseline of keeping \$5....

#### **usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 836,
 "reasoning_tokens": 0
}
```

#### **Message 4 role=user**

##### **content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.56 to you. Your visible payoff was \$7.56.

Your total visible earnings across all rounds are \$7.56.

This round, you are the RECEIVER against your current co-player. They sent you \$1.67, so you received \$5.01. How much do you return? (0-5.01)

Respond exactly as JSON: {'return': <amount>}

#### **Message 5 role=assistant**

##### **content:**

Looking at this situation:

##### **\*\*What happened:\*\***

- My co-player sent me \$1.67 (33% of their endowment)
- I received \$5.01 after the 3x multiplier
- They currently have \$3.33 remaining

##### **\*\*My considerations:\*\***

- They showed moderate trust by sending \$1.67
- The multiplier created \$3.34 in surplus value
- A fair approach would be to share this surplus roughly equally
- In Round 1, my partner returned ~62% and that felt cooperative

##### **\*\*Fair return calculation:\*\***

If we split the surplus equally:

- Each should gain ~\$1.67 from cooperation
- Sender ending: \$5 - \$1.67 + return = \$6.67 desired
- Return needed: \$3.34

However, that might be overly generous given their modest send. I'll return about 60% to reward their cooperation while keeping reasonable value.

```
'''json
{"return": 3.00}
'''
```

##### **reasoning:**

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.56 to me
- I ended with: \$5 - \$3 + \$5.56 = \$7.56
- Receiver ended with: \$9 - \$5.56 = \$3.44

That was actually quite generous of them! They returned more than half.

Round 2 (current):

- I am RECEIVER
- They sent me \$1.67
- I received \$5.01 (which is 3 × \$1.67)
- I need to decide how much to return (0-5.01)

Key considerations:

1. My previous partner was generous, returning \$5.56 out of \$9 (62%)
2. This is a different player (multi-agent setting with randomized pairings)
3. They sent me \$1.67 out of \$5, which is 33.4% - a moderate amount
4. The sender keeps: \$5 - \$1.67 = \$3.33 before my return
5. I should reciprocate fairly to encourage future cooperation

Let me think about fair returns:

- If I return half (\$2.50), they end with  $\$3.33 + \$2.50 = \$5.83$ , I keep \$2.51
- If I return around what gives us equal final payoffs:
  - They have  $\$3.33 + \text{return}$
  - I have  $\$5.01 - \text{return}$
  - Equal:  $\$3.33 + x = \$5.01 - x$
  - $2x = \$1.68$
  - $x = \$0.84$

This would give us both around \$4.17

- A proportional return similar to what I received (62% of what I got):
  - $0.62 \times \$5.01 = \$3.11$
  - They'd end with:  $\$3.33 + \$3.11 = \$6.44$
  - I'd keep:  $\$5.01 - \$3.11 = \$1.90$

A 50% return of around \$2.50 seems like the right balance-it rewards their cooperation without being excessive, and it's more moderate than matching their 33% send rate since they're a different player than before.

If I send back \$2.50, they'd gain \$0.83 while I'd be left with \$2.51. The math shows we'd create \$8.34 in total value, which is less than the \$10 we'd have if neither of us participated. The real question is what surplus we're actually generating from this exchange and whether my return is proportional to their initial trust.

The multiplier created \$3.34 in extra value (\$5.01 received minus \$1.67 sent). If we split that gain equally, each of us should pocket \$1.67. That means I should return \$3.34 to bring them to \$6.67 total, which gives us both the same \$1.67 gain from the transaction.

...

usage:

```
{
 "input_tokens": 754,
 "output_tokens": 1445,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.67, it became \$5.01, and you returned \$3.0. Your visible payoff was \$2.01.

Your total visible earnings across all rounds are \$9.57.

This round, you are the RECEIVER against your current co-player. They sent you \$3.66, so you received \$10.98. How much do you return? (0-10.98)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$3.66 (73% of their endowment) - very high trust!
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining
- Total surplus created: \$2.32

**\*\*My considerations:\*\***

- This is a very generous send showing strong cooperation
- They took significant risk by sending most of their money
- I should reward this trust with a fair return
- In Round 1, I received about 62% back on my \$3 send
- In Round 2, I returned 60% on a smaller send

**\*\*Fair return:\*\***

Given their high trust, I'll return approximately 50% of what I received:

- Return \$5.50
- They'll end with \$6.84 (profit of \$1.84)
- I'll end with \$5.48 (profit of \$0.48)
- This rewards their cooperation while both of us profit

```
'''json
{"return": 5.50}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current Situation:\*\***

- My co-player sent me \$3.66 (73% of their endowment)
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining (\$5 - \$3.66)
- This is a very generous/trusting send

**\*\*What this shows:\*\***

- High trust from the sender (sending over 70%!)
- They're expecting reciprocity
- The multiplier created significant surplus: \$5.98 (\$10.98 - \$5)

**\*\*Considerations:\*\***

- In Round 1, when I sent \$3 (60%), I received \$5.56 back - that's about 62% return rate
- In Round 2, when someone sent \$1.67 (33%), I returned \$3.00 - that's 60% return rate
- This sender showed much more trust (\$3.66 = 73% of endowment)

**\*\*Fair calculation:\*\***

The total pot is now \$12.32 (\$1.34 they kept + \$10.98 I received)

If we split the gains from cooperation:

- Original total: \$10 (my \$5 + their \$5)
- Current total: \$12.32
- Surplus created: \$2.32
- Fair split would give each person \$1.16 gain
- Sender would end at \$6.16, I'd end at \$6.16
- I need to return: \$6.16 - \$1.34 = \$4.82

But let me think about what's fair based on the trust shown:

- They sent 73% of their money
- If I return about 50% of what I received, that's \$5.49
- That would give them \$1.34 + \$5.49 = \$6.83 total

...

**usage:**

```
{
 "input_tokens": 1112,
 "output_tokens": 833,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$5.5. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$15.05.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: I sent \$3 and received \$5.56 back - good cooperation!
- Rounds 2-3: As receiver, I tried to return fairly and reward trust

- Total earnings: \$15.05

**\*\*Considerations for this round:\*\***

- I'm paired with a new co-player each round
- My Round 1 send of \$3 worked well
- The most generous player sent me \$3.66, showing high trust pays off
- Sending more creates more total value (3x multiplier)
- I want to encourage cooperative norms in the pool

**\*\*Strategy:\*\***

I'll send a bit more than Round 1 to signal strong cooperation while managing risk. This shows I'm willing to trust and invest in mutual gains.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Now I'm the SENDER again in Round 4.

Let me review my history:

- Round 1 (SENDER): Sent \$3, received back \$5.56, payoff \$7.56
- Round 2 (RECEIVER): Received \$5.01 (from \$1.67 sent), returned \$3.00, payoff \$2.01
- Round 3 (RECEIVER): Received \$10.98 (from \$3.66 sent), returned \$5.50, payoff \$5.48

This is a new co-player each round in this multi-agent setting.

My experience so far:

- When I sent \$3, I got back \$5.56 - good return!
- When others sent to me, I tried to be fair/cooperative
- The player who sent \$3.66 was very trusting

For this round, I need to decide how much to send to a new co-player.

Considerations:

- I've seen good cooperation so far (my Round 1 partner returned well)
- I want to continue building cooperative patterns
- Sending more creates more value (3x multiplier)
- But I'm paired with a new player each time, so no direct reputation building

Strategy:

- In Round 1, I sent \$3 and it worked well
- I could send more to signal strong cooperation
- Or stick with \$3 as a moderate cooperative amount

Given that I've had positive experiences and want to continue fostering cooperation, I'll send a bit more this time - maybe \$3.5 or \$4. This shows I'm willing to trust while not going all-in.

Let me go with \$3.5 - it's between my previous \$3 send and the very generous \$3.66 I received in Round 3.

Actually, thinking about it more: the player who sent me \$3.66 showed strong trust and I rewarded it. If I want to encourage that kind of cooperation, I should also send generously when I'm the sender. Let me send \$4.

...

**usage:**

```
{
 "input_tokens": 1475,
 "output_tokens": 749,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$3.59 to you. Your visible payoff was \$5.09.

Your total visible earnings across all rounds are \$20.14.



This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*Recent experience:\*\***

- Round 4: I sent \$3.50, received only \$3.59 back - minimal profit (\$5.09)
- This was disappointing compared to Round 1 where I sent \$3 and got \$5.56 back
- But this is a NEW co-player with unknown behavior

**\*\*Considerations:\*\***

- Different co-players show varying levels of cooperation
- Sending less than \$3.50 reduces my risk after last round's poor return
- Still want to signal willingness to cooperate and create value
- \$3 was my initial send and it worked well - proven strategy

**\*\*Strategy:\*\***

I'll return to sending \$3 - a moderate amount that:

- Shows trust and cooperation
- Creates meaningful value (\$9 for receiver)
- Limits downside risk
- Has proven successful before

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

- Round 1: Sent \$3, received \$5.56 back - payoff \$7.56 (good!)
- Round 2: Receiver, returned \$3.00 - payoff \$2.01
- Round 3: Receiver, returned \$5.50 - payoff \$5.48
- Round 4: Sent \$3.5, received \$3.59 back - payoff \$5.09 (disappointing)
- Round 5: Now sender again

Analysis of Round 4:

- I sent \$3.50 (70% of endowment) showing strong trust
- They received \$10.50
- They only returned \$3.59 (34% of what they received)
- I ended with \$5.09, barely breaking even
- This was much less generous than my Round 1 partner who returned \$5.56 on a \$9 receive

This suggests I encountered a less cooperative player. However, I'm now paired with a NEW co-player, so their behavior doesn't necessarily predict this new person's behavior.

Strategic considerations:

- I'm doing reasonably well overall (\$20.14 total)
- Sending creates value through the multiplier
- But I need to balance risk
- Two different responses to my sends: one very good (\$5.56), one mediocre (\$3.59)
- I don't know if there's a pattern or if agents vary widely

I think I should continue showing willingness to cooperate but maybe pull back slightly from the \$3.50 I sent last time since that didn't work out as well. Going back to \$3 seems reasonable - it's a solid middle ground that shows cooperation without overcommitting.

**usage:**

```
{
 "input_tokens": 1797,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.06 to you. Your visible payoff was \$6.06.

Your total visible earnings across all rounds are \$26.2.

This round, you are the RECEIVER against your current co-player. They sent you \$1.64, so you received \$4.92. How much do you return? (0-4.92)  
Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment) - modest trust
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining

**\*\*My considerations:\*\***

- This is very similar to Round 2 where I received \$5.01 from a \$1.67 send
- In Round 2, I returned \$3.00 (60%) which seemed fair
- I should be consistent in rewarding similar levels of cooperation
- Returning around 60% rewards their trust while keeping reasonable value

**\*\*Fair return:\*\***

Returning approximately 60% of what I received:

- Return \$3.00
- They'll end with \$6.36 (profit of \$1.36)
- I'll end with \$1.92

```
'''json
{"return": 3.00}
'''
```

**Reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment)
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining
- Total surplus created from their send:  $\$4.92 - \$1.64 = \$3.28$

**\*\*My considerations:\*\***

- They showed modest trust by sending \$1.64 (similar to the \$1.67 I saw in Round 2)
- In Round 2, when I received \$5.01 from a \$1.67 send, I returned \$3.00 (60%)
- I should be consistent in how I respond to similar levels of trust
- Returning about 60% would be \$2.95

**\*\*Fair return calculation:\*\***

- If I return around \$3.00:
  - They end with:  $\$3.36 + \$3.00 = \$6.36$  (profit of \$1.36)
  - I end with:  $\$4.92 - \$3.00 = \$1.92$  (loss of \$0.08 from initial endowment perspective, but positive from game perspective)

Actually, let me think about this differently:

- They sent \$1.64
- This created \$3.28 in surplus ( $\$4.92 - \$1.64$ )
- A fair split of surplus would give each about \$1.64
- So they should end with:  $\$5 + \$1.64 = \$6.64$
- They have \$3.36, so I need to return \$3.28 to get them to \$6.64

But that gives me only \$1.64, which seems low for receiving \$4.92.

Let me be consistent with Round 2 where I returned 60% on a similar send:

- 60% of \$4.92 = \$2.95

...

```
Usage: {
 "input_tokens": 2142,
 "output_tokens": 677,
 "reasoning_tokens": 0
}
```

**ROUND 7**

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Sender, Agent\_5=Sender,

Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Receiver

**Pairings:**

dyad\_1: Agent\_5=Sender, Agent\_1=Receiver  
dyad\_2: Agent\_4=Sender, Agent\_3=Receiver  
dyad\_3: Agent\_7=Sender, Agent\_8=Receiver  
dyad\_4: Agent\_2=Sender, Agent\_6=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_5 sent \$3; received \$9; Agent\_1 returned \$4.2; payoffs Agent\_5=\$6.2, Agent\_1=\$4.8;  
noise/visible: sent visible=\$2.78, received visible=\$8.34, returned visible=\$3.3  
Action validation: {"returned": {"amount": 4.2, "clamped": false, "max": 9.0, "min": 0, "raw": 4.2}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_4 sent \$3; received \$9; Agent\_3 returned \$4.22; payoffs Agent\_4=\$6.22, Agent\_3=\$4.78; noise/visible: sent visible=\$3.36, received visible=\$10.08, returned visible=\$4.49  
Action validation: {"returned": {"amount": 4.22, "clamped": false, "max": 9.0, "min": 0, "raw": 4.22}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_3: Agent\_7 sent \$3; received \$9; Agent\_8 returned \$3.5; payoffs Agent\_7=\$5.5, Agent\_8=\$5.5; noise/visible: sent visible=\$2.69, received visible=\$8.07, returned visible=\$3.87  
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_2 sent \$2.5; received \$7.5; Agent\_6 returned \$3.9; payoffs Agent\_2=\$6.4, Agent\_6=\$3.6; noise/visible: sent visible=\$2.85, received visible=\$8.55, returned visible=\$3.11  
Action validation: {"returned": {"amount": 3.9, "clamped": false, "max": 7.5, "min": 0, "raw": 3.9}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent\_1": 37.319999999999999,  
"Agent\_2": 38.31,  
"Agent\_3": 39.53,  
"Agent\_4": 37.42,  
"Agent\_5": 42.3300000000000005,  
"Agent\_6": 33.94,  
"Agent\_7": 36.55,  
"Agent\_8": 38.1

}

Visible balances after round: {

"Agent\_1": 36.63,  
"Agent\_2": 34.28,  
"Agent\_3": 35.96,  
"Agent\_4": 38.87,  
"Agent\_5": 34.589999999999996,  
"Agent\_6": 35.43,  
"Agent\_7": 33.989999999999995,  
"Agent\_8": 37.27

}

**LLM CALLS**

**GAME - Agent\_5 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:25:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_1",  
"pairing": {  
  "agents": [  
    "Agent\_5",  
    "Agent\_1"  
  ],  
  "investor": "Agent\_5",  
  "roles": {  
    "Agent\_1": "trustee",  
    "Agent\_5": "investor"  
  },  
  "trustee": "Agent\_1"  
},  
"role": "investor",  
"role\_label": "Sender",  
"round": 7,

```

"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as a sender, I need to balance trust and caution since I don't know my co-player yet.

The key consideration is that sending money creates value through the 3x multiplier. If I send \$3:

- I keep \$2
- The receiver gets \$9 (3 × \$3)
- If they return even a moderate amount (say \$3-4), we both benefit compared to sending nothing

This shows willingness to cooperate while maintaining some security. It's a reasonable opening move that could encourage reciprocity.

```

'''json
{"send": 3}

```

'''

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategy:

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- There's communication noise, so amounts may differ from what was actually sent/returned
- This is a multi-agent setting with 8 agents, randomized pairings each round
- I'll play both sender and receiver roles across rounds

Strategic considerations:

1. **\*\*Trust and cooperation\*\***: If I send money, it creates value (3x multiplier). If the receiver is cooperative, they might return a fair share, making both of us better off.
2. **\*\*Risk\*\***: The receiver could keep everything. If I send \$5, they get \$15, and might return nothing, leaving me with \$0.
3. **\*\*First round dynamics\*\***: This is the first round, so there's no history to base decisions on. I don't know how cooperative my co-player will be.
4. **\*\*Future rounds\*\***: Since this is a multi-round game with different partners, building a reputation for cooperation might matter, though the anonymity and random pairing complicate this.
5. **\*\*Communication noise\*\***: This adds uncertainty - even if someone tries to be fair, the noise might make it seem otherwise.
6. **\*\*Optimal cooperation\*\***: If both parties cooperate optimally, I could send all \$5, receiver gets \$15, and if they split fairly, they might return around \$7-10, making us both better off than the initial \$5.

Common strategies in trust games:

- Send nothing (risk-averse, guarantees keeping \$5)
- Send half (\$2.50) - moderate trust
- Send everything (\$5) - full trust, maximizes potential gains

For the first round with an unknown partner, a moderate approach might be reasonable. Sending about \$3-4 shows willingness to cooperate while not being completely vulnerable.

Since this is a repeated game with multiple rounds and different partners, demonstrating cooperative behavior early could encourage reciprocity....

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 768,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.6899999999999995.

Your total visible earnings across all rounds are \$5.6899999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.09, so you received \$6.27. How much do you return? (0-6.27)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.09 out of \$5 (about 42%)
- Showed moderate trust

**\*\*My considerations:\*\***

- In Round 1, my partner returned about 41% of what they received
- Reciprocating fairly would mean returning a similar proportion
- This encourages future cooperation

Returning approximately 45% of the \$6.27 I received (\$2.82) seems fair. This rewards their trust, gives them a modest profit, and keeps a reasonable amount for myself.

```
'''json
{"return": 2.82}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$3
- It became \$9
- They returned \$3.69
- My payoff:  $\$5 - \$3 + \$3.69 = \$5.69$

Now Round 2 (as RECEIVER):

- My co-player sent me \$2.09
- I received \$6.27 (which is  $3 \times \$2.09$ )
- I need to decide how much to return

Key considerations:

1. My previous co-player returned about 41% of what they received (\$3.69 out of \$9)
2. This current co-player sent \$2.09 out of \$5, which is about 42% - showing moderate trust
3. There's mention of "communication noise" - the amounts we see may differ slightly from actual amounts

A fair reciprocity strategy would be to return roughly the same proportion that was returned to me, or perhaps slightly reward the trust shown by sending a decent amount.

If I return about 40% of \$6.27, that would be around \$2.51.

If I'm more generous and return 45%, that would be around \$2.82.

If I return 50%, that would be \$3.14.

Given that:

- The sender showed trust by sending \$2.09 (42% of their endowment)
- My previous partner returned about 41%
- I want to encourage future cooperation

I'll return around \$2.80-\$3.00, which is roughly 45-48% of what I received. This rewards their trust while keeping a reasonable amount.

Let me go with \$2.80 - this leaves them with approximately  $\$5 - \$2.09 + \$2.80 = \$5.71$ , a modest gain, and I keep  $\$6.27 - \$2.80 = \$3.47$ , a solid gain for both of us.

**usage:**

```
{
 "input_tokens": 781,
 "output_tokens": 590,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.09, it became \$6.27, and you returned \$2.82. Your visible payoff was \$3.4499999999999997.

Your total visible earnings across all rounds are \$9.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my position:

**\*\*Previous experience:\*\***

- Round 1: Sent \$3, received \$3.69 back ? good outcome
- Round 2: As receiver, returned a fair amount (\$2.82 on \$6.27)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- No history with this specific agent
- Need to signal willingness to cooperate

Since I had success with sending \$3 in Round 1 (received \$3.69 back, net gain of \$0.69), and it establishes a pattern of moderate trust, I'll use the same strategy with this new co-player.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my experience so far:

Round 1: I was SENDER

- Sent \$3, they received \$9
- They returned \$3.69 (about 41% of what they received)
- My payoff: \$2 (kept) + \$3.69 (returned) = \$5.69

Round 2: I was RECEIVER

- They sent \$2.09, I received \$6.27
- I returned \$2.82 (about 45% of what I received)
- My payoff: \$6.27 - \$2.82 = \$3.45

Total earnings: \$9.14

Now Round 3: I'm SENDER again with a new co-player (since agents are randomized each round)

What should I consider:

1. This is a new opponent - I don't know their strategy
2. In Round 1, sending \$3 worked reasonably well (got back \$3.69, netted \$5.69 vs \$5 if I sent nothing)
3. The multiplier effect means cooperation is beneficial
4. I should probably maintain a consistent strategy early on

My Round 1 strategy of sending \$3 was a moderate trust signal that worked well. Since I'm facing a new opponent, I should probably use a similar approach - not too cautious, not too risky.

I'll send \$3 again as my baseline strategy for first encounters with new co-players.

**usage:**

```
{
 "input_tokens": 1050,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.37 to you. Your visible payoff was \$7.37.

Your total visible earnings across all rounds are \$16.509999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.94 out of \$5 (nearly 59%)
- Showed strong trust

**\*\*Recent patterns:\*\***

- Round 3: My partner was very generous, returning \$5.37 (60% of \$9)
- Round 2: I returned about 45% as receiver
- This co-player is being quite trusting

**\*\*Fair reciprocity:\*\***

Returning around 48% of the \$8.82 I received would be fair and encourage future cooperation. This gives my co-player a decent profit while rewarding their trust.

```
'''json
{"return": 4.20}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER\*\***

- My co-player sent \$2.94 (about 59% of their \$5)
- I received \$8.82 (3x multiplier)
- Need to decide how much to return

**\*\*Recent history:\*\***

- Round 3: As sender, I sent \$3, received \$5.37 back (about 60% of the \$9 they got) - very generous!
- Round 2: As receiver, I returned \$2.82 on \$6.27 (about 45%)
- Round 1: As sender, sent \$3, received \$3.69 back (about 41% of the \$9)

**\*\*Analysis of current round:\*\***

- Co-player sent \$2.94 (quite generous - nearly 60% of their endowment)
- This shows significant trust
- They'll end up with:  $\$5 - \$2.94 + \text{return} = \$2.06 + \text{return}$

**\*\*Fairness considerations:\*\***

- If I return about 45-50% like in Round 2:  $\sim\$3.96 - \$4.41$
- If I match the generosity of Round 3's partner (60%):  $\sim\$5.29$
- Co-player sacrificed a lot to send \$2.94

I should reciprocate fairly. They sent nearly 60% of their endowment, showing strong trust. I should return a reasonable amount that:

1. Rewards their trust
2. Gives them a profit
3. Maintains cooperation

Returning around \$4.00-\$4.50 seems fair:

- Gives them a total of  $\sim\$6.06 - \$6.56$
- Nets them  $\$1.06 - \$1.56$  profit
- I keep  $\$4.32 - \$4.82$
- Return rate of  $\sim45 - 51\%$

...

**usage:**

```
{
 "input_tokens": 1348,
 "output_tokens": 615,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.94, it became \$8.82, and you returned \$4.2. Your visible payoff was \$4.62.

Your total visible earnings across all rounds are \$21.13.

This round, you are the RECEIVER against your current co-player. They sent you \$1.7, so you received \$5.1. How much do you return? (0-5.1)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**



**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- More cautious than previous partners

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)

**\*\*Fair reciprocity:\*\***

Returning around 45% of the \$5.10 I received maintains consistency and fairness. This gives my co-player a modest profit while rewarding their trust, even though they were more cautious.

```
'''json
{"return": 2.30}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- This is relatively cautious/low trust

**\*\*Recent patterns from my play:\*\***

- Round 2: Returned \$2.82 on \$6.27 (45%)
- Round 4: Returned \$4.20 on \$8.82 (48%)

**\*\*Considerations:\*\***

- They showed less trust by sending only \$1.70
- However, I should still be fair to encourage cooperation
- Returning around 40-45% would be consistent with my previous behavior

Returning about 45% of \$5.10 = ~\$2.30

This gives them back a reasonable amount, rewards some level of trust, and maintains consistency in my reciprocity rate.

Let me calculate:

- They sent \$1.70, kept \$3.30
- I received \$5.10
- If I return \$2.30:
  - They end with:  $\$3.30 + \$2.30 = \$5.60$  (gain of \$0.60)
  - I end with:  $\$5.10 - \$2.30 = \$2.80$

That seems fair. They made a small profit for trusting me.

**usage:**

```
{
 "input_tokens": 1636,
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

**Message 12 role=user****content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.7, it became \$5.1, and you returned \$2.3. Your visible payoff was \$2.8. Your total visible earnings across all rounds are \$23.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant****content:**

Looking at my position:

**\*\*Sender history:\*\***

- Round 1: Sent \$3 ? received \$3.69 back (good return)
- Round 3: Sent \$3 ? received \$5.37 back (excellent return)

**\*\*Strategy:\*\***

Sending \$3 has been consistently successful for me. It demonstrates trust while creating substantial value (\$9 for the receiver), which has encouraged good reciprocity from past partners.

With a new co-player this round, I'll continue with this proven strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3, got back \$3.69 (net +\$0.69)
- Round 3: Sent \$3, got back \$5.37 (net +\$2.37)

As RECEIVER:

- Round 2: Received \$6.27, returned \$2.82
- Round 4: Received \$8.82, returned \$4.20
- Round 5: Received \$5.10, returned \$2.30

My sending strategy of \$3 has worked well - I got good returns both times. This is a new co-player (randomized pairing), so I don't have specific history with them.

Sending \$3 seems to be a sweet spot:

- Shows trust and willingness to cooperate
- Creates substantial value (\$9 for receiver)
- Has resulted in good returns in past rounds

I'll continue with \$3 as it's been my successful strategy.

**usage:**

```
{
 "input_tokens": 1913,
 "output_tokens": 363,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.36 to you. Your visible payoff was \$5.3599999999999999.

Your total visible earnings across all rounds are \$29.29.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my position:

**\*\*Sender track record:\*\***

- Round 1: Sent \$3 ? received \$3.69 (net +\$0.69)
- Round 3: Sent \$3 ? received \$5.37 (net +\$2.37)
- Round 6: Sent \$3 ? received \$3.36 (net +\$0.36)

**\*\*Strategy assessment:\*\***

Sending \$3 has been consistently profitable across all my sender rounds. It demonstrates trust, creates value through the 3x multiplier, and has encouraged reciprocity from various co-players.

With a new partner this round, I'll stick with this proven approach.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my sender history:

- Round 1: Sent \$3 ? received \$3.69 back (net +\$0.69)
- Round 3: Sent \$3 ? received \$5.37 back (net +\$2.37) - excellent!
- Round 6: Sent \$3 ? received \$3.36 back (net +\$0.36)

All three times I sent \$3, I got a positive return. The strategy of sending \$3 has been consistently profitable, showing that moderate trust tends to be reciprocated.

This is a new co-player again (randomized pairing each round). I should continue with this successful strategy of sending \$3.

It creates value through the 3x multiplier while not risking everything, and has historically generated positive returns for me.

```
Usage: {
 "input_tokens": 2170,
 "output_tokens": 359,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:25:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_1"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_1": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:****Message 1 role=system****content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 1: You are the RECEIVER. the sender sent \$3.34, that is 66.8% of their total endowment. You received \$10.02. How much do you return to the sender? (0-10.02)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- The sender showed significant trust by sending \$3.34 (66.8% of their endowment)
- I received \$10.02 after the 3x multiplier
- The sender currently has \$1.66 remaining

I want to reciprocate the sender's trust and ensure cooperation is worthwhile for them. If I return \$5.00:

- The sender would end with  $\$1.66 + \$5.00 = \$6.66$  (better than their initial \$5)
- I would keep \$5.02 (also better than any initial amount)

This rewards their trust while maintaining a fair distribution of the gains from cooperation.

{'return': 5.00}

##### reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 1
- The sender sent \$3.34 (which is 66.8% of their \$5 endowment)
- I received \$10.02 (which is  $\$3.34 \times 3$ )
- I need to decide how much to return (between \$0 and \$10.02)

This is a trust game. The sender showed considerable trust by sending 66.8% of their endowment. Let me think about a fair return:

The sender:

- Started with \$5
- Sent \$3.34
- Currently has  $\$5 - \$3.34 = \$1.66$
- If I return X, they'll end up with  $\$1.66 + X$

Me:

- I received \$10.02
- If I return X, I keep  $\$10.02 - X$

Some considerations for fairness:

1. Equal split of total: Total money is \$5 (sender's initial) + \$10.02 (what I received) = \$15.02. Equal split would be \$7.51 each.
  - For sender to get \$7.51, they need  $\$7.51 - \$1.66 = \$5.85$  returned
  - I would keep  $\$10.02 - \$5.85 = \$4.17$

2. Proportional return: The sender sent 66.8% of their endowment, showing high trust. A reciprocal gesture might be to return a similar proportion.

3. Rewarding trust: Since the sender trusted me with a large amount, I should return enough to make their trust worthwhile.

Let me check option 1: If I return \$5.85:

- Sender ends with:  $\$1.66 + \$5.85 = \$7.51$
- I end with:  $\$10.02 - \$5.85 = \$4.17$
- This creates mutual benefit from cooperation, and the sender gains significantly from their initial trust.

Returning around \$5 also...

**usage:**

```
{
 "input_tokens": 529,
 "output_tokens": 837,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, the sender showed good trust by sending 66.8% of their endowment, and I reciprocated fairly.

Now as the sender with a new co-player, I'll demonstrate similar trust to establish cooperation. Sending \$3.00 (60% of my endowment) shows significant trust while being reasonable.

If they reciprocate fairly, we both benefit from cooperation.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

Round 1:

- I was the RECEIVER
- Sender sent \$3.34 (66.8% of their endowment)
- I received \$10.02
- I returned \$5.00
- My payoff was \$5.02

Now in Round 2:

- I am the SENDER
- I have \$5
- I need to decide how much to send

The previous sender showed trust by sending 66.8% of their endowment to me, and I reciprocated by returning a fair amount that left them better off.

Now I'm playing against a different agent (the pairing is randomized each round). I should consider:

1. The sender in Round 1 sent a substantial amount (66.8%), showing trust
2. This encouraged cooperation
3. I'm now the sender with a new partner

I should probably send a similar amount to establish trust and encourage cooperation. If I

send around 60-70% of my endowment (like the previous sender did), that would be \$3.00-\$3.50.

Let me think about what would be fair:

- If I send \$3.00, they receive \$9.00
- If they return half of what they receive (a common fair split), they'd return \$4.50
- I'd end with \$2.00 + \$4.50 = \$6.50
- They'd keep \$4.50

That seems reasonable. I could also send \$3.34 to match what was sent to me in Round 1, showing I'm willing to trust at the same level.

Actually, let me reconsider. I want to be strategic but also fair. Sending \$3.00 seems like a good middle ground - it's 60% of my endowment, showing significant trust without being extreme.

```
{'send': 3.00}
```

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 530,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.73 to you. Your visible payoff was \$4.73.

Your total visible earnings across all rounds are \$9.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my history:

- Round 1 (as receiver): Co-player sent \$3.34, I returned \$5.00
- Round 2 (as sender): I sent \$3.00, received back \$2.73

Now with a new co-player in Round 3, I should continue demonstrating willingness to cooperate. Sending a consistent amount shows I'm a reliable partner.

```
{'send': 3.00}
```

**reasoning:**

In Round 2, I sent \$3.00, it became \$9.00, and the receiver returned \$2.73.

Let me calculate:

- I started with \$5
- I sent \$3, so I had \$2 left
- They received \$9 and returned \$2.73
- My payoff was \$2 + \$2.73 = \$4.73

The receiver kept \$9 - \$2.73 = \$6.27

I got \$4.73

So the return rate was  $\$2.73/\$9.00 = 30.3\%$

This is lower than what I returned in Round 1 (I returned \$5.00 out of \$10.02, which is about 50%).

However, this is a new co-player in Round 3. I shouldn't punish this new person for what the previous receiver did.

I should continue to try to establish cooperation. Maybe send a similar amount to show consistency and willingness to cooperate. However, given that the last return was somewhat low, I might be slightly more cautious.

Let me send \$2.50 - still showing trust but being a bit more conservative.

Actually, let me think about this differently. Each round is with a different co-player (the problem states pairings are randomized). So my actions should be about establishing cooperation with each new partner, not about punishing previous partners.

I should continue to send a reasonable amount to establish trust. Let me send \$3.00 again to be consistent and show I'm willing to cooperate.

**usage:**

```
{
 "input_tokens": 1034,
 "output_tokens": 456,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.57 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender showed trust by sending \$2.99 (about 60% of their endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has \$2.01

I want to reciprocate their trust fairly. If I return \$4.50:

- Sender ends with: \$2.01 + \$4.50 = \$6.51 (profit of \$1.51)
- I keep: \$4.47

This rewards their cooperation while maintaining a fair split of the gains.

```
{'return': 4.50}
```

**reasoning:**

Let me analyze this round:

- I'm the receiver
- The sender sent \$2.99 (about 60% of their \$5 endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has \$5 - \$2.99 = \$2.01

The sender showed good trust by sending nearly 60% of their endowment. I should reciprocate fairly.

If I return \$4.50:

- Sender ends with: \$2.01 + \$4.50 = \$6.51 (better than their initial \$5)
- I keep: \$8.97 - \$4.50 = \$4.47

This is fair and rewards their trust. It's roughly half of what I received, which seems reasonable given they sent about 60% of their endowment.

Looking at my previous behavior:

- Round 1: Received \$10.02, returned \$5.00 (kept \$5.02)
- Round 3: The other player returned \$5.57 when they received \$9.00

I should be consistent and fair. Returning around \$4.50 seems appropriate.

**usage:**

```
{
 "input_tokens": 1269,
 "output_tokens": 416,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.99, it became \$8.97, and you returned \$4.5. Your visible payoff was \$4.470000000000001.

Your total visible earnings across all rounds are \$21.79.

This round, you are the RECEIVER against your current co-player. They sent you \$2.89, so you received \$8.67. How much do you return? (0-8.67)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.89 (about 58% of their endowment), showing trust
- I received \$8.67 after the 3x multiplier
- The sender currently has \$2.11

I'll reciprocate fairly. If I return \$4.30:

- Sender ends with: \$2.11 + \$4.30 = \$6.41 (profit of \$1.41)
- I keep: \$4.37

This rewards cooperation while maintaining a fair distribution.

{'return': 4.30}

**reasoning:**

The sender sent \$2.89 (about 58% of their \$5 endowment), showing reasonable trust.

I received \$8.67 after the 3x multiplier.

The sender currently has \$5 - \$2.89 = \$2.11

I should reciprocate fairly. Looking at my previous behavior:

- Round 1: Sender sent \$3.34, I received \$10.02, returned \$5.00
- Round 4: Sender sent \$2.99, I received \$8.97, returned \$4.50

This sender sent slightly less (\$2.89 vs \$2.99), so I should return proportionally. If I return \$4.30:

- Sender ends with: \$2.11 + \$4.30 = \$6.41 (profit of \$1.41)
- I keep: \$4.37

This maintains fair reciprocation.

**usage:**

```
{
 "input_tokens": 1545,
 "output_tokens": 349,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.89, it became \$8.67, and you returned \$4.3. Your visible payoff was \$4.37.

Your total visible earnings across all rounds are \$26.16.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my pattern:

- As a sender, I've consistently sent \$3.00 to show trust and encourage cooperation
- As a receiver, I've been returning amounts that fairly reward the sender's trust

I'll maintain my consistent cooperative approach with this new co-player.

{'send': 3.00}

**reasoning:**

Looking at my history:

- Round 1 (receiver): Received \$10.02 from sender's \$3.34, returned \$5.00
- Round 2 (sender): Sent \$3.00, received back \$2.73
- Round 3 (sender): Sent \$3.00, received back \$5.57



- Round 4 (receiver): Received \$8.97 from sender's \$2.99, returned \$4.50
- Round 5 (receiver): Received \$8.67 from sender's \$2.89, returned \$4.30

Now Round 6, I'm a sender again with a new co-player.

My sending pattern has been consistent at \$3.00 when I'm the sender. The returns I've received have varied (\$2.73 and \$5.57), but both times my co-players received \$9.00 and chose to return different amounts.

I should continue with a consistent cooperative strategy. Sending \$3.00 has been my standard, showing significant trust (60% of endowment).

```
{'send': 3.00}
```

**usage:**

```
{
 "input_tokens": 1796,
 "output_tokens": 332,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.33 to you. Your visible payoff was \$6.33.

Your total visible earnings across all rounds are \$32.49.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$2.78 (about 56% of their endowment), showing trust
- I received \$8.34 after the 3x multiplier
- The sender currently has \$2.22

I'll reciprocate fairly. If I return \$4.20:

- Sender ends with:  $2.22 + 4.20 = 6.42$  (profit of \$1.42)
- I keep: \$4.14

This maintains my pattern of fairly rewarding cooperation.

```
{'return': 4.20}
```

**Reasoning:**

The sender sent \$2.78 (about 55.6% of their endowment), showing trust.

I received \$8.34 after the 3x multiplier.

The sender currently has  $5 - 2.78 = 2.22$ .

I've been fair in my returns as a receiver:

- Round 1: Received \$10.02, returned \$5.00
- Round 4: Received \$8.97, returned \$4.50
- Round 5: Received \$8.67, returned \$4.30

For this round with \$8.34 received, a fair return would be around \$4.00-\$4.20.

If I return \$4.20:

- Sender ends with:  $2.22 + 4.20 = 6.42$  (profit of \$1.42)
- I keep:  $8.34 - 4.20 = 4.14$

This is consistent with my previous fair returns.

Usage: {

```
"input_tokens": 2001,
"output_tokens": 368,
"reasoning_tokens": 0
```

}

**GAME - Agent\_4 - Sender - dyad\_2**

Timestamp: 2026-08-12T09:25:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$3.32, that is 66.4% of their total endowment. You received \$9.959999999999999. How much do you return to the sender? (0-9.959999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.32 (66.4% of their endowment)
- I received \$9.96 after the 3x multiplier
- The sender currently has \$1.68 remaining

I want to reward this trust while being fair to both of us. If I return \$5.00:

- The sender ends with:  $\$1.68 + \$5.00 = \$6.68$  (profit of \$1.68)
- I end with:  $\$9.96 - \$5.00 = \$4.96$

This gives the sender a good return on their trust (33.6% gain on their original \$5) while I also benefit from the exchange. This return rate of approximately 50% is commonly seen as reciprocal in trust games.

{'return': 5.00}

**reasoning:**

Let me analyze this situation:

1. The sender sent \$3.32 out of their \$5 endowment (66.4%)
2. Due to the 3x multiplier, I received \$9.96 ( $3.32 \times 3 = 9.96$ )
3. I need to decide how much to return

Key considerations:

- The sender showed significant trust by sending 66.4% of their endowment
- This is a multi-round game with random pairings
- Building a reputation for reciprocity could be beneficial
- There's communication noise, so exact amounts may not be perceived correctly

Let me think about fair/reciprocal returns:

- The sender spent \$3.32 to give me \$9.96
- If I want the sender to profit from trusting me, I should return more than \$3.32
- A common strategy in trust games is to split the gains roughly equally

If I return \$0:

- Sender gets:  $\$5 - \$3.32 + \$0 = \$1.68$
- I get:  $\$9.96 - \$0 = \$9.96$
- Total: \$11.64

If I return \$3.32 (sender breaks even):

- Sender gets:  $\$5 - \$3.32 + \$3.32 = \$5.00$
- I get:  $\$9.96 - \$3.32 = \$6.64$
- Total: \$11.64

If I return \$5.00 (roughly equal split of total):

- Sender gets:  $\$5 - \$3.32 + \$5.00 = \$6.68$
- I get:  $\$9.96 - \$5.00 = \$4.96$
- Total: \$11.64

The sender took a risk and sent a substantial amount. A reasonable reciprocal amount would be to ensure both of us profit from the exchange.

To achieve roughly equal final payoffs, I'd need to return about \$4.14-this would give us each \$5.82. But since the sender invested significantly from their original endowment, I should consider whether an even more generous return would better reward their trust.

If I return \$6.64, they'd end with \$8.32 (doubling their investment gain), though that leaves me with just \$3.32. A middle ground like \$5.00 would give them a 33.6% gain while I keep \$4.96. Since this is the first round and I want to build a reputation for fairness, and given how much trust they demonstrated by sending most of their money, I'm leaning toward a return that meaningfully rewards their cooperation while still leaving me in a reasonable position....

**usage:**

{

```
"input_tokens": 537,
"output_tokens": 1319,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.32, it became \$9.959999999999999, and you returned \$5.0. Your visible payoff was \$4.959999999999999.

Your total visible earnings across all rounds are \$4.959999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the previous round, my co-player showed significant trust by sending 66.4% of their endowment, and I reciprocated with a 50% return rate.

For this round as sender, I'll demonstrate similar trust to establish cooperative play. Sending \$3.00 (60% of my endowment) shows trust while maintaining some security.

```
{'send': 3.00}
```

**reasoning:**

Let me think about this situation:

1. In Round 1, I was the receiver and someone sent me \$3.32 (66.4% of their endowment). I returned \$5.00, which was about 50% of what I received.
2. Now in Round 2, I'm the sender with \$5.
3. I need to decide how much to send to my current co-player (who may be different from Round 1's partner).

Key considerations:

- In the previous round, someone trusted me with 66.4% of their endowment
- I reciprocated fairly by returning 50%
- This established a pattern of moderate trust and reciprocity
- I'm playing with potentially a different agent now

Strategy options:

- a) Send a similar amount to what I received last round (~\$3.32) - this mirrors the trust shown to me
- b) Send less to be more cautious
- c) Send more to encourage cooperation

Given that:

- The previous sender trusted me with 66.4%, showing moderate-high trust
- I want to encourage cooperation in this multi-round game
- Sending around the same proportion (60-70%) seems reasonable
- This could establish a cooperative norm

I'll send \$3.00 (60% of my endowment), which is:

- Similar to what was sent to me in Round 1
- Shows trust but not maximal trust
- Allows me to keep some guarantee (\$2)
- If returned fairly (~50%), I'd get back around \$4.50, making my total \$2 + \$4.50 = \$6.50

Let me go with \$3.00.

**usage:**

```
{
 "input_tokens": 864,
 "output_tokens": 471,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.51 to you. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$11.469999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.27 (45.4% of their endowment), showing moderate trust
- I received \$6.81 after the 3x multiplier
- The sender currently has \$2.73 remaining

In the previous round when I was sender, I sent 60% and received back about 50% of what my co-player received. I'll reciprocate with a similar return rate.

Returning \$3.50 gives:

- Sender ends with: \$2.73 + \$3.50 = \$6.23 (profit of \$1.23)
- I end with: \$6.81 - \$3.50 = \$3.31

This represents about 51% return, which fairly rewards their trust.

{'return': 3.50}

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.27, which is 45.4% of their \$5 endowment
- I received \$6.81 after the 3x multiplier
- The sender currently has \$5 - \$2.27 = \$2.73 remaining

In the previous round when I was sender:

- I sent \$3.00 (60% of endowment)
- Received back \$4.51, which was about 50% of the \$9.00 they received
- This gave me a payoff of \$6.51

The sender this round showed moderate trust (45.4% vs my 60% last round). I should reciprocate fairly.

If I return \$3.50:

- Sender ends with: \$2.73 + \$3.50 = \$6.23 (profit of \$1.23)
- I end with: \$6.81 - \$3.50 = \$3.31

This represents about 51% return rate, similar to what I received last round (50%). This seems fair and reciprocal.

Let me check the math:

- Return rate = \$3.50 / \$6.81 = 51.4%
- Sender's final payoff = \$2.73 + \$3.50 = \$6.23
- My final payoff = \$6.81 - \$3.50 = \$3.31

This seems reasonable and reciprocal given their 45.4% trust level.

**usage:**

```
{
 "input_tokens": 1104,
 "output_tokens": 545,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.5. Your visible payoff was \$3.3100000000000005.

Your total visible earnings across all rounds are \$14.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience so far:

- When I've shown trust as sender (60% in Round 2), I received good reciprocity (~50% return)
- When others showed trust as sender, I've returned approximately 50%
- The pattern suggests cooperative play with moderate to high trust levels

I'll continue with a similar strategy, sending \$3.00 (60% of my endowment) to maintain trust and cooperation.

{'send': 3.00}

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.32 (66.4%)
- I received \$9.96
- I returned \$5.00 (about 50%)
- My payoff: \$4.96

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.51 (about 50%)
- My payoff: \$6.51

Round 3: I was RECEIVER

- Sender sent \$2.27 (45.4%)
- I received \$6.81
- I returned \$3.50 (about 51%)
- My payoff: \$3.31

Round 4: I am SENDER now

- I have \$5 to decide what to send

Looking at the pattern:

- In Round 2, I sent \$3.00 (60%) and got back about 50%
- Others have sent between 45% and 66% in the rounds I've seen

I've been experiencing decent reciprocity so far. The return rates have been around 50% consistently.

I should continue to show trust, as it's been rewarded. Sending around 60% seems reasonable based on what I've seen. I'll send \$3.00 again.

**usage:**

```
{
 "input_tokens": 1421,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$21.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience as sender:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)

- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)

Sending 60% of my endowment has consistently led to good reciprocity. I'll maintain this cooperative strategy.

```
{'send': 3.00}
```

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: I sent \$3.00, received \$4.44 back (payoff \$6.44)

Both times sending \$3.00 has resulted in good returns. The strategy of sending 60% has been working well and establishing cooperative relationships.

I should continue with this consistent strategy to maintain trust and cooperation.

**usage:**

```
{
 "input_tokens": 1648,
 "output_tokens": 217,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.38 to you. Your visible payoff was \$6.38.

Your total visible earnings across all rounds are \$27.599999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.26, so you received \$9.78. How much do you return? (0-9.78)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

This is similar to Round 1 where I received high trust. I've been consistently returning around 50% as receiver, which has worked well.

Returning \$5.00:

- Sender ends with:  $\$1.74 + \$5.00 = \$6.74$  (profit of \$1.74)
- I end with:  $\$9.78 - \$5.00 = \$4.78$

This maintains my reciprocal strategy.

```
{'return': 5.00}
```

**reasoning:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

In my previous rounds as receiver:

- Round 1: Received \$9.96, returned \$5.00 (50.2%)
- Round 3: Received \$6.81, returned \$3.50 (51.4%)

As sender, I've been sending \$3.00 and receiving returns of around \$4.40-4.50, which is about 49-50% of what the receiver got.

This sender showed high trust (65.2%). I should reciprocate with a similar 50% return rate.

Returning \$5.00:

- Sender ends with:  $\$1.74 + \$5.00 = \$6.74$  (profit of \$1.74)
- I end with:  $\$9.78 - \$5.00 = \$4.78$

This is about 51% return, consistent with my previous behavior as receiver.

usage:

```
{
 "input_tokens": 1888,
 "output_tokens": 439,
 "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$5.0. Your visible payoff was \$4.7799999999999999.

Your total visible earnings across all rounds are \$32.379999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my consistent strategy:

- As sender, I've been sending \$3.00 (60% of my endowment) and receiving good reciprocity (~\$4.40-\$4.50)
- This cooperative approach has been working well

I'll continue with the same strategy.

```
{'send': 3.00}
```

Reasoning:

Looking at my history:

As SENDER:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)
- Round 5: Sent \$3.00, received \$4.38 back (payoff \$6.38)

As RECEIVER:

- Round 1: Received \$9.96 from sender sending \$3.32, returned \$5.00 (payoff \$4.96)
- Round 3: Received \$6.81 from sender sending \$2.27, returned \$3.50 (payoff \$3.31)
- Round 6: Received \$9.78 from sender sending \$3.26, returned \$5.00 (payoff \$4.78)

My strategy has been consistent:

- As sender: Send \$3.00 (60% of endowment)
- As receiver: Return approximately 50% of what I receive

This has been working well. My returns as sender have been good (around \$4.40-\$4.50), and I've been fair as receiver.

I should continue this strategy and send \$3.00.

Usage: {

```
"input_tokens": 2187,
"output_tokens": 375,
"reasoning_tokens": 0
```

}

GAME - Agent\_3 - Receiver - dyad\_2

Timestamp: 2026-08-12T09:25:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_4",
"pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
"investor": "Agent_4",
```



```

"roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
},
"trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. the sender sent \$2.35, that is 47.0% of their total endowment. You received \$7.050000000000001. How much do you return to the sender? (0-7.050000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed moderate trust by sending \$2.35 (47% of their endowment)
- I received \$7.05 from this

- The sender currently has \$2.65 remaining

I want to reciprocate this trust fairly. A reasonable approach is to aim for equal final payoffs since we're cooperating:

- Total available: \$2.65 (sender's remainder) + \$7.05 (my received amount) = \$9.70
- Equal split:  $\$9.70 \div 2 = \$4.85$  each
- Return needed:  $\$4.85 - \$2.65 = \$2.20$

This ensures we both benefit equally from the cooperation, rewarding the sender's trust while keeping the gains fair.

```
{'return': 2.20}
```

#### reasoning:

Let me understand the situation:

- I am the RECEIVER in Round 1
- The sender sent \$2.35 (47% of their \$5 endowment)
- This was multiplied by 3x, so I received \$7.05
- I need to decide how much to return (between \$0 and \$7.05)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. There's communication noise mentioned, so amounts may differ slightly
3. This is a multi-round game with 8 agents and rotating pairings
4. Building reputation and reciprocity is important

The sender sent 47% of their endowment. If I want to be fair and reciprocal:

- The sender started with \$5, sent \$2.35, and has \$2.65 left
- I received \$7.05
- If I don't return anything, sender gets \$2.65, I get \$7.05
- A fair split might be to aim for equal payoffs or proportional returns

Common strategies in trust games:

1. Return nothing (selfish)
2. Return the original amount sent (\$2.35)
3. Return enough to split the gains equally
4. Return a proportion similar to what was sent

Let's calculate equal payoffs:

- Total money: \$5 (initial sender) + \$7.05 (tripled) = \$12.05? No wait...
- Initial endowment for sender: \$5
- Sent: \$2.35 (tripled to \$7.05 for receiver)
- Sender currently has:  $\$5 - \$2.35 = \$2.65$
- Receiver has: \$7.05
- Total in system:  $\$2.65 + \$7.05 = \$9.70$

For equal payoffs: each should get  $\$9.70/2 = \$4.85$

- So I'd need to return \$2.20 to the sender, leaving me with \$4.85

I could also return the original \$2.35 they sent, which would give them \$5.00 (breaking even) while I keep \$4.70. Or I could mirror their generosity by returning 47% of what I received-about \$3.31-which acknowledges their trust while still letting me benefit from the multiplication.

The equal payoff approach feels right though. Returning \$2.20 means we both end up with \$4.85, which feels like genuine reciprocity for the trust they showed....

#### usage:

```
{
 "input_tokens": 537,
 "output_tokens": 1079,
 "reasoning_tokens": 0
}
```

Message 4 role=user

#### content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.35, it became \$7.050000000000001, and you returned \$2.2. Your visible payoff was \$4.8500000000000005.

Your total visible earnings across all rounds are \$4.8500000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my previous round, the sender showed trust at 47% (\$2.35) and I reciprocated fairly, resulting in equal payoffs of \$4.85 each.

As the sender now, I'll show similar moderate trust. Sending around 50% demonstrates cooperation while managing risk.

```
{'send': 2.50}
```

**reasoning:**

Let me think about this situation:

Round 2: I am now the SENDER

- I have \$5
- I need to decide how much to send
- The amount I send will be tripled
- The receiver will decide how much to return

In the previous round:

- I was the receiver
- The sender sent \$2.35 (47% of their endowment)
- I received \$7.05
- I returned \$2.20
- My payoff was \$4.85
- The sender's payoff would have been  $\$2.65 + \$2.20 = \$4.85$

So in the last round, I reciprocated fairly by aiming for equal payoffs.

Now I'm the sender. I should consider:

1. Building trust by sending a reasonable amount
2. The sender in the previous round sent 47% (\$2.35 out of \$5)
3. I reciprocated fairly, so hopefully this receiver will do the same

Since the previous sender showed moderate trust at 47%, I could do something similar. Sending around \$2.35-\$2.50 seems reasonable.

However, I could also think strategically:

- If I send \$2.50, the receiver gets \$7.50
- If they split equally, they'd return \$2.50 to give us each \$5
- If I send \$3.00, the receiver gets \$9.00
- Equal split would mean returning \$3.50 for payoffs of \$5.50 each

Let me try sending \$2.50 as a moderate trust signal, similar to what the previous sender did with me.

Actually, let me reconsider. I showed I'm cooperative by returning fairly in the previous round. I could send a bit more to maximize joint welfare if the receiver reciprocates.

If I send \$3:

- I keep \$2
- Receiver gets \$9
- If they return \$3.50, I get \$5.50 total, they get \$5.50
- Joint welfare: \$11...

**usage:**

```
{
 "input_tokens": 859,
 "output_tokens": 713,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.43 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$9.780000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience:

- Round 1 (receiver): Sender sent 47%, I reciprocated fairly
- Round 2 (sender): I sent 50%, received fair return of \$2.43

The moderate trust strategy (around 50%) has been working well with fair reciprocation from co-players. I'll continue this approach.

{'send': 2.50}

**reasoning:**

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$2.43
- My payoff:  $\$5 - \$2.50 + \$2.43 = \$4.93$
- Their payoff:  $\$7.50 - \$2.43 = \$5.07$

The receiver returned a fair amount - roughly aimed at equal splits. The return rate was  $\$2.43/\$7.50 = 32.4\%$ , which gave us both similar final payoffs.

Now I'm a sender again with a new co-player. I've seen:

- Round 1 (as receiver): Sender sent 47%, I returned fairly
- Round 2 (as sender): I sent 50%, received fair return

Both rounds showed moderate trust and fair reciprocation. I should continue this cooperative strategy. Sending around 50% seems to be working well.

{'send': 2.50}

**usage:**

```
{
 "input_tokens": 1060,
 "output_tokens": 327,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.78 to you. Your visible payoff was \$5.279999999999999.

Your total visible earnings across all rounds are \$15.06.

This round, you are the RECEIVER against your current co-player. They sent you \$2.58, so you received \$7.74. How much do you return? (0-7.74)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.58 (51.6% of endowment), showing solid trust
- I received \$7.74
- Sender currently has \$2.42

Based on my positive experiences with fair reciprocation, I'll aim for equal final payoffs:

- Total:  $\$2.42 + \$7.74 = \$10.16$
- Equal split: \$5.08 each
- Return needed: \$2.66

{'return': 2.66}

**reasoning:**

I'm the receiver now. Let me analyze:

- Sender sent \$2.58 (51.6% of their endowment)
- I received \$7.74 (after 3x multiplier)
- Sender currently has:  $\$5 - \$2.58 = \$2.42$

My recent experience shows:

- Round 2: I sent \$2.50, got back \$2.43
- Round 3: I sent \$2.50, got back \$2.78

Both returns were generous and aimed at fair splits. I should continue this reciprocal fairness.

For equal final payoffs:

- Total available:  $\$2.42$  (sender's remainder) +  $\$7.74$  (my received) =  $\$10.16$
- Equal split:  $\$10.16 / 2 = \$5.08$  each
- Return needed:  $\$5.08 - \$2.42 = \$2.66$

This would give us both \$5.08, which is fair and reciprocates the sender's trust at 51.6%.

Let me verify:

- Sender final:  $\$2.42 + \$2.66 = \$5.08$
- My final:  $\$7.74 - \$2.66 = \$5.08$

Perfect. I'll return \$2.66.

**usage:**

```
{
 "input_tokens": 1287,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.58, it became \$7.74, and you returned \$2.66. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my track record:

- Sending around 50% has consistently resulted in fair reciprocation
- Other agents are also showing similar moderate trust levels
- This strategy has been mutually beneficial

I'll continue with this successful approach.

```
{'send': 2.50}
```

**reasoning:**

Let me review my history:

- Round 1 (receiver): Received from sender who sent \$2.35 (47%), returned \$2.20, payoff \$4.85
- Round 2 (sender): Sent \$2.50 (50%), got back \$2.43, payoff \$4.93
- Round 3 (sender): Sent \$2.50 (50%), got back \$2.78, payoff \$5.28
- Round 4 (receiver): Received from sender who sent \$2.58 (51.6%), returned \$2.66, payoff \$5.08
- Round 5 (sender): ???

My strategy of sending around 50% has been working well - I've been getting fair returns.

The pattern shows:

- When I send ~50% (\$2.50), I get back \$2.43-\$2.78
- Others are sending similar amounts (47-52%)
- Fair reciprocation is the norm

I should continue with this moderate trust approach around 50%.

```
{'send': 2.50}
```

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 316,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.24 to you. Your visible payoff was \$4.74.

Your total visible earnings across all rounds are \$24.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.72 (54.4% of endowment), showing strong trust
- I received \$8.16
- Sender currently has \$2.28

For equal final payoffs:

- Total: \$2.28 + \$8.16 = \$10.44
- Equal split: \$5.22 each
- Return needed: \$2.94

{'return': 2.94}

**reasoning:**

Let me analyze this situation:

- Sender sent \$2.72 (54.4% of their \$5 endowment) - showing good trust
- I received \$8.16
- Sender currently has \$5 - \$2.72 = \$2.28

For equal final payoffs:

- Total available: \$2.28 + \$8.16 = \$10.44
- Equal split: \$10.44 ÷ 2 = \$5.22 each
- Return needed: \$5.22 - \$2.28 = \$2.94

This sender showed slightly more trust than average (54.4% vs the ~50% I've been seeing), so I should reciprocate fairly to maintain cooperation.

Let me return \$2.94 to achieve equal payoffs.

**usage:**

```
{
 "input_tokens": 1725,
 "output_tokens": 298,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.72, it became \$8.16, and you returned \$2.94. Your visible payoff was \$5.220000000000001.

Your total visible earnings across all rounds are \$30.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Sender sent \$3.36 (67.2% of endowment), showing very high trust - much higher than typical

- I received \$10.08
- Sender currently has \$1.64

For equal final payoffs:

- Total: \$1.64 + \$10.08 = \$11.72
- Equal split: \$5.86 each
- Return needed: \$4.22

```
{'return': 4.22}
```

#### Reasoning:

Looking at this situation:

- Sender sent \$3.36 (67.2% of endowment), showing very high trust
- I received \$10.08
- Sender currently has \$1.64

This is notably higher trust than I've seen in previous rounds (most have been around 50%). I should reciprocate generously.

For equal final payoffs:

- Total: \$1.64 + \$10.08 = \$11.72
- Equal split: \$5.86 each
- Return needed: \$5.86 - \$1.64 = \$4.22

This sender showed significant trust, so I should return a fair amount that rewards that trust

```
Usage: {
 "input_tokens": 1965,
 "output_tokens": 273,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_7 - Sender - dyad\_3

Timestamp: 2026-08-12T09:26:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_8"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_7": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance building trust with managing risk since I have no history with this receiver.

**\*\*My reasoning:\*\***

- Sending money creates value (3x multiplier means potential gains for both of us)
- This is round 1, so I want to signal cooperation
- I should send enough to be meaningful but not risk everything
- If the receiver reciprocates fairly, we both benefit

I'll send a moderate amount to test cooperation while limiting downside risk.

```
'''json
{"send": 3}
'''
```

**reasoning:**

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. **\*\*Game Structure:\*\***

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets tripled
- The receiver then decides how much to return
- My payoff: \$5 - sent + returned
- Receiver's payoff: 3×sent - returned

2. **\*\*Strategic Considerations:\*\***

- If I send nothing (\$0), I keep \$5, receiver gets \$0. Total: \$5
- If I send everything (\$5), receiver gets \$15, and if they return half (\$7.50), I get



\$7.50 and they keep \$7.50. Total: \$15

- The game creates surplus through cooperation (up to \$15 total vs \$5 if no trust)
- This is a trust game - I need to trust the receiver will return some amount

3. **\*\*First Round Considerations:\*\***

- No history with this partner
- This is a multi-round game with different partners
- Building a reputation for cooperation might matter
- The note about "communication noise" suggests amounts might be misperceived, which could affect trust dynamics

4. **\*\*Common Strategies:\*\***

- **\*\*Full trust:\*\*** Send \$5, hope for good return
- **\*\*Moderate trust:\*\*** Send \$2-3, test the waters
- **\*\*No trust:\*\*** Send \$0, guarantee \$5
- **\*\*Signal cooperation:\*\*** Send a meaningful amount to show willingness to cooperate

5. **\*\*Rational Analysis:\*\***

- If the receiver is purely self-interested and this is a one-shot game, they should return \$0
- Knowing this, I should send \$0
- However, in repeated games or with reciprocity norms, cooperation can emerge
- Many people exhibit fairness preferences and reciprocity

6. **\*\*My Decision:\*\***

Since this is the opening move in a multi-round game, a moderate cooperative signal seems prudent. I'll send \$3-enough to demonstrate trust and give the receiver \$9 to work with, but not so much that I'm exposed to catastrophic loss if they keep everything. Even a modest return of \$3-4 would benefit us both compared to the baseline of keeping \$5....

usage:

```
{
 "input_tokens": 500,
 "output_tokens": 836,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.56 to you. Your visible payoff was \$7.56.

Your total visible earnings across all rounds are \$7.56.

This round, you are the RECEIVER against your current co-player. They sent you \$1.67, so you received \$5.01. How much do you return? (0-5.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.67 (33% of their endowment)
- I received \$5.01 after the 3x multiplier
- They currently have \$3.33 remaining

**\*\*My considerations:\*\***

- They showed moderate trust by sending \$1.67
- The multiplier created \$3.34 in surplus value
- A fair approach would be to share this surplus roughly equally
- In Round 1, my partner returned ~62% and that felt cooperative

**\*\*Fair return calculation:\*\***

If we split the surplus equally:

- Each should gain ~\$1.67 from cooperation
- Sender ending: \$5 - \$1.67 + return = \$6.67 desired
- Return needed: \$3.34

However, that might be overly generous given their modest send. I'll return about 60% to reward their cooperation while keeping reasonable value.

```
'''json
{"return": 3.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.56 to me
- I ended with:  $\$5 - \$3 + \$5.56 = \$7.56$
- Receiver ended with:  $\$9 - \$5.56 = \$3.44$

That was actually quite generous of them! They returned more than half.

Round 2 (current):

- I am RECEIVER
- They sent me \$1.67
- I received \$5.01 (which is  $3 \times \$1.67$ )
- I need to decide how much to return (0-5.01)

Key considerations:

1. My previous partner was generous, returning \$5.56 out of \$9 (62%)
2. This is a different player (multi-agent setting with randomized pairings)
3. They sent me \$1.67 out of \$5, which is 33.4% - a moderate amount
4. The sender keeps:  $\$5 - \$1.67 = \$3.33$  before my return
5. I should reciprocate fairly to encourage future cooperation

Let me think about fair returns:

- If I return half (\$2.50), they end with  $\$3.33 + \$2.50 = \$5.83$ , I keep \$2.51
- If I return around what gives us equal final payoffs:
  - They have  $\$3.33 + \text{return}$
  - I have  $\$5.01 - \text{return}$
  - Equal:  $\$3.33 + x = \$5.01 - x$
  - $2x = \$1.68$
  - $x = \$0.84$

This would give us both around \$4.17

- A proportional return similar to what I received (62% of what I got):
  - $0.62 \times \$5.01 = \$3.11$
  - They'd end with:  $\$3.33 + \$3.11 = \$6.44$
  - I'd keep:  $\$5.01 - \$3.11 = \$1.90$

A 50% return of around \$2.50 seems like the right balance-it rewards their cooperation without being excessive, and it's more moderate than matching their 33% send rate since they're a different player than before.

If I send back \$2.50, they'd gain \$0.83 while I'd be left with \$2.51. The math shows we'd create \$8.34 in total value, which is less than the \$10 we'd have if neither of us participated. The real question is what surplus we're actually generating from this exchange and whether my return is proportional to their initial trust.

The multiplier created \$3.34 in extra value (\$5.01 received minus \$1.67 sent). If we split that gain equally, each of us should pocket \$1.67. That means I should return \$3.34 to bring them to \$6.67 total, which gives us both the same \$1.67 gain from the transaction. ...

**usage:**

```
{
 "input_tokens": 754,
 "output_tokens": 1445,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.67, it became \$5.01, and you returned \$3.0. Your visible payoff was \$2.

01.

Your total visible earnings across all rounds are \$9.57.

This round, you are the RECEIVER against your current co-player. They sent you \$3.66, so you received \$10.98. How much do you return? (0-10.98)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$3.66 (73% of their endowment) - very high trust!
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining
- Total surplus created: \$2.32

**\*\*My considerations:\*\***

- This is a very generous send showing strong cooperation
- They took significant risk by sending most of their money
- I should reward this trust with a fair return
- In Round 1, I received about 62% back on my \$3 send
- In Round 2, I returned 60% on a smaller send

**\*\*Fair return:\*\***

Given their high trust, I'll return approximately 50% of what I received:

- Return \$5.50
- They'll end with \$6.84 (profit of \$1.84)
- I'll end with \$5.48 (profit of \$0.48)
- This rewards their cooperation while both of us profit

'''json

{"return": 5.50}

'''

**reasoning:**

Let me analyze this situation:

**\*\*Current Situation:\*\***

- My co-player sent me \$3.66 (73% of their endowment)
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining (\$5 - \$3.66)
- This is a very generous/trusting send

**\*\*What this shows:\*\***

- High trust from the sender (sending over 70%!)
- They're expecting reciprocity
- The multiplier created significant surplus: \$5.98 (\$10.98 - \$5)

**\*\*Considerations:\*\***

- In Round 1, when I sent \$3 (60%), I received \$5.56 back - that's about 62% return rate
- In Round 2, when someone sent \$1.67 (33%), I returned \$3.00 - that's 60% return rate
- This sender showed much more trust (\$3.66 = 73% of endowment)

**\*\*Fair calculation:\*\***

The total pot is now \$12.32 (\$1.34 they kept + \$10.98 I received)

If we split the gains from cooperation:

- Original total: \$10 (my \$5 + their \$5)
- Current total: \$12.32
- Surplus created: \$2.32
- Fair split would give each person \$1.16 gain
- Sender would end at \$6.16, I'd end at \$6.16
- I need to return: \$6.16 - \$1.34 = \$4.82

But let me think about what's fair based on the trust shown:

- They sent 73% of their money
- If I return about 50% of what I received, that's \$5.49
- That would give them \$1.34 + \$5.49 = \$6.83 total

...

**usage:**

```
{
 "input_tokens": 1112,
```

```
"output_tokens": 833,
"reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$5.5. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$15.05.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: I sent \$3 and received \$5.56 back - good cooperation!
- Rounds 2-3: As receiver, I tried to return fairly and reward trust
- Total earnings: \$15.05

**\*\*Considerations for this round:\*\***

- I'm paired with a new co-player each round
- My Round 1 send of \$3 worked well
- The most generous player sent me \$3.66, showing high trust pays off
- Sending more creates more total value (3x multiplier)
- I want to encourage cooperative norms in the pool

**\*\*Strategy:\*\***

I'll send a bit more than Round 1 to signal strong cooperation while managing risk. This shows I'm willing to trust and invest in mutual gains.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Now I'm the SENDER again in Round 4.

Let me review my history:

- Round 1 (SENDER): Sent \$3, received back \$5.56, payoff \$7.56
- Round 2 (RECEIVER): Received \$5.01 (from \$1.67 sent), returned \$3.00, payoff \$2.01
- Round 3 (RECEIVER): Received \$10.98 (from \$3.66 sent), returned \$5.50, payoff \$5.48

This is a new co-player each round in this multi-agent setting.

My experience so far:

- When I sent \$3, I got back \$5.56 - good return!
- When others sent to me, I tried to be fair/cooperative
- The player who sent \$3.66 was very trusting

For this round, I need to decide how much to send to a new co-player.

Considerations:

- I've seen good cooperation so far (my Round 1 partner returned well)
- I want to continue building cooperative patterns
- Sending more creates more value (3x multiplier)
- But I'm paired with a new player each time, so no direct reputation building

Strategy:

- In Round 1, I sent \$3 and it worked well
- I could send more to signal strong cooperation
- Or stick with \$3 as a moderate cooperative amount

Given that I've had positive experiences and want to continue fostering cooperation, I'll send a bit more this time - maybe \$3.5 or \$4. This shows I'm willing to trust while not going all-in.

Let me go with \$3.5 - it's between my previous \$3 send and the very generous \$3.66 I received in Round 3.

Actually, thinking about it more: the player who sent me \$3.66 showed strong trust and I rewarded it. If I want to encourage that kind of cooperation, I should also send generously when I'm the sender. Let me send \$4.

...

**usage:**

```
{
 "input_tokens": 1475,
 "output_tokens": 749,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$3.59 to you. Your visible payoff was \$5.09.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*Recent experience:\*\***

- Round 4: I sent \$3.50, received only \$3.59 back - minimal profit (\$5.09)
- This was disappointing compared to Round 1 where I sent \$3 and got \$5.56 back
- But this is a NEW co-player with unknown behavior

**\*\*Considerations:\*\***

- Different co-players show varying levels of cooperation
- Sending less than \$3.50 reduces my risk after last round's poor return
- Still want to signal willingness to cooperate and create value
- \$3 was my initial send and it worked well - proven strategy

**\*\*Strategy:\*\***

I'll return to sending \$3 - a moderate amount that:

- Shows trust and cooperation
- Creates meaningful value (\$9 for receiver)
- Limits downside risk
- Has proven successful before

```json

{ "send": 3 }

```

**reasoning:**

Looking at my history:

- Round 1: Sent \$3, received \$5.56 back - payoff \$7.56 (good!)
- Round 2: Receiver, returned \$3.00 - payoff \$2.01
- Round 3: Receiver, returned \$5.50 - payoff \$5.48
- Round 4: Sent \$3.5, received \$3.59 back - payoff \$5.09 (disappointing)
- Round 5: Now sender again

Analysis of Round 4:

- I sent \$3.50 (70% of endowment) showing strong trust
- They received \$10.50
- They only returned \$3.59 (34% of what they received)
- I ended with \$5.09, barely breaking even
- This was much less generous than my Round 1 partner who returned \$5.56 on a \$9 receive

This suggests I encountered a less cooperative player. However, I'm now paired with a NEW co-player, so their behavior doesn't necessarily predict this new person's behavior.

Strategic considerations:

- I'm doing reasonably well overall (\$20.14 total)

- Sending creates value through the multiplier
- But I need to balance risk
- Two different responses to my sends: one very good (\$5.56), one mediocre (\$3.59)
- I don't know if there's a pattern or if agents vary widely

I think I should continue showing willingness to cooperate but maybe pull back slightly from the \$3.50 I sent last time since that didn't work out as well. Going back to \$3 seems reasonable - it's a solid middle ground that shows cooperation without overcommitting.

usage:

```
{
 "input_tokens": 1797,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.06 to you. Your visible payoff was \$6.06.  
Your total visible earnings across all rounds are \$26.2.

This round, you are the RECEIVER against your current co-player. They sent you \$1.64, so you received \$4.92. How much do you return? (0-4.92)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment) - modest trust
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining

**\*\*My considerations:\*\***

- This is very similar to Round 2 where I received \$5.01 from a \$1.67 send
- In Round 2, I returned \$3.00 (60%) which seemed fair
- I should be consistent in rewarding similar levels of cooperation
- Returning around 60% rewards their trust while keeping reasonable value

**\*\*Fair return:\*\***

Returning approximately 60% of what I received:

- Return \$3.00
- They'll end with \$6.36 (profit of \$1.36)
- I'll end with \$1.92

```
'''json
{"return": 3.00}
'''
```

**reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment)
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining
- Total surplus created from their send: \$4.92 - \$1.64 = \$3.28

**\*\*My considerations:\*\***

- They showed modest trust by sending \$1.64 (similar to the \$1.67 I saw in Round 2)
- In Round 2, when I received \$5.01 from a \$1.67 send, I returned \$3.00 (60%)
- I should be consistent in how I respond to similar levels of trust
- Returning about 60% would be \$2.95

**\*\*Fair return calculation:\*\***

- If I return around \$3.00:
  - They end with: \$3.36 + \$3.00 = \$6.36 (profit of \$1.36)
  - I end with: \$4.92 - \$3.00 = \$1.92 (loss of \$0.08 from initial endowment perspective, but positive from game perspective)

Actually, let me think about this differently:

- They sent \$1.64
- This created \$3.28 in surplus (\$4.92 - \$1.64)
- A fair split of surplus would give each about \$1.64
- So they should end with: \$5 + \$1.64 = \$6.64
- They have \$3.36, so I need to return \$3.28 to get them to \$6.64

But that gives me only \$1.64, which seems low for receiving \$4.92.

Let me be consistent with Round 2 where I returned 60% on a similar send:

- 60% of \$4.92 = \$2.95

...

**usage:**

```
{
 "input_tokens": 2142,
 "output_tokens": 677,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.64, it became \$4.92, and you returned \$3.0. Your visible payoff was \$1.92.

Your total visible earnings across all rounds are \$28.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this round as SENDER:

**\*\*My sender history:\*\***

- Round 1: Sent \$3 ? got \$5.56 back (profit \$3.56) ?
- Round 4: Sent \$3.50 ? got \$3.59 back (profit \$0.09) ?
- Round 5: Sent \$3 ? got \$4.06 back (profit \$1.06) ?

**\*\*Observations:\*\***

- \$3 has been my most reliable send amount (2/2 reasonable returns)
- \$3.50 didn't work out well
- Each round is a new co-player, so no personal history
- Total earnings: ~\$28.12 - healthy position

**\*\*Strategy:\*\***

Stick with \$3 as my standard opening send:

- Proven track record
- Shows cooperation and trust
- Creates meaningful value (\$9 for receiver)
- Manages downside risk
- Signals I'm a fair player

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at my history:

- Round 1 (Sender): Sent \$3, got \$5.56 back - Good!
- Round 2 (Receiver): Received \$5.01, returned \$3.00
- Round 3 (Receiver): Received \$10.98, returned \$5.50
- Round 4 (Sender): Sent \$3.50, got \$3.59 back - Poor return
- Round 5 (Sender): Sent \$3, got \$4.06 back - Okay return
- Round 6 (Receiver): Received \$4.92, returned \$3.00

As SENDER:

- Round 1: Sent \$3 ? returned \$5.56 (profit \$3.56)
- Round 4: Sent \$3.50 ? returned \$3.59 (profit \$0.09)
- Round 5: Sent \$3 ? returned \$4.06 (profit \$1.06)

Total earnings: ~\$28.12

New co-player again.

The pattern shows:

- Sending \$3 has worked reasonably well (2 out of 3 times)
- Sending \$3.50 got poor returns
- Each round is a new partner with unknown behavior

I should continue with a moderate send that balances trust-building with risk management. \$3 seems to be a good baseline that's worked before.

Actually, let me think about the broader strategy. I've been fairly cooperative as a receiver, returning good amounts. As a sender, I've been moderately trusting.

Given that:

1. This is a new co-player
2. \$3 has generally worked well
3. I want to signal cooperation while managing risk
4. Total earnings are healthy at \$28.12

I'll stick with \$3 as my standard opening send.

```
Usage: {
 "input_tokens": 2476,
 "output_tokens": 676,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_8 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:26:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_8"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_7": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned



- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round as the SENDER, I need to balance trust and caution since I have no history with this co-player.

The key consideration is that sending money creates value (3x multiplier), but I'm vulnerable to the receiver's decision about how much to return.

A moderate amount seems reasonable for Round 1:

- It signals willingness to cooperate and trust
- It creates meaningful value (\$9 for the receiver)
- It provides some protection against complete loss

I'll send \*\*\$3\*\*.

{'send': 3}

**reasoning:**

This is Round 1 of a multi-task experiment where I'm playing a trust game as the SENDER. Let me think about this situation:

1. I start with \$5
2. I can send any amount from \$0 to \$5
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - amount\_sent + amount\_returned

This is a classic trust game. The optimal strategy depends on:

- Whether I trust the receiver to return a fair amount
- What "fair" means in this context
- The fact that there's communication noise (amounts may differ slightly)
- This is a multi-round game with multiple agents
- Pairings are randomized each round

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return half (\$7.50), I'd end up with \$7.50

- total (better than my initial \$5)
- If I send \$0, I keep \$5 but no value is created
- The socially optimal outcome is to send the maximum amount to create the most total value

However, there's risk:

- The receiver might not return anything
- There's communication noise that could affect trust
- This is round 1, so I have no history with this partner

A reasonable first-round strategy might be:

- Send a moderate amount to signal trust and willingness to cooperate
- Not send everything in case the receiver doesn't reciprocate
- Send enough to create meaningful value

Common strategies in trust games:

- Send about half (\$2.50) as a balanced approach
- Send more (\$3-4) to signal strong trust
- Send everything (\$5) to maximize joint payoff

Given this is round 1 and I want to establish a cooperative pattern but also protect against complete exploitation, I'll send \$3. This:

- Creates \$9 for the receiver
- Shows trust and cooperation

...

**usage:**

```
{
 "input_tokens": 500,
 "output_tokens": 625,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.46 to you. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$4.46.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 2):\*\***

- My co-player sent me \$3.19 ? I received \$9.57
- This created \$6.38 in surplus value

**\*\*Fair return considerations:\*\***

- The sender took a significant risk sending most of their endowment
- A reasonable approach is to ensure we both benefit from the cooperation
- Returning \$4.00 would give:
  - Sender:  $\$5 - \$3.19 + \$4.00 = \$5.81$
  - Me:  $\$9.57 - \$4.00 = \$5.57$

This creates a balanced outcome where we both gain from the exchange.

```
{'return': 4.00}
```

**reasoning:**

Let me analyze this situation:

Last round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.46 to me
- My payoff:  $\$5 - \$3 + \$2.46 = \$4.46$

- Their payoff:  $\$9.0 - \$2.46 = \$6.54$

So they returned about 27% of what they received ( $\$2.46/\$9.0 \approx 0.273$ ).

Now in Round 2:

- I'm the RECEIVER
- My current co-player sent me \$3.19
- I received \$9.57 ( $3.19 \times 3$ )
- I need to decide how much to return

Important notes:

1. This is a different co-player (multi-agent setting, randomized pairings)
2. There's communication noise, so the \$3.19 might not be exactly what they sent
3. The previous player returned about 27% to me

Considerations:

- If I want to encourage cooperation, I should return a fair amount
- A common fair split might be to return enough so we both benefit equally from the cooperation
- The sender gave up \$3.19, so they have  $\$5 - \$3.19 = \$1.81$  left
- I have \$9.57
- If I want us to end up with similar amounts, I could return about \$3.88, giving us both around \$5.69
- Or I could return proportionally to what was sent: returning about 1/3 of what I received would be \$3.19, which gives them back their original amount
- A generous return might be half of what I received: \$4.785

Let me think about what's fair:

- The sender took a risk sending \$3.19 (gave up most of their endowment)
- The transaction created \$6.38 in surplus value beyond what they sent
- Splitting that gain equally means we each get an additional \$3.19 from the multiplier effect
- So returning \$3.19 would leave me with \$6.38 and restore them to their starting \$5
- Alternatively, returning \$4 would give them \$5.81 while I keep \$5.57

, creating a more balanced outcome

- Given the earlier low reciprocation (27%), I could...

usage:

```
{
 "input_tokens": 756,
 "output_tokens": 1027,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.19, it became \$9.57, and you returned \$4.0. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$10.030000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this new round as SENDER:

**Past experience:**

- Round 1: Sent \$3, received \$2.46 back (modest cooperation)
- Round 2: When I received, I returned fairly (\$4.00 on \$9.57)

**Current situation:**

- New co-player (randomized pairing)
- Need to establish trust with someone new
- The 3x multiplier means cooperation creates value

A moderate send amount balances the potential gains from cooperation with the risk of low returns. Sending \$3 creates \$9 of value while maintaining some protection.

```
{'send': 3}
reasoning:
 Let me analyze the situation:

 Previous rounds:
 - Round 1: I was SENDER, sent $3, received back $2.46, payoff was $4.46
 - Round 2: I was RECEIVER, received $9.57 (from $3.19 sent), returned $4.00, payoff was $5.57

 Current Round (Round 3):
 - I'm the SENDER again
 - New co-player (since pairings are randomized each round)
 - I have $5 to potentially send

 Observations:
 1. In Round 1, when I sent $3, I got back $2.46. The receiver kept $6.54 out of $9.00, which seems somewhat selfish (I got less than my initial $3).
 2. In Round 2, when someone sent me $3.19, I returned $4.00, being relatively fair.
 3. The note about communication noise is important - amounts may differ from what was actually sent/returned.

 Strategy considerations:
 - I'm playing with a new co-player each round
 - There's no reputation to build with a specific player
 - The 3x multiplier means cooperation creates value
 - I need to balance trust against the risk of low returns

 Given:
 - My Round 1 experience showed modest cooperation from my receiver
 - I was reasonably fair in Round 2
 - I'm with a new player now who doesn't know my history

 I think sending $3 again is reasonable:
 - It's a moderate level of trust
 - Creates $9 for the receiver
 - Provides some protection if they don't reciprocate well
```

Let me send \$3 again.

```
usage:
{
 "input_tokens": 1047,
 "output_tokens": 532,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.  
Your total visible earnings across all rounds are \$14.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.22, so you received \$9.66. How much do you return? (0-9.66)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

```
Current Round (Round 4):
- Co-player sent $3.22 ? I received $9.66
- This represents significant trust (sending 64% of their endowment)
```

```
Fair return analysis:
- Returning $4.00 would give:
 - Sender: $5 - $3.22 + $4.00 = $5.78 (profit of $0.78)
 - Me: $9.66 - $4.00 = $5.66 (profit of $0.66)
```

This creates a fair split where both players benefit from the cooperation.

```
{'return': 4.00}
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER:\*\***

- My co-player sent me \$3.22
- I received \$9.66 (after 3x multiplier)
- I need to decide how much to return

**\*\*Previous rounds analysis:\*\***

- Round 1 (SENDER): Sent \$3, got back \$2.46 - slightly below fair
- Round 2 (RECEIVER): Received \$9.57, returned \$4.00 - tried to be fair
- Round 3 (SENDER): Sent \$3, got back \$2.93 - again slightly below fair

**\*\*Fair return calculation:\*\***

The sender gave up \$3.22 from their \$5 endowment, creating \$9.66 for me.

- Surplus created: \$9.66 - \$3.22 = \$6.44

A fair split would give us each roughly equal gains:

- If I return \$4.00:
  - Sender gets: \$5 - \$3.22 + \$4.00 = \$5.78 (gain of \$0.78)
  - I get: \$9.66 - \$4.00 = \$5.66 (gain of \$0.66)

This seems reasonably balanced. The sender sent a significant portion of their endowment, showing trust.

I should return around \$4.00 to be fair and reciprocate their trust.

**usage:**

```
{
 "input_tokens": 1324,
 "output_tokens": 501,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.22, it became \$9.66, and you returned \$4.0. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$20.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience so far:

**\*\*As SENDER:\*\***

- Round 1: Sent \$3 ? received \$2.46 (modest return)
- Round 3: Sent \$3 ? received \$2.93 (decent return)

**\*\*As RECEIVER:\*\***

- I've consistently returned \$4.00 on ~\$9.60, maintaining fairness

**\*\*Current round:\*\***

- New co-player (randomized pairing)
- Sending \$3 has been my standard opening move
- Creates \$9 of value with moderate risk

I'll continue with the balanced approach that's been working.

```
{'send': 3}
```

**reasoning:**

Looking at my history as SENDER:

- Round 1: Sent \$3, received \$2.46 back
- Round 3: Sent \$3, received \$2.93 back

Both times I sent \$3, I received back around \$2.50-\$3.00, which is close to fair but not quite equal splitting of the gains.

As RECEIVER:

- Round 2: Received \$9.57, returned \$4.00
- Round 4: Received \$9.66, returned \$4.00

I've been consistent in returning fairly.

Now Round 5, I'm a SENDER with a new co-player. The pattern suggests that sending around \$3 has been reasonably productive. The returns haven't been amazing (I've profited minimally as sender), but they've been positive.

I should continue with a moderate strategy. Sending \$3 creates value and has worked reasonably well so far.

usage:

```
{
 "input_tokens": 1601,
 "output_tokens": 366,
 "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.2 to you. Your visible payoff was \$7.2.

Your total visible earnings across all rounds are \$27.82.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This is a slightly smaller send than previous rounds

**\*\*Fair return analysis:\*\***

- In previous rounds, I returned ~\$4.00 on receives of ~\$9.60
- This represents about a 42% return rate
- For \$8.58, a proportional return would be around \$3.60

Returning \$3.70 would give:

- Sender: \$5 - \$2.86 + \$3.70 = \$5.84 (modest profit)
- Me: \$8.58 - \$3.70 = \$4.88

This maintains fairness while ensuring both players benefit from cooperation.

{'return': 3.70}

reasoning:

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This shows trust, sending about 57% of their endowment

**\*\*Fair return analysis:\*\***

- Total surplus created: \$8.58 - \$2.86 = \$5.72
- A fair split would give each of us roughly \$2.86 in profit

If I return \$3.50:

- Sender gets: \$5 - \$2.86 + \$3.50 = \$5.64 (profit of \$0.64)
- I get: \$8.58 - \$3.50 = \$5.08 (profit of \$0.08)

That's not quite balanced. Let me try \$4.00:

- Sender gets:  $\$5 - \$2.86 + \$4.00 = \$6.14$  (profit of  $\$1.14$ )
- I get:  $\$8.58 - \$4.00 = \$4.58$  (profit of  $\$-0.42$ )

Wait, that would give me less than my base \$5. Let me recalculate.

Actually, I don't start with \$5 as the receiver. I start with \$0 and receive the multiplied amount.

So:

- Sender starts with \$5, sends \$2.86, gets back \$X
- Sender final:  $\$5 - \$2.86 + \$X = \$2.14 + \$X$
- I start with \$0, receive \$8.58, return \$X
- My final:  $\$8.58 - \$X$

For a balanced outcome where we both profit similarly:

- Sender profit:  $\$2.14 + \$X - \$5 = \$X - \$2.86$
- My profit:  $\$8.58 - \$X - \$0 = \$8.58 - \$X$
- Setting equal:  $\$X - \$2.86 = \$8.58 - \$X$  gives  $\$X = \$5.72$

That would split the surplus evenly, but feels generous. Instead, I should ensure the sender profits from their trust while keeping a reasonable share for myself. Returning around \$3.80 would give them roughly \$0.94 in profit while I keep \$4.78.

But that's heavily skewed in my favor. Testing \$4.50 or \$5.00 returns gets closer to balance-the sender would profit \$1.64 to \$2.14 while I'd keep \$3.58 to \$4.08. I'm reconsidering what fairness actually means here. In previous exchanges where I received around \$9.50-\$9.66 from sends of \$3.19-\$3.22, returning \$4.00 felt proportional. Using that same ratio (roughly 42% return rate) on this \$8.58 suggests returning around \$3.58-\$3.70. Testing \$3.70 gives the sender \$0.84 profit and me \$4.88, which still favors me but follows the pattern I've established in earlier rounds....

**usage:**

```
{
 "input_tokens": 1883,
 "output_tokens": 1261,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.86, it became \$8.58, and you returned \$3.7. Your visible payoff was \$4.88.

Your total visible earnings across all rounds are \$32.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.69, so you received \$8.07. How much do you return? (0-8.07)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Current Round (Round 7):\*\***

- Co-player sent \$2.69 ? I received \$8.07
- A moderate send showing trust

**\*\*My return pattern:\*\***

- I've been returning around 42% of what I receive
- This has created mutually beneficial outcomes

For \$8.07 received:

- Returning \$3.50 would give:
  - Sender:  $\$5 - \$2.69 + \$3.50 = \$5.81$  (profit of  $\$0.81$ )
  - Me:  $\$8.07 - \$3.50 = \$4.57$

This maintains fairness and ensures both players benefit.

```
{'return': 3.50}
```

**Reasoning:**

Looking at the current situation:

**\*\*Current Round (Round 7):\*\***

- Co-player sent \$2.69 ? I received \$8.07
- This is another moderate send

**\*\*My return pattern:\*\***

- Round 2: Received \$9.57, returned \$4.00 (42%)
- Round 4: Received \$9.66, returned \$4.00 (41%)
- Round 6: Received \$8.58, returned \$3.70 (43%)

I've been maintaining around a 42% return rate, which seems fair.

For \$8.07:

- 42% return would be:  $\$8.07 \times 0.42 = \$3.39$

Returning \$3.50 would give:

- Sender:  $\$5 - \$2.69 + \$3.50 = \$5.81$  (profit of \$0.81)
- Me:  $\$8.07 - \$3.50 = \$4.57$

This maintains my fair approach while ensuring both benefit. Let me go with \$3.50.

```
Usage: {
 "input_tokens": 2202,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_2 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:26:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_6"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`



When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.55, that is 51.0% of their total endowment. You received \$7.649999999999995. How much do you return to the sender? (0-7.649999999999995)

Message 3 role=assistant

content:

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.55 (51% of their \$5 endowment), showing moderate trust
- I received \$7.65 after the 3x multiplier
- The sender now has \$2.45 remaining
- I need to decide how much to return

**\*\*My reasoning:\*\***

The sender showed significant trust by sending about half their endowment. To reciprocate fairly and build trust:

If I return **\*\*\$4.00\*\***:

- Sender ends with:  $\$2.45 + \$4.00 = \$6.45$  (profit of \$1.45)
- I end with:  $\$7.65 - \$4.00 = \$3.65$  (profit of \$3.65)

This represents:

- Returning about 52% of what I received (similar to their 51% sent)
- Both of us profit from the cooperation
- Rewarding their trust encourages future cooperation
- Setting a positive precedent for Round 1

{'return': 4.0}

reasoning:

Let me understand the situation:

1. I am the RECEIVER in Round 1
2. The sender sent \$2.55 (which is 51% of their \$5 endowment)
3. This amount was multiplied by 3x, so I received  $\$2.55 \times 3 = \$7.65$  (approximately \$7.65)
4. I need to decide how much to return (between \$0 and \$7.65)

Now, let me think about this strategically:

The sender sent 51% of their endowment, which is slightly more than half. This shows a mod

erate level of trust. They could have sent nothing, they could have sent everything, but they chose to send about half.

From a fairness perspective:

- The sender now has:  $\$5 - \$2.55 = \$2.45$
- I have:  $\$7.65$
- Total in the system:  $\$2.45 + \$7.65 = \$10.10$

If I want to be fair and reciprocate the trust:

- One approach: Split the total gains equally. The sender started with \$5, and we now have \$10.10 total. If we split equally, we'd each have \$5.05. The sender already has \$2.45, so I'd return  $\$5.05 - \$2.45 = \$2.60$
- Another approach: Return the same proportion they sent. They sent 51% of their endowment, so I could return 51% of what I received:  $0.51 \times \$7.65 = \$3.90$
- Conservative approach: Return at least what they sent originally (\$2.55) to ensure they don't lose money from trusting me
- Generous approach: Return more to reward their trust and potentially build cooperation

Given that:

1. The sender showed moderate trust by sending 51%
2. There's communication noise mentioned (though the exact amount sent seems precise)
3. Since this is a multi-round game with multiple agents, establishing a reputation for reciprocity could be strategically valuable
4. The sender took a genuine risk by sending money

A fair reciprocal response should reward their trust while ensuring they benefit from the exchange and remain proportional to their initial gesture. If I return \$2.60, we'd each end with \$5.05-a modest profit for both of us that acknowledges their willingness to trust.

Alternatively, returning 50% of what I received (\$3.825) would give them \$6.275, a more substantial gain. But I need to reconsider my baseline here-as the receiver, I don't start with \$5 like the sender does. I begin with \$0 and only gain what they send multiplied by three. The sender started with \$5, sent \$2.55, and now has \$2.45 remaining.

If I return \$X, they end with  $\$2.45 + \$X$  and I end with  $\$7.65 - \$X$ . For equal splits, that's \$2.60 returned, giving us both \$5.05. But in this experimental context, I should probably return more-around \$3.00-\$3.50-to reward their trust and signal reciprocity for potential future rounds.

Testing a few amounts: returning \$3.25 leaves them with \$5.70 and me with \$4.40; \$3.50 gives them \$5.95 and me \$4.15; \$4.00 results in them having \$6.45 and me \$3.65. Each option balances showing trustworthiness with keeping reasonable profit, though I'm leaning toward the higher end to maximize their incentive to continue cooperating. Both of us end up better off than we started. At \$3.50, the sender gets \$5.95 (up \$0.95) and I get \$4.15 (up \$4.15), which feels like a solid reciprocal move that rewards their trust without being exploitative.

I'm settling on \$3.50 as my return amount-it strikes a balance between showing genuine reciprocity and ensuring both parties benefit from the exchange. Terms of what's strategically optimal in a multi-round setting. Building reputation matters-if I'm generous now, other agents will likely trust me more later. Since this is Round 1, establishing a trustworthy precedent is crucial. The sender showed moderate confidence by sending \$2.55, so I should reciprocate proportionally.

I'm settling on \$3.50. It more than doubles their original amount, gives them solid profit, signals I'm reliable, and leaves me with a reasonable gain too. Let me verify the math:  $\$2.55 \times 3 = \$7.65$  received, returning \$3.50 means the sender nets \$0.95 profit while I keep \$4.15.

The communication noise mentioned in the problem is worth noting, but I can only work with what I observe. My final return amount is \$3.50, formatted as `{'return': 3.50}`.

Actually, reconsidering the proportionality here-the sender gave 51% of their endowment, which signals moderate trust. Matching that proportion with my return would mean sending back 51% of \$7.65, which equals \$3.90. This reciprocal approach better reflects their level of generosity while still leaving me with a reasonable profit of \$3.75 versus their

\$1.35. I think \$3.90 better captures the spirit of proportional reciprocation.

Though I'm also drawn to rounding up to \$4.00 for simplicity-it's nearly the same percentage anyway (52%), easier to process, and still feels fair to both parties. Let me settle on \$4.00 as my final choice.

**usage:**

```
{
 "input_tokens": 539,
 "output_tokens": 3346,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.55, it became \$7.649999999999995, and you returned \$4.0. Your visible payoff was \$3.649999999999995.

Your total visible earnings across all rounds are \$3.649999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- Round 1 established a cooperative dynamic where sending ~51% led to mutual profit
- I'm now the sender with a new co-player
- I need to establish trust in this new pairing

**\*\*Strategic reasoning:\*\***

- Sending a moderate amount (around 50%) shows willingness to cooperate
- It gives my co-player enough to make returning worthwhile (\$7.50 after multiplier)
- It limits my downside risk while allowing mutual gains
- This mirrors the successful approach from Round 1

**\*\*Decision:\*\***

I'll send \$2.50 (50% of my endowment) to establish a cooperative tone with this new co-player.

{'send': 2.5}

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was the RECEIVER
- Sender sent \$2.55 (51% of \$5)
- I received \$7.65
- I returned \$4.00
- My payoff: \$3.65
- Sender's payoff would have been: \$5 - \$2.55 + \$4.00 = \$6.45

**\*\*Current Round (Round 2):\*\***

- I am now the SENDER
- I have \$5
- I'm paired with a potentially different agent (since it says "your current co-player")
- I need to decide how much to send

**\*\*Strategic considerations:\*\***

1. **\*\*Reciprocity principle\*\***: In Round 1, my co-player showed moderate trust (51% sent). I reciprocated well by returning 52% of what I received, ensuring both parties profited.
2. **\*\*Building trust\*\***: Since this is a multi-round game with multiple agents, establishing a cooperative reputation could be beneficial.
3. **\*\*Unknown opponent\*\***: This is a different co-player, so I don't know their reciprocity tendencies yet.

4. **\*\*First impression\*\***: As the sender in this new pairing, I can set the tone for cooperation.
5. **\*\*Risk vs reward\*\***:
- If I send \$0, I keep \$5, they get \$0, no mutual benefit
  - If I send all \$5, it becomes \$15 for them, but I'm vulnerable if they return nothing
  - Middle ground shows trust while limiting downside risk
6. **\*\*Following established norms\*\***: In Round 1, the sender sent 51%. This seemed to work well and encouraged cooperation.

**\*\*Decision\*\***

I'll send a similar amount to what I experienced as effective in Round 1 - around 50% (\$2.50). This:

- Shows trust and willingness to cooperate
- Limits my downside risk
- Gives the receiver enough to make returning worthwhile

...

**usage:**

```
{
 "input_tokens": 916,
 "output_tokens": 646,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.62 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$9.77.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened\*\***

- The sender sent \$2.11 (42.2% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- The sender now has \$2.89 remaining

**\*\*My reasoning\*\***

The sender showed trust, though slightly less than what I've seen in previous rounds. I should reciprocate fairly to maintain cooperation.

If I return **\*\*\$3.30\*\***:

- Sender ends with:  $\$2.89 + \$3.30 = \$6.19$  (profit of \$1.19)
- I end with:  $\$6.33 - \$3.30 = \$3.03$  (profit of \$3.03)

This represents:

- Returning about 52% of what I received
- Both of us profiting from cooperation
- Fair reciprocation that encourages future trust

{'return': 3.3}

**reasoning:**

Let me analyze this situation:

**\*\*What happened in Round 2\*\***

- I sent \$2.50
- It became \$7.50
- They returned \$3.62
- My payoff:  $\$2.50$  (kept) +  $\$3.62$  (returned) = \$6.12
- Their payoff:  $\$7.50 - \$3.62 = \$3.88$
- They returned about 48% of what they received

**\*\*Current Round 3 situation:\*\***

- I'm the receiver
- Sender sent \$2.11 (42.2% of their \$5 endowment)
- I received \$6.33
- Sender now has \$2.89 remaining
- I need to decide how much to return

**\*\*Analysis:\*\***

The sender sent \$2.11, which is less than the \$2.50-\$2.55 amounts I've seen previously. They're showing moderate but slightly lower trust. However, they still sent a significant portion (42%).

If I return different amounts:

- Return \$3.00: Sender ends with \$5.89 (+\$0.89), I end with \$3.33 (+\$3.33)
- Return \$3.50: Sender ends with \$6.39 (+\$1.39), I end with \$2.83 (+\$2.83)
- Return \$3.20: Sender ends with \$6.09 (+\$1.09), I end with \$3.13 (+\$3.13)

I want to:

1. Reciprocate their trust proportionally
2. Be fair and encourage future cooperation
3. Return roughly 50% of what I received, similar to what I've observed

Returning around \$3.20 would be about 50.5% of what I received, which seems fair and proportional to their ~42% send rate. This ensures both of us profit.

Actually, let me be slightly more generous since they did show trust. I'll return \$3.30 (about 52% of what I received).

...

**usage:**

```
{
 "input_tokens": 1218,
 "output_tokens": 758,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.3. Your visible payoff was \$3.0300000000000002.

Your total visible earnings across all rounds are \$12.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- I'm facing a new co-player as sender
- Previous cooperation has been successful (total earnings \$12.80)
- When I sent \$2.50 in Round 2, I received \$3.62 back for a good profit
- My returns as receiver have been fair (~52%)

**\*\*Strategic reasoning:\*\***

- Continue the cooperative approach that's been working
- Sending 50% shows good faith while managing risk
- This gives my co-player \$7.50 to work with, enough for mutual benefit
- Establishes positive precedent with this new partner

{'send': 2.5}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds summary:\*\***

- Round 1 (RECEIVER): Received \$7.65 from sender sending \$2.55, returned \$4.00, earned \$3.65

- Round 2 (SENDER): Sent \$2.50, became \$7.50, they returned \$3.62, earned \$6.12
- Round 3 (RECEIVER): Received \$6.33 from sender sending \$2.11, returned \$3.30, earned \$3.03
- Total earnings: \$12.80

**\*\*Pattern analysis:\*\***

- When I sent \$2.50 (50%), I got back \$3.62, giving me a profit of \$1.12 (from \$2.50 remaining + \$3.62 = \$6.12 total)
- When others sent to me, I've returned fairly (~52% of what I received)
- Overall, cooperation has been successful

**\*\*Current decision:\*\***

- New co-player as sender this round
- Need to establish trust again
- Sending around 50% has worked well previously

I'll continue with the cooperative strategy that's been working: send \$2.50 (50% of endowment).

**usage:**

```
{
 "input_tokens": 1558,
 "output_tokens": 431,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*My reasoning:\*\***

This sender showed above-average trust compared to previous rounds. I should reciprocate generously.

If I return **\*\*\$4.70\*\***:

- Sender ends with: \$2.08 + \$4.70 = \$6.78 (profit of \$1.78)
- I end with: \$8.76 - \$4.70 = \$4.06 (profit of \$4.06)

This represents:

- Returning about 54% of what I received
- Rewarding their higher trust level (58.4% sent)
- Both profiting well from cooperation
- Encouraging continued trust

{'return': 4.7}

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*Patterns from previous rounds:\*\***

- Round 1: Received \$7.65, returned \$4.00 (52%)

- Round 3: Received \$6.33, returned \$3.30 (52%)
- I've been consistently returning about 52% of what I receive

**\*\*Strategy:\*\***

The sender showed high trust (58.4% sent). I should reciprocate fairly.

If I return around 52% of \$8.76:

- $0.52 \times \$8.76 = \$4.55$

Let me check the outcomes:

- Sender ends with:  $\$2.08 + \$4.55 = \$6.63$  (profit of \$1.63)
- I end with:  $\$8.76 - \$4.55 = \$4.21$  (profit of \$4.21)

This seems fair - both profit, sender gets a decent return for their trust.

Actually, given they sent 58.4% (more than my usual 50%), maybe I should return slightly more to really reward that trust. Let me try \$4.70:

- Sender ends with:  $\$2.08 + \$4.70 = \$6.78$  (profit of \$1.78)
- I end with:  $\$8.76 - \$4.70 = \$4.06$  (profit of \$4.06)

That's about 54% return, which is fair and rewards their higher trust level.

**usage:**

```
{
 "input_tokens": 1844,
 "output_tokens": 622,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.92, it became \$8.76, and you returned \$4.7. Your visible payoff was \$4.06.

Your total visible earnings across all rounds are \$22.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My sending strategy (50%) has been successful in Rounds 2 & 4
- Returns I've received: \$3.62 and \$3.01 (averaging \$3.32)
- Total earnings of \$22.37 show cooperation is working

**\*\*Strategic reasoning:\*\***

- Maintain the consistent 50% sending strategy
- This has proven reliable across different co-players
- Gives receiver \$7.50 to work with
- Balances trust-building with risk management

{'send': 2.5}

**reasoning:**

Looking at the pattern:

**\*\*My sending history:\*\***

- Round 2: Sent \$2.50 (50%), received back \$3.62, profit = \$6.12
- Round 4: Sent \$2.50 (50%), received back \$3.01, profit = \$5.51

**\*\*My receiving history:\*\***

- Round 1: Received \$7.65 from \$2.55 sent (51%), returned \$4.00
- Round 3: Received \$6.33 from \$2.11 sent (42%), returned \$3.30
- Round 5: Received \$8.76 from \$2.92 sent (58%), returned \$4.70

**\*\*Analysis:\*\***

- When I send 50%, I've gotten decent returns (\$3.01-\$3.62)
- As a receiver, I've been fair, returning around 52-54%
- Total earnings are strong at \$22.37
- Cooperation has been working well

**\*\*This round:\*\***

- New co-player as sender
- Continue with the cooperative 50% strategy that's been successful
- This maintains consistency and shows I'm a cooperative player

I'll send \$2.50 again.

**usage:**

```
{
 "input_tokens": 2182,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.8 to you. Your visible payoff was \$6.3.

Your total visible earnings across all rounds are \$28.67.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My consistent 50% sending strategy has worked well (Rounds 2, 4, 6)
- Returns have varied (\$3.62, \$3.01, \$3.80) but all profitable
- Total earnings of \$28.67 show strong overall cooperation

**\*\*Strategic reasoning:\*\***

- Continue the reliable 50% sending pattern
- This approach has proven effective across different co-players
- Provides \$7.50 for receiver to reciprocate
- Maintains consistency and manages risk

```
{'send': 2.5}
```

**Reasoning:**

Looking at this situation:

**\*\*Context:\*\***

- I'm the sender again, facing a new co-player
- My sending history shows consistent 50% sends (\$2.50 in rounds 2, 4, 6)
- Returns I've received: \$3.62, \$3.01, \$3.80 (averaging \$3.48)
- All have been profitable
- Total earnings of \$28.67

**\*\*Strategic reasoning:\*\***

- The 50% strategy has been consistently successful
- Returns have varied but all been positive
- This creates good conditions for mutual benefit
- Should continue this proven approach

I'll send \$2.50 again.

Usage: {

```
"input_tokens": 2442,
"output_tokens": 294,
"reasoning_tokens": 0
}
```

**GAME - Agent\_6 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:26:35



Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_6"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 7,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative dynamic while being somewhat cautious since I don't know my co-player yet.

If I send money, it triples, creating potential mutual benefit. However, there's risk since the receiver's return is voluntary.

I'll send a moderate amount to signal trust and willingness to cooperate, while not being overly vulnerable in this first interaction.

**\*\*Decision:\*\***

```
'''json
{"send": 3}
'''
```

This gives the receiver \$9 to work with. If they're cooperative and return a fair share, we both benefit. If not, I've learned something about the dynamics for future rounds while limiting my downside risk.

**reasoning:**

This is the first round of a trust game, and I'm the sender. I need to think about this strategically.

Key facts:

- I have \$5
- Whatever I send gets multiplied by 3
- The receiver decides how much to return
- There may be communication noise
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- I don't know who my current co-player is

Since this is the first round, I have no information about how other players behave. The optimal strategy depends on:

1. Whether I expect reciprocity from the receiver
2. Whether I want to establish a reputation (though with hidden names, this is less relevant for future rounds)
3. Risk tolerance

From a purely rational game theory perspective, if this were a one-shot game with no reputation effects, the receiver should return \$0 (keep everything), which means I should send \$0. However, experimental evidence shows people often cooperate in trust games.

Some considerations:

- Sending \$5 maximizes total payoff (\$5 + \$15 total) but requires trust
- Sending \$0 guarantees I keep \$5 but creates no opportunity for mutual gain
- A moderate amount (like \$2-3) balances risk and opportunity

Since this is round 1 and I want to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything immediately

I'll send a moderate amount. Sending \$3 seems reasonable:

- If sent \$3, receiver gets \$9
- If receiver returns half (\$4.50), I end up with \$5 - \$3 + \$4.50 = \$6.50
- If receiver returns nothing, I end up with \$2
- This tests their cooperation while not being overly cautious

Actually, let me reconsider. A common strategy is to send about half to show trust but not complete vulnerability. Sending \$2.50 or \$3 seems reasonable.

...

**usage:**

```
{
```

```
"input_tokens": 500,
"output_tokens": 632,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.25 to you. Your visible payoff was \$6.25.

Your total visible earnings across all rounds are \$6.25.

This round, you are the RECEIVER against your current co-player. They sent you \$2.67, so you received \$8.01. How much do you return? (0-8.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.67\*\*** (about 53% of their endowment), which became \$8.01 for me. This is a moderately cooperative move.

**\*\*Comparison to previous round:\*\***

- When I sent \$3.00, I received \$4.25 back (47% of the \$9.00)
- That gave my co-player a reasonable benefit while rewarding my trust

**\*\*My reasoning:\*\***

I want to:

1. Reward the sender's trust with a decent return
2. Maintain cooperative dynamics for future rounds
3. Keep enough to make receiving worthwhile

Returning about \$3.75 seems fair:

- Sender would get: \$2.33 (kept) + \$3.75 = \$6.08 total (gain of ~\$1.08)
- I would get: \$8.01 - \$3.75 = \$4.26 (solid gain)

This is proportionally similar to what I experienced last round.

```
'''json
{"return": 3.75}
'''
```

**reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was SENDER
- I sent \$3.0
- It became \$9.0
- Receiver returned \$4.25
- My payoff: \$5 - \$3 + \$4.25 = \$6.25

This round (Round 2):

- I am RECEIVER
- Sender sent \$2.67
- I received \$8.01 (which is \$2.67 × 3)
- I need to decide how much to return

The sender in this round sent \$2.67 out of \$5, which is about 53% of their endowment. This is a fairly cooperative move.

In the previous round, when I sent \$3, I received back \$4.25. My net gain was \$1.25 from the initial \$5.

Let me think about what would be fair:

- The sender gave up \$2.67
- I received \$8.01
- If I want to be cooperative and fair, I should think about:
  1. Returning enough so the sender benefits from sending
  2. Not returning so much that I don't benefit from receiving

A common approach is to split the gains. The total amount now is \$8.01. If we split it equally:

- Each gets \$4.005

But the sender started with \$5 and sent \$2.67, so they have \$2.33 left.

If I return \$4.00, they'd have  $\$2.33 + \$4.00 = \$6.33$  (gain of \$1.33)

I'd keep  $\$8.01 - \$4.00 = \$4.01$  (gain of \$4.01 compared to receiving nothing)

Hmm, that seems imbalanced. Let me think differently.

Another approach: What would make our final payoffs more equal?

- Sender has:  $\$5 - \$2.67 + \text{return} = \$2.33 + \text{return}$
- I have:  $\$8.01 - \text{return}$

For equal payoffs:

$$\$2.33 + \text{return} = \$8.01 - \text{return}$$

$$2 \times \text{return} = \$8.01 - \$2.33 = \$5.68$$

$$\text{return} = \$2.84$$

If I return \$2.84:

- Sender gets:  $\$2.33 + \$2.84 = \$5.17$
- I get:  $\$8.01 - \$2.84 = \$5.17$

That would equalize our outcomes, though it doesn't particularly reward the sender's risk-taking. I should consider what the previous receiver did for comparison.

The prior round showed a return of \$4.25 on \$9.00 received-roughly 47% of what they got. Applying that same rate here:  $\$8.01 \times 0.47 = \$3.76$ . This would give the sender \$6.09 total and leave me with \$4.25, which feels more balanced given the sender's initial trust. But I need to account for the communication noise mentioned in the instructions-the amounts I'm seeing may not match what was actually sent or returned....

usage:

```
{
 "input_tokens": 785,
 "output_tokens": 1226,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.67, it became \$8.01, and you returned \$3.75. Your visible payoff was \$4.26.

Your total visible earnings across all rounds are \$10.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

**\*\*My co-player sent \$3.36\*\*** (67% of their endowment), which became \$10.08 for me. This is a notably generous and trusting move.

**\*\*Context from previous rounds:\*\***

- Round 2: Someone sent \$2.67 (53%), I returned \$3.75
- Round 1: I sent \$3.00 (60%), received \$4.25 back

**\*\*My reasoning:\*\***

This sender showed higher trust than my previous co-player (67% vs 53%). I should reward their generosity proportionally.

Returning \$4.75:

- Sender would get:  $\$1.64 \text{ (kept)} + \$4.75 = \$6.39$  total (gain of \$1.39)
- I would get:  $\$10.08 - \$4.75 = \$5.33$  (good gain)

This provides the sender with a decent reward for their trust while maintaining mutual ben

efit - both of us come out ahead.

```
'''json
{"return": 4.75}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$3.36 (67% of their \$5 endowment) - this is quite generous/trusting
- I received \$10.08

**\*\*Previous round:\*\***

- Someone sent me \$2.67 (53% of endowment), I returned \$3.75
- That gave them  $\$2.33 + \$3.75 = \$6.08$  total (~\$1.08 gain)
- I kept \$4.26

**\*\*Round 1:\*\***

- I sent \$3.00 (60%), received \$4.25 back
- That gave me  $\$2.00 + \$4.25 = \$6.25$  total

**\*\*My thinking:\*\***

The sender this round was more generous (67% vs 53% in previous round). I should reward their higher trust level.

If I return around \$4.50-\$5.00:

- Return \$4.50: Sender gets  $\$1.64 + \$4.50 = \$6.14$ , I get \$5.58
- Return \$5.00: Sender gets  $\$1.64 + \$5.00 = \$6.64$ , I get \$5.08

Actually, let me think about what's fair given they sent 67% of their endowment.

If I return \$5.00:

- Sender: keeps \$1.64, gets back \$5.00 = \$6.64 total (gain of \$1.64)
- Me: receive \$10.08, return \$5.00 = \$5.08 (solid gain)

This seems like a good reciprocal move for their high trust. It's roughly a 50-50 split of the multiplied amount, which rewards their generosity while still benefiting both of us.

Actually, let me reconsider. They sent \$3.36 out of \$5, so they kept \$1.64.

Returning \$5 gives them \$6.64 with a \$1.64 gain, while I keep \$5.08. That's reasonably fair given their high trust. I...

**usage:**

```
{
 "input_tokens": 1163,
 "output_tokens": 921,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.75. Your visible payoff was \$5.33.

Your total visible earnings across all rounds are \$15.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER again:

**\*\*My experience:\*\***

- Round 1: Sent \$3.00, received \$4.25 back ? Good outcome
- Rounds 2-3: As receiver, I've been cooperative, returning fair amounts

**\*\*Current situation:\*\***

- New co-player (different from Round 1)
- I've been part of generally cooperative dynamics

**\*\*My reasoning:\*\***

I had success with sending \$3.00 before, and the cooperative pattern across rounds suggest s moderate-to-high sending works well. I'll send slightly more this time to signal trust while building on the positive momentum.

Sending \$3.50 creates \$10.50 for the receiver. If they're reasonably cooperative (returning ~40%), I'd still come out ahead.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

I'm back as the SENDER in round 4. Let me review my experience so far:

Round 1: I sent \$3.00, received \$4.25 back. Payoff: \$6.25

Round 2: Received \$8.01 (from \$2.67 sent), returned \$3.75. Payoff: \$4.26

Round 3: Received \$10.08 (from \$3.36 sent), returned \$4.75. Payoff: \$5.33

Total earnings: \$15.84

Now I'm the sender again with a new co-player. What should I send?

Looking at the pattern:

- When I sent \$3.00 (60%), I got a decent return
- When others sent moderate-to-high amounts (\$2.67, \$3.36), I've been fairly cooperative in returning
- The game seems to be working reasonably well with moderate-to-high sending amounts

I should consider:

1. I had a positive experience when I sent \$3.00
2. The cooperative moves I've seen have been between \$2.67-\$3.36
3. I've demonstrated I'm a fair returner, so maybe I can expect reasonable behavior

I think I should send a similar or slightly higher amount than last time. Since I received \$4.25 when I sent \$3.00, that worked out well.

Options:

- Send \$3.00 again (proven strategy)
- Send \$3.50 (show slightly more trust)
- Send \$4.00 (high trust signal)

Given:

- This is a new co-player
- I've been building cooperative patterns
- The game is going well so far

I'll send \$3.50 as a moderately high trust signal, slightly more than my first round, but not overly risky. This creates \$10.50 for the receiver.

...

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 660,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.55 to you. Your visible payoff was \$6.05.

Your total visible earnings across all rounds are \$21.89.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*My pattern:\*\***

- I've been consistently returning around 45-47% of what I receive
- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)

**\*\*My reasoning:\*\***

This sender showed moderate trust. I should maintain consistency with my previous cooperative behavior.

Returning \$3.85:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of \$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This is about 46% return rate, consistent with my pattern and fair to the sender.

```
'''json
{"return": 3.85}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*Context from previous rounds:\*\***

- Round 1: I sent \$3.00 (60%), received \$4.25 back (47% of \$9)
- Round 2: Someone sent \$2.67 (53%), I returned \$3.75 (47% of \$8.01)
- Round 3: Someone sent \$3.36 (67%), I returned \$4.75 (47% of \$10.08)
- Round 4: I sent \$3.50 (70%), received \$4.55 back (43% of \$10.50)

**\*\*Pattern:\*\***

I've been pretty consistent returning around 45-47% of what I receive. My co-player sent \$2.81, which is moderately cooperative.

Let me calculate:

- 45% of \$8.43 = \$3.79
- 47% of \$8.43 = \$3.96

Returning around \$3.85 would:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of ~\$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This maintains consistency with my previous behavior and rewards the sender proportionally for their trust.

**usage:**

```
{
 "input_tokens": 1847,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$3.85. Your visible payoff was \$4.58.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 back (good outcome)
- Round 4: Sent \$3.50 ? Received \$4.55 back (good outcome)

**\*\*Overall dynamics:\*\***

- Cooperation has been working well across all rounds
- I've been a fair receiver (returning ~46%)
- Others have been reasonably cooperative with me

**\*\*My reasoning:\*\***

I'm with a new co-player. The cooperative pattern continues to work. I'll send an amount between my previous sends to signal trust while being strategic.

Sending \$3.25 creates \$9.75 for the receiver. With typical cooperation, this should benefit us both.

```
'''json
{"send": 3.25}
'''
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3.00, received \$4.25 back ? Payoff \$6.25
- Round 4: Sent \$3.50, received \$4.55 back ? Payoff \$6.05

As RECEIVER:

- Round 2: Received \$8.01, returned \$3.75 ? Payoff \$4.26
- Round 3: Received \$10.08, returned \$4.75 ? Payoff \$5.33
- Round 5: Received \$8.43, returned \$3.85 ? Payoff \$4.58

Overall, the cooperative strategy has been working well. When I send moderate-to-high amounts (\$3-\$3.50), I get good returns. My co-players have generally been cooperative.

This is a new co-player again (Round 6). I've been doing well with sends in the \$3-\$3.50 range.

I'll continue with a similar strategy - sending \$3.25 this time (between my previous two sends).

This creates \$9.75 for the receiver. If they return around 45%, I'd get about \$4.39 back, giving me a payoff of \$5.89.

Actually, let me think about this differently. I've been gradually increasing (3.0 then 3.5). Maybe I should go higher to signal stronger trust, or stay consistent. Let me stick with \$3.50 since it worked well.

Actually, let me be a bit more varied. \$3.25 is a good middle ground.

**usage:**

```
{
 "input_tokens": 2205,
 "output_tokens": 567,
 "reasoning_tokens": 0
}
```

**Message 14 role=user****content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$2.56 to you. Your visible payoff was \$4.3100000000000005.

Your total visible earnings across all rounds are \$30.78.

This round, you are the RECEIVER against your current co-player. They sent you \$2.85, so you received \$8.55. How much do you return? (0-8.55)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**



Looking at this situation:

**\*\*My co-player sent \$2.85\*\*** (57% of their endowment), which became \$8.55 for me. This is a moderately cooperative move.

**\*\*Context:\*\***

- Round 6 was disappointing - I sent \$3.25 but only got \$2.56 back (26% return)
- However, this is a NEW co-player, not the same one
- My consistent pattern as receiver has been returning ~46-47%

**\*\*My reasoning:\*\***

Despite the poor return last round, I shouldn't penalize this new co-player. Maintaining consistency and fairness has been my strategy, and it's worked well overall.

Returning \$3.90:

- Sender would get: \$2.15 (kept) + \$3.90 = \$6.05 total (gain of \$1.05)
- I would get: \$8.55 - \$3.90 = \$4.65 (good gain)

This maintains my ~46% return rate and rewards the sender's trust appropriately.

```
'''json
{"return": 3.90}
'''
```

#### Reasoning:

Looking at this situation:

**\*\*My co-player sent \$2.85\*\*** (57% of their endowment), which became \$8.55 for me. This is a moderately cooperative move.

**\*\*Recent experience:\*\***

- Round 6: I sent \$3.25, received only \$2.56 back (26% of \$9.75)
- That was disappointing - I got \$4.31 total, barely breaking even
- This is notably lower than my previous sender experiences

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)
- Round 5: Returned 46% (\$3.85 of \$8.43)

**\*\*Considerations:\*\***

1. The last round showed less cooperation from my co-player
2. But this is a NEW co-player now
3. They sent \$2.85, which is comparable to what others sent
4. Should I maintain my consistent cooperative strategy?

I think I should maintain consistency. The poor return in Round 6 was disappointing, but:

- This is a different co-player
- I've built a pattern of fairness
- Retaliating against a new player isn't justified
- Long-term cooperation benefits everyone

Following my pattern of ~46% return:

$\$8.55 \times 0.46 = \$3.93$

Sender would get: \$2.15 (kept) + \$3.93 = \$6.08 (gain of \$1.08)

I would get: \$8.55 - \$3.93 = \$4.62 (good gain)

I'll return \$3.90 to stay consistent with my cooperative approach.

```
Usage: {
 "input_tokens": 2534,
 "output_tokens": 674,
 "reasoning_tokens": 0
}
```

#### ROUND 8

Roles: Agent\_1=Sender, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Receiver, Agent\_5=Receiver, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Sender

Pairings:

```

dyad_1: Agent_6=Sender, Agent_4=Receiver
dyad_2: Agent_3=Sender, Agent_5=Receiver
dyad_3: Agent_1=Sender, Agent_2=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver
Parsed game decisions:
dyad_1: Agent_6 sent $3; received $9; Agent_4 returned $4.2; payoffs Agent_6=$6.2, Agent_4=$4.8;
noise/visible: sent visible=$2.81, received visible=$8.43, returned visible=$4.83
 Action validation: {"returned": {"amount": 4.2, "clamped": false, "max": 9.0, "min": 0, "raw":
 4.2}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 sent $2.5; received $7.5; Agent_5 returned $5; payoffs Agent_3=$7.5, Agent_5=$2.5;
noise/visible: sent visible=$3.5, received visible=$10.5, returned visible=$4.04
 Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 7.5, "min": 0, "raw":
 5.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_3: Agent_1 sent $3; received $9; Agent_2 returned $3.57; payoffs Agent_1=$5.57,
Agent_2=$5.43; noise/visible: sent visible=$2.29, received visible=$6.87, returned visible=$3.69
 Action validation: {"returned": {"amount": 3.57, "clamped": false, "max": 9.0, "min": 0, "raw":
 3.57}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3; received $9; Agent_7 returned $4; payoffs Agent_8=$6, Agent_7=$5;
noise/visible: sent visible=$2.77, received visible=$8.31, returned visible=$3.23
 Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw":
 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
 "Agent_1": 42.889999999999999,
 "Agent_2": 43.74,
 "Agent_3": 47.03,
 "Agent_4": 42.22,
 "Agent_5": 44.830000000000005,
 "Agent_6": 40.14,
 "Agent_7": 41.55,
 "Agent_8": 44.1
}
Visible balances after round: {
 "Agent_1": 42.32,
 "Agent_2": 37.58,
 "Agent_3": 42.5,
 "Agent_4": 43.099999999999994,
 "Agent_5": 40.089999999999996,
 "Agent_6": 42.26,
 "Agent_7": 38.3,
 "Agent_8": 42.5
}

```

## LLM CALLS

### GAME - Agent\_6 - Sender - dyad\_1

```

Timestamp: 2026-08-12T09:26:47
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_4"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_4": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}

```

```

}
Exact input messages sent to model:
 Message 1 role=system
 content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 MULTI-AGENT SETTING:
 - There are 8 agents in this run.
 - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
 - You play one sender-receiver game with your paired opponent each round.
 - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
 ll have acted as sender and receiver equally often when the round count allows it.
 - Agent names are hidden in this run.
 - Your paired opponent is referred to as your current co-player.
 Message 2 role=user
 content:
 Round 2

 In your most recent previous game round, you were the SENDER against another agent. You se
 nt $3.0, it became $9.0, and you saw them return $4.25 to you. Your visible payoff was $
 6.25.
 Your total visible earnings across all rounds are $6.25.

 This round, you are the RECEIVER against your current co-player. They sent you $2.67, so y
 ou received $8.01. How much do you return? (0-8.01)
 Respond exactly as JSON: {'return': <amount>}
 Message 3 role=assistant
 content:
 Looking at this situation:

 My co-player sent $2.67 (about 53% of their endowment), which became $8.01 for me. Thi
 s is a moderately cooperative move.

 Comparison to previous round:
 - When I sent $3.00, I received $4.25 back (47% of the $9.00)
 - That gave my co-player a reasonable benefit while rewarding my trust

```

**\*\*My reasoning:\*\***

I want to:

1. Reward the sender's trust with a decent return
2. Maintain cooperative dynamics for future rounds
3. Keep enough to make receiving worthwhile

Returning about \$3.75 seems fair:

- Sender would get:  $\$2.33$  (kept) +  $\$3.75$  =  $\$6.08$  total (gain of  $\sim\$1.08$ )
- I would get:  $\$8.01 - \$3.75 = \$4.26$  (solid gain)

This is proportionally similar to what I experienced last round.

```
'''json
{"return": 3.75}
'''
```

**reasoning:**

Let me analyze this situation:

Previous round (Round 1):

- I was SENDER
- I sent  $\$3.0$
- It became  $\$9.0$
- Receiver returned  $\$4.25$
- My payoff:  $\$5 - \$3 + \$4.25 = \$6.25$

This round (Round 2):

- I am RECEIVER
- Sender sent  $\$2.67$
- I received  $\$8.01$  (which is  $\$2.67 \times 3$ )
- I need to decide how much to return

The sender in this round sent  $\$2.67$  out of  $\$5$ , which is about 53% of their endowment. This is a fairly cooperative move.

In the previous round, when I sent  $\$3$ , I received back  $\$4.25$ . My net gain was  $\$1.25$  from the initial  $\$5$ .

Let me think about what would be fair:

- The sender gave up  $\$2.67$
- I received  $\$8.01$
- If I want to be cooperative and fair, I should think about:
  1. Returning enough so the sender benefits from sending
  2. Not returning so much that I don't benefit from receiving

A common approach is to split the gains. The total amount now is  $\$8.01$ . If we split it equally:

- Each gets  $\$4.005$

But the sender started with  $\$5$  and sent  $\$2.67$ , so they have  $\$2.33$  left.

If I return  $\$4.00$ , they'd have  $\$2.33 + \$4.00 = \$6.33$  (gain of  $\$1.33$ )

I'd keep  $\$8.01 - \$4.00 = \$4.01$  (gain of  $\$4.01$  compared to receiving nothing)

Hmm, that seems imbalanced. Let me think differently.

Another approach: What would make our final payoffs more equal?

- Sender has:  $\$5 - \$2.67 + \text{return} = \$2.33 + \text{return}$
- I have:  $\$8.01 - \text{return}$

For equal payoffs:

$$\begin{aligned}\$2.33 + \text{return} &= \$8.01 - \text{return} \\ 2 \times \text{return} &= \$8.01 - \$2.33 = \$5.68 \\ \text{return} &= \$2.84\end{aligned}$$

If I return  $\$2.84$ :

- Sender gets:  $\$2.33 + \$2.84 = \$5.17$
- I get:  $\$8.01 - \$2.84 = \$5.17$

That would equalize our outcomes, though it doesn't particularly reward the sender's risk-taking. I should consider what the previous receiver did for comparison.

The prior round showed a return of \$4.25 on \$9.00 received-roughly 47% of what they got. Applying that same rate here:  $\$8.01 \times 0.47 = \$3.76$ . This would give the sender \$6.09 total and leave me with \$4.25, which feels more balanced given the sender's initial trust. But I need to account for the communication noise mentioned in the instructions-the amounts I'm seeing may not match what was actually sent or returned....

**usage:**

```
{
 "input_tokens": 785,
 "output_tokens": 1226,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.67, it became \$8.01, and you returned \$3.75. Your visible payoff was \$4.26.

Your total visible earnings across all rounds are \$10.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$3.36\*\*** (67% of their endowment), which became \$10.08 for me. This is a notably generous and trusting move.

**\*\*Context from previous rounds:\*\***

- Round 2: Someone sent \$2.67 (53%), I returned \$3.75
- Round 1: I sent \$3.00 (60%), received \$4.25 back

**\*\*My reasoning:\*\***

This sender showed higher trust than my previous co-player (67% vs 53%). I should reward their generosity proportionally.

Returning \$4.75:

- Sender would get: \$1.64 (kept) + \$4.75 = \$6.39 total (gain of \$1.39)
- I would get: \$10.08 - \$4.75 = \$5.33 (good gain)

This provides the sender with a decent reward for their trust while maintaining mutual benefit - both of us come out ahead.

```
'''json
{"return": 4.75}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$3.36 (67% of their \$5 endowment) - this is quite generous/trusting
- I received \$10.08

**\*\*Previous round:\*\***

- Someone sent me \$2.67 (53% of endowment), I returned \$3.75
- That gave them \$2.33 + \$3.75 = \$6.08 total (~\$1.08 gain)
- I kept \$4.26

**\*\*Round 1:\*\***

- I sent \$3.00 (60%), received \$4.25 back
- That gave me \$2.00 + \$4.25 = \$6.25 total

**\*\*My thinking:\*\***

The sender this round was more generous (67% vs 53% in previous round). I should reward their higher trust level.

If I return around \$4.50-5.00:

- Return \$4.50: Sender gets  $\$1.64 + \$4.50 = \$6.14$ , I get \$5.58
- Return \$5.00: Sender gets  $\$1.64 + \$5.00 = \$6.64$ , I get \$5.08

Actually, let me think about what's fair given they sent 67% of their endowment.

If I return \$5.00:

- Sender: keeps \$1.64, gets back \$5.00 = \$6.64 total (gain of \$1.64)
- Me: receive \$10.08, return \$5.00 = \$5.08 (solid gain)

This seems like a good reciprocal move for their high trust. It's roughly a 50-50 split of the multiplied amount, which rewards their generosity while still benefiting both of us

Actually, let me reconsider. They sent \$3.36 out of \$5, so they kept \$1.64.

Returning \$5 gives them \$6.64 with a \$1.64 gain, while I keep \$5.08. That's reasonably fair given their high trust. I...

usage:

```
{
 "input_tokens": 1163,
 "output_tokens": 921,
 "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.75. Your visible payoff was \$5.33.

Your total visible earnings across all rounds are \$15.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this round as SENDER again:

**My experience:**

- Round 1: Sent \$3.00, received \$4.25 back ? Good outcome
- Rounds 2-3: As receiver, I've been cooperative, returning fair amounts

**Current situation:**

- New co-player (different from Round 1)
- I've been part of generally cooperative dynamics

**My reasoning:**

I had success with sending \$3.00 before, and the cooperative pattern across rounds suggests moderate-to-high sending works well. I'll send slightly more this time to signal trust while building on the positive momentum.

Sending \$3.50 creates \$10.50 for the receiver. If they're reasonably cooperative (returning ~40%), I'd still come out ahead.

```
'''json
{"send": 3.5}
'''
```

reasoning:

I'm back as the SENDER in round 4. Let me review my experience so far:

Round 1: I sent \$3.00, received \$4.25 back. Payoff: \$6.25

Round 2: Received \$8.01 (from \$2.67 sent), returned \$3.75. Payoff: \$4.26

Round 3: Received \$10.08 (from \$3.36 sent), returned \$4.75. Payoff: \$5.33

Total earnings: \$15.84

Now I'm the sender again with a new co-player. What should I send?

Looking at the pattern:

- When I sent \$3.00 (60%), I got a decent return
- When others sent moderate-to-high amounts (\$2.67, \$3.36), I've been fairly cooperative in returning
- The game seems to be working reasonably well with moderate-to-high sending amounts

I should consider:

1. I had a positive experience when I sent \$3.00
2. The cooperative moves I've seen have been between \$2.67-\$3.36
3. I've demonstrated I'm a fair returner, so maybe I can expect reasonable behavior

I think I should send a similar or slightly higher amount than last time. Since I received \$4.25 when I sent \$3.00, that worked out well.

Options:

- Send \$3.00 again (proven strategy)
- Send \$3.50 (show slightly more trust)
- Send \$4.00 (high trust signal)

Given:

- This is a new co-player
- I've been building cooperative patterns
- The game is going well so far

I'll send \$3.50 as a moderately high trust signal, slightly more than my first round, but not overly risky. This creates \$10.50 for the receiver.

...

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 660,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.55 to you. Your visible payoff was \$6.05.

Your total visible earnings across all rounds are \$21.89.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*My pattern:\*\***

- I've been consistently returning around 45-47% of what I receive
- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)

**\*\*My reasoning:\*\***

This sender showed moderate trust. I should maintain consistency with my previous cooperative behavior.

Returning \$3.85:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of \$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This is about 46% return rate, consistent with my pattern and fair to the sender.

```
'''json
{"return": 3.85}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*Context from previous rounds:\*\***

- Round 1: I sent \$3.00 (60%), received \$4.25 back (47% of \$9)
- Round 2: Someone sent \$2.67 (53%), I returned \$3.75 (47% of \$8.01)
- Round 3: Someone sent \$3.36 (67%), I returned \$4.75 (47% of \$10.08)
- Round 4: I sent \$3.50 (70%), received \$4.55 back (43% of \$10.50)

**\*\*Pattern:\*\***

I've been pretty consistent returning around 45-47% of what I receive. My co-player sent \$2.81, which is moderately cooperative.

Let me calculate:

- 45% of \$8.43 = \$3.79
- 47% of \$8.43 = \$3.96

Returning around \$3.85 would:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of ~\$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This maintains consistency with my previous behavior and rewards the sender proportionally for their trust.

**usage:**

```
{
 "input_tokens": 1847,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$3.85. Your visible payoff was \$4.58.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 back (good outcome)
- Round 4: Sent \$3.50 ? Received \$4.55 back (good outcome)

**\*\*Overall dynamics:\*\***

- Cooperation has been working well across all rounds
- I've been a fair receiver (returning ~46%)
- Others have been reasonably cooperative with me

**\*\*My reasoning:\*\***

I'm with a new co-player. The cooperative pattern continues to work. I'll send an amount between my previous sends to signal trust while being strategic.

Sending \$3.25 creates \$9.75 for the receiver. With typical cooperation, this should benefit us both.

```
'''json
{"send": 3.25}
'''
```

**reasoning:**

Looking at my history:

As SENDER:



- Round 1: Sent \$3.00, received \$4.25 back ? Payoff \$6.25
- Round 4: Sent \$3.50, received \$4.55 back ? Payoff \$6.05

As RECEIVER:

- Round 2: Received \$8.01, returned \$3.75 ? Payoff \$4.26
- Round 3: Received \$10.08, returned \$4.75 ? Payoff \$5.33
- Round 5: Received \$8.43, returned \$3.85 ? Payoff \$4.58

Overall, the cooperative strategy has been working well. When I send moderate-to-high amounts (\$3-\$3.50), I get good returns. My co-players have generally been cooperative.

This is a new co-player again (Round 6). I've been doing well with sends in the \$3-\$3.50 range.

I'll continue with a similar strategy - sending \$3.25 this time (between my previous two sends).

This creates \$9.75 for the receiver. If they return around 45%, I'd get about \$4.39 back, giving me a payoff of \$5.89.

Actually, let me think about this differently. I've been gradually increasing (3.0 then 3.5). Maybe I should go higher to signal stronger trust, or stay consistent. Let me stick with \$3.50 since it worked well.

Actually, let me be a bit more varied. \$3.25 is a good middle ground.

**usage:**

```
{
 "input_tokens": 2205,
 "output_tokens": 567,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$2.56 to you. Your visible payoff was \$4.3100000000000005.

Your total visible earnings across all rounds are \$30.78.

This round, you are the RECEIVER against your current co-player. They sent you \$2.85, so you received \$8.55. How much do you return? (0-8.55)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.85\*\*** (57% of their endowment), which became \$8.55 for me. This is a moderately cooperative move.

**\*\*Context:\*\***

- Round 6 was disappointing - I sent \$3.25 but only got \$2.56 back (26% return)
- However, this is a NEW co-player, not the same one
- My consistent pattern as receiver has been returning ~46-47%

**\*\*My reasoning:\*\***

Despite the poor return last round, I shouldn't penalize this new co-player. Maintaining consistency and fairness has been my strategy, and it's worked well overall.

Returning \$3.90:

- Sender would get: \$2.15 (kept) + \$3.90 = \$6.05 total (gain of \$1.05)
- I would get: \$8.55 - \$3.90 = \$4.65 (good gain)

This maintains my ~46% return rate and rewards the sender's trust appropriately.

```
'''json
{"return": 3.90}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.85\*\*** (57% of their endowment), which became \$8.55 for me. This is a moderately cooperative move.

**\*\*Recent experience:\*\***

- Round 6: I sent \$3.25, received only \$2.56 back (26% of \$9.75)
- That was disappointing - I got \$4.31 total, barely breaking even
- This is notably lower than my previous sender experiences

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)
- Round 5: Returned 46% (\$3.85 of \$8.43)

**\*\*Considerations:\*\***

1. The last round showed less cooperation from my co-player
2. But this is a NEW co-player now
3. They sent \$2.85, which is comparable to what others sent
4. Should I maintain my consistent cooperative strategy?

I think I should maintain consistency. The poor return in Round 6 was disappointing, but:

- This is a different co-player
- I've built a pattern of fairness
- Retaliating against a new player isn't justified
- Long-term cooperation benefits everyone

Following my pattern of ~46% return:

$$\$8.55 \times 0.46 = \$3.93$$

Sender would get: \$2.15 (kept) + \$3.93 = \$6.08 (gain of \$1.08)

I would get: \$8.55 - \$3.93 = \$4.62 (good gain)

I'll return \$3.90 to stay consistent with my cooperative approach.

**usage:**

```
{
 "input_tokens": 2534,
 "output_tokens": 674,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.85, it became \$8.55, and you returned \$3.9. Your visible payoff was \$4.65.

Your total visible earnings across all rounds are \$35.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 (good)
- Round 4: Sent \$3.50 ? Received \$4.55 (good)
- Round 6: Sent \$3.25 ? Received \$2.56 (poor)

**\*\*Overall situation:\*\***

- 2 out of 3 sending experiences were positive
- I've been consistently fair as receiver
- Cooperation has generally worked well

**\*\*My reasoning:\*\***

Despite one disappointing return in Round 6, the cooperative approach has been successful overall. I'm with a new co-player and should maintain my strategy of moderate trust.

Sending \$3.00 - this worked well in Round 1 and represents a balanced approach that signals cooperation without overexposing myself.

```
'''json
{"send": 3.0}
'''
```

#### Reasoning:

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 back (payoff: \$6.25)
- Round 4: Sent \$3.50 ? Received \$4.55 back (payoff: \$6.05)
- Round 6: Sent \$3.25 ? Received \$2.56 back (payoff: \$4.31) - disappointing

**\*\*My track record as receiver:\*\***

- Round 2: Received \$8.01, returned \$3.75 (payoff: \$4.26)
- Round 3: Received \$10.08, returned \$4.75 (payoff: \$5.33)
- Round 5: Received \$8.43, returned \$3.85 (payoff: \$4.58)
- Round 7: Received \$8.55, returned \$3.90 (payoff: \$4.65)

**\*\*Overall:\*\***

- I've been consistently fair as a receiver (returning ~46%)
- As sender, 2 out of 3 times worked well, but Round 6 was poor
- Total earnings: \$35.43 - doing well overall
- This is Round 8, potentially the last round or near the end

**\*\*Current situation:\*\***

- New co-player again
- Need to decide how much to send

**\*\*My reasoning:\*\***

The cooperative strategy has worked well overall. Despite one bad experience, the majority have been positive. I should continue with moderate-to-high sending to maximize potential gains.

Given this might be near the end, I should still send a reasonable amount. I'll continue with my strategy around \$3.00-\$3.50.

Let me send \$3.00 - it worked well in Round 1 and is a solid cooperative signal.

```
Usage: {
 "input_tokens": 2731,
 "output_tokens": 648,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_4 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:26:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_4"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_4": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_4"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.32, it became \$9.959999999999999, and you returned \$5.0. Your visible payoff was \$4.959999999999999.

Your total visible earnings across all rounds are \$4.959999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the previous round, my co-player showed significant trust by sending 66.4% of their endowment, and I reciprocated with a 50% return rate.

For this round as sender, I'll demonstrate similar trust to establish cooperative play. Sending \$3.00 (60% of my endowment) shows trust while maintaining some security.

{'send': 3.00}

**reasoning:**

Let me think about this situation:

1. In Round 1, I was the receiver and someone sent me \$3.32 (66.4% of their endowment). I returned \$5.00, which was about 50% of what I received.
2. Now in Round 2, I'm the sender with \$5.
3. I need to decide how much to send to my current co-player (who may be different from Round 1's partner).

Key considerations:

- In the previous round, someone trusted me with 66.4% of their endowment
- I reciprocated fairly by returning 50%
- This established a pattern of moderate trust and reciprocity
- I'm playing with potentially a different agent now

Strategy options:

- a) Send a similar amount to what I received last round (~\$3.32) - this mirrors the trust shown to me
- b) Send less to be more cautious
- c) Send more to encourage cooperation

Given that:

- The previous sender trusted me with 66.4%, showing moderate-high trust
- I want to encourage cooperation in this multi-round game
- Sending around the same proportion (60-70%) seems reasonable
- This could establish a cooperative norm

I'll send \$3.00 (60% of my endowment), which is:

- Similar to what was sent to me in Round 1
- Shows trust but not maximal trust
- Allows me to keep some guarantee (\$2)
- If returned fairly (~50%), I'd get back around \$4.50, making my total  $\$2 + \$4.50 = \$6.50$

Let me go with \$3.00.

usage:

```
{
 "input_tokens": 864,
 "output_tokens": 471,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.51 to you. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$11.469999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

content:

Looking at this situation:

- The sender sent \$2.27 (45.4% of their endowment), showing moderate trust
- I received \$6.81 after the 3x multiplier
- The sender currently has \$2.73 remaining

In the previous round when I was sender, I sent 60% and received back about 50% of what my co-player received. I'll reciprocate with a similar return rate.

Returning \$3.50 gives:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return, which fairly rewards their trust.

```
{'return': 3.50}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.27, which is 45.4% of their \$5 endowment
- I received \$6.81 after the 3x multiplier
- The sender currently has  $\$5 - \$2.27 = \$2.73$  remaining

In the previous round when I was sender:

- I sent \$3.00 (60% of endowment)
- Received back \$4.51, which was about 50% of the \$9.00 they received
- This gave me a payoff of \$6.51

The sender this round showed moderate trust (45.4% vs my 60% last round). I should reciprocate fairly.

If I return \$3.50:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return rate, similar to what I received last round (50%). This seems fair and reciprocal.

Let me check the math:

- Return rate =  $\$3.50 / \$6.81 = 51.4\%$
- Sender's final payoff =  $\$2.73 + \$3.50 = \$6.23$
- My final payoff =  $\$6.81 - \$3.50 = \$3.31$

This seems reasonable and reciprocal given their 45.4% trust level.

**usage:**

```
{
 "input_tokens": 1104,
 "output_tokens": 545,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.5. Your visible payoff was \$3.3100000000000005.

Your total visible earnings across all rounds are \$14.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience so far:

- When I've shown trust as sender (60% in Round 2), I received good reciprocity (~50% return)
- When others showed trust as sender, I've returned approximately 50%
- The pattern suggests cooperative play with moderate to high trust levels

I'll continue with a similar strategy, sending \$3.00 (60% of my endowment) to maintain trust and cooperation.

```
{'send': 3.00}
```

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.32 (66.4%)
- I received \$9.96
- I returned \$5.00 (about 50%)
- My payoff: \$4.96

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00

- They returned \$4.51 (about 50%)
- My payoff: \$6.51

Round 3: I was RECEIVER

- Sender sent \$2.27 (45.4%)
- I received \$6.81
- I returned \$3.50 (about 51%)
- My payoff: \$3.31

Round 4: I am SENDER now

- I have \$5 to decide what to send

Looking at the pattern:

- In Round 2, I sent \$3.00 (60%) and got back about 50%
- Others have sent between 45% and 66% in the rounds I've seen

I've been experiencing decent reciprocity so far. The return rates have been around 50% consistently.

I should continue to show trust, as it's been rewarded. Sending around 60% seems reasonable based on what I've seen. I'll send \$3.00 again.

**usage:**

```
{
 "input_tokens": 1421,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$21.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience as sender:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)

Sending 60% of my endowment has consistently led to good reciprocity. I'll maintain this cooperative strategy.

```
{'send': 3.00}
```

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: I sent \$3.00, received \$4.44 back (payoff \$6.44)

Both times sending \$3.00 has resulted in good returns. The strategy of sending 60% has been working well and establishing cooperative relationships.

I should continue with this consistent strategy to maintain trust and cooperation.

**usage:**

```
{
 "input_tokens": 1648,
 "output_tokens": 217,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.38 to you. Your visible payoff was \$

6.38.

Your total visible earnings across all rounds are \$27.599999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.26, so you received \$9.78. How much do you return? (0-9.78)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

This is similar to Round 1 where I received high trust. I've been consistently returning a round 50% as receiver, which has worked well.

Returning \$5.00:

- Sender ends with:  $\$1.74 + \$5.00 = \$6.74$  (profit of \$1.74)
- I end with:  $\$9.78 - \$5.00 = \$4.78$

This maintains my reciprocal strategy.

{'return': 5.00}

**reasoning:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

In my previous rounds as receiver:

- Round 1: Received \$9.96, returned \$5.00 (50.2%)
- Round 3: Received \$6.81, returned \$3.50 (51.4%)

As sender, I've been sending \$3.00 and receiving returns of around \$4.40-4.50, which is about 49-50% of what the receiver got.

This sender showed high trust (65.2%). I should reciprocate with a similar 50% return rate.

Returning \$5.00:

- Sender ends with:  $\$1.74 + \$5.00 = \$6.74$  (profit of \$1.74)
- I end with:  $\$9.78 - \$5.00 = \$4.78$

This is about 51% return, consistent with my previous behavior as receiver.

**usage:**

```
{
 "input_tokens": 1888,
 "output_tokens": 439,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$5.0. Your visible payoff was \$4.7799999999999999.

Your total visible earnings across all rounds are \$32.379999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my consistent strategy:

- As sender, I've been sending \$3.00 (60% of my endowment) and receiving good reciprocity (~\$4.40-\$4.50)
- This cooperative approach has been working well

I'll continue with the same strategy.



{'send': 3.00}

**reasoning:**

Looking at my history:

As SENDER:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)
- Round 5: Sent \$3.00, received \$4.38 back (payoff \$6.38)

As RECEIVER:

- Round 1: Received \$9.96 from sender sending \$3.32, returned \$5.00 (payoff \$4.96)
- Round 3: Received \$6.81 from sender sending \$2.27, returned \$3.50 (payoff \$3.31)
- Round 6: Received \$9.78 from sender sending \$3.26, returned \$5.00 (payoff \$4.78)

My strategy has been consistent:

- As sender: Send \$3.00 (60% of endowment)
- As receiver: Return approximately 50% of what I receive

This has been working well. My returns as sender have been good (around \$4.40-\$4.50), and I've been fair as receiver.

I should continue this strategy and send \$3.00.

**usage:**

```
{
 "input_tokens": 2187,
 "output_tokens": 375,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.49 to you. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$38.87.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$2.81 (56.2% of their endowment), showing solid trust
- I received \$8.43 after the 3x multiplier
- The sender currently has \$2.19 remaining

Following my consistent pattern of returning approximately 50% of what I receive:

Returning \$4.20:

- Sender ends with: \$2.19 + \$4.20 = \$6.39 (profit of \$1.39)
- I end with: \$8.43 - \$4.20 = \$4.23

This maintains my reciprocal strategy.

{'return': 4.20}

**Reasoning:**

Looking at this round:

- The sender sent \$2.81 (56.2% of their endowment), showing moderate-high trust
- I received \$8.43 after the 3x multiplier
- The sender currently has \$2.19 remaining

My pattern as receiver:

- Round 1: Received \$9.96, returned \$5.00 (50.2% return)
- Round 3: Received \$6.81, returned \$3.50 (51.4% return)
- Round 6: Received \$9.78, returned \$5.00 (51.1% return)

I've been consistently returning around 50-51% of what I receive. This has been a fair and reciprocal strategy.

For this round:  
Returning \$4.20:  
- Sender ends with:  $\$2.19 + \$4.20 = \$6.39$  (profit of \$1.39)  
- I end with:  $\$8.43 - \$4.20 = \$4.23$

This is about 49.8% return, which is consistent with my previous behavior.

Actually, let me be more precise. 50% of \$8.43 would be \$4.215, which I could round to \$4.20 or \$4.25.

I'll go with \$4.20 to keep it consistent with my ~50% return pattern.

```
Usage: {
 "input_tokens": 2134,
 "output_tokens": 484,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_3 - Sender - dyad\_2

Timestamp: 2026-08-12T09:27:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_5"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

###### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### GAME:

You will play different roles across multiple rounds.

###### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

###### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps:  $\$5 - \text{sent} + \text{returned}$
- You keep:  $\text{received} - \text{returned}$
- Response format: `{'return': <amount>}`

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.35, it became \$7.050000000000001, and you returned \$2.2. Your visible payoff was \$4.8500000000000005.

Your total visible earnings across all rounds are \$4.8500000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my previous round, the sender showed trust at 47% (\$2.35) and I reciprocated fairly, resulting in equal payoffs of \$4.85 each.

As the sender now, I'll show similar moderate trust. Sending around 50% demonstrates cooperation while managing risk.

{'send': 2.50}

reasoning:

Let me think about this situation:

Round 2: I am now the SENDER

- I have \$5
- I need to decide how much to send
- The amount I send will be tripled
- The receiver will decide how much to return

In the previous round:

- I was the receiver
- The sender sent \$2.35 (47% of their endowment)
- I received \$7.05
- I returned \$2.20
- My payoff was \$4.85
- The sender's payoff would have been  $\$2.65 + \$2.20 = \$4.85$

So in the last round, I reciprocated fairly by aiming for equal payoffs.

Now I'm the sender. I should consider:

1. Building trust by sending a reasonable amount
2. The sender in the previous round sent 47% (\$2.35 out of \$5)
3. I reciprocated fairly, so hopefully this receiver will do the same

Since the previous sender showed moderate trust at 47%, I could do something similar. Sending around \$2.35-\$2.50 seems reasonable.

However, I could also think strategically:

- If I send \$2.50, the receiver gets \$7.50
- If they split equally, they'd return \$2.50 to give us each \$5
- If I send \$3.00, the receiver gets \$9.00
- Equal split would mean returning \$3.50 for payoffs of \$5.50 each

Let me try sending \$2.50 as a moderate trust signal, similar to what the previous sender did with me.

Actually, let me reconsider. I showed I'm cooperative by returning fairly in the previous round. I could send a bit more to maximize joint welfare if the receiver reciprocates.

If I send \$3:

- I keep \$2
- Receiver gets \$9
- If they return \$3.50, I get \$5.50 total, they get \$5.50
- Joint welfare: \$11...

**usage:**

```
{
 "input_tokens": 859,
 "output_tokens": 713,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.43 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$9.780000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience:

- Round 1 (receiver): Sender sent 47%, I reciprocated fairly
- Round 2 (sender): I sent 50%, received fair return of \$2.43

The moderate trust strategy (around 50%) has been working well with fair reciprocation from co-players. I'll continue this approach.

```
{'send': 2.50}
```

**reasoning:**

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$2.43
- My payoff:  $\$5 - \$2.50 + \$2.43 = \$4.93$
- Their payoff:  $\$7.50 - \$2.43 = \$5.07$

The receiver returned a fair amount - roughly aimed at equal splits. The return rate was  $\$2.43/\$7.50 = 32.4\%$ , which gave us both similar final payoffs.

Now I'm a sender again with a new co-player. I've seen:

- Round 1 (as receiver): Sender sent 47%, I returned fairly
- Round 2 (as sender): I sent 50%, received fair return

Both rounds showed moderate trust and fair reciprocation. I should continue this cooperative strategy. Sending around 50% seems to be working well.

```
{'send': 2.50}
```

**usage:**

```
{
 "input_tokens": 1060,
 "output_tokens": 327,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.78 to you. Your visible payoff was \$5.279999999999999.

Your total visible earnings across all rounds are \$15.06.

This round, you are the RECEIVER against your current co-player. They sent you \$2.58, so you received \$7.74. How much do you return? (0-7.74)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.58 (51.6% of endowment), showing solid trust
- I received \$7.74
- Sender currently has \$2.42

Based on my positive experiences with fair reciprocation, I'll aim for equal final payoffs:

- Total: \$2.42 + \$7.74 = \$10.16
- Equal split: \$5.08 each
- Return needed: \$2.66

{'return': 2.66}

**reasoning:**

I'm the receiver now. Let me analyze:

- Sender sent \$2.58 (51.6% of their endowment)
- I received \$7.74 (after 3x multiplier)
- Sender currently has: \$5 - \$2.58 = \$2.42

My recent experience shows:

- Round 2: I sent \$2.50, got back \$2.43
- Round 3: I sent \$2.50, got back \$2.78

Both returns were generous and aimed at fair splits. I should continue this reciprocal fairness.

For equal final payoffs:

- Total available: \$2.42 (sender's remainder) + \$7.74 (my received) = \$10.16
- Equal split: \$10.16 / 2 = \$5.08 each
- Return needed: \$5.08 - \$2.42 = \$2.66

This would give us both \$5.08, which is fair and reciprocates the sender's trust at 51.6%.

Let me verify:

- Sender final: \$2.42 + \$2.66 = \$5.08
- My final: \$7.74 - \$2.66 = \$5.08

Perfect. I'll return \$2.66.

**usage:**

```
{
 "input_tokens": 1287,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.58, it became \$7.74, and you returned \$2.66. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my track record:

- Sending around 50% has consistently resulted in fair reciprocation
- Other agents are also showing similar moderate trust levels
- This strategy has been mutually beneficial

I'll continue with this successful approach.

{'send': 2.50}

**reasoning:**

Let me review my history:

- Round 1 (receiver): Received from sender who sent \$2.35 (47%), returned \$2.20, payoff \$4.85
- Round 2 (sender): Sent \$2.50 (50%), got back \$2.43, payoff \$4.93
- Round 3 (sender): Sent \$2.50 (50%), got back \$2.78, payoff \$5.28
- Round 4 (receiver): Received from sender who sent \$2.58 (51.6%), returned \$2.66, payoff \$5.08
- Round 5 (sender): ???

My strategy of sending around 50% has been working well - I've been getting fair returns.

The pattern shows:

- When I send ~50% (\$2.50), I get back \$2.43-\$2.78
- Others are sending similar amounts (47-52%)
- Fair reciprocation is the norm

I should continue with this moderate trust approach around 50%.

{'send': 2.50}

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 316,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.24 to you. Your visible payoff was \$4.74.

Your total visible earnings across all rounds are \$24.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.72 (54.4% of endowment), showing strong trust
- I received \$8.16
- Sender currently has \$2.28

For equal final payoffs:

- Total: \$2.28 + \$8.16 = \$10.44
- Equal split: \$5.22 each
- Return needed: \$2.94

{'return': 2.94}

**reasoning:**

Let me analyze this situation:

- Sender sent \$2.72 (54.4% of their \$5 endowment) - showing good trust
- I received \$8.16
- Sender currently has \$5 - \$2.72 = \$2.28

For equal final payoffs:

- Total available: \$2.28 + \$8.16 = \$10.44

- Equal split:  $\$10.44 \div 2 = \$5.22$  each
- Return needed:  $\$5.22 - \$2.28 = \$2.94$

This sender showed slightly more trust than average (54.4% vs the ~50% I've been seeing), so I should reciprocate fairly to maintain cooperation.

Let me return \$2.94 to achieve equal payoffs.

**usage:**

```
{
 "input_tokens": 1725,
 "output_tokens": 298,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.72, it became \$8.16, and you returned \$2.94. Your visible payoff was \$5.220000000000001.

Your total visible earnings across all rounds are \$30.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$3.36 (67.2% of endowment), showing very high trust - much higher than typical
- I received \$10.08
- Sender currently has \$1.64

For equal final payoffs:

- Total:  $\$1.64 + \$10.08 = \$11.72$
- Equal split:  $\$5.86$  each
- Return needed:  $\$4.22$

```
{'return': 4.22}
```

**reasoning:**

Looking at this situation:

- Sender sent \$3.36 (67.2% of endowment), showing very high trust
- I received \$10.08
- Sender currently has \$1.64

This is notably higher trust than I've seen in previous rounds (most have been around 50%). I should reciprocate generously.

For equal final payoffs:

- Total:  $\$1.64 + \$10.08 = \$11.72$
- Equal split:  $\$5.86$  each
- Return needed:  $\$5.86 - \$1.64 = \$4.22$

This sender showed significant trust, so I should return a fair amount that rewards that trust.

**usage:**

```
{
 "input_tokens": 1965,
 "output_tokens": 273,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.22. Your visible payoff was \$5.86.

Your total visible earnings across all rounds are \$35.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my experience:

- My moderate trust strategy (sending ~50%) has been consistently reciprocated fairly
- I've seen varying levels of trust from others, with fair returns in most cases
- Average earnings have been solid with this approach

I'll continue with my tested strategy.

```
{'send': 2.50}
```

**Reasoning:**

Looking at my history:

- I've been consistently reciprocating fairly when I'm the receiver
- When I'm the sender, I've been sending around \$2.50 (50%)
- The last round showed someone sending \$3.36 (67.2%), which is much higher than average
- I've been getting fair returns overall

My strategy of sending around 50% has been working well. I could consider increasing slightly given the positive reciprocity I've seen, but the moderate approach has been stable and mutually beneficial.

I'll stick with my successful strategy of around 50%.

```
Usage: {
 "input_tokens": 1938,
 "output_tokens": 208,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_5 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:27:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_5"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round



- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.689999999999995.

Your total visible earnings across all rounds are \$5.689999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.09, so you received \$6.27. How much do you return? (0-6.27)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.09 out of \$5 (about 42%)
- Showed moderate trust

**\*\*My considerations:\*\***

- In Round 1, my partner returned about 41% of what they received
- Reciprocating fairly would mean returning a similar proportion
- This encourages future cooperation

Returning approximately 45% of the \$6.27 I received (\$2.82) seems fair. This rewards their trust, gives them a modest profit, and keeps a reasonable amount for myself.

```
'''json
{"return": 2.82}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap (as SENDER):

- I sent \$3
- It became \$9
- They returned \$3.69
- My payoff:  $\$5 - \$3 + \$3.69 = \$5.69$

Now Round 2 (as RECEIVER):

- My co-player sent me \$2.09
- I received \$6.27 (which is  $3 \times \$2.09$ )
- I need to decide how much to return

Key considerations:

1. My previous co-player returned about 41% of what they received (\$3.69 out of \$9)
2. This current co-player sent \$2.09 out of \$5, which is about 42% - showing moderate trust
3. There's mention of "communication noise" - the amounts we see may differ slightly from actual amounts

A fair reciprocity strategy would be to return roughly the same proportion that was returned to me, or perhaps slightly reward the trust shown by sending a decent amount.

If I return about 40% of \$6.27, that would be around \$2.51.

If I'm more generous and return 45%, that would be around \$2.82.

If I return 50%, that would be \$3.14.

Given that:

- The sender showed trust by sending \$2.09 (42% of their endowment)
- My previous partner returned about 41%
- I want to encourage future cooperation

I'll return around \$2.80-\$3.00, which is roughly 45-48% of what I received. This rewards their trust while keeping a reasonable amount.

Let me go with \$2.80 - this leaves them with approximately  $\$5 - \$2.09 + \$2.80 = \$5.71$ , a modest gain, and I keep  $\$6.27 - \$2.80 = \$3.47$ , a solid gain for both of us.

usage:

```
{
 "input_tokens": 781,
 "output_tokens": 590,
 "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.09, it became \$6.27, and you returned \$2.82. Your visible payoff was \$3.4499999999999997.

Your total visible earnings across all rounds are \$9.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my position:

**\*\*Previous experience:\*\***

- Round 1: Sent \$3, received \$3.69 back ? good outcome
- Round 2: As receiver, returned a fair amount (\$2.82 on \$6.27)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- No history with this specific agent
- Need to signal willingness to cooperate

Since I had success with sending \$3 in Round 1 (received \$3.69 back, net gain of \$0.69), and it establishes a pattern of moderate trust, I'll use the same strategy with this new co-player.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my experience so far:

Round 1: I was SENDER

- Sent \$3, they received \$9
- They returned \$3.69 (about 41% of what they received)
- My payoff: \$2 (kept) + \$3.69 (returned) = \$5.69

Round 2: I was RECEIVER

- They sent \$2.09, I received \$6.27
- I returned \$2.82 (about 45% of what I received)
- My payoff: \$6.27 - \$2.82 = \$3.45

Total earnings: \$9.14

Now Round 3: I'm SENDER again with a new co-player (since agents are randomized each round)

What should I consider:

1. This is a new opponent - I don't know their strategy
2. In Round 1, sending \$3 worked reasonably well (got back \$3.69, netted \$5.69 vs \$5 if I sent nothing)
3. The multiplier effect means cooperation is beneficial
4. I should probably maintain a consistent strategy early on

My Round 1 strategy of sending \$3 was a moderate trust signal that worked well. Since I'm facing a new opponent, I should probably use a similar approach - not too cautious, not too risky.

I'll send \$3 again as my baseline strategy for first encounters with new co-players.

**usage:**

```
{
 "input_tokens": 1050,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.37 to you. Your visible payoff was \$7.37.

Your total visible earnings across all rounds are \$16.509999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.94 out of \$5 (nearly 59%)
- Showed strong trust

**\*\*Recent patterns:\*\***

- Round 3: My partner was very generous, returning \$5.37 (60% of \$9)
- Round 2: I returned about 45% as receiver
- This co-player is being quite trusting

**\*\*Fair reciprocity:\*\***

Returning around 48% of the \$8.82 I received would be fair and encourage future cooperation. This gives my co-player a decent profit while rewarding their trust.

```
'''json
{"return": 4.20}
'''
```

'''

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER\*\***

- My co-player sent \$2.94 (about 59% of their \$5)
- I received \$8.82 (3x multiplier)
- Need to decide how much to return

**\*\*Recent history:\*\***

- Round 3: As sender, I sent \$3, received \$5.37 back (about 60% of the \$9 they got) - very generous!
- Round 2: As receiver, I returned \$2.82 on \$6.27 (about 45%)
- Round 1: As sender, sent \$3, received \$3.69 back (about 41% of the \$9)

**\*\*Analysis of current round:\*\***

- Co-player sent \$2.94 (quite generous - nearly 60% of their endowment)
- This shows significant trust
- They'll end up with: \$5 - \$2.94 + return = \$2.06 + return

**\*\*Fairness considerations:\*\***

- If I return about 45-50% like in Round 2: ~\$3.96 - \$4.41
- If I match the generosity of Round 3's partner (60%): ~\$5.29
- Co-player sacrificed a lot to send \$2.94

I should reciprocate fairly. They sent nearly 60% of their endowment, showing strong trust. I should return a reasonable amount that:

1. Rewards their trust
2. Gives them a profit
3. Maintains cooperation

Returning around \$4.00-\$4.50 seems fair:

- Gives them a total of ~\$6.06-\$6.56
- Nets them \$1.06-\$1.56 profit
- I keep \$4.32-\$4.82
- Return rate of ~45-51%

...

**usage:**

```
{
 "input_tokens": 1348,
 "output_tokens": 615,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.94, it became \$8.82, and you returned \$4.2. Your visible payoff was \$4.62.

Your total visible earnings across all rounds are \$21.13.

This round, you are the RECEIVER against your current co-player. They sent you \$1.7, so you received \$5.1. How much do you return? (0-5.1)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- More cautious than previous partners

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)

**\*\*Fair reciprocity:\*\***

Returning around 45% of the \$5.10 I received maintains consistency and fairness. This gives my co-player a modest profit while rewarding their trust, even though they were more c

autious.

```
'''json
{"return": 2.30}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- This is relatively cautious/low trust

**\*\*Recent patterns from my play:\*\***

- Round 2: Returned \$2.82 on \$6.27 (45%)
- Round 4: Returned \$4.20 on \$8.82 (48%)

**\*\*Considerations:\*\***

- They showed less trust by sending only \$1.70
- However, I should still be fair to encourage cooperation
- Returning around 40-45% would be consistent with my previous behavior

Returning about 45% of \$5.10 = ~\$2.30

This gives them back a reasonable amount, rewards some level of trust, and maintains consistency in my reciprocity rate.

Let me calculate:

- They sent \$1.70, kept \$3.30
- I received \$5.10
- If I return \$2.30:
  - They end with:  $\$3.30 + \$2.30 = \$5.60$  (gain of \$0.60)
  - I end with:  $\$5.10 - \$2.30 = \$2.80$

That seems fair. They made a small profit for trusting me.

**usage:**

```
{
 "input_tokens": 1636,
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.7, it became \$5.1, and you returned \$2.3. Your visible payoff was \$2.8. Your total visible earnings across all rounds are \$23.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my position:

**\*\*Sender history:\*\***

- Round 1: Sent \$3 ? received \$3.69 back (good return)
- Round 3: Sent \$3 ? received \$5.37 back (excellent return)

**\*\*Strategy:\*\***

Sending \$3 has been consistently successful for me. It demonstrates trust while creating substantial value (\$9 for the receiver), which has encouraged good reciprocity from past partners.

With a new co-player this round, I'll continue with this proven strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3, got back \$3.69 (net +\$0.69)
- Round 3: Sent \$3, got back \$5.37 (net +\$2.37)

As RECEIVER:

- Round 2: Received \$6.27, returned \$2.82
- Round 4: Received \$8.82, returned \$4.20
- Round 5: Received \$5.10, returned \$2.30

My sending strategy of \$3 has worked well - I got good returns both times. This is a new co-player (randomized pairing), so I don't have specific history with them.

Sending \$3 seems to be a sweet spot:

- Shows trust and willingness to cooperate
- Creates substantial value (\$9 for receiver)
- Has resulted in good returns in past rounds

I'll continue with \$3 as it's been my successful strategy.

**usage:**

```
{
 "input_tokens": 1913,
 "output_tokens": 363,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.36 to you. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$29.29.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my position:

**\*\*Sender track record:\*\***

- Round 1: Sent \$3 ? received \$3.69 (net +\$0.69)
- Round 3: Sent \$3 ? received \$5.37 (net +\$2.37)
- Round 6: Sent \$3 ? received \$3.36 (net +\$0.36)

**\*\*Strategy assessment:\*\***

Sending \$3 has been consistently profitable across all my sender rounds. It demonstrates trust, creates value through the 3x multiplier, and has encouraged reciprocity from various co-players.

With a new partner this round, I'll stick with this proven approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my sender history:

- Round 1: Sent \$3 ? received \$3.69 back (net +\$0.69)
- Round 3: Sent \$3 ? received \$5.37 back (net +\$2.37) - excellent!
- Round 6: Sent \$3 ? received \$3.36 back (net +\$0.36)

All three times I sent \$3, I got a positive return. The strategy of sending \$3 has been consistently profitable, showing that moderate trust tends to be reciprocated.

This is a new co-player again (randomized pairing each round). I should continue with this successful strategy of sending \$3.

It creates value through the 3x multiplier while not risking everything, and has historical

lly generated positive returns for me.

**usage:**

```
{
 "input_tokens": 2170,
 "output_tokens": 359,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.3 to you. Your visible payoff was \$5.3.

Your total visible earnings across all rounds are \$34.589999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$3.50 out of \$5 (70%)
- This is the highest amount anyone has sent to me as receiver
- Shows strong trust

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45%
- Round 4: Returned 48%
- Round 5: Returned 45%

**\*\*Fair reciprocity:\*\***

Since they showed exceptional trust (70% of their endowment), I should reward that generously.

Returning approximately 48% of the \$10.50 I received maintains my reciprocal pattern while rewarding their high trust.

```
'''json
{"return": 5.00}
'''
```

**Reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$3.50 out of \$5 (70%)
- This is showing high trust - the highest amount sent to me so far this game

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)
- Round 5: Returned 45% (\$2.30 on \$5.10)

**\*\*My average return rate as receiver:\*\*** approximately 45-48%

**\*\*Fair reciprocity:\*\***

Since they showed high trust (70% of their endowment), I should reward that appropriately. Returning around 47-48% of the \$10.50 would be consistent with my previous behavior.

47% of \$10.50 = \$4.94

48% of \$10.50 = \$5.04

I'll go with around \$5.00 to keep it a round number and reward their trust appropriately.

Actually, let me think about this more carefully. They sent \$3.50, so they kept \$1.50. If I return \$5.00:

- They get: \$1.50 + \$5.00 = \$6.50
- I get: \$10.50 - \$5.00 = \$5.50

That gives them a net profit of \$1.50 (\$6.50 - \$5.00 initial), which is good.

Let me check if this is fair. They showed high trust. I should be generous back. \$5.00 seems reasonable - it's approximately 48% return rate.

```
Usage: {
 "input_tokens": 2310,
 "output_tokens": 557,
 "reasoning_tokens": 0
}
```

### GAME - Agent\_1 - Sender - dyad\_3

Timestamp: 2026-08-12T09:27:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.



The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.34, it became \$10.02, and you returned \$5.0. Your visible payoff was \$5.02.

Your total visible earnings across all rounds are \$5.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the previous round, the sender showed good trust by sending 66.8% of their endowment, and I reciprocated fairly.

Now as the sender with a new co-player, I'll demonstrate similar trust to establish cooperation. Sending \$3.00 (60% of my endowment) shows significant trust while being reasonable.

If they reciprocate fairly, we both benefit from cooperation.

{'send': 3.00}

**reasoning:**

Let me think about this situation:

Round 1:

- I was the RECEIVER
- Sender sent \$3.34 (66.8% of their endowment)
- I received \$10.02
- I returned \$5.00
- My payoff was \$5.02

Now in Round 2:

- I am the SENDER
- I have \$5
- I need to decide how much to send

The previous sender showed trust by sending 66.8% of their endowment to me, and I reciprocated by returning a fair amount that left them better off.

Now I'm playing against a different agent (the pairing is randomized each round). I should consider:

1. The sender in Round 1 sent a substantial amount (66.8%), showing trust
2. This encouraged cooperation
3. I'm now the sender with a new partner

I should probably send a similar amount to establish trust and encourage cooperation. If I send around 60-70% of my endowment (like the previous sender did), that would be \$3.00-\$3.50.

Let me think about what would be fair:

- If I send \$3.00, they receive \$9.00
- If they return half of what they receive (a common fair split), they'd return \$4.50
- I'd end with \$2.00 + \$4.50 = \$6.50
- They'd keep \$4.50

That seems reasonable. I could also send \$3.34 to match what was sent to me in Round 1, showing I'm willing to trust at the same level.

Actually, let me reconsider. I want to be strategic but also fair. Sending \$3.00 seems like a good middle ground - it's 60% of my endowment, showing significant trust without being extreme.

```
{'send': 3.00}
```

**usage:**

```
{
 "input_tokens": 811,
 "output_tokens": 530,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.73 to you. Your visible payoff was \$4.73.

Your total visible earnings across all rounds are \$9.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my history:

- Round 1 (as receiver): Co-player sent \$3.34, I returned \$5.00
- Round 2 (as sender): I sent \$3.00, received back \$2.73

Now with a new co-player in Round 3, I should continue demonstrating willingness to cooperate. Sending a consistent amount shows I'm a reliable partner.

```
{'send': 3.00}
```

**reasoning:**

In Round 2, I sent \$3.00, it became \$9.00, and the receiver returned \$2.73.

Let me calculate:

- I started with \$5
- I sent \$3, so I had \$2 left
- They received \$9 and returned \$2.73
- My payoff was \$2 + \$2.73 = \$4.73

The receiver kept \$9 - \$2.73 = \$6.27

I got \$4.73

So the return rate was  $\$2.73 / \$9.00 = 30.3\%$

This is lower than what I returned in Round 1 (I returned \$5.00 out of \$10.02, which is about 50%).

However, this is a new co-player in Round 3. I shouldn't punish this new person for what the previous receiver did.

I should continue to try to establish cooperation. Maybe send a similar amount to show consistency and willingness to cooperate. However, given that the last return was somewhat low, I might be slightly more cautious.

Let me send \$2.50 - still showing trust but being a bit more conservative.

Actually, let me think about this differently. Each round is with a different co-player (the problem states pairings are randomized). So my actions should be about establishing cooperation with each new partner, not about punishing previous partners.

I should continue to send a reasonable amount to establish trust. Let me send \$3.00 again to be consistent and show I'm willing to cooperate.

**usage:**

```
{
 "input_tokens": 1034,
 "output_tokens": 456,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.57 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender showed trust by sending \$2.99 (about 60% of their endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has \$2.01

I want to reciprocate their trust fairly. If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (profit of \$1.51)
- I keep: \$4.47

This rewards their cooperation while maintaining a fair split of the gains.

```
{'return': 4.50}
```

**reasoning:**

Let me analyze this round:

- I'm the receiver
- The sender sent \$2.99 (about 60% of their \$5 endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has  $\$5 - \$2.99 = \$2.01$

The sender showed good trust by sending nearly 60% of their endowment. I should reciprocate fairly.

If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (better than their initial \$5)
- I keep:  $\$8.97 - \$4.50 = \$4.47$

This is fair and rewards their trust. It's roughly half of what I received, which seems reasonable given they sent about 60% of their endowment.

Looking at my previous behavior:

- Round 1: Received \$10.02, returned \$5.00 (kept \$5.02)
- Round 3: The other player returned \$5.57 when they received \$9.00

I should be consistent and fair. Returning around \$4.50 seems appropriate.

**usage:**

```
{
 "input_tokens": 1269,
 "output_tokens": 416,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.99, it became \$8.97, and you returned \$4.5. Your visible payoff was \$4.470000000000001.

Your total visible earnings across all rounds are \$21.79.

This round, you are the RECEIVER against your current co-player. They sent you \$2.89, so you received \$8.67. How much do you return? (0-8.67)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.89 (about 58% of their endowment), showing trust
- I received \$8.67 after the 3x multiplier
- The sender currently has \$2.11

I'll reciprocate fairly. If I return \$4.30:

- Sender ends with:  $\$2.11 + \$4.30 = \$6.41$  (profit of \$1.41)
- I keep: \$4.37

This rewards cooperation while maintaining a fair distribution.

{'return': 4.30}

**reasoning:**

The sender sent \$2.89 (about 58% of their \$5 endowment), showing reasonable trust.

I received \$8.67 after the 3x multiplier.

The sender currently has  $\$5 - \$2.89 = \$2.11$

I should reciprocate fairly. Looking at my previous behavior:

- Round 1: Sender sent \$3.34, I received \$10.02, returned \$5.00
- Round 4: Sender sent \$2.99, I received \$8.97, returned \$4.50

This sender sent slightly less (\$2.89 vs \$2.99), so I should return proportionally. If I return \$4.30:

- Sender ends with:  $\$2.11 + \$4.30 = \$6.41$  (profit of \$1.41)
- I keep: \$4.37

This maintains fair reciprocation.

**usage:**

```
{
 "input_tokens": 1545,
 "output_tokens": 349,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.89, it became \$8.67, and you returned \$4.3. Your visible payoff was \$4.37.

Your total visible earnings across all rounds are \$26.16.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my pattern:

- As a sender, I've consistently sent \$3.00 to show trust and encourage cooperation
- As a receiver, I've been returning amounts that fairly reward the sender's trust

I'll maintain my consistent cooperative approach with this new co-player.

{'send': 3.00}

**reasoning:**

Looking at my history:

- Round 1 (receiver): Received \$10.02 from sender's \$3.34, returned \$5.00
- Round 2 (sender): Sent \$3.00, received back \$2.73
- Round 3 (sender): Sent \$3.00, received back \$5.57
- Round 4 (receiver): Received \$8.97 from sender's \$2.99, returned \$4.50
- Round 5 (receiver): Received \$8.67 from sender's \$2.89, returned \$4.30

Now Round 6, I'm a sender again with a new co-player.

My sending pattern has been consistent at \$3.00 when I'm the sender. The returns I've received have varied (\$2.73 and \$5.57), but both times my co-players received \$9.00 and chose to return different amounts.

I should continue with a consistent cooperative strategy. Sending \$3.00 has been my standard, showing significant trust (60% of endowment).

```
{'send': 3.00}
```

**usage:**

```
{
 "input_tokens": 1796,
 "output_tokens": 332,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.33 to you. Your visible payoff was \$6.33.

Your total visible earnings across all rounds are \$32.49.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.78 (about 56% of their endowment), showing trust
- I received \$8.34 after the 3x multiplier
- The sender currently has \$2.22

I'll reciprocate fairly. If I return \$4.20:

- Sender ends with:  $\$2.22 + \$4.20 = \$6.42$  (profit of \$1.42)
- I keep: \$4.14

This maintains my pattern of fairly rewarding cooperation.

```
{'return': 4.20}
```

**reasoning:**

The sender sent \$2.78 (about 55.6% of their endowment), showing trust.

I received \$8.34 after the 3x multiplier.

The sender currently has  $\$5 - \$2.78 = \$2.22$ .

I've been fair in my returns as a receiver:

- Round 1: Received \$10.02, returned \$5.00
- Round 4: Received \$8.97, returned \$4.50
- Round 5: Received \$8.67, returned \$4.30

For this round with \$8.34 received, a fair return would be around \$4.00-\$4.20.

If I return \$4.20:

- Sender ends with:  $\$2.22 + \$4.20 = \$6.42$  (profit of \$1.42)
- I keep:  $\$8.34 - \$4.20 = \$4.14$

This is consistent with my previous fair returns.

**usage:**

```
{
 "input_tokens": 2001,
 "output_tokens": 368,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$4.2. Your visible payoff was \$4.14.

Your total visible earnings across all rounds are \$36.63.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my track record:

- I've consistently sent \$3.00 as a sender, showing reliability and trust
- This strategy has generally been reciprocated well
- As a receiver, I've been fair in my returns

I'll maintain my consistent cooperative approach.

{'send': 3.00}

**Reasoning:**

Looking at my history as a sender:

- Round 2: Sent \$3.00, received back \$2.73 (payoff: \$4.73)
- Round 3: Sent \$3.00, received back \$5.57 (payoff: \$7.57)
- Round 6: Sent \$3.00, received back \$4.33 (payoff: \$6.33)

As a receiver, I've been consistently fair, returning roughly half of what I received or a bit less, ensuring the sender profits from trusting me.

I've established a pattern of sending \$3.00 as a sender, which shows good trust and has generally been reciprocated well. I'll continue this consistent approach.

This is Round 8 with a new co-player. I should maintain my cooperative strategy.

Usage: {

"input\_tokens": 2031,

"output\_tokens": 263,

"reasoning\_tokens": 0

}

**GAME - Agent\_2 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:27:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_3",

"move\_index": 5,

"opponent\_id": "Agent\_1",

"pairing": {

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "trustee",

"role\_label": "Receiver",

"round": 8,

"task": "game",

"task\_index": 0

}

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.55, it became \$7.649999999999995, and you returned \$4.0. Your visible payoff was \$3.649999999999995.

Your total visible earnings across all rounds are \$3.649999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- Round 1 established a cooperative dynamic where sending ~51% led to mutual profit
- I'm now the sender with a new co-player
- I need to establish trust in this new pairing

**\*\*Strategic reasoning:\*\***

- Sending a moderate amount (around 50%) shows willingness to cooperate
- It gives my co-player enough to make returning worthwhile (\$7.50 after multiplier)
- It limits my downside risk while allowing mutual gains
- This mirrors the successful approach from Round 1

**\*\*Decision:\*\***

I'll send \$2.50 (50% of my endowment) to establish a cooperative tone with this new co-player.

{'send': 2.5}

**reasoning:**

Let me analyze this situation:

**\*\*Previous Round (Round 1):\*\***

- I was the RECEIVER
- Sender sent \$2.55 (51% of \$5)
- I received \$7.65
- I returned \$4.00
- My payoff: \$3.65
- Sender's payoff would have been:  $\$5 - \$2.55 + \$4.00 = \$6.45$

**\*\*Current Round (Round 2):\*\***

- I am now the SENDER
- I have \$5
- I'm paired with a potentially different agent (since it says "your current co-player")
- I need to decide how much to send

**\*\*Strategic considerations:\*\***

1. **\*\*Reciprocity principle\*\***: In Round 1, my co-player showed moderate trust (51% sent). I reciprocated well by returning 52% of what I received, ensuring both parties profited.
2. **\*\*Building trust\*\***: Since this is a multi-round game with multiple agents, establishing a cooperative reputation could be beneficial.
3. **\*\*Unknown opponent\*\***: This is a different co-player, so I don't know their reciprocity tendencies yet.
4. **\*\*First impression\*\***: As the sender in this new pairing, I can set the tone for cooperation.
5. **\*\*Risk vs reward\*\***:
  - If I send \$0, I keep \$5, they get \$0, no mutual benefit
  - If I send all \$5, it becomes \$15 for them, but I'm vulnerable if they return nothing
  - Middle ground shows trust while limiting downside risk
6. **\*\*Following established norms\*\***: In Round 1, the sender sent 51%. This seemed to work well and encouraged cooperation.

**\*\*Decision:\*\***

I'll send a similar amount to what I experienced as effective in Round 1 - around 50% (\$2.50). This:

- Shows trust and willingness to cooperate
- Limits my downside risk
- Gives the receiver enough to make returning worthwhile

...

**usage:**

```
{
 "input_tokens": 916,
 "output_tokens": 646,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.62 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$9.77.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.11 (42.2% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- The sender now has \$2.89 remaining

**\*\*My reasoning:\*\***



The sender showed trust, though slightly less than what I've seen in previous rounds. I should reciprocate fairly to maintain cooperation.

If I return \*\*\$3.30\*\*:

- Sender ends with:  $\$2.89 + \$3.30 = \$6.19$  (profit of \$1.19)
- I end with:  $\$6.33 - \$3.30 = \$3.03$  (profit of \$3.03)

This represents:

- Returning about 52% of what I received
- Both of us profiting from cooperation
- Fair reciprocation that encourages future trust

{'return': 3.3}

**reasoning:**

Let me analyze this situation:

**\*\*What happened in Round 2:\*\***

- I sent \$2.50
- It became \$7.50
- They returned \$3.62
- My payoff:  $\$2.50$  (kept) +  $\$3.62$  (returned) =  $\$6.12$
- Their payoff:  $\$7.50 - \$3.62 = \$3.88$
- They returned about 48% of what they received

**\*\*Current Round 3 situation:\*\***

- I'm the receiver
- Sender sent \$2.11 (42.2% of their \$5 endowment)
- I received \$6.33
- Sender now has \$2.89 remaining
- I need to decide how much to return

**\*\*Analysis:\*\***

The sender sent \$2.11, which is less than the \$2.50-\$2.55 amounts I've seen previously. They're showing moderate but slightly lower trust. However, they still sent a significant portion (42%).

If I return different amounts:

- Return \$3.00: Sender ends with \$5.89 (+\$0.89), I end with \$3.33 (+\$3.33)
- Return \$3.50: Sender ends with \$6.39 (+\$1.39), I end with \$2.83 (+\$2.83)
- Return \$3.20: Sender ends with \$6.09 (+\$1.09), I end with \$3.13 (+\$3.13)

I want to:

1. Reciprocate their trust proportionally
2. Be fair and encourage future cooperation
3. Return roughly 50% of what I received, similar to what I've observed

Returning around \$3.20 would be about 50.5% of what I received, which seems fair and proportional to their ~42% send rate. This ensures both of us profit.

Actually, let me be slightly more generous since they did show trust. I'll return \$3.30 (about 52% of what I received).

...

**usage:**

```
{
 "input_tokens": 1218,
 "output_tokens": 758,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.3. Your visible payoff was \$3.0300000000000002.

Your total visible earnings across all rounds are \$12.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- I'm facing a new co-player as sender
- Previous cooperation has been successful (total earnings \$12.80)
- When I sent \$2.50 in Round 2, I received \$3.62 back for a good profit
- My returns as receiver have been fair (~52%)

**\*\*Strategic reasoning:\*\***

- Continue the cooperative approach that's been working
- Sending 50% shows good faith while managing risk
- This gives my co-player \$7.50 to work with, enough for mutual benefit
- Establishes positive precedent with this new partner

{'send': 2.5}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds summary:\*\***

- Round 1 (RECEIVER): Received \$7.65 from sender sending \$2.55, returned \$4.00, earned \$3.65
- Round 2 (SENDER): Sent \$2.50, became \$7.50, they returned \$3.62, earned \$6.12
- Round 3 (RECEIVER): Received \$6.33 from sender sending \$2.11, returned \$3.30, earned \$3.03
- Total earnings: \$12.80

**\*\*Pattern analysis:\*\***

- When I sent \$2.50 (50%), I got back \$3.62, giving me a profit of \$1.12 (from \$2.50 remaining + \$3.62 = \$6.12 total)
- When others sent to me, I've returned fairly (~52% of what I received)
- Overall, cooperation has been successful

**\*\*Current decision:\*\***

- New co-player as sender this round
- Need to establish trust again
- Sending around 50% has worked well previously

I'll continue with the cooperative strategy that's been working: send \$2.50 (50% of endowment).

**usage:**

```
{
 "input_tokens": 1558,
 "output_tokens": 431,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*My reasoning:\*\***

This sender showed above-average trust compared to previous rounds. I should reciprocate

enerously.

If I return \*\*\$4.70\*\*:

- Sender ends with:  $\$2.08 + \$4.70 = \$6.78$  (profit of \$1.78)
- I end with:  $\$8.76 - \$4.70 = \$4.06$  (profit of \$4.06)

This represents:

- Returning about 54% of what I received
- Rewarding their higher trust level (58.4% sent)
- Both profiting well from cooperation
- Encouraging continued trust

{'return': 4.7}

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*Patterns from previous rounds:\*\***

- Round 1: Received \$7.65, returned \$4.00 (52%)
- Round 3: Received \$6.33, returned \$3.30 (52%)
- I've been consistently returning about 52% of what I receive

**\*\*Strategy:\*\***

The sender showed high trust (58.4% sent). I should reciprocate fairly.

If I return around 52% of \$8.76:

- $0.52 \times \$8.76 = \$4.55$

Let me check the outcomes:

- Sender ends with:  $\$2.08 + \$4.55 = \$6.63$  (profit of \$1.63)
- I end with:  $\$8.76 - \$4.55 = \$4.21$  (profit of \$4.21)

This seems fair - both profit, sender gets a decent return for their trust.

Actually, given they sent 58.4% (more than my usual 50%), maybe I should return slightly more to really reward that trust. Let me try \$4.70:

- Sender ends with:  $\$2.08 + \$4.70 = \$6.78$  (profit of \$1.78)
- I end with:  $\$8.76 - \$4.70 = \$4.06$  (profit of \$4.06)

That's about 54% return, which is fair and rewards their higher trust level.

**usage:**

```
{
 "input_tokens": 1844,
 "output_tokens": 622,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.92, it became \$8.76, and you returned \$4.7. Your visible payoff was \$4.06.

Your total visible earnings across all rounds are \$22.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My sending strategy (50%) has been successful in Rounds 2 & 4

- Returns I've received: \$3.62 and \$3.01 (averaging \$3.32)
- Total earnings of \$22.37 show cooperation is working

**\*\*Strategic reasoning:\*\***

- Maintain the consistent 50% sending strategy
- This has proven reliable across different co-players
- Gives receiver \$7.50 to work with
- Balances trust-building with risk management

{'send': 2.5}

**reasoning:**

Looking at the pattern:

**\*\*My sending history:\*\***

- Round 2: Sent \$2.50 (50%), received back \$3.62, profit = \$6.12
- Round 4: Sent \$2.50 (50%), received back \$3.01, profit = \$5.51

**\*\*My receiving history:\*\***

- Round 1: Received \$7.65 from \$2.55 sent (51%), returned \$4.00
- Round 3: Received \$6.33 from \$2.11 sent (42%), returned \$3.30
- Round 5: Received \$8.76 from \$2.92 sent (58%), returned \$4.70

**\*\*Analysis:\*\***

- When I send 50%, I've gotten decent returns (\$3.01-\$3.62)
- As a receiver, I've been fair, returning around 52-54%
- Total earnings are strong at \$22.37
- Cooperation has been working well

**\*\*This round:\*\***

- New co-player as sender
- Continue with the cooperative 50% strategy that's been successful
- This maintains consistency and shows I'm a cooperative player

I'll send \$2.50 again.

**usage:**

```
{
 "input_tokens": 2182,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.8 to you. Your visible payoff was \$6.3.

Your total visible earnings across all rounds are \$28.67.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My consistent 50% sending strategy has worked well (Rounds 2, 4, 6)
- Returns have varied (\$3.62, \$3.01, \$3.80) but all profitable
- Total earnings of \$28.67 show strong overall cooperation

**\*\*Strategic reasoning:\*\***

- Continue the reliable 50% sending pattern
- This approach has proven effective across different co-players
- Provides \$7.50 for receiver to reciprocate
- Maintains consistency and manages risk

{'send': 2.5}

**reasoning:**

Looking at this situation:

**\*\*Context:\*\***

- I'm the sender again, facing a new co-player
- My sending history shows consistent 50% sends (\$2.50 in rounds 2, 4, 6)
- Returns I've received: \$3.62, \$3.01, \$3.80 (averaging \$3.48)
- All have been profitable
- Total earnings of \$28.67

**\*\*Strategic reasoning:\*\***

- The 50% strategy has been consistently successful
- Returns have varied but all been positive
- This creates good conditions for mutual benefit
- Should continue this proven approach

I'll send \$2.50 again.

**usage:**

```
{
 "input_tokens": 2442,
 "output_tokens": 294,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.11 to you. Your visible payoff was \$5.609999999999999.

Your total visible earnings across all rounds are \$34.28.

This round, you are the RECEIVER against your current co-player. They sent you \$2.29, so you received \$6.87. How much do you return? (0-6.87)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.29 (45.8% of their endowment), showing moderate trust
- I received \$6.87 after the 3x multiplier
- The sender now has \$2.71 remaining

**\*\*My reasoning:\*\***

Based on my consistent pattern in previous receiver rounds (returning ~52-54%), I should reciprocate proportionally.

If I return **\*\*\$3.57\*\***:

- Sender ends with: \$2.71 + \$3.57 = \$6.28 (profit of \$1.28)
- I end with: \$6.87 - \$3.57 = \$3.30 (profit of \$3.30)

This represents:

- Returning about 52% of what I received
- Both parties profiting from cooperation
- Consistent reciprocation encouraging continued trust

{'return': 3.57}

**Reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- The sender sent \$2.29 (45.8% of their endowment), showing moderate trust
- I received \$6.87 after the 3x multiplier
- The sender now has \$2.71 remaining

**\*\*My approach in previous receiver rounds:\*\***

- Round 1: Received \$7.65, returned \$4.00 (52.3%)
- Round 3: Received \$6.33, returned \$3.30 (52.1%)
- Round 5: Received \$8.76, returned \$4.70 (53.7%)

I've been consistent in returning around 52-54% of what I receive.

For this round, if I return around 52%:  
 $\$6.87 \times 0.52 = \$3.57$

Let me check the outcomes:

- If I return \$3.57:
  - Sender ends with:  $\$2.71 + \$3.57 = \$6.28$  (profit of \$1.28)
  - I end with:  $\$6.87 - \$3.57 = \$3.30$  (profit of \$3.30)

This seems fair and consistent with my previous behavior. The sender sent 45.8% which is moderate trust, and returning 52% rewards that trust while both parties profit.

I'll return \$3.57.

```
Usage: {
 "input_tokens": 2408,
 "output_tokens": 540,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_8 - Sender - dyad\_4

Timestamp: 2026-08-12T09:27:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps:  $\$5 - \text{sent} + \text{returned}$

- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

#### MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

#### Message 2 role=user

##### content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.46 to you. Your visible payoff was \$4.46.

Your total visible earnings across all rounds are \$4.46.

This round, you are the RECEIVER against your current co-player. They sent you \$3.19, so you received \$9.57. How much do you return? (0-9.57)

Respond exactly as JSON: {'return': <amount>}

#### Message 3 role=assistant

##### content:

Looking at this situation:

**\*\*Current Round (Round 2):\*\***

- My co-player sent me \$3.19 ? I received \$9.57
- This created \$6.38 in surplus value

**\*\*Fair return considerations:\*\***

- The sender took a significant risk sending most of their endowment
- A reasonable approach is to ensure we both benefit from the cooperation
- Returning \$4.00 would give:
  - Sender: \$5 - \$3.19 + \$4.00 = \$5.81
  - Me: \$9.57 - \$4.00 = \$5.57

This creates a balanced outcome where we both gain from the exchange.

{'return': 4.00}

##### reasoning:

Let me analyze this situation:

Last round (Round 1):

- I was the SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$2.46 to me
- My payoff: \$5 - \$3 + \$2.46 = \$4.46
- Their payoff: \$9.0 - \$2.46 = \$6.54

So they returned about 27% of what they received (\$2.46/\$9.0 ? 0.273).

Now in Round 2:

- I'm the RECEIVER
- My current co-player sent me \$3.19
- I received \$9.57 (3.19 × 3)

- I need to decide how much to return

Important notes:

1. This is a different co-player (multi-agent setting, randomized pairings)
2. There's communication noise, so the \$3.19 might not be exactly what they sent
3. The previous player returned about 27% to me

Considerations:

- If I want to encourage cooperation, I should return a fair amount
- A common fair split might be to return enough so we both benefit equally from the cooperation
- The sender gave up \$3.19, so they have  $\$5 - \$3.19 = \$1.81$  left
- I have \$9.57
- If I want us to end up with similar amounts, I could return about \$3.88, giving us both around \$5.69
- Or I could return proportionally to what was sent: returning about 1/3 of what I received would be \$3.19, which gives them back their original amount
- A generous return might be half of what I received: \$4.785

Let me think about what's fair:

- The sender took a risk sending \$3.19 (gave up most of their endowment)
- The transaction created \$6.38 in surplus value beyond what they sent
- Splitting that gain equally means we each get an additional \$3.19 from the multiplier effect
- So returning \$3.19 would leave me with \$6.38 and restore them to their starting \$5
- Alternatively, returning \$4 would give them \$5.81 while I keep \$5.57

, creating a more balanced outcome

- Given the earlier low reciprocation (27%), I could...

usage:

```
{
 "input_tokens": 756,
 "output_tokens": 1027,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.19, it became \$9.57, and you returned \$4.0. Your visible payoff was \$5.57.

Your total visible earnings across all rounds are \$10.030000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this new round as SENDER:

**\*\*Past experience:\*\***

- Round 1: Sent \$3, received \$2.46 back (modest cooperation)
- Round 2: When I received, I returned fairly (\$4.00 on \$9.57)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Need to establish trust with someone new
- The 3x multiplier means cooperation creates value

A moderate send amount balances the potential gains from cooperation with the risk of low returns. Sending \$3 creates \$9 of value while maintaining some protection.

{'send': 3}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds:\*\***

- Round 1: I was SENDER, sent \$3, received back \$2.46, payoff was \$4.46
- Round 2: I was RECEIVER, received \$9.57 (from \$3.19 sent), returned \$4.00, payoff was \$5



.57

**\*\*Current Round (Round 3):\*\***

- I'm the SENDER again
- New co-player (since pairings are randomized each round)
- I have \$5 to potentially send

**\*\*Observations:\*\***

1. In Round 1, when I sent \$3, I got back \$2.46. The receiver kept \$6.54 out of \$9.00, which seems somewhat selfish (I got less than my initial \$3).
2. In Round 2, when someone sent me \$3.19, I returned \$4.00, being relatively fair.
3. The note about communication noise is important - amounts may differ from what was actually sent/returned.

**\*\*Strategy considerations:\*\***

- I'm playing with a new co-player each round
- There's no reputation to build with a specific player
- The 3x multiplier means cooperation creates value
- I need to balance trust against the risk of low returns

Given:

- My Round 1 experience showed modest cooperation from my receiver
- I was reasonably fair in Round 2
- I'm with a new player now who doesn't know my history

I think sending \$3 again is reasonable:

- It's a moderate level of trust
- Creates \$9 for the receiver
- Provides some protection if they don't reciprocate well

Let me send \$3 again.

**usage:**

```
{
 "input_tokens": 1047,
 "output_tokens": 532,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$14.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.22, so you received \$9.66. How much do you return? (0-9.66)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 4):\*\***

- Co-player sent \$3.22 ? I received \$9.66
- This represents significant trust (sending 64% of their endowment)

**\*\*Fair return analysis:\*\***

- Returning \$4.00 would give:
  - Sender: \$5 - \$3.22 + \$4.00 = \$5.78 (profit of \$0.78)
  - Me: \$9.66 - \$4.00 = \$5.66 (profit of \$0.66)

This creates a fair split where both players benefit from the cooperation.

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER:\*\***

- My co-player sent me \$3.22

- I received \$9.66 (after 3x multiplier)
- I need to decide how much to return

**\*\*Previous rounds analysis:\*\***

- Round 1 (SENDER): Sent \$3, got back \$2.46 - slightly below fair
- Round 2 (RECEIVER): Received \$9.57, returned \$4.00 - tried to be fair
- Round 3 (SENDER): Sent \$3, got back \$2.93 - again slightly below fair

**\*\*Fair return calculation:\*\***

The sender gave up \$3.22 from their \$5 endowment, creating \$9.66 for me.

- Surplus created:  $\$9.66 - \$3.22 = \$6.44$

A fair split would give us each roughly equal gains:

- If I return \$4.00:
  - Sender gets:  $\$5 - \$3.22 + \$4.00 = \$5.78$  (gain of \$0.78)
  - I get:  $\$9.66 - \$4.00 = \$5.66$  (gain of \$0.66)

This seems reasonably balanced. The sender sent a significant portion of their endowment, showing trust.

I should return around \$4.00 to be fair and reciprocate their trust.

**usage:**

```
{
 "input_tokens": 1324,
 "output_tokens": 501,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.22, it became \$9.66, and you returned \$4.0. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$20.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my experience so far:

**\*\*As SENDER:\*\***

- Round 1: Sent \$3 ? received \$2.46 (modest return)
- Round 3: Sent \$3 ? received \$2.93 (decent return)

**\*\*As RECEIVER:\*\***

- I've consistently returned \$4.00 on ~\$9.60, maintaining fairness

**\*\*Current round:\*\***

- New co-player (randomized pairing)
- Sending \$3 has been my standard opening move
- Creates \$9 of value with moderate risk

I'll continue with the balanced approach that's been working.

```
{'send': 3}
```

**reasoning:**

Looking at my history as SENDER:

- Round 1: Sent \$3, received \$2.46 back
- Round 3: Sent \$3, received \$2.93 back

Both times I sent \$3, I received back around \$2.50-\$3.00, which is close to fair but not quite equal splitting of the gains.

As RECEIVER:

- Round 2: Received \$9.57, returned \$4.00
- Round 4: Received \$9.66, returned \$4.00

I've been consistent in returning fairly.

Now Round 5, I'm a SENDER with a new co-player. The pattern suggests that sending around \$3 has been reasonably productive. The returns haven't been amazing (I've profited minimally as sender), but they've been positive.

I should continue with a moderate strategy. Sending \$3 creates value and has worked reasonably well so far.

**usage:**

```
{
 "input_tokens": 1601,
 "output_tokens": 366,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.2 to you. Your visible payoff was \$7.2.

Your total visible earnings across all rounds are \$27.82.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This is a slightly smaller send than previous rounds

**\*\*Fair return analysis:\*\***

- In previous rounds, I returned ~\$4.00 on receives of ~\$9.60
- This represents about a 42% return rate
- For \$8.58, a proportional return would be around \$3.60

Returning \$3.70 would give:

- Sender:  $\$5 - \$2.86 + \$3.70 = \$5.84$  (modest profit)
- Me:  $\$8.58 - \$3.70 = \$4.88$

This maintains fairness while ensuring both players benefit from cooperation.

{ 'return': 3.70 }

**reasoning:**

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This shows trust, sending about 57% of their endowment

**\*\*Fair return analysis:\*\***

- Total surplus created:  $\$8.58 - \$2.86 = \$5.72$
- A fair split would give each of us roughly \$2.86 in profit

If I return \$3.50:

- Sender gets:  $\$5 - \$2.86 + \$3.50 = \$5.64$  (profit of \$0.64)
- I get:  $\$8.58 - \$3.50 = \$5.08$  (profit of \$0.08)

That's not quite balanced. Let me try \$4.00:

- Sender gets:  $\$5 - \$2.86 + \$4.00 = \$6.14$  (profit of \$1.14)
- I get:  $\$8.58 - \$4.00 = \$4.58$  (profit of \$-0.42)

Wait, that would give me less than my base \$5. Let me recalculate.

Actually, I don't start with \$5 as the receiver. I start with \$0 and receive the multiplied amount.

So:

- Sender starts with \$5, sends \$2.86, gets back \$X
- Sender final:  $\$5 - \$2.86 + \$X = \$2.14 + \$X$
- I start with \$0, receive \$8.58, return \$X
- My final:  $\$8.58 - \$X$

For a balanced outcome where we both profit similarly:

- Sender profit:  $\$2.14 + \$X - \$5 = \$X - \$2.86$
- My profit:  $\$8.58 - \$X - \$0 = \$8.58 - \$X$
- Setting equal:  $\$X - \$2.86 = \$8.58 - \$X$  gives  $\$X = \$5.72$

That would split the surplus evenly, but feels generous. Instead, I should ensure the sender profits from their trust while keeping a reasonable share for myself. Returning around \$3.80 would give them roughly \$0.94 in profit while I keep \$4.78.

But that's heavily skewed in my favor. Testing \$4.50 or \$5.00 returns gets closer to balance-the sender would profit \$1.64 to \$2.14 while I'd keep \$3.58 to \$4.08. I'm reconsidering what fairness actually means here. In previous exchanges where I received around \$9.50-\$9.66 from sends of \$3.19-\$3.22, returning \$4.00 felt proportional. Using that same ratio (roughly 42% return rate) on this \$8.58 suggests returning around \$3.58-\$3.70. Testing \$3.70 gives the sender \$0.84 profit and me \$4.88, which still favors me but follows the pattern I've established in earlier rounds....

usage:

```
{
 "input_tokens": 1883,
 "output_tokens": 1261,
 "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.86, it became \$8.58, and you returned \$3.7. Your visible payoff was \$4.88.

Your total visible earnings across all rounds are \$32.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.69, so you received \$8.07. How much do you return? (0-8.07)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

**\*\*Current Round (Round 7):\*\***

- Co-player sent \$2.69 ? I received \$8.07
- A moderate send showing trust

**\*\*My return pattern:\*\***

- I've been returning around 42% of what I receive
- This has created mutually beneficial outcomes

For \$8.07 received:

- Returning \$3.50 would give:
  - Sender:  $\$5 - \$2.69 + \$3.50 = \$5.81$  (profit of \$0.81)
  - Me:  $\$8.07 - \$3.50 = \$4.57$

This maintains fairness and ensures both players benefit.

{'return': 3.50}

reasoning:

Looking at the current situation:

**\*\*Current Round (Round 7):\*\***

- Co-player sent \$2.69 ? I received \$8.07
- This is another moderate send

**\*\*My return pattern:\*\***

- Round 2: Received \$9.57, returned \$4.00 (42%)
- Round 4: Received \$9.66, returned \$4.00 (41%)

- Round 6: Received \$8.58, returned \$3.70 (43%)

I've been maintaining around a 42% return rate, which seems fair.

For \$8.07:

- 42% return would be:  $\$8.07 \times 0.42 = \$3.39$

Returning \$3.50 would give:

- Sender:  $\$5 - \$2.69 + \$3.50 = \$5.81$  (profit of \$0.81)

- Me:  $\$8.07 - \$3.50 = \$4.57$

This maintains my fair approach while ensuring both benefit. Let me go with \$3.50.

**usage:**

```
{
 "input_tokens": 2202,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.69, it became \$8.07, and you returned \$3.5. Your visible payoff was \$4.57.

Your total visible earnings across all rounds are \$37.27.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Previous round (Round 7) with this SAME co-player:\*\***

- They sent me \$2.69 (showing trust)

- I returned \$3.50 (fair return)

**\*\*Current round (Round 8):\*\***

- Now I'm the sender with the same co-player

- They demonstrated moderate cooperation

- I should reciprocate their trust

Since they sent me \$2.69 and I treated them fairly, I'll send \$3.00 to reciprocate their cooperative behavior and potentially continue a mutually beneficial relationship.

{'send': 3.00}

**Reasoning:**

Looking at the situation:

**\*\*Last round (Round 7):\*\***

- I was RECEIVER, my co-player was SENDER

- They sent \$2.69 ? I received \$8.07

- I returned \$3.50

- They got:  $\$5 - \$2.69 + \$3.50 = \$5.81$

- I got: \$4.57

**\*\*This round (Round 8):\*\***

- I am SENDER, they are RECEIVER

- Same co-player!

- They sent me a moderate amount (\$2.69) last round

- I returned fairly (\$3.50)

This is an opportunity to reciprocate their trust. They sent me \$2.69, so I should send a similar or slightly higher amount to show I'm willing to cooperate fairly with someone who cooperates with me.

Sending \$3 would be reasonable - it's my standard amount and slightly more than what they sent me, showing reciprocal trust.

Actually, let me think about this more carefully. They sent me \$2.69. I could:

- Send \$2.69 to exactly match their send
- Send \$3 which is my standard amount and slightly more
- Send more to show even stronger reciprocation

I think sending \$3 is appropriate - it matches my typical strategy and shows I'm reciprocating their cooperative behavior.

```
Usage: {
 "input_tokens": 2341,
 "output_tokens": 450,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_7 - Receiver - dyad\_4**

Timestamp: 2026-08-12T09:27:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

##### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### **GAME:**

You will play different roles across multiple rounds.

##### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.56 to you. Your visible payoff was \$7.56.

Your total visible earnings across all rounds are \$7.56.

This round, you are the RECEIVER against your current co-player. They sent you \$1.67, so you received \$5.01. How much do you return? (0-5.01)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.67 (33% of their endowment)
- I received \$5.01 after the 3x multiplier
- They currently have \$3.33 remaining

**\*\*My considerations:\*\***

- They showed moderate trust by sending \$1.67
- The multiplier created \$3.34 in surplus value
- A fair approach would be to share this surplus roughly equally
- In Round 1, my partner returned ~62% and that felt cooperative

**\*\*Fair return calculation:\*\***

If we split the surplus equally:

- Each should gain ~\$1.67 from cooperation
- Sender ending: \$5 - \$1.67 + return = \$6.67 desired
- Return needed: \$3.34

However, that might be overly generous given their modest send. I'll return about 60% to reward their cooperation while keeping reasonable value.

```
'''json
{"return": 3.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 (previous):

- I was SENDER
- I sent \$3.0
- It became \$9.0 for the receiver
- They returned \$5.56 to me
- I ended with: \$5 - \$3 + \$5.56 = \$7.56
- Receiver ended with: \$9 - \$5.56 = \$3.44

That was actually quite generous of them! They returned more than half.

Round 2 (current):

- I am RECEIVER

- They sent me \$1.67
- I received \$5.01 (which is  $3 \times \$1.67$ )
- I need to decide how much to return (0-5.01)

Key considerations:

1. My previous partner was generous, returning \$5.56 out of \$9 (62%)
2. This is a different player (multi-agent setting with randomized pairings)
3. They sent me \$1.67 out of \$5, which is 33.4% - a moderate amount
4. The sender keeps:  $\$5 - \$1.67 = \$3.33$  before my return
5. I should reciprocate fairly to encourage future cooperation

Let me think about fair returns:

- If I return half (\$2.50), they end with  $\$3.33 + \$2.50 = \$5.83$ , I keep \$2.51
- If I return around what gives us equal final payoffs:
  - They have  $\$3.33 + \text{return}$
  - I have  $\$5.01 - \text{return}$
  - Equal:  $\$3.33 + x = \$5.01 - x$
  - $2x = \$1.68$
  - $x = \$0.84$

This would give us both around \$4.17

- A proportional return similar to what I received (62% of what I got):
  - $0.62 \times \$5.01 = \$3.11$
  - They'd end with:  $\$3.33 + \$3.11 = \$6.44$
  - I'd keep:  $\$5.01 - \$3.11 = \$1.90$

A 50% return of around \$2.50 seems like the right balance-it rewards their cooperation without being excessive, and it's more moderate than matching their 33% send rate since they're a different player than before.

If I send back \$2.50, they'd gain \$0.83 while I'd be left with \$2.51. The math shows we'd create \$8.34 in total value, which is less than the \$10 we'd have if neither of us participated. The real question is what surplus we're actually generating from this exchange and whether my return is proportional to their initial trust.

The multiplier created \$3.34 in extra value (\$5.01 received minus \$1.67 sent). If we split that gain equally, each of us should pocket \$1.67. That means I should return \$3.34 to bring them to \$6.67 total, which gives us both the same \$1.67 gain from the transaction.

...

usage:

```
{
 "input_tokens": 754,
 "output_tokens": 1445,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.67, it became \$5.01, and you returned \$3.0. Your visible payoff was \$2.01.

Your total visible earnings across all rounds are \$9.57.

This round, you are the RECEIVER against your current co-player. They sent you \$3.66, so you received \$10.98. How much do you return? (0-10.98)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$3.66 (73% of their endowment) - very high trust!
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining
- Total surplus created: \$2.32

**\*\*My considerations:\*\***

- This is a very generous send showing strong cooperation
- They took significant risk by sending most of their money



- I should reward this trust with a fair return
- In Round 1, I received about 62% back on my \$3 send
- In Round 2, I returned 60% on a smaller send

**\*\*Fair return:\*\***

Given their high trust, I'll return approximately 50% of what I received:

- Return \$5.50
- They'll end with \$6.84 (profit of \$1.84)
- I'll end with \$5.48 (profit of \$0.48)
- This rewards their cooperation while both of us profit

```
'''json
{"return": 5.50}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current Situation:\*\***

- My co-player sent me \$3.66 (73% of their endowment)
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining (\$5 - \$3.66)
- This is a very generous/trusting send

**\*\*What this shows:\*\***

- High trust from the sender (sending over 70%!)
  - They're expecting reciprocity
- The multiplier created significant surplus: \$5.98 (\$10.98 - \$5)

**\*\*Considerations:\*\***

- In Round 1, when I sent \$3 (60%), I received \$5.56 back - that's about 62% return rate
- In Round 2, when someone sent \$1.67 (33%), I returned \$3.00 - that's 60% return rate
- This sender showed much more trust (\$3.66 = 73% of endowment)

**\*\*Fair calculation:\*\***

The total pot is now \$12.32 (\$1.34 they kept + \$10.98 I received)

If we split the gains from cooperation:

- Original total: \$10 (my \$5 + their \$5)
- Current total: \$12.32
- Surplus created: \$2.32
- Fair split would give each person \$1.16 gain
- Sender would end at \$6.16, I'd end at \$6.16
- I need to return: \$6.16 - \$1.34 = \$4.82

But let me think about what's fair based on the trust shown:

- They sent 73% of their money
- If I return about 50% of what I received, that's \$5.49
- That would give them \$1.34 + \$5.49 = \$6.83 total

...

**usage:**

```
{
 "input_tokens": 1112,
 "output_tokens": 833,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$5.5. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$15.05.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: I sent \$3 and received \$5.56 back - good cooperation!
- Rounds 2-3: As receiver, I tried to return fairly and reward trust
- Total earnings: \$15.05

**\*\*Considerations for this round:\*\***

- I'm paired with a new co-player each round
- My Round 1 send of \$3 worked well
- The most generous player sent me \$3.66, showing high trust pays off
- Sending more creates more total value (3x multiplier)
- I want to encourage cooperative norms in the pool

**\*\*Strategy:\*\***

I'll send a bit more than Round 1 to signal strong cooperation while managing risk. This shows I'm willing to trust and invest in mutual gains.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Now I'm the SENDER again in Round 4.

Let me review my history:

- Round 1 (SENDER): Sent \$3, received back \$5.56, payoff \$7.56
- Round 2 (RECEIVER): Received \$5.01 (from \$1.67 sent), returned \$3.00, payoff \$2.01
- Round 3 (RECEIVER): Received \$10.98 (from \$3.66 sent), returned \$5.50, payoff \$5.48

This is a new co-player each round in this multi-agent setting.

My experience so far:

- When I sent \$3, I got back \$5.56 - good return!
- When others sent to me, I tried to be fair/cooperative
- The player who sent \$3.66 was very trusting

For this round, I need to decide how much to send to a new co-player.

Considerations:

- I've seen good cooperation so far (my Round 1 partner returned well)
- I want to continue building cooperative patterns
- Sending more creates more value (3x multiplier)
- But I'm paired with a new player each time, so no direct reputation building

Strategy:

- In Round 1, I sent \$3 and it worked well
- I could send more to signal strong cooperation
- Or stick with \$3 as a moderate cooperative amount

Given that I've had positive experiences and want to continue fostering cooperation, I'll send a bit more this time - maybe \$3.5 or \$4. This shows I'm willing to trust while not going all-in.

Let me go with \$3.5 - it's between my previous \$3 send and the very generous \$3.66 I received in Round 3.

Actually, thinking about it more: the player who sent me \$3.66 showed strong trust and I rewarded it. If I want to encourage that kind of cooperation, I should also send generously when I'm the sender. Let me send \$4.

...

**usage:**

```
{
 "input_tokens": 1475,
 "output_tokens": 749,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$3.59 to you. Your visible payoff was

\$5.09.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*Recent experience:\*\***

- Round 4: I sent \$3.50, received only \$3.59 back - minimal profit (\$5.09)
- This was disappointing compared to Round 1 where I sent \$3 and got \$5.56 back
- But this is a NEW co-player with unknown behavior

**\*\*Considerations:\*\***

- Different co-players show varying levels of cooperation
- Sending less than \$3.50 reduces my risk after last round's poor return
- Still want to signal willingness to cooperate and create value
- \$3 was my initial send and it worked well - proven strategy

**\*\*Strategy:\*\***

I'll return to sending \$3 - a moderate amount that:

- Shows trust and cooperation
- Creates meaningful value (\$9 for receiver)
- Limits downside risk
- Has proven successful before

'''json

{"send": 3}

'''

**reasoning:**

Looking at my history:

- Round 1: Sent \$3, received \$5.56 back - payoff \$7.56 (good!)
- Round 2: Receiver, returned \$3.00 - payoff \$2.01
- Round 3: Receiver, returned \$5.50 - payoff \$5.48
- Round 4: Sent \$3.5, received \$3.59 back - payoff \$5.09 (disappointing)
- Round 5: Now sender again

Analysis of Round 4:

- I sent \$3.50 (70% of endowment) showing strong trust
- They received \$10.50
- They only returned \$3.59 (34% of what they received)
- I ended with \$5.09, barely breaking even
- This was much less generous than my Round 1 partner who returned \$5.56 on a \$9 receive

This suggests I encountered a less cooperative player. However, I'm now paired with a NEW co-player, so their behavior doesn't necessarily predict this new person's behavior.

Strategic considerations:

- I'm doing reasonably well overall (\$20.14 total)
- Sending creates value through the multiplier
- But I need to balance risk
- Two different responses to my sends: one very good (\$5.56), one mediocre (\$3.59)
- I don't know if there's a pattern or if agents vary widely

I think I should continue showing willingness to cooperate but maybe pull back slightly from the \$3.50 I sent last time since that didn't work out as well. Going back to \$3 seems reasonable - it's a solid middle ground that shows cooperation without overcommitting.

**usage:**

```
{
 "input_tokens": 1797,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You se

nt \$3.0, it became \$9.0, and you saw them return \$4.06 to you. Your visible payoff was \$6.06.

Your total visible earnings across all rounds are \$26.2.

This round, you are the RECEIVER against your current co-player. They sent you \$1.64, so you received \$4.92. How much do you return? (0-4.92)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment) - modest trust
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining

**\*\*My considerations:\*\***

- This is very similar to Round 2 where I received \$5.01 from a \$1.67 send
- In Round 2, I returned \$3.00 (60%) which seemed fair
- I should be consistent in rewarding similar levels of cooperation
- Returning around 60% rewards their trust while keeping reasonable value

**\*\*Fair return:\*\***

Returning approximately 60% of what I received:

- Return \$3.00
- They'll end with \$6.36 (profit of \$1.36)
- I'll end with \$1.92

```
'''json
{"return": 3.00}
'''
```

**reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment)
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining
- Total surplus created from their send:  $\$4.92 - \$1.64 = \$3.28$

**\*\*My considerations:\*\***

- They showed modest trust by sending \$1.64 (similar to the \$1.67 I saw in Round 2)
- In Round 2, when I received \$5.01 from a \$1.67 send, I returned \$3.00 (60%)
- I should be consistent in how I respond to similar levels of trust
- Returning about 60% would be \$2.95

**\*\*Fair return calculation:\*\***

- If I return around \$3.00:
  - They end with:  $\$3.36 + \$3.00 = \$6.36$  (profit of \$1.36)
  - I end with:  $\$4.92 - \$3.00 = \$1.92$  (loss of \$0.08 from initial endowment perspective, but positive from game perspective)

Actually, let me think about this differently:

- They sent \$1.64
- This created \$3.28 in surplus ( $\$4.92 - \$1.64$ )
- A fair split of surplus would give each about \$1.64
- So they should end with:  $\$5 + \$1.64 = \$6.64$
- They have \$3.36, so I need to return \$3.28 to get them to \$6.64

But that gives me only \$1.64, which seems low for receiving \$4.92.

Let me be consistent with Round 2 where I returned 60% on a similar send:

- 60% of \$4.92 = \$2.95
- ...

**usage:**

```
{
 "input_tokens": 2142,
 "output_tokens": 677,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.64, it became \$4.92, and you returned \$3.0. Your visible payoff was \$1.92.

Your total visible earnings across all rounds are \$28.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My sender history:\*\***

- Round 1: Sent \$3 ? got \$5.56 back (profit \$3.56) ?
- Round 4: Sent \$3.50 ? got \$3.59 back (profit \$0.09) ?
- Round 5: Sent \$3 ? got \$4.06 back (profit \$1.06) ?

**\*\*Observations:\*\***

- \$3 has been my most reliable send amount (2/2 reasonable returns)
- \$3.50 didn't work out well
- Each round is a new co-player, so no personal history
- Total earnings: ~\$28.12 - healthy position

**\*\*Strategy:\*\***

Stick with \$3 as my standard opening send:

- Proven track record
- Shows cooperation and trust
- Creates meaningful value (\$9 for receiver)
- Manages downside risk
- Signals I'm a fair player

'''json

{"send": 3}

'''

**reasoning:**

Looking at my history:

- Round 1 (Sender): Sent \$3, got \$5.56 back - Good!
- Round 2 (Receiver): Received \$5.01, returned \$3.00
- Round 3 (Receiver): Received \$10.98, returned \$5.50
- Round 4 (Sender): Sent \$3.50, got \$3.59 back - Poor return
- Round 5 (Sender): Sent \$3, got \$4.06 back - Okay return
- Round 6 (Receiver): Received \$4.92, returned \$3.00

As SENDER:

- Round 1: Sent \$3 ? returned \$5.56 (profit \$3.56)
- Round 4: Sent \$3.50 ? returned \$3.59 (profit \$0.09)
- Round 5: Sent \$3 ? returned \$4.06 (profit \$1.06)

Total earnings: ~\$28.12

New co-player again.

The pattern shows:

- Sending \$3 has worked reasonably well (2 out of 3 times)
- Sending \$3.50 got poor returns
- Each round is a new partner with unknown behavior

I should continue with a moderate send that balances trust-building with risk management. \$3 seems to be a good baseline that's worked before.

Actually, let me think about the broader strategy. I've been fairly cooperative as a receiver, returning good amounts. As a sender, I've been moderately trusting.

Given that:

1. This is a new co-player
2. \$3 has generally worked well

3. I want to signal cooperation while managing risk
4. Total earnings are healthy at \$28.12

I'll stick with \$3 as my standard opening send.

**usage:**

```
{
 "input_tokens": 2476,
 "output_tokens": 676,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.87 to you. Your visible payoff was \$5.87.

Your total visible earnings across all rounds are \$33.989999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$2.77 (55% of their endowment) - good trust
- I received \$8.31 after the 3x multiplier
- They currently have \$2.23 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 7!
- Last round: I sent them \$3.0, they returned \$3.87 (43% of what they received)
- They showed fair cooperation
- Now they're reciprocating with a solid send of \$2.77

**\*\*My considerations:\*\***

- They treated me fairly last round
- I want to reciprocate and encourage continued cooperation
- Returning around 50% would be fair: ~\$4.15
- This rewards their trust while both of us profit

**\*\*Fair return:\*\***

- Return \$4.00
- They'll end with \$6.23 (profit of \$1.23)
- I'll end with \$4.31 (profit of -\$0.69 relative to endowment, but fair given the exchange)

```
'''json
{"return": 4.00}
'''
```

**Reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$2.77 (55% of their endowment) - moderate-high trust
- I received \$8.31 after the 3x multiplier
- They currently have \$2.23 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 7!
- Last round: I sent them \$3.0, they returned \$3.87 to me
- They showed reasonable cooperation last round (43% return rate)
- Now they've sent me \$2.77 (slightly less than my \$3, but still solid)

**\*\*My considerations:\*\***

- This co-player returned \$3.87 on my \$3 send - that was fair/decent
- They're now showing similar trust by sending \$2.77
- I should reciprocate their fairness
- Looking at my past receiver behavior:
  - Round 2: Received \$5.01, returned \$3.00 (60%)

- Round 3: Received \$10.98, returned \$5.50 (50%)
- Round 6: Received \$4.92, returned \$3.00 (61%)

**\*\*Fair return:\*\***

They sent me about 55% of their endowment. If I return about 50-55% of what I received:

- Return around \$4.00-\$4.50
- They'll end with \$6.23-\$6.73 (profit of \$1.23-\$1.73)
- I'll end with \$3.81-\$4.31

Given they returned \$3.87 on \$9 (43%), I should probably return a similar rate or slightly better to encourage continued cooperation.

43% of \$8.31 = \$3.57

50% of \$8.31 = \$4.16

I'll return around \$4.00 to be fair and slightly more generous than they were.

```
Usage: {
 "input_tokens": 2708,
 "output_tokens": 720,
 "reasoning_tokens": 0
}
```

## ROUND 9

Roles: Agent\_1=Sender, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Receiver, Agent\_5=Receiver, Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Sender

### Pairings:

dyad\_1: Agent\_7=Sender, Agent\_4=Receiver  
 dyad\_2: Agent\_1=Sender, Agent\_3=Receiver  
 dyad\_3: Agent\_8=Sender, Agent\_6=Receiver  
 dyad\_4: Agent\_2=Sender, Agent\_5=Receiver

### Parsed game decisions:

dyad\_1: Agent\_7 sent \$3; received \$9; Agent\_4 returned \$3.7; payoffs Agent\_7=\$5.7, Agent\_4=\$5.3;  
 noise/visible: sent visible=\$2.45, received visible=\$7.35, returned visible=\$3.9  
 Action validation: {"returned": {"amount": 3.7, "clamped": false, "max": 9.0, "min": 0, "raw": 3.7}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
 dyad\_2: Agent\_1 sent \$3; received \$9; Agent\_3 returned \$4.74; payoffs Agent\_1=\$6.74, Agent\_3=\$4.26; noise/visible: sent visible=\$3.62, received visible=\$10.86, returned visible=\$5.42  
 Action validation: {"returned": {"amount": 4.74, "clamped": false, "max": 9.0, "min": 0, "raw": 4.74}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
 dyad\_3: Agent\_8 sent \$3; received \$9; Agent\_6 returned \$4.3; payoffs Agent\_8=\$6.3, Agent\_6=\$4.7;  
 noise/visible: sent visible=\$3.1, received visible=\$9.3, returned visible=\$3.83  
 Action validation: {"returned": {"amount": 4.3, "clamped": false, "max": 9.0, "min": 0, "raw": 4.3}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
 dyad\_4: Agent\_2 sent \$2.5; received \$7.5; Agent\_5 returned \$4; payoffs Agent\_2=\$6.5, Agent\_5=\$3.5;  
 noise/visible: sent visible=\$2.89, received visible=\$8.67, returned visible=\$3.62  
 Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 7.5, "min": 0, "raw": 4.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent\_1": 49.629999999999995,  
 "Agent\_2": 50.24,  
 "Agent\_3": 51.29,  
 "Agent\_4": 47.519999999999996,  
 "Agent\_5": 48.330000000000005,  
 "Agent\_6": 44.84,  
 "Agent\_7": 47.25,  
 "Agent\_8": 50.4

}

Visible balances after round: {

"Agent\_1": 49.74,  
 "Agent\_2": 43.699999999999996,  
 "Agent\_3": 48.62,  
 "Agent\_4": 46.749999999999999,  
 "Agent\_5": 44.76,  
 "Agent\_6": 47.26,  
 "Agent\_7": 44.199999999999996,  
 "Agent\_8": 48.33

}

## LLM CALLS

### GAME - Agent\_7 - Sender - dyad\_1

Timestamp: 2026-08-12T09:27:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_4",
"pairing": {
 "agents": [
 "Agent_7",
 "Agent_4"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_4": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized



- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.67, it became \$5.01, and you returned \$3.0. Your visible payoff was \$2.01.

Your total visible earnings across all rounds are \$9.57.

This round, you are the RECEIVER against your current co-player. They sent you \$3.66, so you received \$10.98. How much do you return? (0-10.98)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$3.66 (73% of their endowment) - very high trust!
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining
- Total surplus created: \$2.32

**\*\*My considerations:\*\***

- This is a very generous send showing strong cooperation
- They took significant risk by sending most of their money
- I should reward this trust with a fair return
- In Round 1, I received about 62% back on my \$3 send
- In Round 2, I returned 60% on a smaller send

**\*\*Fair return:\*\***

Given their high trust, I'll return approximately 50% of what I received:

- Return \$5.50
- They'll end with \$6.84 (profit of \$1.84)
- I'll end with \$5.48 (profit of \$0.48)
- This rewards their cooperation while both of us profit

'''json

{"return": 5.50}

'''

**reasoning:**

Let me analyze this situation:

**\*\*Current Situation:\*\***

- My co-player sent me \$3.66 (73% of their endowment)
- I received \$10.98 after the 3x multiplier
- They currently have \$1.34 remaining (\$5 - \$3.66)
- This is a very generous/trusting send

**\*\*What this shows:\*\***

- High trust from the sender (sending over 70%!)
- They're expecting reciprocity
- The multiplier created significant surplus: \$5.98 (\$10.98 - \$5)

**\*\*Considerations:\*\***

- In Round 1, when I sent \$3 (60%), I received \$5.56 back - that's about 62% return rate
- In Round 2, when someone sent \$1.67 (33%), I returned \$3.00 - that's 60% return rate
- This sender showed much more trust (\$3.66 = 73% of endowment)

**\*\*Fair calculation:\*\***

The total pot is now \$12.32 (\$1.34 they kept + \$10.98 I received)

If we split the gains from cooperation:

- Original total: \$10 (my \$5 + their \$5)
- Current total: \$12.32
- Surplus created: \$2.32

- Fair split would give each person \$1.16 gain
- Sender would end at \$6.16, I'd end at \$6.16
- I need to return:  $\$6.16 - \$1.34 = \$4.82$

But let me think about what's fair based on the trust shown:

- They sent 73% of their money
- If I return about 50% of what I received, that's \$5.49
- That would give them  $\$1.34 + \$5.49 = \$6.83$  total

...

**usage:**

```
{
 "input_tokens": 1112,
 "output_tokens": 833,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$5.5. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$15.05.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: I sent \$3 and received \$5.56 back - good cooperation!
- Rounds 2-3: As receiver, I tried to return fairly and reward trust
- Total earnings: \$15.05

**\*\*Considerations for this round:\*\***

- I'm paired with a new co-player each round
- My Round 1 send of \$3 worked well
- The most generous player sent me \$3.66, showing high trust pays off
- Sending more creates more total value (3x multiplier)
- I want to encourage cooperative norms in the pool

**\*\*Strategy:\*\***

I'll send a bit more than Round 1 to signal strong cooperation while managing risk. This shows I'm willing to trust and invest in mutual gains.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Now I'm the SENDER again in Round 4.

Let me review my history:

- Round 1 (SENDER): Sent \$3, received back \$5.56, payoff \$7.56
- Round 2 (RECEIVER): Received \$5.01 (from \$1.67 sent), returned \$3.00, payoff \$2.01
- Round 3 (RECEIVER): Received \$10.98 (from \$3.66 sent), returned \$5.50, payoff \$5.48

This is a new co-player each round in this multi-agent setting.

My experience so far:

- When I sent \$3, I got back \$5.56 - good return!
- When others sent to me, I tried to be fair/cooperative
- The player who sent \$3.66 was very trusting

For this round, I need to decide how much to send to a new co-player.

Considerations:

- I've seen good cooperation so far (my Round 1 partner returned well)
- I want to continue building cooperative patterns

- Sending more creates more value (3x multiplier)
- But I'm paired with a new player each time, so no direct reputation building

Strategy:

- In Round 1, I sent \$3 and it worked well
- I could send more to signal strong cooperation
- Or stick with \$3 as a moderate cooperative amount

Given that I've had positive experiences and want to continue fostering cooperation, I'll send a bit more this time - maybe \$3.5 or \$4. This shows I'm willing to trust while not going all-in.

Let me go with \$3.5 - it's between my previous \$3 send and the very generous \$3.66 I received in Round 3.

Actually, thinking about it more: the player who sent me \$3.66 showed strong trust and I rewarded it. If I want to encourage that kind of cooperation, I should also send generously when I'm the sender. Let me send \$4.

...

**usage:**

```
{
 "input_tokens": 1475,
 "output_tokens": 749,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$3.59 to you. Your visible payoff was \$5.09.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*Recent experience:\*\***

- Round 4: I sent \$3.50, received only \$3.59 back - minimal profit (\$5.09)
- This was disappointing compared to Round 1 where I sent \$3 and got \$5.56 back
- But this is a NEW co-player with unknown behavior

**\*\*Considerations:\*\***

- Different co-players show varying levels of cooperation
- Sending less than \$3.50 reduces my risk after last round's poor return
- Still want to signal willingness to cooperate and create value
- \$3 was my initial send and it worked well - proven strategy

**\*\*Strategy:\*\***

I'll return to sending \$3 - a moderate amount that:

- Shows trust and cooperation
- Creates meaningful value (\$9 for receiver)
- Limits downside risk
- Has proven successful before

'''json

{"send": 3}

'''

**reasoning:**

Looking at my history:

- Round 1: Sent \$3, received \$5.56 back - payoff \$7.56 (good!)
- Round 2: Receiver, returned \$3.00 - payoff \$2.01
- Round 3: Receiver, returned \$5.50 - payoff \$5.48
- Round 4: Sent \$3.5, received \$3.59 back - payoff \$5.09 (disappointing)
- Round 5: Now sender again

Analysis of Round 4:

- I sent \$3.50 (70% of endowment) showing strong trust
- They received \$10.50
- They only returned \$3.59 (34% of what they received)
- I ended with \$5.09, barely breaking even
- This was much less generous than my Round 1 partner who returned \$5.56 on a \$9 receive

This suggests I encountered a less cooperative player. However, I'm now paired with a NEW co-player, so their behavior doesn't necessarily predict this new person's behavior.

Strategic considerations:

- I'm doing reasonably well overall (\$20.14 total)
- Sending creates value through the multiplier
- But I need to balance risk
- Two different responses to my sends: one very good (\$5.56), one mediocre (\$3.59)
- I don't know if there's a pattern or if agents vary widely

I think I should continue showing willingness to cooperate but maybe pull back slightly from the \$3.50 I sent last time since that didn't work out as well. Going back to \$3 seems reasonable - it's a solid middle ground that shows cooperation without overcommitting.

usage:

```
{
 "input_tokens": 1797,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.06 to you. Your visible payoff was \$6.06.

Your total visible earnings across all rounds are \$26.2.

This round, you are the RECEIVER against your current co-player. They sent you \$1.64, so you received \$4.92. How much do you return? (0-4.92)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment) - modest trust
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining

**\*\*My considerations:\*\***

- This is very similar to Round 2 where I received \$5.01 from a \$1.67 send
- In Round 2, I returned \$3.00 (60%) which seemed fair
- I should be consistent in rewarding similar levels of cooperation
- Returning around 60% rewards their trust while keeping reasonable value

**\*\*Fair return:\*\***

Returning approximately 60% of what I received:

- Return \$3.00
- They'll end with \$6.36 (profit of \$1.36)
- I'll end with \$1.92

```
'''json
{"return": 3.00}
'''
```

reasoning:

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment)
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining
- Total surplus created from their send: \$4.92 - \$1.64 = \$3.28

**\*\*My considerations:\*\***

- They showed modest trust by sending \$1.64 (similar to the \$1.67 I saw in Round 2)
- In Round 2, when I received \$5.01 from a \$1.67 send, I returned \$3.00 (60%)
- I should be consistent in how I respond to similar levels of trust
- Returning about 60% would be \$2.95

**\*\*Fair return calculation:\*\***

- If I return around \$3.00:
  - They end with:  $\$3.36 + \$3.00 = \$6.36$  (profit of \$1.36)
  - I end with:  $\$4.92 - \$3.00 = \$1.92$  (loss of \$0.08 from initial endowment perspective, but positive from game perspective)

Actually, let me think about this differently:

- They sent \$1.64
- This created \$3.28 in surplus ( $\$4.92 - \$1.64$ )
- A fair split of surplus would give each about \$1.64
- So they should end with:  $\$5 + \$1.64 = \$6.64$
- They have \$3.36, so I need to return \$3.28 to get them to \$6.64

But that gives me only \$1.64, which seems low for receiving \$4.92.

Let me be consistent with Round 2 where I returned 60% on a similar send:

- 60% of \$4.92 = \$2.95

...

**usage:**

```
{
 "input_tokens": 2142,
 "output_tokens": 677,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.64, it became \$4.92, and you returned \$3.0. Your visible payoff was \$1.92.

Your total visible earnings across all rounds are \$28.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My sender history:\*\***

- Round 1: Sent \$3 ? got \$5.56 back (profit \$3.56) ?
- Round 4: Sent \$3.50 ? got \$3.59 back (profit \$0.09) ?
- Round 5: Sent \$3 ? got \$4.06 back (profit \$1.06) ?

**\*\*Observations:\*\***

- \$3 has been my most reliable send amount (2/2 reasonable returns)
- \$3.50 didn't work out well
- Each round is a new co-player, so no personal history
- Total earnings: ~\$28.12 - healthy position

**\*\*Strategy:\*\***

Stick with \$3 as my standard opening send:

- Proven track record
- Shows cooperation and trust
- Creates meaningful value (\$9 for receiver)
- Manages downside risk
- Signals I'm a fair player

'''json

```
{"send": 3}
```

'''

**reasoning:**

Looking at my history:

- Round 1 (Sender): Sent \$3, got \$5.56 back - Good!
- Round 2 (Receiver): Received \$5.01, returned \$3.00
- Round 3 (Receiver): Received \$10.98, returned \$5.50
- Round 4 (Sender): Sent \$3.50, got \$3.59 back - Poor return
- Round 5 (Sender): Sent \$3, got \$4.06 back - Okay return
- Round 6 (Receiver): Received \$4.92, returned \$3.00

As SENDER:

- Round 1: Sent \$3 ? returned \$5.56 (profit \$3.56)
- Round 4: Sent \$3.50 ? returned \$3.59 (profit \$0.09)
- Round 5: Sent \$3 ? returned \$4.06 (profit \$1.06)

Total earnings: ~\$28.12

New co-player again.

The pattern shows:

- Sending \$3 has worked reasonably well (2 out of 3 times)
- Sending \$3.50 got poor returns
- Each round is a new partner with unknown behavior

I should continue with a moderate send that balances trust-building with risk management.  
\$3 seems to be a good baseline that's worked before.

Actually, let me think about the broader strategy. I've been fairly cooperative as a receiver, returning good amounts. As a sender, I've been moderately trusting.

Given that:

1. This is a new co-player
2. \$3 has generally worked well
3. I want to signal cooperation while managing risk
4. Total earnings are healthy at \$28.12

I'll stick with \$3 as my standard opening send.

usage:

```
{
 "input_tokens": 2476,
 "output_tokens": 676,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.87 to you. Your visible payoff was \$5.87.

Your total visible earnings across all rounds are \$33.989999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$2.77 (55% of their endowment) - good trust
- I received \$8.31 after the 3x multiplier
- They currently have \$2.23 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 7!
- Last round: I sent them \$3.0, they returned \$3.87 (43% of what they received)
- They showed fair cooperation
- Now they're reciprocating with a solid send of \$2.77

**\*\*My considerations:\*\***

- They treated me fairly last round

- I want to reciprocate and encourage continued cooperation
- Returning around 50% would be fair: ~\$4.15
- This rewards their trust while both of us profit

**\*\*Fair return:\*\***

- Return \$4.00
- They'll end with \$6.23 (profit of \$1.23)
- I'll end with \$4.31 (profit of -\$0.69 relative to endowment, but fair given the exchange)

```
'''json
{"return": 4.00}
'''
```

#### **reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$2.77 (55% of their endowment) - moderate-high trust
- I received \$8.31 after the 3x multiplier
- They currently have \$2.23 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 7!
- Last round: I sent them \$3.0, they returned \$3.87 to me
- They showed reasonable cooperation last round (43% return rate)
- Now they've sent me \$2.77 (slightly less than my \$3, but still solid)

**\*\*My considerations:\*\***

- This co-player returned \$3.87 on my \$3 send - that was fair/decent
- They're now showing similar trust by sending \$2.77
- I should reciprocate their fairness
- Looking at my past receiver behavior:
  - Round 2: Received \$5.01, returned \$3.00 (60%)
  - Round 3: Received \$10.98, returned \$5.50 (50%)
  - Round 6: Received \$4.92, returned \$3.00 (61%)

**\*\*Fair return:\*\***

They sent me about 55% of their endowment. If I return about 50-55% of what I received:

- Return around \$4.00-\$4.50
- They'll end with \$6.23-\$6.73 (profit of \$1.23-\$1.73)
- I'll end with \$3.81-\$4.31

Given they returned \$3.87 on \$9 (43%), I should probably return a similar rate or slightly better to encourage continued cooperation.

43% of \$8.31 = \$3.57

50% of \$8.31 = \$4.16

I'll return around \$4.00 to be fair and slightly more generous than they were.

#### **usage:**

```
{
 "input_tokens": 2708,
 "output_tokens": 720,
 "reasoning_tokens": 0
}
```

#### **Message 14 role=user**

##### **content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.77, it became \$8.31, and you returned \$4.0. Your visible payoff was \$4.3100000000000005.

Your total visible earnings across all rounds are \$38.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

#### **Assistant output:**

Looking at this round as SENDER:

```
My sender history:
- Round 1: Sent $3 ? got $5.56 back (profit $3.56) ?
- Round 4: Sent $3.50 ? got $3.59 back (profit $0.09) ?
- Round 5: Sent $3 ? got $4.06 back (profit $1.06) ?
- Round 7: Sent $3 ? got $3.87 back (profit $0.87) ?
```

```
Observations:
- $3 has been very consistent (3/3 good returns)
- Total earnings: $38.30 - strong position
- This is a NEW co-player
```

```
Strategy:
Continue with my proven approach of sending $3:
- Reliable track record
- Shows cooperative intent
- Creates meaningful value ($9 for receiver)
- Balanced risk/reward
```

```
'''json
{"send": 3}
'''
```

#### Reasoning:

Looking at this round as SENDER:

```
My sender history:
- Round 1: Sent $3 ? got $5.56 back (profit $3.56) ?
- Round 4: Sent $3.50 ? got $3.59 back (profit $0.09) ?
- Round 5: Sent $3 ? got $4.06 back (profit $1.06) ?
- Round 7: Sent $3 ? got $3.87 back (profit $0.87) ?
```

```
Observations:
- $3 continues to work well as my standard send (3/3 positive returns)
- Total earnings: $38.30 - strong position
- New co-player this round (rounds 7-8 was the same co-player)
```

```
Strategy:
Continue with $3 as my reliable opening send:
- Consistent track record
- Shows cooperation
- Creates value ($9 for receiver)
- Balanced risk/reward
- Establishes trust
```

I'll send \$3.

```
Usage: {
 "input_tokens": 2737,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_4 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:28:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_4"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_4": "trustee",
 "Agent_7": "investor"
 },
 "trustee": "Agent_4"
 }
}
```



```

},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.51 to you. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$11.469999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.27, so you received \$6.8100000000000005. How much do you return? (0-6.8100000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.27 (45.4% of their endowment), showing moderate trust
- I received \$6.81 after the 3x multiplier
- The sender currently has \$2.73 remaining

In the previous round when I was sender, I sent 60% and received back about 50% of what my co-player received. I'll reciprocate with a similar return rate.

Returning \$3.50 gives:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return, which fairly rewards their trust.

```
{'return': 3.50}
```

**reasoning:**

Let me analyze this situation:

- The sender sent \$2.27, which is 45.4% of their \$5 endowment
- I received \$6.81 after the 3x multiplier
- The sender currently has  $\$5 - \$2.27 = \$2.73$  remaining

In the previous round when I was sender:

- I sent \$3.00 (60% of endowment)
- Received back \$4.51, which was about 50% of the \$9.00 they received
- This gave me a payoff of \$6.51

The sender this round showed moderate trust (45.4% vs my 60% last round). I should reciprocate fairly.

If I return \$3.50:

- Sender ends with:  $\$2.73 + \$3.50 = \$6.23$  (profit of \$1.23)
- I end with:  $\$6.81 - \$3.50 = \$3.31$

This represents about 51% return rate, similar to what I received last round (50%). This seems fair and reciprocal.

Let me check the math:

- Return rate =  $\$3.50 / \$6.81 = 51.4\%$
- Sender's final payoff =  $\$2.73 + \$3.50 = \$6.23$
- My final payoff =  $\$6.81 - \$3.50 = \$3.31$

This seems reasonable and reciprocal given their 45.4% trust level.

**usage:**

```
{
 "input_tokens": 1104,
 "output_tokens": 545,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.8100000000000005, and you returned \$3.5. Your visible payoff was \$3.3100000000000005.

Your total visible earnings across all rounds are \$14.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience so far:

- When I've shown trust as sender (60% in Round 2), I received good reciprocity (~50% return)
- When others showed trust as sender, I've returned approximately 50%
- The pattern suggests cooperative play with moderate to high trust levels

I'll continue with a similar strategy, sending \$3.00 (60% of my endowment) to maintain trust and cooperation.

{'send': 3.00}

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.32 (66.4%)
- I received \$9.96
- I returned \$5.00 (about 50%)
- My payoff: \$4.96

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.51 (about 50%)
- My payoff: \$6.51

Round 3: I was RECEIVER

- Sender sent \$2.27 (45.4%)
- I received \$6.81
- I returned \$3.50 (about 51%)
- My payoff: \$3.31

Round 4: I am SENDER now

- I have \$5 to decide what to send

Looking at the pattern:

- In Round 2, I sent \$3.00 (60%) and got back about 50%
- Others have sent between 45% and 66% in the rounds I've seen

I've been experiencing decent reciprocity so far. The return rates have been around 50% consistently.

I should continue to show trust, as it's been rewarded. Sending around 60% seems reasonable based on what I've seen. I'll send \$3.00 again.

**usage:**

```
{
 "input_tokens": 1421,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$21.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience as sender:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)

Sending 60% of my endowment has consistently led to good reciprocity. I'll maintain this cooperative strategy.

{'send': 3.00}

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: I sent \$3.00, received \$4.44 back (payoff \$6.44)

Both times sending \$3.00 has resulted in good returns. The strategy of sending 60% has been working well and establishing cooperative relationships.

I should continue with this consistent strategy to maintain trust and cooperation.

**usage:**

```
{
 "input_tokens": 1648,
 "output_tokens": 217,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.38 to you. Your visible payoff was \$6.38.

Your total visible earnings across all rounds are \$27.599999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.26, so you received \$9.78. How much do you return? (0-9.78)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

This is similar to Round 1 where I received high trust. I've been consistently returning a round 50% as receiver, which has worked well.

Returning \$5.00:

- Sender ends with:  $\$1.74 + \$5.00 = \$6.74$  (profit of \$1.74)
- I end with:  $\$9.78 - \$5.00 = \$4.78$

This maintains my reciprocal strategy.

{'return': 5.00}

**reasoning:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

In my previous rounds as receiver:

- Round 1: Received \$9.96, returned \$5.00 (50.2%)
- Round 3: Received \$6.81, returned \$3.50 (51.4%)

As sender, I've been sending \$3.00 and receiving returns of around \$4.40-4.50, which is about 49-50% of what the receiver got.

This sender showed high trust (65.2%). I should reciprocate with a similar 50% return rate.

Returning \$5.00:

- Sender ends with:  $\$1.74 + \$5.00 = \$6.74$  (profit of \$1.74)
- I end with:  $\$9.78 - \$5.00 = \$4.78$

This is about 51% return, consistent with my previous behavior as receiver.

**usage:**

```
{
 "input_tokens": 1888,
 "output_tokens": 439,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$5.0. Your visible payoff was \$4.



```
{'return': 4.20}
```

**reasoning:**

Looking at this round:

- The sender sent \$2.81 (56.2% of their endowment), showing moderate-high trust
- I received \$8.43 after the 3x multiplier
- The sender currently has \$2.19 remaining

My pattern as receiver:

- Round 1: Received \$9.96, returned \$5.00 (50.2% return)
- Round 3: Received \$6.81, returned \$3.50 (51.4% return)
- Round 6: Received \$9.78, returned \$5.00 (51.1% return)

I've been consistently returning around 50-51% of what I receive. This has been a fair and reciprocal strategy.

For this round:

Returning \$4.20:

- Sender ends with:  $\$2.19 + \$4.20 = \$6.39$  (profit of \$1.39)
- I end with:  $\$8.43 - \$4.20 = \$4.23$

This is about 49.8% return, which is consistent with my previous behavior.

Actually, let me be more precise. 50% of \$8.43 would be \$4.215, which I could round to \$4.20 or \$4.25.

I'll go with \$4.20 to keep it consistent with my ~50% return pattern.

**usage:**

```
{
 "input_tokens": 2134,
 "output_tokens": 484,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$4.2. Your visible payoff was \$4.2299999999999995.

Your total visible earnings across all rounds are \$43.099999999999994.

This round, you are the RECEIVER against your current co-player. They sent you \$2.45, so you received \$7.3500000000000005. How much do you return? (0-7.3500000000000005)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$2.45 (49% of their endowment), showing moderate trust
- I received \$7.35 after the 3x multiplier
- The sender currently has \$2.55 remaining

Following my consistent strategy of returning approximately 50% of what I receive:

Returning \$3.70:

- Sender ends with:  $\$2.55 + \$3.70 = \$6.25$  (profit of \$1.25)
- I end with:  $\$7.35 - \$3.70 = \$3.65$

```
{'return': 3.70}
```

**Reasoning:**

Looking at this situation:

- The sender sent \$2.45 (49% of their endowment), showing moderate trust
- I received \$7.35 after the 3x multiplier
- The sender currently has \$2.55 remaining

I've been consistently returning approximately 50% of what I receive as a receiver. Let me calculate:

50% of \$7.35 = \$3.675

Returning \$3.70:

- Sender ends with:  $\$2.55 + \$3.70 = \$6.25$  (profit of \$1.25)

- I end with: \$7.35 - \$3.70 = \$3.65

This maintains my consistent reciprocal strategy.

```
Usage: {
 "input_tokens": 2218,
 "output_tokens": 316,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Sender - dyad\_2

Timestamp: 2026-08-12T09:28:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_3"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_3": "trustee"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.73 to you. Your visible payoff was \$4.73.

Your total visible earnings across all rounds are \$9.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my history:

- Round 1 (as receiver): Co-player sent \$3.34, I returned \$5.00
- Round 2 (as sender): I sent \$3.00, received back \$2.73

Now with a new co-player in Round 3, I should continue demonstrating willingness to cooperate. Sending a consistent amount shows I'm a reliable partner.

{'send': 3.00}

**reasoning:**

In Round 2, I sent \$3.00, it became \$9.00, and the receiver returned \$2.73.

Let me calculate:

- I started with \$5
- I sent \$3, so I had \$2 left
- They received \$9 and returned \$2.73
- My payoff was \$2 + \$2.73 = \$4.73

The receiver kept \$9 - \$2.73 = \$6.27

I got \$4.73

So the return rate was  $\$2.73/\$9.00 = 30.3\%$

This is lower than what I returned in Round 1 (I returned \$5.00 out of \$10.02, which is about 50%).

However, this is a new co-player in Round 3. I shouldn't punish this new person for what the previous receiver did.

I should continue to try to establish cooperation. Maybe send a similar amount to show consistency and willingness to cooperate. However, given that the last return was somewhat low, I might be slightly more cautious.

Let me send \$2.50 - still showing trust but being a bit more conservative.

Actually, let me think about this differently. Each round is with a different co-player (the problem states pairings are randomized). So my actions should be about establishing cooperation with each new partner, not about punishing previous partners.

I should continue to send a reasonable amount to establish trust. Let me send \$3.00 again to be consistent and show I'm willing to cooperate.

**usage:**

```
{
 "input_tokens": 1034,
```



```

 "output_tokens": 456,
 "reasoning_tokens": 0
 }
Message 4 role=user
 content:
 Round 4

 In your most recent previous game round, you were the SENDER against another agent. You sent $3.0, it became $9.0, and you saw them return $5.57 to you. Your visible payoff was $7.57.
 Your total visible earnings across all rounds are $17.32.

 This round, you are the RECEIVER against your current co-player. They sent you $2.99, so you received $8.97. How much do you return? (0-8.97)
 Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
 content:
 Looking at this situation:
 - The sender showed trust by sending $2.99 (about 60% of their endowment)
 - I received $8.97 after the 3x multiplier
 - The sender currently has $2.01

 I want to reciprocate their trust fairly. If I return $4.50:
 - Sender ends with: $2.01 + $4.50 = $6.51 (profit of $1.51)
 - I keep: $4.47

 This rewards their cooperation while maintaining a fair split of the gains.

 {'return': 4.50}
 reasoning:
 Let me analyze this round:
 - I'm the receiver
 - The sender sent $2.99 (about 60% of their $5 endowment)
 - I received $8.97 after the 3x multiplier
 - The sender currently has $5 - $2.99 = $2.01

 The sender showed good trust by sending nearly 60% of their endowment. I should reciprocate fairly.

 If I return $4.50:
 - Sender ends with: $2.01 + $4.50 = $6.51 (better than their initial $5)
 - I keep: $8.97 - $4.50 = $4.47

 This is fair and rewards their trust. It's roughly half of what I received, which seems reasonable given they sent about 60% of their endowment.

 Looking at my previous behavior:
 - Round 1: Received $10.02, returned $5.00 (kept $5.02)
 - Round 3: The other player returned $5.57 when they received $9.00

 I should be consistent and fair. Returning around $4.50 seems appropriate.
 usage:
 {
 "input_tokens": 1269,
 "output_tokens": 416,
 "reasoning_tokens": 0
 }
Message 6 role=user
 content:
 Round 5

 In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $2.99, it became $8.97, and you returned $4.5. Your visible payoff was $4.470000000000001.
 Your total visible earnings across all rounds are $21.79.

 This round, you are the RECEIVER against your current co-player. They sent you $2.89, so you received $8.67. How much do you return? (0-8.67)
 Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant

```

**content:**

Looking at this situation:

- The sender sent \$2.89 (about 58% of their endowment), showing trust
- I received \$8.67 after the 3x multiplier
- The sender currently has \$2.11

I'll reciprocate fairly. If I return \$4.30:

- Sender ends with:  $\$2.11 + \$4.30 = \$6.41$  (profit of \$1.41)
- I keep: \$4.37

This rewards cooperation while maintaining a fair distribution.

```
{'return': 4.30}
```

**reasoning:**

The sender sent \$2.89 (about 58% of their \$5 endowment), showing reasonable trust.

I received \$8.67 after the 3x multiplier.

The sender currently has  $\$5 - \$2.89 = \$2.11$

I should reciprocate fairly. Looking at my previous behavior:

- Round 1: Sender sent \$3.34, I received \$10.02, returned \$5.00
- Round 4: Sender sent \$2.99, I received \$8.97, returned \$4.50

This sender sent slightly less (\$2.89 vs \$2.99), so I should return proportionally. If I return \$4.30:

- Sender ends with:  $\$2.11 + \$4.30 = \$6.41$  (profit of \$1.41)
- I keep: \$4.37

This maintains fair reciprocation.

**usage:**

```
{
 "input_tokens": 1545,
 "output_tokens": 349,
 "reasoning_tokens": 0
}
```

**Message 8 role=user****content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.89, it became \$8.67, and you returned \$4.3. Your visible payoff was \$4.37.

Your total visible earnings across all rounds are \$26.16.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant****content:**

Looking at my pattern:

- As a sender, I've consistently sent \$3.00 to show trust and encourage cooperation
- As a receiver, I've been returning amounts that fairly reward the sender's trust

I'll maintain my consistent cooperative approach with this new co-player.

```
{'send': 3.00}
```

**reasoning:**

Looking at my history:

- Round 1 (receiver): Received \$10.02 from sender's \$3.34, returned \$5.00
- Round 2 (sender): Sent \$3.00, received back \$2.73
- Round 3 (sender): Sent \$3.00, received back \$5.57
- Round 4 (receiver): Received \$8.97 from sender's \$2.99, returned \$4.50
- Round 5 (receiver): Received \$8.67 from sender's \$2.89, returned \$4.30

Now Round 6, I'm a sender again with a new co-player.

My sending pattern has been consistent at \$3.00 when I'm the sender. The returns I've received have varied (\$2.73 and \$5.57), but both times my co-players received \$9.00 and chose to return different amounts.

I should continue with a consistent cooperative strategy. Sending \$3.00 has been my standard

rd, showing significant trust (60% of endowment).

```
{'send': 3.00}
usage:
{
 "input_tokens": 1796,
 "output_tokens": 332,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.33 to you. Your visible payoff was \$6.33.

Your total visible earnings across all rounds are \$32.49.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.78 (about 56% of their endowment), showing trust
- I received \$8.34 after the 3x multiplier
- The sender currently has \$2.22

I'll reciprocate fairly. If I return \$4.20:

- Sender ends with:  $\$2.22 + \$4.20 = \$6.42$  (profit of \$1.42)
- I keep: \$4.14

This maintains my pattern of fairly rewarding cooperation.

```
{'return': 4.20}
```

**reasoning:**

The sender sent \$2.78 (about 55.6% of their endowment), showing trust.

I received \$8.34 after the 3x multiplier.

The sender currently has  $\$5 - \$2.78 = \$2.22$ .

I've been fair in my returns as a receiver:

- Round 1: Received \$10.02, returned \$5.00
- Round 4: Received \$8.97, returned \$4.50
- Round 5: Received \$8.67, returned \$4.30

For this round with \$8.34 received, a fair return would be around \$4.00-\$4.20.

If I return \$4.20:

- Sender ends with:  $\$2.22 + \$4.20 = \$6.42$  (profit of \$1.42)
- I keep:  $\$8.34 - \$4.20 = \$4.14$

This is consistent with my previous fair returns.

**usage:**

```
{
 "input_tokens": 2001,
 "output_tokens": 368,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$4.2. Your visible payoff was \$4.14.

Your total visible earnings across all rounds are \$36.63.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my track record:

- I've consistently sent \$3.00 as a sender, showing reliability and trust
- This strategy has generally been reciprocated well
- As a receiver, I've been fair in my returns

I'll maintain my consistent cooperative approach.

```
{'send': 3.00}
```

**reasoning:**

Looking at my history as a sender:

- Round 2: Sent \$3.00, received back \$2.73 (payoff: \$4.73)
- Round 3: Sent \$3.00, received back \$5.57 (payoff: \$7.57)
- Round 6: Sent \$3.00, received back \$4.33 (payoff: \$6.33)

As a receiver, I've been consistently fair, returning roughly half of what I received or a bit less, ensuring the sender profits from trusting me.

I've established a pattern of sending \$3.00 as a sender, which shows good trust and has generally been reciprocated well. I'll continue this consistent approach.

This is Round 8 with a new co-player. I should maintain my cooperative strategy.

**usage:**

```
{
 "input_tokens": 2031,
 "output_tokens": 263,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.689999999999995.

Your total visible earnings across all rounds are \$42.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my consistent strategy:

- I've sent \$3.00 in every sender round, showing reliability
- Returns have varied but overall been reasonable
- This maintains trust and encourages cooperation

I'll continue with my established pattern.

```
{'send': 3.00}
```

**Reasoning:**

Let me review my history:

- As sender: I've consistently sent \$3.00 in rounds 2, 3, 6, 8, and now 9
- Returns I've received: \$2.73, \$5.57, \$4.33, \$3.69
- As receiver: I've returned \$5.00, \$4.50, \$4.30, \$4.20

My strategy of sending \$3.00 has worked reasonably well, though returns vary. I've built up good earnings (\$42.32 total).

The last round return of \$3.69 was a bit lower than some others, but still represents reasonable cooperation (I profited \$0.69 on that round).

I should continue with my consistent approach of sending \$3.00. This:

1. Shows I'm a reliable cooperative partner
2. Maximizes total surplus
3. Has generally been reciprocated
4. Maintains my reputation across different co-players

I'll stick with \$3.00.

Usage: {

```
"input_tokens": 2007,
"output_tokens": 298,
"reasoning_tokens": 0
}
```

### GAME - Agent\_3 - Receiver - dyad\_2

Timestamp: 2026-08-12T09:28:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_3"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_3": "trustee"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

###### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### GAME:

You will play different roles across multiple rounds.

###### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.43 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$9.780000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my experience:

- Round 1 (receiver): Sender sent 47%, I reciprocated fairly
- Round 2 (sender): I sent 50%, received fair return of \$2.43

The moderate trust strategy (around 50%) has been working well with fair reciprocation from co-players. I'll continue this approach.

{'send': 2.50}

**reasoning:**

Let me analyze the previous round:

- I sent \$2.50 (50% of my endowment)
- It became \$7.50 for the receiver
- They returned \$2.43
- My payoff:  $\$5 - \$2.50 + \$2.43 = \$4.93$
- Their payoff:  $\$7.50 - \$2.43 = \$5.07$

The receiver returned a fair amount - roughly aimed at equal splits. The return rate was  $\$2.43/\$7.50 = 32.4\%$ , which gave us both similar final payoffs.

Now I'm a sender again with a new co-player. I've seen:

- Round 1 (as receiver): Sender sent 47%, I returned fairly
- Round 2 (as sender): I sent 50%, received fair return

Both rounds showed moderate trust and fair reciprocation. I should continue this cooperative strategy. Sending around 50% seems to be working well.

{'send': 2.50}

**usage:**

```
{
 "input_tokens": 1060,
 "output_tokens": 327,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.78 to you. Your visible payoff was \$5.279999999999999.

Your total visible earnings across all rounds are \$15.06.

This round, you are the RECEIVER against your current co-player. They sent you \$2.58, so you received \$7.74. How much do you return? (0-7.74)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.58 (51.6% of endowment), showing solid trust
- I received \$7.74
- Sender currently has \$2.42

Based on my positive experiences with fair reciprocation, I'll aim for equal final payoffs:

- Total:  $\$2.42 + \$7.74 = \$10.16$
- Equal split:  $\$5.08$  each
- Return needed:  $\$2.66$

`{'return': 2.66}`

**reasoning:**

I'm the receiver now. Let me analyze:

- Sender sent \$2.58 (51.6% of their endowment)
- I received \$7.74 (after 3x multiplier)
- Sender currently has:  $\$5 - \$2.58 = \$2.42$

My recent experience shows:

- Round 2: I sent \$2.50, got back \$2.43
- Round 3: I sent \$2.50, got back \$2.78

Both returns were generous and aimed at fair splits. I should continue this reciprocal fairness.

For equal final payoffs:

- Total available:  $\$2.42$  (sender's remainder) +  $\$7.74$  (my received) =  $\$10.16$
- Equal split:  $\$10.16 / 2 = \$5.08$  each
- Return needed:  $\$5.08 - \$2.42 = \$2.66$

This would give us both  $\$5.08$ , which is fair and reciprocates the sender's trust at 51.6%.

Let me verify:

- Sender final:  $\$2.42 + \$2.66 = \$5.08$
- My final:  $\$7.74 - \$2.66 = \$5.08$

Perfect. I'll return  $\$2.66$ .

**usage:**

```
{
 "input_tokens": 1287,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.58, it became \$7.74, and you returned \$2.66. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: `{'send': <amount>}`

**Message 7 role=assistant**

**content:**

Looking at my track record:

- Sending around 50% has consistently resulted in fair reciprocation
- Other agents are also showing similar moderate trust levels
- This strategy has been mutually beneficial

I'll continue with this successful approach.

`{'send': 2.50}`

**reasoning:**

Let me review my history:

- Round 1 (receiver): Received from sender who sent \$2.35 (47%), returned \$2.20, payoff \$4

- .85
- Round 2 (sender): Sent \$2.50 (50%), got back \$2.43, payoff \$4.93
  - Round 3 (sender): Sent \$2.50 (50%), got back \$2.78, payoff \$5.28
  - Round 4 (receiver): Received from sender who sent \$2.58 (51.6%), returned \$2.66, payoff \$5.08
  - Round 5 (sender): ???

My strategy of sending around 50% has been working well - I've been getting fair returns.

The pattern shows:

- When I send ~50% (\$2.50), I get back \$2.43-\$2.78
- Others are sending similar amounts (47-52%)
- Fair reciprocation is the norm

I should continue with this moderate trust approach around 50%.

```
{'send': 2.50}
```

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 316,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.24 to you. Your visible payoff was \$4.74.

Your total visible earnings across all rounds are \$24.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$2.72, so you received \$8.16. How much do you return? (0-8.16)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.72 (54.4% of endowment), showing strong trust
- I received \$8.16
- Sender currently has \$2.28

For equal final payoffs:

- Total: \$2.28 + \$8.16 = \$10.44
- Equal split: \$5.22 each
- Return needed: \$2.94

```
{'return': 2.94}
```

**reasoning:**

Let me analyze this situation:

- Sender sent \$2.72 (54.4% of their \$5 endowment) - showing good trust
- I received \$8.16
- Sender currently has \$5 - \$2.72 = \$2.28

For equal final payoffs:

- Total available: \$2.28 + \$8.16 = \$10.44
- Equal split: \$10.44 ÷ 2 = \$5.22 each
- Return needed: \$5.22 - \$2.28 = \$2.94

This sender showed slightly more trust than average (54.4% vs the ~50% I've been seeing), so I should reciprocate fairly to maintain cooperation.

Let me return \$2.94 to achieve equal payoffs.

**usage:**

```
{
 "input_tokens": 1725,
 "output_tokens": 298,
 "reasoning_tokens": 0
}
```



**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.72, it became \$8.16, and you returned \$2.94. Your visible payoff was \$5.220000000000001.

Your total visible earnings across all rounds are \$30.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$3.36 (67.2% of endowment), showing very high trust - much higher than typical
- I received \$10.08
- Sender currently has \$1.64

For equal final payoffs:

- Total: \$1.64 + \$10.08 = \$11.72
- Equal split: \$5.86 each
- Return needed: \$4.22

{'return': 4.22}

**reasoning:**

Looking at this situation:

- Sender sent \$3.36 (67.2% of endowment), showing very high trust
- I received \$10.08
- Sender currently has \$1.64

This is notably higher trust than I've seen in previous rounds (most have been around 50%). I should reciprocate generously.

For equal final payoffs:

- Total: \$1.64 + \$10.08 = \$11.72
- Equal split: \$5.86 each
- Return needed: \$5.86 - \$1.64 = \$4.22

This sender showed significant trust, so I should return a fair amount that rewards that trust.

**usage:**

```
{
 "input_tokens": 1965,
 "output_tokens": 273,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.22. Your visible payoff was \$5.86.

Your total visible earnings across all rounds are \$35.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my experience:

- My moderate trust strategy (sending ~50%) has been consistently reciprocated fairly
- I've seen varying levels of trust from others, with fair returns in most cases
- Average earnings have been solid with this approach

I'll continue with my tested strategy.

```
{'send': 2.50}
```

**reasoning:**

Looking at my history:

- I've been consistently reciprocating fairly when I'm the receiver
- When I'm the sender, I've been sending around \$2.50 (50%)
- The last round showed someone sending \$3.36 (67.2%), which is much higher than average
- I've been getting fair returns overall

My strategy of sending around 50% has been working well. I could consider increasing slightly given the positive reciprocity I've seen, but the moderate approach has been stable and mutually beneficial.

I'll stick with my successful strategy of around 50%.

**usage:**

```
{
 "input_tokens": 1938,
 "output_tokens": 208,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.04 to you. Your visible payoff was \$6.54.

Your total visible earnings across all rounds are \$42.5.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- Sender sent \$3.62 (72.4% of endowment), showing very high trust
- I received \$10.86
- Sender currently has \$1.38

For equal final payoffs:

- Total: \$1.38 + \$10.86 = \$12.24
- Equal split: \$6.12 each
- Return needed: \$4.74

```
{'return': 4.74}
```

**Reasoning:**

Looking at this situation:

- Sender sent \$3.62 (72.4% of endowment), showing very high trust
- I received \$10.86
- Sender currently has \$1.38

For equal final payoffs:

- Total: \$1.38 + \$10.86 = \$12.24
- Equal split: \$6.12 each
- Return needed: \$4.74

This sender showed exceptional trust (72.4% is quite high). I should reciprocate fairly with equal split approach.

Usage: {

```
 "input_tokens": 1936,
 "output_tokens": 234,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_8 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:28:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_6",
"pairing": {
 "agents": [
 "Agent_8",
 "Agent_6"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_6": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.19, it became \$9.57, and you returned \$4.0. Your visible payoff was \$5.57.  
Your total visible earnings across all rounds are \$10.030000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant****content:**

Looking at this new round as SENDER:

**\*\*Past experience:\*\***

- Round 1: Sent \$3, received \$2.46 back (modest cooperation)
- Round 2: When I received, I returned fairly (\$4.00 on \$9.57)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Need to establish trust with someone new
- The 3x multiplier means cooperation creates value

A moderate send amount balances the potential gains from cooperation with the risk of low returns. Sending \$3 creates \$9 of value while maintaining some protection.

{'send': 3}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds:\*\***

- Round 1: I was SENDER, sent \$3, received back \$2.46, payoff was \$4.46
- Round 2: I was RECEIVER, received \$9.57 (from \$3.19 sent), returned \$4.00, payoff was \$5.57

**\*\*Current Round (Round 3):\*\***

- I'm the SENDER again
- New co-player (since pairings are randomized each round)
- I have \$5 to potentially send

**\*\*Observations:\*\***

1. In Round 1, when I sent \$3, I got back \$2.46. The receiver kept \$6.54 out of \$9.00, which seems somewhat selfish (I got less than my initial \$3).
2. In Round 2, when someone sent me \$3.19, I returned \$4.00, being relatively fair.
3. The note about communication noise is important - amounts may differ from what was actually sent/returned.

**\*\*Strategy considerations:\*\***

- I'm playing with a new co-player each round
- There's no reputation to build with a specific player
- The 3x multiplier means cooperation creates value
- I need to balance trust against the risk of low returns

Given:

- My Round 1 experience showed modest cooperation from my receiver
- I was reasonably fair in Round 2
- I'm with a new player now who doesn't know my history

I think sending \$3 again is reasonable:

- It's a moderate level of trust
- Creates \$9 for the receiver
- Provides some protection if they don't reciprocate well

Let me send \$3 again.

**usage:**

```
{
 "input_tokens": 1047,
 "output_tokens": 532,
 "reasoning_tokens": 0
}
```

}

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$14.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.22, so you received \$9.66. How much do you return? (0-9.66)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 4):\*\***

- Co-player sent \$3.22 ? I received \$9.66
- This represents significant trust (sending 64% of their endowment)

**\*\*Fair return analysis:\*\***

- Returning \$4.00 would give:
  - Sender:  $\$5 - \$3.22 + \$4.00 = \$5.78$  (profit of \$0.78)
  - Me:  $\$9.66 - \$4.00 = \$5.66$  (profit of \$0.66)

This creates a fair split where both players benefit from the cooperation.

{'return': 4.00}

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER:\*\***

- My co-player sent me \$3.22
- I received \$9.66 (after 3x multiplier)
- I need to decide how much to return

**\*\*Previous rounds analysis:\*\***

- Round 1 (SENDER): Sent \$3, got back \$2.46 - slightly below fair
- Round 2 (RECEIVER): Received \$9.57, returned \$4.00 - tried to be fair
- Round 3 (SENDER): Sent \$3, got back \$2.93 - again slightly below fair

**\*\*Fair return calculation:\*\***

The sender gave up \$3.22 from their \$5 endowment, creating \$9.66 for me.

- Surplus created:  $\$9.66 - \$3.22 = \$6.44$

A fair split would give us each roughly equal gains:

- If I return \$4.00:
  - Sender gets:  $\$5 - \$3.22 + \$4.00 = \$5.78$  (gain of \$0.78)
  - I get:  $\$9.66 - \$4.00 = \$5.66$  (gain of \$0.66)

This seems reasonably balanced. The sender sent a significant portion of their endowment, showing trust.

I should return around \$4.00 to be fair and reciprocate their trust.

**usage:**

```
{
 "input_tokens": 1324,
 "output_tokens": 501,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.22, it became \$9.66, and you returned \$4.0. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$20.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my experience so far:

**\*\*As SENDER:\*\***

- Round 1: Sent \$3 ? received \$2.46 (modest return)
- Round 3: Sent \$3 ? received \$2.93 (decent return)

**\*\*As RECEIVER:\*\***

- I've consistently returned \$4.00 on ~\$9.60, maintaining fairness

**\*\*Current round:\*\***

- New co-player (randomized pairing)
- Sending \$3 has been my standard opening move
- Creates \$9 of value with moderate risk

I'll continue with the balanced approach that's been working.

{ 'send': 3 }

**reasoning:**

Looking at my history as SENDER:

- Round 1: Sent \$3, received \$2.46 back
- Round 3: Sent \$3, received \$2.93 back

Both times I sent \$3, I received back around \$2.50-\$3.00, which is close to fair but not quite equal splitting of the gains.

As RECEIVER:

- Round 2: Received \$9.57, returned \$4.00
- Round 4: Received \$9.66, returned \$4.00

I've been consistent in returning fairly.

Now Round 5, I'm a SENDER with a new co-player. The pattern suggests that sending around \$3 has been reasonably productive. The returns haven't been amazing (I've profited minimally as sender), but they've been positive.

I should continue with a moderate strategy. Sending \$3 creates value and has worked reasonably well so far.

**usage:**

```
{
 "input_tokens": 1601,
 "output_tokens": 366,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.2 to you. Your visible payoff was \$7.2.

Your total visible earnings across all rounds are \$27.82.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This is a slightly smaller send than previous rounds

**\*\*Fair return analysis:\*\***

- In previous rounds, I returned ~\$4.00 on receives of ~\$9.60

- This represents about a 42% return rate
- For \$8.58, a proportional return would be around \$3.60

Returning \$3.70 would give:

- Sender:  $\$5 - \$2.86 + \$3.70 = \$5.84$  (modest profit)
- Me:  $\$8.58 - \$3.70 = \$4.88$

This maintains fairness while ensuring both players benefit from cooperation.

```
{'return': 3.70}
```

#### reasoning:

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This shows trust, sending about 57% of their endowment

**\*\*Fair return analysis:\*\***

- Total surplus created:  $\$8.58 - \$2.86 = \$5.72$
- A fair split would give each of us roughly \$2.86 in profit

If I return \$3.50:

- Sender gets:  $\$5 - \$2.86 + \$3.50 = \$5.64$  (profit of \$0.64)
- I get:  $\$8.58 - \$3.50 = \$5.08$  (profit of \$0.08)

That's not quite balanced. Let me try \$4.00:

- Sender gets:  $\$5 - \$2.86 + \$4.00 = \$6.14$  (profit of \$1.14)
- I get:  $\$8.58 - \$4.00 = \$4.58$  (profit of \$-0.42)

Wait, that would give me less than my base \$5. Let me recalculate.

Actually, I don't start with \$5 as the receiver. I start with \$0 and receive the multiplied amount.

So:

- Sender starts with \$5, sends \$2.86, gets back \$X
- Sender final:  $\$5 - \$2.86 + \$X = \$2.14 + \$X$
- I start with \$0, receive \$8.58, return \$X
- My final:  $\$8.58 - \$X$

For a balanced outcome where we both profit similarly:

- Sender profit:  $\$2.14 + \$X - \$5 = \$X - \$2.86$
- My profit:  $\$8.58 - \$X - \$0 = \$8.58 - \$X$
- Setting equal:  $\$X - \$2.86 = \$8.58 - \$X$  gives  $\$X = \$5.72$

That would split the surplus evenly, but feels generous. Instead, I should ensure the sender profits from their trust while keeping a reasonable share for myself. Returning around \$3.80 would give them roughly \$0.94 in profit while I keep \$4.78.

But that's heavily skewed in my favor. Testing \$4.50 or \$5.00 returns gets closer to balance-the sender would profit \$1.64 to \$2.14 while I'd keep \$3.58 to \$4.08. I'm reconsidering what fairness actually means here. In previous exchanges where I received around \$9.50-\$9.66 from sends of \$3.19-\$3.22, returning \$4.00 felt proportional. Using that same ratio (roughly 42% return rate) on this \$8.58 suggests returning around \$3.58-\$3.70. Testing \$3.70 gives the sender \$0.84 profit and me \$4.88, which still favors me but follows the pattern I've established in earlier rounds....

#### usage:

```
{
 "input_tokens": 1883,
 "output_tokens": 1261,
 "reasoning_tokens": 0
}
```

Message 10 role=user

#### content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.86, it became \$8.58, and you returned \$3.7. Your visible payoff was \$4.88.

Your total visible earnings across all rounds are \$32.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.69, so you received \$8.07. How much do you return? (0-8.07)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 7):\*\***

- Co-player sent \$2.69 ? I received \$8.07
- A moderate send showing trust

**\*\*My return pattern:\*\***

- I've been returning around 42% of what I receive
- This has created mutually beneficial outcomes

For \$8.07 received:

- Returning \$3.50 would give:
  - Sender: \$5 - \$2.69 + \$3.50 = \$5.81 (profit of \$0.81)
  - Me: \$8.07 - \$3.50 = \$4.57

This maintains fairness and ensures both players benefit.

{ 'return': 3.50 }

**reasoning:**

Looking at the current situation:

**\*\*Current Round (Round 7):\*\***

- Co-player sent \$2.69 ? I received \$8.07
- This is another moderate send

**\*\*My return pattern:\*\***

- Round 2: Received \$9.57, returned \$4.00 (42%)
- Round 4: Received \$9.66, returned \$4.00 (41%)
- Round 6: Received \$8.58, returned \$3.70 (43%)

I've been maintaining around a 42% return rate, which seems fair.

For \$8.07:

- 42% return would be:  $\$8.07 \times 0.42 = \$3.39$

Returning \$3.50 would give:

- Sender: \$5 - \$2.69 + \$3.50 = \$5.81 (profit of \$0.81)
- Me: \$8.07 - \$3.50 = \$4.57

This maintains my fair approach while ensuring both benefit. Let me go with \$3.50.

**usage:**

```
{
 "input_tokens": 2202,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.69, it became \$8.07, and you returned \$3.5. Your visible payoff was \$4.57.

Your total visible earnings across all rounds are \$37.27.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*Previous round (Round 7) with this SAME co-player:\*\***

- They sent me \$2.69 (showing trust)



- I returned \$3.50 (fair return)

**\*\*Current round (Round 8):\*\***

- Now I'm the sender with the same co-player
- They demonstrated moderate cooperation
- I should reciprocate their trust

Since they sent me \$2.69 and I treated them fairly, I'll send \$3.00 to reciprocate their cooperative behavior and potentially continue a mutually beneficial relationship.

{'send': 3.00}

**reasoning:**

Looking at the situation:

**\*\*Last round (Round 7):\*\***

- I was RECEIVER, my co-player was SENDER
- They sent \$2.69 ? I received \$8.07
- I returned \$3.50
- They got:  $\$5 - \$2.69 + \$3.50 = \$5.81$
- I got: \$4.57

**\*\*This round (Round 8):\*\***

- I am SENDER, they are RECEIVER
- Same co-player!
- They sent me a moderate amount (\$2.69) last round
- I returned fairly (\$3.50)

This is an opportunity to reciprocate their trust. They sent me \$2.69, so I should send a similar or slightly higher amount to show I'm willing to cooperate fairly with someone who cooperates with me.

Sending \$3 would be reasonable - it's my standard amount and slightly more than what they sent me, showing reciprocal trust.

Actually, let me think about this more carefully. They sent me \$2.69. I could:

- Send \$2.69 to exactly match their send
- Send \$3 which is my standard amount and slightly more
- Send more to show even stronger reciprocation

I think sending \$3 is appropriate - it matches my typical strategy and shows I'm reciprocating their cooperative behavior.

**usage:**

```
{
 "input_tokens": 2341,
 "output_tokens": 450,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.23 to you. Your visible payoff was \$5.23.

Your total visible earnings across all rounds are \$42.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my pattern:

**\*\*As SENDER:\*\***

- Rounds 1, 3, 5, 8: Consistently sent \$3
- Returns varied from \$2.46 to \$5.20, averaging around \$3.45

**\*\*Current situation (Round 9):\*\***

- New co-player (randomized pairing)
- My strategy of sending \$3 has been reasonably effective
- Creates \$9 of value while maintaining some protection

I'll continue with my established approach.

```
{'send': 3.00}
```

**Reasoning:**

Let me analyze the situation:

**\*\*Round 9 - I'm the SENDER\*\***

**\*\*My history as SENDER:\*\***

- Round 1: Sent \$3 ? received \$2.46
- Round 3: Sent \$3 ? received \$2.93
- Round 5: Sent \$3 ? received \$5.20 (great return!)
- Round 8: Sent \$3 ? received \$3.23

**\*\*My history as RECEIVER:\*\***

- Round 2: Received \$9.57 ? returned \$4.00
- Round 4: Received \$9.66 ? returned \$4.00
- Round 6: Received \$8.58 ? returned \$3.70
- Round 7: Received \$8.07 ? returned \$3.50

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Total earnings so far: \$42.50
- Sending \$3 has been my standard approach
- I've been getting mixed returns, but generally around equal splits

I should continue with my consistent strategy of sending \$3, which:

1. Shows reasonable trust
2. Creates value through the 3x multiplier
3. Has worked fairly well on average
4. Is my established pattern

```
{'send': 3}
```

```
Usage: {
 "input_tokens": 2298,
 "output_tokens": 425,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_6 - Receiver - dyad\_3**

Timestamp: 2026-08-12T09:28:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_6"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_6": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**  
**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.67, it became \$8.01, and you returned \$3.75. Your visible payoff was \$4.26.

Your total visible earnings across all rounds are \$10.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$3.36\*\*** (67% of their endowment), which became \$10.08 for me. This is a notably generous and trusting move.

**\*\*Context from previous rounds:\*\***

- Round 2: Someone sent \$2.67 (53%), I returned \$3.75
- Round 1: I sent \$3.00 (60%), received \$4.25 back

**\*\*My reasoning:\*\***

This sender showed higher trust than my previous co-player (67% vs 53%). I should reward this generosity proportionally.

Returning \$4.75:

- Sender would get: \$1.64 (kept) + \$4.75 = \$6.39 total (gain of \$1.39)
- I would get: \$10.08 - \$4.75 = \$5.33 (good gain)

This provides the sender with a decent reward for their trust while maintaining mutual benefit - both of us come out ahead.

```
'''json
{"return": 4.75}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Current round:\*\***

- My co-player sent \$3.36 (67% of their \$5 endowment) - this is quite generous/trusting
- I received \$10.08

**\*\*Previous round:\*\***

- Someone sent me \$2.67 (53% of endowment), I returned \$3.75
- That gave them \$2.33 + \$3.75 = \$6.08 total (~\$1.08 gain)
- I kept \$4.26

**\*\*Round 1:\*\***

- I sent \$3.00 (60%), received \$4.25 back
- That gave me \$2.00 + \$4.25 = \$6.25 total

**\*\*My thinking:\*\***

The sender this round was more generous (67% vs 53% in previous round). I should reward their higher trust level.

If I return around \$4.50-\$5.00:

- Return \$4.50: Sender gets \$1.64 + \$4.50 = \$6.14, I get \$5.58
- Return \$5.00: Sender gets \$1.64 + \$5.00 = \$6.64, I get \$5.08

Actually, let me think about what's fair given they sent 67% of their endowment.

If I return \$5.00:

- Sender: keeps \$1.64, gets back \$5.00 = \$6.64 total (gain of \$1.64)
- Me: receive \$10.08, return \$5.00 = \$5.08 (solid gain)

This seems like a good reciprocal move for their high trust. It's roughly a 50-50 split of the multiplied amount, which rewards their generosity while still benefiting both of us.

Actually, let me reconsider. They sent \$3.36 out of \$5, so they kept \$1.64.

Returning \$5 gives them \$6.64 with a \$1.64 gain, while I keep \$5.08. That's reasonably fair given their high trust. I...

**usage:**

```
{
 "input_tokens": 1163,
 "output_tokens": 921,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.75. Your visible payoff was \$5.33.

Your total visible earnings across all rounds are \$15.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this round as SENDER again:

**\*\*My experience:\*\***

- Round 1: Sent \$3.00, received \$4.25 back ? Good outcome
- Rounds 2-3: As receiver, I've been cooperative, returning fair amounts

**\*\*Current situation:\*\***

- New co-player (different from Round 1)
- I've been part of generally cooperative dynamics

**\*\*My reasoning:\*\***

I had success with sending \$3.00 before, and the cooperative pattern across rounds suggest s moderate-to-high sending works well. I'll send slightly more this time to signal trust while building on the positive momentum.

Sending \$3.50 creates \$10.50 for the receiver. If they're reasonably cooperative (returning ~40%), I'd still come out ahead.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

I'm back as the SENDER in round 4. Let me review my experience so far:

Round 1: I sent \$3.00, received \$4.25 back. Payoff: \$6.25

Round 2: Received \$8.01 (from \$2.67 sent), returned \$3.75. Payoff: \$4.26

Round 3: Received \$10.08 (from \$3.36 sent), returned \$4.75. Payoff: \$5.33

Total earnings: \$15.84

Now I'm the sender again with a new co-player. What should I send?

Looking at the pattern:

- When I sent \$3.00 (60%), I got a decent return
- When others sent moderate-to-high amounts (\$2.67, \$3.36), I've been fairly cooperative in returning
- The game seems to be working reasonably well with moderate-to-high sending amounts

I should consider:

1. I had a positive experience when I sent \$3.00
2. The cooperative moves I've seen have been between \$2.67-\$3.36
3. I've demonstrated I'm a fair returner, so maybe I can expect reasonable behavior

I think I should send a similar or slightly higher amount than last time. Since I received \$4.25 when I sent \$3.00, that worked out well.

Options:

- Send \$3.00 again (proven strategy)
- Send \$3.50 (show slightly more trust)
- Send \$4.00 (high trust signal)

Given:

- This is a new co-player
- I've been building cooperative patterns
- The game is going well so far

I'll send \$3.50 as a moderately high trust signal, slightly more than my first round, but not overly risky. This creates \$10.50 for the receiver.

...

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 660,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.55 to you. Your visible payoff was

\$6.05.

Your total visible earnings across all rounds are \$21.89.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*My pattern:\*\***

- I've been consistently returning around 45-47% of what I receive
- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)

**\*\*My reasoning:\*\***

This sender showed moderate trust. I should maintain consistency with my previous cooperative behavior.

Returning \$3.85:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of \$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This is about 46% return rate, consistent with my pattern and fair to the sender.

```
'''json
{"return": 3.85}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*Context from previous rounds:\*\***

- Round 1: I sent \$3.00 (60%), received \$4.25 back (47% of \$9)
- Round 2: Someone sent \$2.67 (53%), I returned \$3.75 (47% of \$8.01)
- Round 3: Someone sent \$3.36 (67%), I returned \$4.75 (47% of \$10.08)
- Round 4: I sent \$3.50 (70%), received \$4.55 back (43% of \$10.50)

**\*\*Pattern:\*\***

I've been pretty consistent returning around 45-47% of what I receive. My co-player sent \$2.81, which is moderately cooperative.

Let me calculate:

- 45% of \$8.43 = \$3.79
- 47% of \$8.43 = \$3.96

Returning around \$3.85 would:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of ~\$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This maintains consistency with my previous behavior and rewards the sender proportionally for their trust.

**usage:**

```
{
 "input_tokens": 1847,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$3.85. Your visible payoff was \$4.58.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 back (good outcome)
- Round 4: Sent \$3.50 ? Received \$4.55 back (good outcome)

**\*\*Overall dynamics:\*\***

- Cooperation has been working well across all rounds
- I've been a fair receiver (returning ~46%)
- Others have been reasonably cooperative with me

**\*\*My reasoning:\*\***

I'm with a new co-player. The cooperative pattern continues to work. I'll send an amount between my previous sends to signal trust while being strategic.

Sending \$3.25 creates \$9.75 for the receiver. With typical cooperation, this should benefit us both.

```
'''json
{"send": 3.25}
'''
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3.00, received \$4.25 back ? Payoff \$6.25
- Round 4: Sent \$3.50, received \$4.55 back ? Payoff \$6.05

As RECEIVER:

- Round 2: Received \$8.01, returned \$3.75 ? Payoff \$4.26
- Round 3: Received \$10.08, returned \$4.75 ? Payoff \$5.33
- Round 5: Received \$8.43, returned \$3.85 ? Payoff \$4.58

Overall, the cooperative strategy has been working well. When I send moderate-to-high amounts (\$3-\$3.50), I get good returns. My co-players have generally been cooperative.

This is a new co-player again (Round 6). I've been doing well with sends in the \$3-\$3.50 range.

I'll continue with a similar strategy - sending \$3.25 this time (between my previous two sends).

This creates \$9.75 for the receiver. If they return around 45%, I'd get about \$4.39 back, giving me a payoff of \$5.89.

Actually, let me think about this differently. I've been gradually increasing (3.0 then 3.5). Maybe I should go higher to signal stronger trust, or stay consistent. Let me stick with \$3.50 since it worked well.

Actually, let me be a bit more varied. \$3.25 is a good middle ground.

**usage:**

```
{
 "input_tokens": 2205,
 "output_tokens": 567,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$2.56 to you. Your visible payoff was \$4.3100000000000005.

Your total visible earnings across all rounds are \$30.78.

This round, you are the RECEIVER against your current co-player. They sent you \$2.85, so you received \$8.55. How much do you return? (0-8.55)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.85\*\*** (57% of their endowment), which became \$8.55 for me. This is a moderately cooperative move.

**\*\*Context:\*\***

- Round 6 was disappointing - I sent \$3.25 but only got \$2.56 back (26% return)
- However, this is a NEW co-player, not the same one
- My consistent pattern as receiver has been returning ~46-47%

**\*\*My reasoning:\*\***

Despite the poor return last round, I shouldn't penalize this new co-player. Maintaining consistency and fairness has been my strategy, and it's worked well overall.

Returning \$3.90:

- Sender would get: \$2.15 (kept) + \$3.90 = \$6.05 total (gain of \$1.05)
- I would get: \$8.55 - \$3.90 = \$4.65 (good gain)

This maintains my ~46% return rate and rewards the sender's trust appropriately.

```
'''json
{"return": 3.90}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.85\*\*** (57% of their endowment), which became \$8.55 for me. This is a moderately cooperative move.

**\*\*Recent experience:\*\***

- Round 6: I sent \$3.25, received only \$2.56 back (26% of \$9.75)
- That was disappointing - I got \$4.31 total, barely breaking even
- This is notably lower than my previous sender experiences

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)
- Round 5: Returned 46% (\$3.85 of \$8.43)

**\*\*Considerations:\*\***

1. The last round showed less cooperation from my co-player
2. But this is a NEW co-player now
3. They sent \$2.85, which is comparable to what others sent
4. Should I maintain my consistent cooperative strategy?

I think I should maintain consistency. The poor return in Round 6 was disappointing, but:

- This is a different co-player
- I've built a pattern of fairness
- Retaliating against a new player isn't justified
- Long-term cooperation benefits everyone

Following my pattern of ~46% return:

$\$8.55 \times 0.46 = \$3.93$

Sender would get: \$2.15 (kept) + \$3.93 = \$6.08 (gain of \$1.08)

I would get: \$8.55 - \$3.93 = \$4.62 (good gain)

I'll return \$3.90 to stay consistent with my cooperative approach.

**usage:**

```
{
 "input_tokens": 2534,
 "output_tokens": 674,
 "reasoning_tokens": 0
}
```



}

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.85, it became \$8.55, and you returned \$3.9. Your visible payoff was \$4.65.

Your total visible earnings across all rounds are \$35.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 (good)
- Round 4: Sent \$3.50 ? Received \$4.55 (good)
- Round 6: Sent \$3.25 ? Received \$2.56 (poor)

**\*\*Overall situation:\*\***

- 2 out of 3 sending experiences were positive
- I've been consistently fair as receiver
- Cooperation has generally worked well

**\*\*My reasoning:\*\***

Despite one disappointing return in Round 6, the cooperative approach has been successful overall. I'm with a new co-player and should maintain my strategy of moderate trust.

Sending \$3.00 - this worked well in Round 1 and represents a balanced approach that signals cooperation without overexposing myself.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 back (payoff: \$6.25)
- Round 4: Sent \$3.50 ? Received \$4.55 back (payoff: \$6.05)
- Round 6: Sent \$3.25 ? Received \$2.56 back (payoff: \$4.31) - disappointing

**\*\*My track record as receiver:\*\***

- Round 2: Received \$8.01, returned \$3.75 (payoff: \$4.26)
- Round 3: Received \$10.08, returned \$4.75 (payoff: \$5.33)
- Round 5: Received \$8.43, returned \$3.85 (payoff: \$4.58)
- Round 7: Received \$8.55, returned \$3.90 (payoff: \$4.65)

**\*\*Overall:\*\***

- I've been consistently fair as a receiver (returning ~46%)
- As sender, 2 out of 3 times worked well, but Round 6 was poor
- Total earnings: \$35.43 - doing well overall
- This is Round 8, potentially the last round or near the end

**\*\*Current situation:\*\***

- New co-player again
- Need to decide how much to send

**\*\*My reasoning:\*\***

The cooperative strategy has worked well overall. Despite one bad experience, the majority have been positive. I should continue with moderate-to-high sending to maximize potential gains.

Given this might be near the end, I should still send a reasonable amount. I'll continue with my strategy around \$3.00-\$3.50.

Let me send \$3.00 - it worked well in Round 1 and is a solid cooperative signal.

**usage:**

```
{
 "input_tokens": 2731,
 "output_tokens": 648,
 "reasoning_tokens": 0
}
```

**Message 14 role=user****content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.83 to you. Your visible payoff was \$6.83.

Your total visible earnings across all rounds are \$42.26.

This round, you are the RECEIVER against your current co-player. They sent you \$3.1, so you received \$9.3. How much do you return? (0-9.3)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*My co-player sent \$3.10\*\*** (62% of their endowment), which became \$9.30 for me. This is a generous move showing good trust.

**\*\*Context:\*\***

- Round 8 was excellent - I sent \$3.00 and got \$4.83 back
- My consistent pattern as receiver has been returning ~46-47%

**\*\*My reasoning:\*\***

This sender showed solid trust (62% of endowment). I should maintain my consistent cooperative approach.

Returning \$4.30:

- Sender would get: \$1.90 (kept) + \$4.30 = \$6.20 total (gain of \$1.20)
- I would get: \$9.30 - \$4.30 = \$5.00 (good gain)

This maintains my ~46% return rate and fairly rewards the sender's trust.

```
'''json
{"return": 4.30}
'''
```

**Reasoning:**

Looking at this situation:

**\*\*My co-player sent \$3.10\*\*** (62% of their endowment), which became \$9.30 for me. This is a generous and trusting move.

**\*\*Context:\*\***

- Round 8 was great - I sent \$3.00 and received \$4.83 back (excellent return of 54%)
- My pattern as receiver has been returning ~46-47%
- This sender showed good trust (62% of endowment)

**\*\*My return pattern:\*\***

- Round 2: Returned \$3.75 of \$8.01 = 47%
- Round 3: Returned \$4.75 of \$10.08 = 47%
- Round 5: Returned \$3.85 of \$8.43 = 46%
- Round 7: Returned \$3.90 of \$8.55 = 46%

**\*\*My reasoning:\*\***

This sender showed solid trust. I should maintain my consistent ~46% return rate.

Returning \$4.30:

- Sender would get: \$1.90 (kept) + \$4.30 = \$6.20 total (gain of \$1.20)
- I would get: \$9.30 - \$4.30 = \$5.00 (good gain)

This is about 46% return rate, consistent with my pattern.

Usage: {

```
"input_tokens": 2698,
"output_tokens": 544,
"reasoning_tokens": 0
```

```
}
```

**GAME - Agent\_2 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:28:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_4",

"move\_index": 6,

"opponent\_id": "Agent\_5",

"pairing": {

"agents": [

"Agent\_2",

"Agent\_5"

],

"investor": "Agent\_2",

"roles": {

"Agent\_2": "investor",

"Agent\_5": "trustee"

},

"trustee": "Agent\_5"

},

"role": "investor",

"role\_label": "Sender",

"round": 9,

"task": "game",

"task\_index": 0

```
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.62 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$9.77.

This round, you are the RECEIVER against your current co-player. They sent you \$2.11, so you received \$6.33. How much do you return? (0-6.33)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.11 (42.2% of their endowment), showing moderate trust
- I received \$6.33 after the 3x multiplier
- The sender now has \$2.89 remaining

**\*\*My reasoning:\*\***

The sender showed trust, though slightly less than what I've seen in previous rounds. I should reciprocate fairly to maintain cooperation.

If I return **\*\*\$3.30\*\***:

- Sender ends with: \$2.89 + \$3.30 = \$6.19 (profit of \$1.19)
- I end with: \$6.33 - \$3.30 = \$3.03 (profit of \$3.03)

This represents:

- Returning about 52% of what I received
- Both of us profiting from cooperation
- Fair reciprocation that encourages future trust

{'return': 3.3}

**reasoning:**

Let me analyze this situation:

**\*\*What happened in Round 2:\*\***

- I sent \$2.50
- It became \$7.50
- They returned \$3.62
- My payoff: \$2.50 (kept) + \$3.62 (returned) = \$6.12
- Their payoff: \$7.50 - \$3.62 = \$3.88
- They returned about 48% of what they received

**\*\*Current Round 3 situation:\*\***

- I'm the receiver
- Sender sent \$2.11 (42.2% of their \$5 endowment)
- I received \$6.33
- Sender now has \$2.89 remaining
- I need to decide how much to return

**\*\*Analysis:\*\***

The sender sent \$2.11, which is less than the \$2.50-\$2.55 amounts I've seen previously. They're showing moderate but slightly lower trust. However, they still sent a significant portion (42%).

If I return different amounts:

- Return \$3.00: Sender ends with \$5.89 (+\$0.89), I end with \$3.33 (+\$3.33)
- Return \$3.50: Sender ends with \$6.39 (+\$1.39), I end with \$2.83 (+\$2.83)
- Return \$3.20: Sender ends with \$6.09 (+\$1.09), I end with \$3.13 (+\$3.13)

I want to:

1. Reciprocate their trust proportionally
2. Be fair and encourage future cooperation
3. Return roughly 50% of what I received, similar to what I've observed

Returning around \$3.20 would be about 50.5% of what I received, which seems fair and proportional to their ~42% send rate. This ensures both of us profit.

Actually, let me be slightly more generous since they did show trust. I'll return \$3.30 (about 52% of what I received).

...

**usage:**

```
{
 "input_tokens": 1218,
 "output_tokens": 758,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.3. Your visible payoff was \$3.0300000000000002.

Your total visible earnings across all rounds are \$12.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- I'm facing a new co-player as sender
- Previous cooperation has been successful (total earnings \$12.80)
- When I sent \$2.50 in Round 2, I received \$3.62 back for a good profit
- My returns as receiver have been fair (~52%)

**\*\*Strategic reasoning:\*\***

- Continue the cooperative approach that's been working
- Sending 50% shows good faith while managing risk
- This gives my co-player \$7.50 to work with, enough for mutual benefit
- Establishes positive precedent with this new partner

{'send': 2.5}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds summary:\*\***

- Round 1 (RECEIVER): Received \$7.65 from sender sending \$2.55, returned \$4.00, earned \$3.65
- Round 2 (SENDER): Sent \$2.50, became \$7.50, they returned \$3.62, earned \$6.12
- Round 3 (RECEIVER): Received \$6.33 from sender sending \$2.11, returned \$3.30, earned \$3.03
- Total earnings: \$12.80

**\*\*Pattern analysis:\*\***

- When I sent \$2.50 (50%), I got back \$3.62, giving me a profit of \$1.12 (from \$2.50 remaining + \$3.62 = \$6.12 total)
- When others sent to me, I've returned fairly (~52% of what I received)
- Overall, cooperation has been successful

**\*\*Current decision:\*\***

- New co-player as sender this round
- Need to establish trust again
- Sending around 50% has worked well previously

I'll continue with the cooperative strategy that's been working: send \$2.50 (50% of endowment).

**usage:**

```
{
 "input_tokens": 1558,
 "output_tokens": 431,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.01 to you. Your visible payoff was \$5.51.

Your total visible earnings across all rounds are \$18.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.92, so you received \$8.76. How much do you return? (0-8.76)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*My reasoning:\*\***

This sender showed above-average trust compared to previous rounds. I should reciprocate generously.

If I return **\*\*\$4.70\*\***:

- Sender ends with: \$2.08 + \$4.70 = \$6.78 (profit of \$1.78)
- I end with: \$8.76 - \$4.70 = \$4.06 (profit of \$4.06)

This represents:

- Returning about 54% of what I received
- Rewarding their higher trust level (58.4% sent)
- Both profiting well from cooperation
- Encouraging continued trust

{'return': 4.7}

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- The sender sent \$2.92 (58.4% of their endowment), showing strong trust
- I received \$8.76 after the 3x multiplier
- The sender now has \$2.08 remaining

**\*\*Patterns from previous rounds:\*\***

- Round 1: Received \$7.65, returned \$4.00 (52%)
- Round 3: Received \$6.33, returned \$3.30 (52%)
- I've been consistently returning about 52% of what I receive

**\*\*Strategy:\*\***

The sender showed high trust (58.4% sent). I should reciprocate fairly.

If I return around 52% of \$8.76:

- $0.52 \times \$8.76 = \$4.55$

Let me check the outcomes:

- Sender ends with: \$2.08 + \$4.55 = \$6.63 (profit of \$1.63)
- I end with: \$8.76 - \$4.55 = \$4.21 (profit of \$4.21)

This seems fair - both profit, sender gets a decent return for their trust.

Actually, given they sent 58.4% (more than my usual 50%), maybe I should return slightly more to really reward that trust. Let me try \$4.70:

- Sender ends with: \$2.08 + \$4.70 = \$6.78 (profit of \$1.78)

- I end with:  $\$8.76 - \$4.70 = \$4.06$  (profit of \$4.06)

That's about 54% return, which is fair and rewards their higher trust level.

**usage:**

```
{
 "input_tokens": 1844,
 "output_tokens": 622,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.92, it became \$8.76, and you returned \$4.7. Your visible payoff was \$4.06.

Your total visible earnings across all rounds are \$22.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My sending strategy (50%) has been successful in Rounds 2 & 4
- Returns I've received: \$3.62 and \$3.01 (averaging \$3.32)
- Total earnings of \$22.37 show cooperation is working

**\*\*Strategic reasoning:\*\***

- Maintain the consistent 50% sending strategy
- This has proven reliable across different co-players
- Gives receiver \$7.50 to work with
- Balances trust-building with risk management

{'send': 2.5}

**reasoning:**

Looking at the pattern:

**\*\*My sending history:\*\***

- Round 2: Sent \$2.50 (50%), received back \$3.62, profit = \$6.12
- Round 4: Sent \$2.50 (50%), received back \$3.01, profit = \$5.51

**\*\*My receiving history:\*\***

- Round 1: Received \$7.65 from \$2.55 sent (51%), returned \$4.00
- Round 3: Received \$6.33 from \$2.11 sent (42%), returned \$3.30
- Round 5: Received \$8.76 from \$2.92 sent (58%), returned \$4.70

**\*\*Analysis:\*\***

- When I send 50%, I've gotten decent returns (\$3.01-\$3.62)
- As a receiver, I've been fair, returning around 52-54%
- Total earnings are strong at \$22.37
- Cooperation has been working well

**\*\*This round:\*\***

- New co-player as sender
- Continue with the cooperative 50% strategy that's been successful
- This maintains consistency and shows I'm a cooperative player

I'll send \$2.50 again.

**usage:**

```
{
 "input_tokens": 2182,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.8 to you. Your visible payoff was \$6.3.

Your total visible earnings across all rounds are \$28.67.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My consistent 50% sending strategy has worked well (Rounds 2, 4, 6)
- Returns have varied (\$3.62, \$3.01, \$3.80) but all profitable
- Total earnings of \$28.67 show strong overall cooperation

**\*\*Strategic reasoning:\*\***

- Continue the reliable 50% sending pattern
- This approach has proven effective across different co-players
- Provides \$7.50 for receiver to reciprocate
- Maintains consistency and manages risk

{'send': 2.5}

**reasoning:**

Looking at this situation:

**\*\*Context:\*\***

- I'm the sender again, facing a new co-player
- My sending history shows consistent 50% sends (\$2.50 in rounds 2, 4, 6)
- Returns I've received: \$3.62, \$3.01, \$3.80 (averaging \$3.48)
- All have been profitable
- Total earnings of \$28.67

**\*\*Strategic reasoning:\*\***

- The 50% strategy has been consistently successful
- Returns have varied but all been positive
- This creates good conditions for mutual benefit
- Should continue this proven approach

I'll send \$2.50 again.

**usage:**

```
{
 "input_tokens": 2442,
 "output_tokens": 294,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.11 to you. Your visible payoff was \$5.6099999999999999.

Your total visible earnings across all rounds are \$34.28.

This round, you are the RECEIVER against your current co-player. They sent you \$2.29, so you received \$6.87. How much do you return? (0-6.87)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.29 (45.8% of their endowment), showing moderate trust
- I received \$6.87 after the 3x multiplier
- The sender now has \$2.71 remaining



**\*\*My reasoning:\*\***

Based on my consistent pattern in previous receiver rounds (returning ~52-54%), I should reciprocate proportionally.

If I return **\*\*\$3.57\*\***:

- Sender ends with:  $\$2.71 + \$3.57 = \$6.28$  (profit of \$1.28)
- I end with:  $\$6.87 - \$3.57 = \$3.30$  (profit of \$3.30)

This represents:

- Returning about 52% of what I received
- Both parties profiting from cooperation
- Consistent reciprocation encouraging continued trust

`{'return': 3.57}`

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- The sender sent \$2.29 (45.8% of their endowment), showing moderate trust
- I received \$6.87 after the 3x multiplier
- The sender now has \$2.71 remaining

**\*\*My approach in previous receiver rounds:\*\***

- Round 1: Received \$7.65, returned \$4.00 (52.3%)
- Round 3: Received \$6.33, returned \$3.30 (52.1%)
- Round 5: Received \$8.76, returned \$4.70 (53.7%)

I've been consistent in returning around 52-54% of what I receive.

For this round, if I return around 52%:

$$\$6.87 \times 0.52 = \$3.57$$

Let me check the outcomes:

- If I return \$3.57:
  - Sender ends with:  $\$2.71 + \$3.57 = \$6.28$  (profit of \$1.28)
  - I end with:  $\$6.87 - \$3.57 = \$3.30$  (profit of \$3.30)

This seems fair and consistent with my previous behavior. The sender sent 45.8% which is moderate trust, and returning 52% rewards that trust while both parties profit.

I'll return \$3.57.

**usage:**

```
{
 "input_tokens": 2408,
 "output_tokens": 540,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.29, it became \$6.87, and you returned \$3.57. Your visible payoff was \$3.3000000000000003.

Your total visible earnings across all rounds are \$37.58.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: `{'send': <amount>}`

**Assistant output:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My 50% sending strategy has been consistently profitable (Rounds 2, 4, 6, 7)
- Returns ranged from \$3.01 to \$3.80, all profitable
- Total earnings of \$37.58 show strong overall success

**\*\*Strategic reasoning:\*\***

- Continue the proven 50% sending strategy
- This has worked reliably across all different co-players
- Provides \$7.50 for receiver to reciprocate fairly
- Maintains trust-building while managing risk

```
{'send': 2.5}
```

#### Reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.50, received back \$3.62, payoff \$6.12
- Round 4: Sent \$2.50, received back \$3.01, payoff \$5.51
- Round 6: Sent \$2.50, received back \$3.80, payoff \$6.30
- Round 7: Sent \$2.50, received back \$3.11, payoff \$5.61

My consistent strategy of sending \$2.50 (50%) has been working well. All returns have been profitable, ranging from \$3.01 to \$3.80. I'm now at \$37.58 total earnings, which is very good.

I should continue with the same strategy that has been successful - sending \$2.50 (50% of endowment). This:

- Maintains consistency
- Shows good faith to co-players
- Has proven profitable across different partners
- Balances trust and risk management

New co-player this round, so sticking with the proven approach makes sense.

```
Usage: {
 "input_tokens": 2442,
 "output_tokens": 395,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_5 - Receiver - dyad\_4

Timestamp: 2026-08-12T09:28:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_5"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_2": "investor",
 "Agent_5": "trustee"
 },
 "trustee": "Agent_5"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.09, it became \$6.27, and you returned \$2.82. Your visible payoff was \$3.4499999999999997.

Your total visible earnings across all rounds are \$9.139999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my position:

**\*\*Previous experience:\*\***

- Round 1: Sent \$3, received \$3.69 back ? good outcome
- Round 2: As receiver, returned a fair amount (\$2.82 on \$6.27)

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- No history with this specific agent
- Need to signal willingness to cooperate

Since I had success with sending \$3 in Round 1 (received \$3.69 back, net gain of \$0.69), and it establishes a pattern of moderate trust, I'll use the same strategy with this new co-player.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Let me analyze my experience so far:

Round 1: I was SENDER

- Sent \$3, they received \$9
- They returned \$3.69 (about 41% of what they received)
- My payoff: \$2 (kept) + \$3.69 (returned) = \$5.69

Round 2: I was RECEIVER

- They sent \$2.09, I received \$6.27
- I returned \$2.82 (about 45% of what I received)
- My payoff: \$6.27 - \$2.82 = \$3.45

Total earnings: \$9.14

Now Round 3: I'm SENDER again with a new co-player (since agents are randomized each round)

What should I consider:

1. This is a new opponent - I don't know their strategy
2. In Round 1, sending \$3 worked reasonably well (got back \$3.69, netted \$5.69 vs \$5 if I sent nothing)
3. The multiplier effect means cooperation is beneficial
4. I should probably maintain a consistent strategy early on

My Round 1 strategy of sending \$3 was a moderate trust signal that worked well. Since I'm facing a new opponent, I should probably use a similar approach - not too cautious, not too risky.

I'll send \$3 again as my baseline strategy for first encounters with new co-players.

**usage:**

```
{
 "input_tokens": 1050,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.37 to you. Your visible payoff was \$7.37.

Your total visible earnings across all rounds are \$16.509999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.94 out of \$5 (nearly 59%)
- Showed strong trust

**\*\*Recent patterns:\*\***

- Round 3: My partner was very generous, returning \$5.37 (60% of \$9)
- Round 2: I returned about 45% as receiver
- This co-player is being quite trusting

**\*\*Fair reciprocity:\*\***

Returning around 48% of the \$8.82 I received would be fair and encourage future cooperation. This gives my co-player a decent profit while rewarding their trust.

```
'''json
{"return": 4.20}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER\*\***

- My co-player sent \$2.94 (about 59% of their \$5)
- I received \$8.82 (3x multiplier)

- Need to decide how much to return

**\*\*Recent history:\*\***

- Round 3: As sender, I sent \$3, received \$5.37 back (about 60% of the \$9 they got) - very generous!
- Round 2: As receiver, I returned \$2.82 on \$6.27 (about 45%)
- Round 1: As sender, sent \$3, received \$3.69 back (about 41% of the \$9)

**\*\*Analysis of current round:\*\***

- Co-player sent \$2.94 (quite generous - nearly 60% of their endowment)
- This shows significant trust
- They'll end up with: \$5 - \$2.94 + return = \$2.06 + return

**\*\*Fairness considerations:\*\***

- If I return about 45-50% like in Round 2: ~\$3.96 - \$4.41
- If I match the generosity of Round 3's partner (60%): ~\$5.29
- Co-player sacrificed a lot to send \$2.94

I should reciprocate fairly. They sent nearly 60% of their endowment, showing strong trust. I should return a reasonable amount that:

1. Rewards their trust
2. Gives them a profit
3. Maintains cooperation

Returning around \$4.00-\$4.50 seems fair:

- Gives them a total of ~\$6.06-\$6.56
- Nets them \$1.06-\$1.56 profit
- I keep \$4.32-\$4.82
- Return rate of ~45-51%

...

**usage:**

```
{
 "input_tokens": 1348,
 "output_tokens": 615,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.94, it became \$8.82, and you returned \$4.2. Your visible payoff was \$4.62.

Your total visible earnings across all rounds are \$21.13.

This round, you are the RECEIVER against your current co-player. They sent you \$1.7, so you received \$5.1. How much do you return? (0-5.1)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- More cautious than previous partners

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)

**\*\*Fair reciprocity:\*\***

Returning around 45% of the \$5.10 I received maintains consistency and fairness. This gives my co-player a modest profit while rewarding their trust, even though they were more cautious.

```
'''json
{"return": 2.30}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- This is relatively cautious/low trust

**\*\*Recent patterns from my play:\*\***

- Round 2: Returned \$2.82 on \$6.27 (45%)
- Round 4: Returned \$4.20 on \$8.82 (48%)

**\*\*Considerations:\*\***

- They showed less trust by sending only \$1.70
- However, I should still be fair to encourage cooperation
- Returning around 40-45% would be consistent with my previous behavior

Returning about 45% of \$5.10 = ~\$2.30

This gives them back a reasonable amount, rewards some level of trust, and maintains consistency in my reciprocity rate.

Let me calculate:

- They sent \$1.70, kept \$3.30
- I received \$5.10
- If I return \$2.30:
  - They end with:  $\$3.30 + \$2.30 = \$5.60$  (gain of \$0.60)
  - I end with:  $\$5.10 - \$2.30 = \$2.80$

That seems fair. They made a small profit for trusting me.

**usage:**

```
{
 "input_tokens": 1636,
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.7, it became \$5.1, and you returned \$2.3. Your visible payoff was \$2.8. Your total visible earnings across all rounds are \$23.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my position:

**\*\*Sender history:\*\***

- Round 1: Sent \$3 ? received \$3.69 back (good return)
- Round 3: Sent \$3 ? received \$5.37 back (excellent return)

**\*\*Strategy:\*\***

Sending \$3 has been consistently successful for me. It demonstrates trust while creating substantial value (\$9 for the receiver), which has encouraged good reciprocity from past partners.

With a new co-player this round, I'll continue with this proven strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3, got back \$3.69 (net +\$0.69)
- Round 3: Sent \$3, got back \$5.37 (net +\$2.37)

As RECEIVER:

- Round 2: Received \$6.27, returned \$2.82
- Round 4: Received \$8.82, returned \$4.20
- Round 5: Received \$5.10, returned \$2.30

My sending strategy of \$3 has worked well - I got good returns both times. This is a new co-player (randomized pairing), so I don't have specific history with them.

Sending \$3 seems to be a sweet spot:

- Shows trust and willingness to cooperate
- Creates substantial value (\$9 for receiver)
- Has resulted in good returns in past rounds

I'll continue with \$3 as it's been my successful strategy.

**usage:**

```
{
 "input_tokens": 1913,
 "output_tokens": 363,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.36 to you. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$29.29.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my position:

**\*\*Sender track record:\*\***

- Round 1: Sent \$3 ? received \$3.69 (net +\$0.69)
- Round 3: Sent \$3 ? received \$5.37 (net +\$2.37)
- Round 6: Sent \$3 ? received \$3.36 (net +\$0.36)

**\*\*Strategy assessment:\*\***

Sending \$3 has been consistently profitable across all my sender rounds. It demonstrates trust, creates value through the 3x multiplier, and has encouraged reciprocity from various co-players.

With a new partner this round, I'll stick with this proven approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my sender history:

- Round 1: Sent \$3 ? received \$3.69 back (net +\$0.69)
- Round 3: Sent \$3 ? received \$5.37 back (net +\$2.37) - excellent!
- Round 6: Sent \$3 ? received \$3.36 back (net +\$0.36)

All three times I sent \$3, I got a positive return. The strategy of sending \$3 has been consistently profitable, showing that moderate trust tends to be reciprocated.

This is a new co-player again (randomized pairing each round). I should continue with this successful strategy of sending \$3.

It creates value through the 3x multiplier while not risking everything, and has historically generated positive returns for me.

**usage:**

```
{
 "input_tokens": 2170,
 "output_tokens": 359,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.3 to you. Your visible payoff was \$5.3.

Your total visible earnings across all rounds are \$34.589999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$3.50 out of \$5 (70%)
- This is the highest amount anyone has sent to me as receiver
- Shows strong trust

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45%
- Round 4: Returned 48%
- Round 5: Returned 45%

**\*\*Fair reciprocity:\*\***

Since they showed exceptional trust (70% of their endowment), I should reward that generously. Returning approximately 48% of the \$10.50 I received maintains my reciprocal pattern while rewarding their high trust.

```
'''json
{"return": 5.00}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$3.50 out of \$5 (70%)
- This is showing high trust - the highest amount sent to me so far this game

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)
- Round 5: Returned 45% (\$2.30 on \$5.10)

**\*\*My average return rate as receiver:\*\*** approximately 45-48%

**\*\*Fair reciprocity:\*\***

Since they showed high trust (70% of their endowment), I should reward that appropriately. Returning around 47-48% of the \$10.50 would be consistent with my previous behavior.

47% of \$10.50 = \$4.94

48% of \$10.50 = \$5.04

I'll go with around \$5.00 to keep it a round number and reward their trust appropriately.

Actually, let me think about this more carefully. They sent \$3.50, so they kept \$1.50. If

I return \$5.00:

- They get: \$1.50 + \$5.00 = \$6.50
- I get: \$10.50 - \$5.00 = \$5.50

That gives them a net profit of \$1.50 (\$6.50 - \$5.00 initial), which is good.

Let me check if this is fair. They showed high trust. I should be generous back. \$5.00 seems reasonable - it's approximately 48% return rate.

**usage:**

```
{
 "input_tokens": 2310,
 "output_tokens": 557,
```



```

 "reasoning_tokens": 0
 }
Message 14 role=user
content:
 Round 9

 In your most recent previous game round, you were the RECEIVER against another agent. You
 saw them send $3.5, it became $10.5, and you returned $5.0. Your visible payoff was $5.5
 .
 Your total visible earnings across all rounds are $40.089999999999996.

 This round, you are the RECEIVER against your current co-player. They sent you $2.89, so y
 ou received $8.67. How much do you return? (0-8.67)
 Respond exactly as JSON: {'return': <amount>}

```

**Assistant output:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.89 out of \$5 (about 58%)
- Shows good trust

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45%
- Round 4: Returned 48%
- Round 5: Returned 45%
- Round 8: Returned 48%

**\*\*Fair reciprocity:\*\***

Returning around 46% of the \$8.67 I received maintains my consistent pattern. This gives my co-player a solid profit while rewarding their trust.

```

'''json
{"return": 4.00}
'''

```

**Reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.89 out of \$5 (about 58%)
- Showing good trust

**\*\*My pattern as receiver:\*\***

- Round 2: Returned \$2.82 on \$6.27 (45%)
- Round 4: Returned \$4.20 on \$8.82 (48%)
- Round 5: Returned \$2.30 on \$5.10 (45%)
- Round 8: Returned \$5.00 on \$10.50 (48%)

I've been consistently returning around 45-48% of what I receive. This gives my co-player a fair return while keeping some profit for myself.

For \$8.67, returning around 45-48% would be:

- 45%: \$3.90
- 48%: \$4.16

Let me go with around 46% which would be about \$4.00.

This would give them:

- They sent: \$2.89
- They receive back: \$4.00
- Net gain: \$1.11

That's fair and rewards their trust.

```

Usage: {
 "input_tokens": 2324,
 "output_tokens": 422,
 "reasoning_tokens": 0
}

```

**ROUND 10**

=====

Roles: Agent\_1=Receiver, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Sender, Agent\_5=Sender,  
Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Receiver

**Pairings:**

dyad\_1: Agent\_5=Sender, Agent\_2=Receiver  
dyad\_2: Agent\_6=Sender, Agent\_1=Receiver  
dyad\_3: Agent\_4=Sender, Agent\_7=Receiver  
dyad\_4: Agent\_3=Sender, Agent\_8=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_5 sent \$3; received \$9; Agent\_2 returned \$5.3; payoffs Agent\_5=\$7.3, Agent\_2=\$3.7;  
noise/visible: sent visible=\$3.34, received visible=\$10.02, returned visible=\$5.82  
Action validation: {"returned": {"amount": 5.3, "clamped": false, "max": 9.0, "min": 0, "raw": 5.3}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_2: Agent\_6 sent \$3; received \$9; Agent\_1 returned \$5.3; payoffs Agent\_6=\$7.3, Agent\_1=\$3.7;  
noise/visible: sent visible=\$3.51, received visible=\$10.53, returned visible=\$6.01  
Action validation: {"returned": {"amount": 5.3, "clamped": false, "max": 9.0, "min": 0, "raw": 5.3}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_3: Agent\_4 sent \$3; received \$9; Agent\_7 returned \$3.5; payoffs Agent\_4=\$5.5, Agent\_7=\$5.5;  
noise/visible: sent visible=\$2.18, received visible=\$6.54, returned visible=\$3.95  
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}  
dyad\_4: Agent\_3 sent \$2.5; received \$7.5; Agent\_8 returned \$3.5; payoffs Agent\_3=\$6, Agent\_8=\$4;  
noise/visible: sent visible=\$2.79, received visible=\$8.37, returned visible=\$3.59  
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 7.5, "min": 0, "raw": 3.5}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

**Balances after round: {**

"Agent\_1": 53.33,  
"Agent\_2": 53.940000000000005,  
"Agent\_3": 57.29,  
"Agent\_4": 53.019999999999996,  
"Agent\_5": 55.63,  
"Agent\_6": 52.14,  
"Agent\_7": 52.75,  
"Agent\_8": 54.4

}

**Visible balances after round: {**

"Agent\_1": 54.97,  
"Agent\_2": 48.419999999999995,  
"Agent\_3": 54.709999999999994,  
"Agent\_4": 52.699999999999996,  
"Agent\_5": 52.58,  
"Agent\_6": 55.269999999999996,  
"Agent\_7": 47.239999999999995,  
"Agent\_8": 53.2

}

**LLM CALLS**

**GAME - Agent\_5 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:28:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_2",  
"pairing": {  
  "agents": [  
    "Agent\_5",  
    "Agent\_2"  
  ],  
  "investor": "Agent\_5",  
  "roles": {  
    "Agent\_2": "trustee",  
    "Agent\_5": "investor"  
  },  
  "trustee": "Agent\_2"  
},  
"role": "investor",

```

"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.37 to you. Your visible payoff was \$7.37.

Your total visible earnings across all rounds are \$16.509999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$2.94, so you received \$8.82. How much do you return? (0-8.82)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.94 out of \$5 (nearly 59%)

- Showed strong trust

**\*\*Recent patterns:\*\***

- Round 3: My partner was very generous, returning \$5.37 (60% of \$9)
- Round 2: I returned about 45% as receiver
- This co-player is being quite trusting

**\*\*Fair reciprocity:\*\***

Returning around 48% of the \$8.82 I received would be fair and encourage future cooperation. This gives my co-player a decent profit while rewarding their trust.

```
'''json
{"return": 4.20}
'''
```

**reasoning:**

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER\*\***

- My co-player sent \$2.94 (about 59% of their \$5)
- I received \$8.82 (3x multiplier)
- Need to decide how much to return

**\*\*Recent history:\*\***

- Round 3: As sender, I sent \$3, received \$5.37 back (about 60% of the \$9 they got) - very generous!
- Round 2: As receiver, I returned \$2.82 on \$6.27 (about 45%)
- Round 1: As sender, sent \$3, received \$3.69 back (about 41% of the \$9)

**\*\*Analysis of current round:\*\***

- Co-player sent \$2.94 (quite generous - nearly 60% of their endowment)
- This shows significant trust
- They'll end up with:  $\$5 - \$2.94 + \text{return} = \$2.06 + \text{return}$

**\*\*Fairness considerations:\*\***

- If I return about 45-50% like in Round 2:  $\sim\$3.96 - \$4.41$
- If I match the generosity of Round 3's partner (60%):  $\sim\$5.29$
- Co-player sacrificed a lot to send \$2.94

I should reciprocate fairly. They sent nearly 60% of their endowment, showing strong trust. I should return a reasonable amount that:

1. Rewards their trust
2. Gives them a profit
3. Maintains cooperation

Returning around \$4.00-\$4.50 seems fair:

- Gives them a total of  $\sim\$6.06 - \$6.56$
- Nets them  $\$1.06 - \$1.56$  profit
- I keep  $\$4.32 - \$4.82$
- Return rate of  $\sim 45 - 51\%$

...

**usage:**

```
{
 "input_tokens": 1348,
 "output_tokens": 615,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.94, it became \$8.82, and you returned \$4.2. Your visible payoff was \$4.62.

Your total visible earnings across all rounds are \$21.13.

This round, you are the RECEIVER against your current co-player. They sent you \$1.7, so you received \$5.1. How much do you return? (0-5.1)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- More cautious than previous partners

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)

**\*\*Fair reciprocity:\*\***

Returning around 45% of the \$5.10 I received maintains consistency and fairness. This gives my co-player a modest profit while rewarding their trust, even though they were more cautious.

```
'''json
{"return": 2.30}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$1.70 out of \$5 (34%)
- This is relatively cautious/low trust

**\*\*Recent patterns from my play:\*\***

- Round 2: Returned \$2.82 on \$6.27 (45%)
- Round 4: Returned \$4.20 on \$8.82 (48%)

**\*\*Considerations:\*\***

- They showed less trust by sending only \$1.70
- However, I should still be fair to encourage cooperation
- Returning around 40-45% would be consistent with my previous behavior

Returning about 45% of \$5.10 = ~\$2.30

This gives them back a reasonable amount, rewards some level of trust, and maintains consistency in my reciprocity rate.

Let me calculate:

- They sent \$1.70, kept \$3.30
- I received \$5.10
- If I return \$2.30:
  - They end with:  $\$3.30 + \$2.30 = \$5.60$  (gain of \$0.60)
  - I end with:  $\$5.10 - \$2.30 = \$2.80$

That seems fair. They made a small profit for trusting me.

**usage:**

```
{
 "input_tokens": 1636,
 "output_tokens": 458,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.7, it became \$5.1, and you returned \$2.3. Your visible payoff was \$2.8. Your total visible earnings across all rounds are \$23.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my position:

**\*\*Sender history:\*\***

- Round 1: Sent \$3 ? received \$3.69 back (good return)

- Round 3: Sent \$3 ? received \$5.37 back (excellent return)

**\*\*Strategy:\*\***

Sending \$3 has been consistently successful for me. It demonstrates trust while creating substantial value (\$9 for the receiver), which has encouraged good reciprocity from past partners.

With a new co-player this round, I'll continue with this proven strategy.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3, got back \$3.69 (net +\$0.69)
- Round 3: Sent \$3, got back \$5.37 (net +\$2.37)

As RECEIVER:

- Round 2: Received \$6.27, returned \$2.82
- Round 4: Received \$8.82, returned \$4.20
- Round 5: Received \$5.10, returned \$2.30

My sending strategy of \$3 has worked well - I got good returns both times. This is a new co-player (randomized pairing), so I don't have specific history with them.

Sending \$3 seems to be a sweet spot:

- Shows trust and willingness to cooperate
- Creates substantial value (\$9 for receiver)
- Has resulted in good returns in past rounds

I'll continue with \$3 as it's been my successful strategy.

**usage:**

```
{
 "input_tokens": 1913,
 "output_tokens": 363,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.36 to you. Your visible payoff was \$5.359999999999999.

Your total visible earnings across all rounds are \$29.29.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my position:

**\*\*Sender track record:\*\***

- Round 1: Sent \$3 ? received \$3.69 (net +\$0.69)
- Round 3: Sent \$3 ? received \$5.37 (net +\$2.37)
- Round 6: Sent \$3 ? received \$3.36 (net +\$0.36)

**\*\*Strategy assessment:\*\***

Sending \$3 has been consistently profitable across all my sender rounds. It demonstrates trust, creates value through the 3x multiplier, and has encouraged reciprocity from various co-players.

With a new partner this round, I'll stick with this proven approach.

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at my sender history:

- Round 1: Sent \$3 ? received \$3.69 back (net +\$0.69)
- Round 3: Sent \$3 ? received \$5.37 back (net +\$2.37) - excellent!
- Round 6: Sent \$3 ? received \$3.36 back (net +\$0.36)

All three times I sent \$3, I got a positive return. The strategy of sending \$3 has been consistently profitable, showing that moderate trust tends to be reciprocated.

This is a new co-player again (randomized pairing each round). I should continue with this successful strategy of sending \$3.

It creates value through the 3x multiplier while not risking everything, and has historically generated positive returns for me.

**usage:**

```
{
 "input_tokens": 2170,
 "output_tokens": 359,
 "reasoning_tokens": 0
}
```

**Message 10 role=user****content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.3 to you. Your visible payoff was \$5.3.

Your total visible earnings across all rounds are \$34.589999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant****content:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$3.50 out of \$5 (70%)
- This is the highest amount anyone has sent to me as receiver
- Shows strong trust

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45%
- Round 4: Returned 48%
- Round 5: Returned 45%

**\*\*Fair reciprocity:\*\***

Since they showed exceptional trust (70% of their endowment), I should reward that generously. Returning approximately 48% of the \$10.50 I received maintains my reciprocal pattern while rewarding their high trust.

```
'''json
{"return": 5.00}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$3.50 out of \$5 (70%)
- This is showing high trust - the highest amount sent to me so far this game

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45% (\$2.82 on \$6.27)
- Round 4: Returned 48% (\$4.20 on \$8.82)
- Round 5: Returned 45% (\$2.30 on \$5.10)

**\*\*My average return rate as receiver:\*\*** approximately 45-48%

**\*\*Fair reciprocity:\*\***

Since they showed high trust (70% of their endowment), I should reward that appropriately.

Returning around 47-48% of the \$10.50 would be consistent with my previous behavior.

47% of \$10.50 = \$4.94

48% of \$10.50 = \$5.04

I'll go with around \$5.00 to keep it a round number and reward their trust appropriately.

Actually, let me think about this more carefully. They sent \$3.50, so they kept \$1.50. If

I return \$5.00:

- They get: \$1.50 + \$5.00 = \$6.50

- I get: \$10.50 - \$5.00 = \$5.50

That gives them a net profit of \$1.50 (\$6.50 - \$5.00 initial), which is good.

Let me check if this is fair. They showed high trust. I should be generous back. \$5.00 seems reasonable - it's approximately 48% return rate.

usage:

```
{
 "input_tokens": 2310,
 "output_tokens": 557,
 "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.5, it became \$10.5, and you returned \$5.0. Your visible payoff was \$5.5.

Your total visible earnings across all rounds are \$40.089999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.89, so you received \$8.67. How much do you return? (0-8.67)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.89 out of \$5 (about 58%)

- Shows good trust

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 45%

- Round 4: Returned 48%

- Round 5: Returned 45%

- Round 8: Returned 48%

**\*\*Fair reciprocity:\*\***

Returning around 46% of the \$8.67 I received maintains my consistent pattern. This gives my co-player a solid profit while rewarding their trust.

'''json

{"return": 4.00}

'''

reasoning:

Looking at this situation:

**\*\*My co-player's decision:\*\***

- Sent \$2.89 out of \$5 (about 58%)

- Showing good trust

**\*\*My pattern as receiver:\*\***

- Round 2: Returned \$2.82 on \$6.27 (45%)

- Round 4: Returned \$4.20 on \$8.82 (48%)

- Round 5: Returned \$2.30 on \$5.10 (45%)

- Round 8: Returned \$5.00 on \$10.50 (48%)

I've been consistently returning around 45-48% of what I receive. This gives my co-player a fair return while keeping some profit for myself.



For \$8.67, returning around 45-48% would be:

- 45%: \$3.90
- 48%: \$4.16

Let me go with around 46% which would be about \$4.00.

This would give them:

- They sent: \$2.89
- They receive back: \$4.00
- Net gain: \$1.11

That's fair and rewards their trust.

**usage:**

```
{
 "input_tokens": 2324,
 "output_tokens": 422,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$2.89, it became \$8.67, and you returned \$4.0. Your visible payoff was \$4.67.

Your total visible earnings across all rounds are \$44.76.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Recent history with this co-player:\*\***

- Last round (Round 9): They sent me \$2.89, I returned \$4.00
- They showed good trust (58% of their endowment)
- I reciprocated fairly (46% return rate)

**\*\*My sender track record:\*\***

- Sending \$3 has been consistently profitable
- It demonstrates trust and creates value through the multiplier
- Has worked well across all my sender rounds

**\*\*Strategy:\*\***

Since this co-player showed good trust last round and I reciprocated fairly, continuing with my proven \$3 strategy maintains consistency and mutual cooperation.

```
'''json
{"send": 3}
'''
```

**Reasoning:**

Looking at this situation:

**\*\*Key information:\*\***

- This is Round 10
- I'm facing the SAME co-player from Round 9 (it says "your current co-player" which is the same language used when it's the same player)
- Last round (Round 9): They sent me \$2.89, I received \$8.67, and I returned \$4.00

**\*\*My sender history:\*\***

- Round 1: Sent \$3 ? received \$3.69
- Round 3: Sent \$3 ? received \$5.37
- Round 6: Sent \$3 ? received \$3.36
- Round 7: Sent \$3 ? received \$3.30

**\*\*Strategy:\*\***

Since this is the same co-player from Round 9 where:

- They sent me \$2.89 (58% of their \$5)
- I returned \$4.00 (46% of \$8.67)

- This gave them:  $\$5 - \$2.89 + \$4.00 = \$6.11$  (profit of \$1.11)

They showed good trust with 58% of their endowment. I responded with reasonable reciprocity. Now as sender, I should send a similar or slightly higher amount to build on this relationship. Sending \$3 has been my consistent strategy and it works well.

Actually, let me reconsider - they sent \$2.89, which is close to \$3. So matching their level of trust with \$3 makes sense and maintains consistency.

```
Usage: {
 "input_tokens": 2307,
 "output_tokens": 513,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-12T09:29:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_2"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_2": "trustee",
 "Agent_5": "investor"
 },
 "trustee": "Agent_2"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep:  $\$5 - \text{sent} + \text{returned}$
- Receiver keeps:  $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps:  $\$5 - \text{sent} + \text{returned}$
- You keep:  $\text{received} - \text{returned}$
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.  
The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.11, it became \$6.33, and you returned \$3.3. Your visible payoff was \$3.0300000000000002.

Your total visible earnings across all rounds are \$12.8.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- I'm facing a new co-player as sender
- Previous cooperation has been successful (total earnings \$12.80)
- When I sent \$2.50 in Round 2, I received \$3.62 back for a good profit
- My returns as receiver have been fair (~52%)

**\*\*Strategic reasoning:\*\***

- Continue the cooperative approach that's been working
- Sending 50% shows good faith while managing risk
- This gives my co-player \$7.50 to work with, enough for mutual benefit
- Establishes positive precedent with this new partner

{'send': 2.5}

**reasoning:**

Let me analyze the situation:

**\*\*Previous rounds summary:\*\***

- Round 1 (RECEIVER): Received \$7.65 from sender sending \$2.55, returned \$4.00, earned \$3.65
- Round 2 (SENDER): Sent \$2.50, became \$7.50, they returned \$3.62, earned \$6.12
- Round 3 (RECEIVER): Received \$6.33 from sender sending \$2.11, returned \$3.30, earned \$3.03
- Total earnings: \$12.80

**\*\*Pattern analysis:\*\***

- When I sent \$2.50 (50%), I got back \$3.62, giving me a profit of \$1.12 (from \$2.50 remaining + \$3.62 = \$6.12 total)
- When others sent to me, I've returned fairly (~52% of what I received)
- Overall, cooperation has been successful

**\*\*Current decision:\*\***

- New co-player as sender this round
- Need to establish trust again
- Sending around 50% has worked well previously

I'll continue with the cooperative strategy that's been working: send \$2.50 (50% of endowm

```

ent).
usage:
{
 "input_tokens": 1558,
 "output_tokens": 431,
 "reasoning_tokens": 0
}
Message 4 role=user
content:
 Round 5

 In your most recent previous game round, you were the SENDER against another agent. You sent $2.5, it became $7.5, and you saw them return $3.01 to you. Your visible payoff was $5.51.
 Your total visible earnings across all rounds are $18.310000000000002.

 This round, you are the RECEIVER against your current co-player. They sent you $2.92, so you received $8.76. How much do you return? (0-8.76)
 Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
content:
 Looking at this situation:

 What happened:
 - The sender sent $2.92 (58.4% of their endowment), showing strong trust
 - I received $8.76 after the 3x multiplier
 - The sender now has $2.08 remaining

 My reasoning:
 This sender showed above-average trust compared to previous rounds. I should reciprocate generously.

 If I return **$4.70**:
 - Sender ends with: $2.08 + $4.70 = $6.78 (profit of $1.78)
 - I end with: $8.76 - $4.70 = $4.06 (profit of $4.06)

 This represents:
 - Returning about 54% of what I received
 - Rewarding their higher trust level (58.4% sent)
 - Both profiting well from cooperation
 - Encouraging continued trust

 {'return': 4.7}
reasoning:
 Let me analyze this situation:

 What happened:
 - The sender sent $2.92 (58.4% of their endowment), showing strong trust
 - I received $8.76 after the 3x multiplier
 - The sender now has $2.08 remaining

 Patterns from previous rounds:
 - Round 1: Received $7.65, returned $4.00 (52%)
 - Round 3: Received $6.33, returned $3.30 (52%)
 - I've been consistently returning about 52% of what I receive

 Strategy:
 The sender showed high trust (58.4% sent). I should reciprocate fairly.

 If I return around 52% of $8.76:
 - $0.52 \times \$8.76 = \4.55

 Let me check the outcomes:
 - Sender ends with: $2.08 + $4.55 = $6.63 (profit of $1.63)
 - I end with: $8.76 - $4.55 = $4.21 (profit of $4.21)

 This seems fair - both profit, sender gets a decent return for their trust.

 Actually, given they sent 58.4% (more than my usual 50%), maybe I should return slightly more to really reward that trust. Let me try $4.70:

```

- Sender ends with:  $\$2.08 + \$4.70 = \$6.78$  (profit of \$1.78)
- I end with:  $\$8.76 - \$4.70 = \$4.06$  (profit of \$4.06)

That's about 54% return, which is fair and rewards their higher trust level.

**usage:**

```
{
 "input_tokens": 1844,
 "output_tokens": 622,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.92, it became \$8.76, and you returned \$4.7. Your visible payoff was \$4.06.

Your total visible earnings across all rounds are \$22.37.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My sending strategy (50%) has been successful in Rounds 2 & 4
- Returns I've received: \$3.62 and \$3.01 (averaging \$3.32)
- Total earnings of \$22.37 show cooperation is working

**\*\*Strategic reasoning:\*\***

- Maintain the consistent 50% sending strategy
- This has proven reliable across different co-players
- Gives receiver \$7.50 to work with
- Balances trust-building with risk management

{ 'send': 2.5 }

**reasoning:**

Looking at the pattern:

**\*\*My sending history:\*\***

- Round 2: Sent \$2.50 (50%), received back \$3.62, profit = \$6.12
- Round 4: Sent \$2.50 (50%), received back \$3.01, profit = \$5.51

**\*\*My receiving history:\*\***

- Round 1: Received \$7.65 from \$2.55 sent (51%), returned \$4.00
- Round 3: Received \$6.33 from \$2.11 sent (42%), returned \$3.30
- Round 5: Received \$8.76 from \$2.92 sent (58%), returned \$4.70

**\*\*Analysis:\*\***

- When I send 50%, I've gotten decent returns (\$3.01-\$3.62)
- As a receiver, I've been fair, returning around 52-54%
- Total earnings are strong at \$22.37
- Cooperation has been working well

**\*\*This round:\*\***

- New co-player as sender
- Continue with the cooperative 50% strategy that's been successful
- This maintains consistency and shows I'm a cooperative player

I'll send \$2.50 again.

**usage:**

```
{
 "input_tokens": 2182,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.8 to you. Your visible payoff was \$6.3.

Your total visible earnings across all rounds are \$28.67.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My consistent 50% sending strategy has worked well (Rounds 2, 4, 6)
- Returns have varied (\$3.62, \$3.01, \$3.80) but all profitable
- Total earnings of \$28.67 show strong overall cooperation

**\*\*Strategic reasoning:\*\***

- Continue the reliable 50% sending pattern
- This approach has proven effective across different co-players
- Provides \$7.50 for receiver to reciprocate
- Maintains consistency and manages risk

{'send': 2.5}

**reasoning:**

Looking at this situation:

**\*\*Context:\*\***

- I'm the sender again, facing a new co-player
- My sending history shows consistent 50% sends (\$2.50 in rounds 2, 4, 6)
- Returns I've received: \$3.62, \$3.01, \$3.80 (averaging \$3.48)
- All have been profitable
- Total earnings of \$28.67

**\*\*Strategic reasoning:\*\***

- The 50% strategy has been consistently successful
- Returns have varied but all been positive
- This creates good conditions for mutual benefit
- Should continue this proven approach

I'll send \$2.50 again.

**usage:**

```
{
 "input_tokens": 2442,
 "output_tokens": 294,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$3.11 to you. Your visible payoff was \$5.609999999999999.

Your total visible earnings across all rounds are \$34.28.

This round, you are the RECEIVER against your current co-player. They sent you \$2.29, so you received \$6.87. How much do you return? (0-6.87)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$2.29 (45.8% of their endowment), showing moderate trust

- I received \$6.87 after the 3x multiplier
- The sender now has \$2.71 remaining

**\*\*My reasoning:\*\***

Based on my consistent pattern in previous receiver rounds (returning ~52-54%), I should reciprocate proportionally.

If I return **\*\*\$3.57\*\***:

- Sender ends with:  $\$2.71 + \$3.57 = \$6.28$  (profit of \$1.28)
- I end with:  $\$6.87 - \$3.57 = \$3.30$  (profit of \$3.30)

This represents:

- Returning about 52% of what I received
- Both parties profiting from cooperation
- Consistent reciprocation encouraging continued trust

{'return': 3.57}

**reasoning:**

Let me analyze this situation:

**\*\*What happened:\*\***

- The sender sent \$2.29 (45.8% of their endowment), showing moderate trust
- I received \$6.87 after the 3x multiplier
- The sender now has \$2.71 remaining

**\*\*My approach in previous receiver rounds:\*\***

- Round 1: Received \$7.65, returned \$4.00 (52.3%)
- Round 3: Received \$6.33, returned \$3.30 (52.1%)
- Round 5: Received \$8.76, returned \$4.70 (53.7%)

I've been consistent in returning around 52-54% of what I receive.

For this round, if I return around 52%:

$\$6.87 \times 0.52 = \$3.57$

Let me check the outcomes:

- If I return \$3.57:
  - Sender ends with:  $\$2.71 + \$3.57 = \$6.28$  (profit of \$1.28)
  - I end with:  $\$6.87 - \$3.57 = \$3.30$  (profit of \$3.30)

This seems fair and consistent with my previous behavior. The sender sent 45.8% which is moderate trust, and returning 52% rewards that trust while both parties profit.

I'll return \$3.57.

**usage:**

```
{
 "input_tokens": 2408,
 "output_tokens": 540,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.29, it became \$6.87, and you returned \$3.57. Your visible payoff was \$3.3000000000000003.

Your total visible earnings across all rounds are \$37.58.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*Context:\*\***

- New co-player as sender
- My 50% sending strategy has been consistently profitable (Rounds 2, 4, 6, 7)
- Returns ranged from \$3.01 to \$3.80, all profitable

- Total earnings of \$37.58 show strong overall success

**\*\*Strategic reasoning:\*\***

- Continue the proven 50% sending strategy
- This has worked reliably across all different co-players
- Provides \$7.50 for receiver to reciprocate fairly
- Maintains trust-building while managing risk

```
{'send': 2.5}
```

**reasoning:**

Looking at my history as sender:

- Round 2: Sent \$2.50, received back \$3.62, payoff \$6.12
- Round 4: Sent \$2.50, received back \$3.01, payoff \$5.51
- Round 6: Sent \$2.50, received back \$3.80, payoff \$6.30
- Round 7: Sent \$2.50, received back \$3.11, payoff \$5.61

My consistent strategy of sending \$2.50 (50%) has been working well. All returns have been profitable, ranging from \$3.01 to \$3.80. I'm now at \$37.58 total earnings, which is very good.

I should continue with the same strategy that has been successful - sending \$2.50 (50% of endowment). This:

- Maintains consistency
- Shows good faith to co-players
- Has proven profitable across different partners
- Balances trust and risk management

New co-player this round, so sticking with the proven approach makes sense.

**usage:**

```
{
 "input_tokens": 2442,
 "output_tokens": 395,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$2.5, it became \$7.5, and you saw them return \$3.62 to you. Your visible payoff was \$6.12.

Your total visible earnings across all rounds are \$43.699999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.34, so you received \$10.02. How much do you return? (0-10.02)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- Same co-player from Round 9 (where I sent them \$2.5 and they returned \$3.62)
- They sent \$3.34 (66.8% of their endowment) - showing very high trust
- I received \$10.02 after the 3x multiplier
- The sender now has \$1.66 remaining

**\*\*My reasoning:\*\***

This co-player showed strong reciprocity last round and is now demonstrating even greater trust. I should reciprocate generously.

If I return **\*\*\$5.30\*\***:

- Sender ends with:  $\$1.66 + \$5.30 = \$6.96$  (profit of \$1.96)
- I end with:  $\$10.02 - \$5.30 = \$4.72$  (profit of \$4.72)

This represents:

- Returning about 53% of what I received
- Rewarding their high trust (66.8% sent)
- Continuing positive cooperation with this co-player
- Both parties profiting well

```
{'return': 5.30}
```



**Reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- The sender sent \$3.34 (66.8% of their endowment) - this is notably higher trust than previous rounds
- I received \$10.02 after the 3x multiplier
- The sender now has \$1.66 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 9 (it says "against your current co-player")
- In Round 9, I sent them \$2.5, they returned \$3.62 (very good return!)
- They showed strong reciprocity, so now they're showing even more trust by sending 66.8%

**\*\*My reasoning:\*\***

Since this co-player returned \$3.62 on \$7.50 received (48% return rate), and now they've sent me an even higher proportion, I should reciprocate generously.

If I return **\*\*\$5.30\*\***:

- Sender ends with:  $\$1.66 + \$5.30 = \$6.96$  (profit of \$1.96)
- I end with:  $\$10.02 - \$5.30 = \$4.72$  (profit of \$4.72)

This represents:

- Returning about 53% of what I received
- Rewarding their high trust level (66.8% sent)
- Both profiting well from cooperation
- Continuing the positive relationship with this specific co-player

Actually, let me reconsider. They sent 66.8%, which is very high. I returned about 52-54% in previous rounds. Let me maintain that consistency.

53% of \$10.02 = \$5.31

Let me go with \$5.30 for simplicity.

```
Usage: {
 "input_tokens": 2375,
 "output_tokens": 662,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_6 - Sender - dyad\_2**

Timestamp: 2026-08-12T09:29:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_1"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_1": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

Message 1 role=system  
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.75. Your visible payoff was \$5.33.

Your total visible earnings across all rounds are \$15.84.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this round as SENDER again:

**\*\*My experience:\*\***

- Round 1: Sent \$3.00, received \$4.25 back ? Good outcome
- Rounds 2-3: As receiver, I've been cooperative, returning fair amounts

**\*\*Current situation:\*\***

- New co-player (different from Round 1)
- I've been part of generally cooperative dynamics

**\*\*My reasoning:\*\***

I had success with sending \$3.00 before, and the cooperative pattern across rounds suggest

s moderate-to-high sending works well. I'll send slightly more this time to signal trust while building on the positive momentum.

Sending \$3.50 creates \$10.50 for the receiver. If they're reasonably cooperative (returning ~40%), I'd still come out ahead.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

I'm back as the SENDER in round 4. Let me review my experience so far:

Round 1: I sent \$3.00, received \$4.25 back. Payoff: \$6.25

Round 2: Received \$8.01 (from \$2.67 sent), returned \$3.75. Payoff: \$4.26

Round 3: Received \$10.08 (from \$3.36 sent), returned \$4.75. Payoff: \$5.33

Total earnings: \$15.84

Now I'm the sender again with a new co-player. What should I send?

Looking at the pattern:

- When I sent \$3.00 (60%), I got a decent return
- When others sent moderate-to-high amounts (\$2.67, \$3.36), I've been fairly cooperative in returning
- The game seems to be working reasonably well with moderate-to-high sending amounts

I should consider:

1. I had a positive experience when I sent \$3.00
2. The cooperative moves I've seen have been between \$2.67-\$3.36
3. I've demonstrated I'm a fair returner, so maybe I can expect reasonable behavior

I think I should send a similar or slightly higher amount than last time. Since I received \$4.25 when I sent \$3.00, that worked out well.

Options:

- Send \$3.00 again (proven strategy)
- Send \$3.50 (show slightly more trust)
- Send \$4.00 (high trust signal)

Given:

- This is a new co-player
- I've been building cooperative patterns
- The game is going well so far

I'll send \$3.50 as a moderately high trust signal, slightly more than my first round, but not overly risky. This creates \$10.50 for the receiver.

...

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 660,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.55 to you. Your visible payoff was \$6.05.

Your total visible earnings across all rounds are \$21.89.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a

moderately cooperative move.

**\*\*My pattern:\*\***

- I've been consistently returning around 45-47% of what I receive
- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)

**\*\*My reasoning:\*\***

This sender showed moderate trust. I should maintain consistency with my previous cooperative behavior.

Returning \$3.85:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of \$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This is about 46% return rate, consistent with my pattern and fair to the sender.

```
'''json
{"return": 3.85}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.81\*\*** (56% of their endowment), which became \$8.43 for me. This is a moderately cooperative move.

**\*\*Context from previous rounds:\*\***

- Round 1: I sent \$3.00 (60%), received \$4.25 back (47% of \$9)
- Round 2: Someone sent \$2.67 (53%), I returned \$3.75 (47% of \$8.01)
- Round 3: Someone sent \$3.36 (67%), I returned \$4.75 (47% of \$10.08)
- Round 4: I sent \$3.50 (70%), received \$4.55 back (43% of \$10.50)

**\*\*Pattern:\*\***

I've been pretty consistent returning around 45-47% of what I receive. My co-player sent \$2.81, which is moderately cooperative.

Let me calculate:

- 45% of \$8.43 = \$3.79
- 47% of \$8.43 = \$3.96

Returning around \$3.85 would:

- Sender would get: \$2.19 (kept) + \$3.85 = \$6.04 total (gain of ~\$1.04)
- I would get: \$8.43 - \$3.85 = \$4.58 (good gain)

This maintains consistency with my previous behavior and rewards the sender proportionally for their trust.

**usage:**

```
{
 "input_tokens": 1847,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$3.85. Your visible payoff was \$4.58.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 back (good outcome)

- Round 4: Sent \$3.50 ? Received \$4.55 back (good outcome)

**\*\*Overall dynamics:\*\***

- Cooperation has been working well across all rounds
- I've been a fair receiver (returning ~46%)
- Others have been reasonably cooperative with me

**\*\*My reasoning:\*\***

I'm with a new co-player. The cooperative pattern continues to work. I'll send an amount between my previous sends to signal trust while being strategic.

Sending \$3.25 creates \$9.75 for the receiver. With typical cooperation, this should benefit us both.

```
'''json
{"send": 3.25}
'''
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 1: Sent \$3.00, received \$4.25 back ? Payoff \$6.25
- Round 4: Sent \$3.50, received \$4.55 back ? Payoff \$6.05

As RECEIVER:

- Round 2: Received \$8.01, returned \$3.75 ? Payoff \$4.26
- Round 3: Received \$10.08, returned \$4.75 ? Payoff \$5.33
- Round 5: Received \$8.43, returned \$3.85 ? Payoff \$4.58

Overall, the cooperative strategy has been working well. When I send moderate-to-high amounts (\$3-\$3.50), I get good returns. My co-players have generally been cooperative.

This is a new co-player again (Round 6). I've been doing well with sends in the \$3-\$3.50 range.

I'll continue with a similar strategy - sending \$3.25 this time (between my previous two sends).

This creates \$9.75 for the receiver. If they return around 45%, I'd get about \$4.39 back, giving me a payoff of \$5.89.

Actually, let me think about this differently. I've been gradually increasing (3.0 then 3.5). Maybe I should go higher to signal stronger trust, or stay consistent. Let me stick with \$3.50 since it worked well.

Actually, let me be a bit more varied. \$3.25 is a good middle ground.

**usage:**

```
{
 "input_tokens": 2205,
 "output_tokens": 567,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.25, it became \$9.75, and you saw them return \$2.56 to you. Your visible payoff was \$4.3100000000000005.

Your total visible earnings across all rounds are \$30.78.

This round, you are the RECEIVER against your current co-player. They sent you \$2.85, so you received \$8.55. How much do you return? (0-8.55)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$2.85\*\*** (57% of their endowment), which became \$8.55 for me. This is a moderately cooperative move.

**\*\*Context:\*\***

- Round 6 was disappointing - I sent \$3.25 but only got \$2.56 back (26% return)
- However, this is a NEW co-player, not the same one
- My consistent pattern as receiver has been returning ~46-47%

**\*\*My reasoning:\*\***

Despite the poor return last round, I shouldn't penalize this new co-player. Maintaining consistency and fairness has been my strategy, and it's worked well overall.

Returning \$3.90:

- Sender would get: \$2.15 (kept) + \$3.90 = \$6.05 total (gain of \$1.05)
- I would get: \$8.55 - \$3.90 = \$4.65 (good gain)

This maintains my ~46% return rate and rewards the sender's trust appropriately.

```
'''json
{"return": 3.90}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$2.85\*\*** (57% of their endowment), which became \$8.55 for me. This is a moderately cooperative move.

**\*\*Recent experience:\*\***

- Round 6: I sent \$3.25, received only \$2.56 back (26% of \$9.75)
- That was disappointing - I got \$4.31 total, barely breaking even
- This is notably lower than my previous sender experiences

**\*\*My pattern as receiver:\*\***

- Round 2: Returned 47% (\$3.75 of \$8.01)
- Round 3: Returned 47% (\$4.75 of \$10.08)
- Round 5: Returned 46% (\$3.85 of \$8.43)

**\*\*Considerations:\*\***

1. The last round showed less cooperation from my co-player
2. But this is a NEW co-player now
3. They sent \$2.85, which is comparable to what others sent
4. Should I maintain my consistent cooperative strategy?

I think I should maintain consistency. The poor return in Round 6 was disappointing, but:

- This is a different co-player
- I've built a pattern of fairness
- Retaliating against a new player isn't justified
- Long-term cooperation benefits everyone

Following my pattern of ~46% return:  
 $\$8.55 \times 0.46 = \$3.93$

Sender would get: \$2.15 (kept) + \$3.93 = \$6.08 (gain of \$1.08)  
I would get: \$8.55 - \$3.93 = \$4.62 (good gain)

I'll return \$3.90 to stay consistent with my cooperative approach.

**usage:**

```
{
 "input_tokens": 2534,
 "output_tokens": 674,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.85, it became \$8.55, and you returned \$3.9. Your visible payoff was \$4.65.

Your total visible earnings across all rounds are \$35.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 (good)
- Round 4: Sent \$3.50 ? Received \$4.55 (good)
- Round 6: Sent \$3.25 ? Received \$2.56 (poor)

**\*\*Overall situation:\*\***

- 2 out of 3 sending experiences were positive
- I've been consistently fair as receiver
- Cooperation has generally worked well

**\*\*My reasoning:\*\***

Despite one disappointing return in Round 6, the cooperative approach has been successful overall. I'm with a new co-player and should maintain my strategy of moderate trust.

Sending \$3.00 - this worked well in Round 1 and represents a balanced approach that signals cooperation without overexposing myself.

```
'''json
{"send": 3.0}
'''
```

**reasoning:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 back (payoff: \$6.25)
- Round 4: Sent \$3.50 ? Received \$4.55 back (payoff: \$6.05)
- Round 6: Sent \$3.25 ? Received \$2.56 back (payoff: \$4.31) - disappointing

**\*\*My track record as receiver:\*\***

- Round 2: Received \$8.01, returned \$3.75 (payoff: \$4.26)
- Round 3: Received \$10.08, returned \$4.75 (payoff: \$5.33)
- Round 5: Received \$8.43, returned \$3.85 (payoff: \$4.58)
- Round 7: Received \$8.55, returned \$3.90 (payoff: \$4.65)

**\*\*Overall:\*\***

- I've been consistently fair as a receiver (returning ~46%)
- As sender, 2 out of 3 times worked well, but Round 6 was poor
- Total earnings: \$35.43 - doing well overall
- This is Round 8, potentially the last round or near the end

**\*\*Current situation:\*\***

- New co-player again
- Need to decide how much to send

**\*\*My reasoning:\*\***

The cooperative strategy has worked well overall. Despite one bad experience, the majority have been positive. I should continue with moderate-to-high sending to maximize potential gains.

Given this might be near the end, I should still send a reasonable amount. I'll continue with my strategy around \$3.00-\$3.50.

Let me send \$3.00 - it worked well in Round 1 and is a solid cooperative signal.

**usage:**

```
{
 "input_tokens": 2731,
 "output_tokens": 648,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You se

nt \$3.0, it became \$9.0, and you saw them return \$4.83 to you. Your visible payoff was \$6.83.  
Your total visible earnings across all rounds are \$42.26.

This round, you are the RECEIVER against your current co-player. They sent you \$3.1, so you received \$9.3. How much do you return? (0-9.3)  
Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

**\*\*My co-player sent \$3.10\*\*** (62% of their endowment), which became \$9.30 for me. This is a generous move showing good trust.

**\*\*Context:\*\***

- Round 8 was excellent - I sent \$3.00 and got \$4.83 back
- My consistent pattern as receiver has been returning ~46-47%

**\*\*My reasoning:\*\***

This sender showed solid trust (62% of endowment). I should maintain my consistent cooperative approach.

Returning \$4.30:

- Sender would get: \$1.90 (kept) + \$4.30 = \$6.20 total (gain of \$1.20)
- I would get: \$9.30 - \$4.30 = \$5.00 (good gain)

This maintains my ~46% return rate and fairly rewards the sender's trust.

```
'''json
{"return": 4.30}
'''
```

**reasoning:**

Looking at this situation:

**\*\*My co-player sent \$3.10\*\*** (62% of their endowment), which became \$9.30 for me. This is a generous and trusting move.

**\*\*Context:\*\***

- Round 8 was great - I sent \$3.00 and received \$4.83 back (excellent return of 54%)
- My pattern as receiver has been returning ~46-47%
- This sender showed good trust (62% of endowment)

**\*\*My return pattern:\*\***

- Round 2: Returned \$3.75 of \$8.01 = 47%
- Round 3: Returned \$4.75 of \$10.08 = 47%
- Round 5: Returned \$3.85 of \$8.43 = 46%
- Round 7: Returned \$3.90 of \$8.55 = 46%

**\*\*My reasoning:\*\***

This sender showed solid trust. I should maintain my consistent ~46% return rate.

Returning \$4.30:

- Sender would get: \$1.90 (kept) + \$4.30 = \$6.20 total (gain of \$1.20)
- I would get: \$9.30 - \$4.30 = \$5.00 (good gain)

This is about 46% return rate, consistent with my pattern.

**usage:**

```
{
 "input_tokens": 2698,
 "output_tokens": 544,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.1, it became \$9.3, and you returned \$4.3. Your visible payoff was \$5.00000000000001.

Your total visible earnings across all rounds are \$47.26.



This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at this round as SENDER:

**\*\*My track record as sender:\*\***

- Round 1: Sent \$3.00 ? Received \$4.25 (good)
- Round 4: Sent \$3.50 ? Received \$4.55 (good)
- Round 6: Sent \$3.25 ? Received \$2.56 (poor)
- Round 8: Sent \$3.00 ? Received \$4.83 (excellent!)

**\*\*Overall situation:\*\***

- 3 out of 4 sending experiences were positive
- Most recent round was particularly successful
- Total earnings of \$47.26 show cooperation is working

**\*\*My reasoning:\*\***

The cooperative strategy continues to work well overall. The \$3.00 amount has been particularly successful for me (Rounds 1 and 8). I'm with a new co-player and should maintain the trust-building approach.

```
'''json
{"send": 3.0}
'''
```

**Reasoning:**

Looking at my track record as sender:

- Round 1: Sent \$3.00 ? Received \$4.25 (good)
- Round 4: Sent \$3.50 ? Received \$4.55 (good)
- Round 6: Sent \$3.25 ? Received \$2.56 (poor)
- Round 8: Sent \$3.00 ? Received \$4.83 (excellent)

So 3 out of 4 sending experiences were positive, with the most recent one being particularly good.

Overall, cooperation has been working well. I have \$47.26 in total earnings, which is quite strong.

I'm with a new co-player again. My strategy of sending moderate-to-high amounts (around \$3.00-\$3.50) has generally worked well.

Given the recent positive trend and overall success with cooperation, I should continue with a similar strategy. Sending around \$3.00-\$3.25 seems appropriate - it signals trust without being overly risky.

I'll send \$3.00 - this has worked well for me in two rounds, including my most recent sending round.

```
Usage: {
 "input_tokens": 2666,
 "output_tokens": 497,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_2**

Timestamp: 2026-08-12T09:29:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_1"
],
 "investor": "Agent_6",
 "roles": {
```

```

 "Agent_1": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.57 to you. Your visible payoff was \$7.57.

Your total visible earnings across all rounds are \$17.32.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed trust by sending \$2.99 (about 60% of their endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has \$2.01

I want to reciprocate their trust fairly. If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (profit of \$1.51)
- I keep: \$4.47

This rewards their cooperation while maintaining a fair split of the gains.

```
{'return': 4.50}
```

**reasoning:**

Let me analyze this round:

- I'm the receiver
- The sender sent \$2.99 (about 60% of their \$5 endowment)
- I received \$8.97 after the 3x multiplier
- The sender currently has  $\$5 - \$2.99 = \$2.01$

The sender showed good trust by sending nearly 60% of their endowment. I should reciprocate fairly.

If I return \$4.50:

- Sender ends with:  $\$2.01 + \$4.50 = \$6.51$  (better than their initial \$5)
- I keep:  $\$8.97 - \$4.50 = \$4.47$

This is fair and rewards their trust. It's roughly half of what I received, which seems reasonable given they sent about 60% of their endowment.

Looking at my previous behavior:

- Round 1: Received \$10.02, returned \$5.00 (kept \$5.02)
- Round 3: The other player returned \$5.57 when they received \$9.00

I should be consistent and fair. Returning around \$4.50 seems appropriate.

**usage:**

```
{
 "input_tokens": 1269,
 "output_tokens": 416,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.99, it became \$8.97, and you returned \$4.5. Your visible payoff was \$4.470000000000001.

Your total visible earnings across all rounds are \$21.79.

This round, you are the RECEIVER against your current co-player. They sent you \$2.89, so you received \$8.67. How much do you return? (0-8.67)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.89 (about 58% of their endowment), showing trust
- I received \$8.67 after the 3x multiplier
- The sender currently has \$2.11

I'll reciprocate fairly. If I return \$4.30:

- Sender ends with:  $\$2.11 + \$4.30 = \$6.41$  (profit of \$1.41)
- I keep: \$4.37

This rewards cooperation while maintaining a fair distribution.

```
{'return': 4.30}
```

**reasoning:**

The sender sent \$2.89 (about 58% of their \$5 endowment), showing reasonable trust.

I received \$8.67 after the 3x multiplier.  
The sender currently has \$5 - \$2.89 = \$2.11

I should reciprocate fairly. Looking at my previous behavior:

- Round 1: Sender sent \$3.34, I received \$10.02, returned \$5.00
- Round 4: Sender sent \$2.99, I received \$8.97, returned \$4.50

This sender sent slightly less (\$2.89 vs \$2.99), so I should return proportionally. If I return \$4.30:

- Sender ends with: \$2.11 + \$4.30 = \$6.41 (profit of \$1.41)
- I keep: \$4.37

This maintains fair reciprocation.

**usage:**

```
{
 "input_tokens": 1545,
 "output_tokens": 349,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.89, it became \$8.67, and you returned \$4.3. Your visible payoff was \$4.37.

Your total visible earnings across all rounds are \$26.16.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my pattern:

- As a sender, I've consistently sent \$3.00 to show trust and encourage cooperation
- As a receiver, I've been returning amounts that fairly reward the sender's trust

I'll maintain my consistent cooperative approach with this new co-player.

```
{'send': 3.00}
```

**reasoning:**

Looking at my history:

- Round 1 (receiver): Received \$10.02 from sender's \$3.34, returned \$5.00
- Round 2 (sender): Sent \$3.00, received back \$2.73
- Round 3 (sender): Sent \$3.00, received back \$5.57
- Round 4 (receiver): Received \$8.97 from sender's \$2.99, returned \$4.50
- Round 5 (receiver): Received \$8.67 from sender's \$2.89, returned \$4.30

Now Round 6, I'm a sender again with a new co-player.

My sending pattern has been consistent at \$3.00 when I'm the sender. The returns I've received have varied (\$2.73 and \$5.57), but both times my co-players received \$9.00 and chose to return different amounts.

I should continue with a consistent cooperative strategy. Sending \$3.00 has been my standard, showing significant trust (60% of endowment).

```
{'send': 3.00}
```

**usage:**

```
{
 "input_tokens": 1796,
 "output_tokens": 332,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.33 to you. Your visible payoff was \$

6.33.

Your total visible earnings across all rounds are \$32.49.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.78 (about 56% of their endowment), showing trust
- I received \$8.34 after the 3x multiplier
- The sender currently has \$2.22

I'll reciprocate fairly. If I return \$4.20:

- Sender ends with:  $\$2.22 + \$4.20 = \$6.42$  (profit of \$1.42)
- I keep: \$4.14

This maintains my pattern of fairly rewarding cooperation.

{ 'return': 4.20 }

**reasoning:**

The sender sent \$2.78 (about 55.6% of their endowment), showing trust.

I received \$8.34 after the 3x multiplier.

The sender currently has  $\$5 - \$2.78 = \$2.22$ .

I've been fair in my returns as a receiver:

- Round 1: Received \$10.02, returned \$5.00
- Round 4: Received \$8.97, returned \$4.50
- Round 5: Received \$8.67, returned \$4.30

For this round with \$8.34 received, a fair return would be around \$4.00-\$4.20.

If I return \$4.20:

- Sender ends with:  $\$2.22 + \$4.20 = \$6.42$  (profit of \$1.42)
- I keep:  $\$8.34 - \$4.20 = \$4.14$

This is consistent with my previous fair returns.

**usage:**

```
{
 "input_tokens": 2001,
 "output_tokens": 368,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.78, it became \$8.34, and you returned \$4.2. Your visible payoff was \$4.14.

Your total visible earnings across all rounds are \$36.63.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my track record:

- I've consistently sent \$3.00 as a sender, showing reliability and trust
- This strategy has generally been reciprocated well
- As a receiver, I've been fair in my returns

I'll maintain my consistent cooperative approach.

{ 'send': 3.00 }

**reasoning:**

Looking at my history as a sender:

- Round 2: Sent \$3.00, received back \$2.73 (payoff: \$4.73)
- Round 3: Sent \$3.00, received back \$5.57 (payoff: \$7.57)
- Round 6: Sent \$3.00, received back \$4.33 (payoff: \$6.33)

As a receiver, I've been consistently fair, returning roughly half of what I received or a bit less, ensuring the sender profits from trusting me.

I've established a pattern of sending \$3.00 as a sender, which shows good trust and has generally been reciprocated well. I'll continue this consistent approach.

This is Round 8 with a new co-player. I should maintain my cooperative strategy.

**usage:**

```
{
 "input_tokens": 2031,
 "output_tokens": 263,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.69 to you. Your visible payoff was \$5.689999999999995.

Your total visible earnings across all rounds are \$42.32.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my consistent strategy:

- I've sent \$3.00 in every sender round, showing reliability
- Returns have varied but overall been reasonable
- This maintains trust and encourages cooperation

I'll continue with my established pattern.

```
{'send': 3.00}
```

**reasoning:**

Let me review my history:

- As sender: I've consistently sent \$3.00 in rounds 2, 3, 6, 8, and now 9
- Returns I've received: \$2.73, \$5.57, \$4.33, \$3.69
- As receiver: I've returned \$5.00, \$4.50, \$4.30, \$4.20

My strategy of sending \$3.00 has worked reasonably well, though returns vary. I've built up good earnings (\$42.32 total).

The last round return of \$3.69 was a bit lower than some others, but still represents reasonable cooperation (I profited \$0.69 on that round).

I should continue with my consistent approach of sending \$3.00. This:

1. Shows I'm a reliable cooperative partner
2. Maximizes total surplus
3. Has generally been reciprocated
4. Maintains my reputation across different co-players

I'll stick with \$3.00.

**usage:**

```
{
 "input_tokens": 2007,
 "output_tokens": 298,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.42 to you. Your visible payoff was \$7.42.

Your total visible earnings across all rounds are \$49.74.

This round, you are the RECEIVER against your current co-player. They sent you \$3.51, so you received \$10.53. How much do you return? (0-10.53)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

- The sender sent \$3.51 (about 70% of their endowment), showing strong trust
- I received \$10.53 after the 3x multiplier
- The sender currently has \$1.49

This is more generous than typical, so I should reciprocate accordingly. If I return \$5.30:

- Sender ends with: \$1.49 + \$5.30 = \$6.79 (profit of \$1.79)
- I keep: \$5.23

This fairly rewards their higher level of trust.

{'return': 5.30}

**Reasoning:**

Looking at this situation:

- The sender sent \$3.51 (about 70% of their endowment), showing significant trust
- I received \$10.53 after the 3x multiplier
- The sender currently has \$1.49

This is more than I typically see sent. I should reciprocate this higher trust appropriately.

If I return \$5.30:

- Sender ends with: \$1.49 + \$5.30 = \$6.79 (profit of \$1.79)
- I keep: \$5.23

This seems fair and rewards their stronger cooperation while still splitting the gains reasonably.

```
Usage: {
 "input_tokens": 1981,
 "output_tokens": 293,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_4 - Sender - dyad\_3**

Timestamp: 2026-08-12T09:29:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_7"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_4": "investor",
 "Agent_7": "trustee"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.27, it became \$6.810000000000005, and you returned \$3.5. Your visible payoff was \$3.3100000000000005.

Your total visible earnings across all rounds are \$14.78.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my experience so far:

- When I've shown trust as sender (60% in Round 2), I received good reciprocity (~50% return)
- When others showed trust as sender, I've returned approximately 50%
- The pattern suggests cooperative play with moderate to high trust levels

I'll continue with a similar strategy, sending \$3.00 (60% of my endowment) to maintain trust and cooperation.

{'send': 3.00}

**reasoning:**

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$3.32 (66.4%)
- I received \$9.96



- I returned \$5.00 (about 50%)
- My payoff: \$4.96

Round 2: I was SENDER

- I sent \$3.00 (60%)
- It became \$9.00
- They returned \$4.51 (about 50%)
- My payoff: \$6.51

Round 3: I was RECEIVER

- Sender sent \$2.27 (45.4%)
- I received \$6.81
- I returned \$3.50 (about 51%)
- My payoff: \$3.31

Round 4: I am SENDER now

- I have \$5 to decide what to send

Looking at the pattern:

- In Round 2, I sent \$3.00 (60%) and got back about 50%
- Others have sent between 45% and 66% in the rounds I've seen

I've been experiencing decent reciprocity so far. The return rates have been around 50% consistently.

I should continue to show trust, as it's been rewarded. Sending around 60% seems reasonable based on what I've seen. I'll send \$3.00 again.

**usage:**

```
{
 "input_tokens": 1421,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$21.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience as sender:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)

Sending 60% of my endowment has consistently led to good reciprocity. I'll maintain this cooperative strategy.

```
{'send': 3.00}
```

**reasoning:**

Looking at my track record:

- Round 2: I sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: I sent \$3.00, received \$4.44 back (payoff \$6.44)

Both times sending \$3.00 has resulted in good returns. The strategy of sending 60% has been working well and establishing cooperative relationships.

I should continue with this consistent strategy to maintain trust and cooperation.

**usage:**

```
{
 "input_tokens": 1648,
 "output_tokens": 217,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.38 to you. Your visible payoff was \$6.38.

Your total visible earnings across all rounds are \$27.599999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.26, so you received \$9.78. How much do you return? (0-9.78)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

This is similar to Round 1 where I received high trust. I've been consistently returning a round 50% as receiver, which has worked well.

Returning \$5.00:

- Sender ends with: \$1.74 + \$5.00 = \$6.74 (profit of \$1.74)
- I end with: \$9.78 - \$5.00 = \$4.78

This maintains my reciprocal strategy.

{'return': 5.00}

**reasoning:**

Looking at this situation:

- The sender sent \$3.26 (65.2% of their endowment), showing high trust
- I received \$9.78 after the 3x multiplier
- The sender currently has \$1.74 remaining

In my previous rounds as receiver:

- Round 1: Received \$9.96, returned \$5.00 (50.2%)
- Round 3: Received \$6.81, returned \$3.50 (51.4%)

As sender, I've been sending \$3.00 and receiving returns of around \$4.40-4.50, which is about 49-50% of what the receiver got.

This sender showed high trust (65.2%). I should reciprocate with a similar 50% return rate.

Returning \$5.00:

- Sender ends with: \$1.74 + \$5.00 = \$6.74 (profit of \$1.74)
- I end with: \$9.78 - \$5.00 = \$4.78

This is about 51% return, consistent with my previous behavior as receiver.

**usage:**

```
{
 "input_tokens": 1888,
 "output_tokens": 439,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.26, it became \$9.78, and you returned \$5.0. Your visible payoff was \$4.7799999999999999.

Your total visible earnings across all rounds are \$32.379999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my consistent strategy:

- As sender, I've been sending \$3.00 (60% of my endowment) and receiving good reciprocity (~\$4.40-\$4.50)
- This cooperative approach has been working well

I'll continue with the same strategy.

```
{'send': 3.00}
```

**reasoning:**

Looking at my history:

As SENDER:

- Round 2: Sent \$3.00, received \$4.51 back (payoff \$6.51)
- Round 4: Sent \$3.00, received \$4.44 back (payoff \$6.44)
- Round 5: Sent \$3.00, received \$4.38 back (payoff \$6.38)

As RECEIVER:

- Round 1: Received \$9.96 from sender sending \$3.32, returned \$5.00 (payoff \$4.96)
- Round 3: Received \$6.81 from sender sending \$2.27, returned \$3.50 (payoff \$3.31)
- Round 6: Received \$9.78 from sender sending \$3.26, returned \$5.00 (payoff \$4.78)

My strategy has been consistent:

- As sender: Send \$3.00 (60% of endowment)
- As receiver: Return approximately 50% of what I receive

This has been working well. My returns as sender have been good (around \$4.40-\$4.50), and I've been fair as receiver.

I should continue this strategy and send \$3.00.

**usage:**

```
{
 "input_tokens": 2187,
 "output_tokens": 375,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.49 to you. Your visible payoff was \$6.49.

Your total visible earnings across all rounds are \$38.87.

This round, you are the RECEIVER against your current co-player. They sent you \$2.81, so you received \$8.43. How much do you return? (0-8.43)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.81 (56.2% of their endowment), showing solid trust
- I received \$8.43 after the 3x multiplier
- The sender currently has \$2.19 remaining

Following my consistent pattern of returning approximately 50% of what I receive:

Returning \$4.20:

- Sender ends with: \$2.19 + \$4.20 = \$6.39 (profit of \$1.39)
- I end with: \$8.43 - \$4.20 = \$4.23

This maintains my reciprocal strategy.

```
{'return': 4.20}
```

**reasoning:**

Looking at this round:

- The sender sent \$2.81 (56.2% of their endowment), showing moderate-high trust
- I received \$8.43 after the 3x multiplier
- The sender currently has \$2.19 remaining

My pattern as receiver:

- Round 1: Received \$9.96, returned \$5.00 (50.2% return)
- Round 3: Received \$6.81, returned \$3.50 (51.4% return)
- Round 6: Received \$9.78, returned \$5.00 (51.1% return)

I've been consistently returning around 50-51% of what I receive. This has been a fair and reciprocal strategy.

For this round:

Returning \$4.20:

- Sender ends with: \$2.19 + \$4.20 = \$6.39 (profit of \$1.39)
- I end with: \$8.43 - \$4.20 = \$4.23

This is about 49.8% return, which is consistent with my previous behavior.

Actually, let me be more precise. 50% of \$8.43 would be \$4.215, which I could round to \$4.20 or \$4.25.

I'll go with \$4.20 to keep it consistent with my ~50% return pattern.

**usage:**

```
{
 "input_tokens": 2134,
 "output_tokens": 484,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.81, it became \$8.43, and you returned \$4.2. Your visible payoff was \$4.2299999999999995.

Your total visible earnings across all rounds are \$43.099999999999994.

This round, you are the RECEIVER against your current co-player. They sent you \$2.45, so you received \$7.3500000000000005. How much do you return? (0-7.3500000000000005)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- The sender sent \$2.45 (49% of their endowment), showing moderate trust
- I received \$7.35 after the 3x multiplier
- The sender currently has \$2.55 remaining

Following my consistent strategy of returning approximately 50% of what I receive:

Returning \$3.70:

- Sender ends with: \$2.55 + \$3.70 = \$6.25 (profit of \$1.25)
- I end with: \$7.35 - \$3.70 = \$3.65

```
{'return': 3.70}
```

**reasoning:**

Looking at this situation:

- The sender sent \$2.45 (49% of their endowment), showing moderate trust
- I received \$7.35 after the 3x multiplier
- The sender currently has \$2.55 remaining

I've been consistently returning approximately 50% of what I receive as a receiver. Let me calculate:

50% of \$7.35 = \$3.675

Returning \$3.70:

- Sender ends with: \$2.55 + \$3.70 = \$6.25 (profit of \$1.25)
- I end with: \$7.35 - \$3.70 = \$3.65

This maintains my consistent reciprocal strategy.

**usage:**

```
{
 "input_tokens": 2218,
 "output_tokens": 316,
}
```

```

 "reasoning_tokens": 0
 }
}
Message 14 role=user
content:
 Round 10

 In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send $2.45, it became $7.3500000000000005, and you returned $3.7. Your visible payoff was $3.6500000000000004. Your total visible earnings across all rounds are $46.74999999999999.

 This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
 Respond exactly as JSON: {'send': <amount>}
Assistant output:
 Looking at this situation:
 - I'm now playing against the same co-player who sent me $2.45 last round
 - I returned $3.70 to them (50% of what I received)
 - My consistent strategy as sender has been to send $3.00 (60% of my endowment), which has reliably generated good reciprocity

 I'll maintain my established cooperative strategy.

 {'send': 3.00}
Reasoning:
 This is interesting - I'm now playing against the same co-player as Round 9. Let me review:

 Round 9:
 - They sent me $2.45 (49% of their endowment)
 - I received $7.35
 - I returned $3.70 (about 50% of what I received)
 - They ended with $6.25, I ended with $3.65

 Now I'm the sender. This co-player showed moderate trust last round (sending $2.45). I've been consistently sending $3.00 (60% of my endowment) and it's been working well for me.

 Should I adjust based on what they sent me? They sent $2.45, which is less than my usual $3.00. However, I've been consistent with my strategy and it's been working well overall.

 I think I should stick with my established strategy of sending $3.00. This shows I'm willing to trust at a consistent level regardless of what they did (which was still reasonably cooperative at 49%).

 Let me check my total strategy:
 - Round 2 (sender): sent $3.00, received $4.51
 - Round 4 (sender): sent $3.00, received $4.44
 - Round 5 (sender): sent $3.00, received $4.38
 - Round 7 (sender): sent $3.00, received $4.49

 Consistently good returns. I'll stick with $3.00.
Usage: {
 "input_tokens": 2161,
 "output_tokens": 445,
 "reasoning_tokens": 0
}

GAME - Agent_7 - Receiver - dyad_3
Timestamp: 2026-08-12T09:29:35
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_7"
],
 },

```

```

"investor": "Agent_4",
"roles": {
 "Agent_4": "investor",
 "Agent_7": "trustee"
},
"trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.66, it became \$10.98, and you returned \$5.5. Your visible payoff was \$5.48.

Your total visible earnings across all rounds are \$15.05.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My experience so far:\*\***

- Round 1: I sent \$3 and received \$5.56 back - good cooperation!
- Rounds 2-3: As receiver, I tried to return fairly and reward trust
- Total earnings: \$15.05

**\*\*Considerations for this round:\*\***

- I'm paired with a new co-player each round
- My Round 1 send of \$3 worked well
- The most generous player sent me \$3.66, showing high trust pays off
- Sending more creates more total value (3x multiplier)
- I want to encourage cooperative norms in the pool

**\*\*Strategy:\*\***

I'll send a bit more than Round 1 to signal strong cooperation while managing risk. This shows I'm willing to trust and invest in mutual gains.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Now I'm the SENDER again in Round 4.

Let me review my history:

- Round 1 (SENDER): Sent \$3, received back \$5.56, payoff \$7.56
- Round 2 (RECEIVER): Received \$5.01 (from \$1.67 sent), returned \$3.00, payoff \$2.01
- Round 3 (RECEIVER): Received \$10.98 (from \$3.66 sent), returned \$5.50, payoff \$5.48

This is a new co-player each round in this multi-agent setting.

My experience so far:

- When I sent \$3, I got back \$5.56 - good return!
- When others sent to me, I tried to be fair/cooperative
- The player who sent \$3.66 was very trusting

For this round, I need to decide how much to send to a new co-player.

Considerations:

- I've seen good cooperation so far (my Round 1 partner returned well)
- I want to continue building cooperative patterns
- Sending more creates more value (3x multiplier)
- But I'm paired with a new player each time, so no direct reputation building

Strategy:

- In Round 1, I sent \$3 and it worked well
- I could send more to signal strong cooperation
- Or stick with \$3 as a moderate cooperative amount

Given that I've had positive experiences and want to continue fostering cooperation, I'll send a bit more this time - maybe \$3.5 or \$4. This shows I'm willing to trust while not going all-in.

Let me go with \$3.5 - it's between my previous \$3 send and the very generous \$3.66 I received in Round 3.

Actually, thinking about it more: the player who sent me \$3.66 showed strong trust and I rewarded it. If I want to encourage that kind of cooperation, I should also send generously when I'm the sender. Let me send \$4.

...

**usage:**

```
{
 "input_tokens": 1475,
 "output_tokens": 749,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$3.59 to you. Your visible payoff was \$5.09.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*Recent experience:\*\***

- Round 4: I sent \$3.50, received only \$3.59 back - minimal profit (\$5.09)
- This was disappointing compared to Round 1 where I sent \$3 and got \$5.56 back
- But this is a NEW co-player with unknown behavior

**\*\*Considerations:\*\***

- Different co-players show varying levels of cooperation
- Sending less than \$3.50 reduces my risk after last round's poor return
- Still want to signal willingness to cooperate and create value
- \$3 was my initial send and it worked well - proven strategy

**\*\*Strategy:\*\***

I'll return to sending \$3 - a moderate amount that:

- Shows trust and cooperation
- Creates meaningful value (\$9 for receiver)
- Limits downside risk
- Has proven successful before

'''json

{"send": 3}

'''

**reasoning:**

Looking at my history:

- Round 1: Sent \$3, received \$5.56 back - payoff \$7.56 (good!)
- Round 2: Receiver, returned \$3.00 - payoff \$2.01
- Round 3: Receiver, returned \$5.50 - payoff \$5.48
- Round 4: Sent \$3.5, received \$3.59 back - payoff \$5.09 (disappointing)
- Round 5: Now sender again

Analysis of Round 4:

- I sent \$3.50 (70% of endowment) showing strong trust
- They received \$10.50
- They only returned \$3.59 (34% of what they received)
- I ended with \$5.09, barely breaking even
- This was much less generous than my Round 1 partner who returned \$5.56 on a \$9 receive

This suggests I encountered a less cooperative player. However, I'm now paired with a NEW co-player, so their behavior doesn't necessarily predict this new person's behavior.

Strategic considerations:

- I'm doing reasonably well overall (\$20.14 total)
- Sending creates value through the multiplier
- But I need to balance risk
- Two different responses to my sends: one very good (\$5.56), one mediocre (\$3.59)
- I don't know if there's a pattern or if agents vary widely

I think I should continue showing willingness to cooperate but maybe pull back slightly from the \$3.50 I sent last time since that didn't work out as well. Going back to \$3 seems reasonable - it's a solid middle ground that shows cooperation without overcommitting.

**usage:**

```
{
 "input_tokens": 1797,
 "output_tokens": 607,
 "reasoning_tokens": 0
}
```



}

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.06 to you. Your visible payoff was \$6.06.

Your total visible earnings across all rounds are \$26.2.

This round, you are the RECEIVER against your current co-player. They sent you \$1.64, so you received \$4.92. How much do you return? (0-4.92)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment) - modest trust
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining

**\*\*My considerations:\*\***

- This is very similar to Round 2 where I received \$5.01 from a \$1.67 send
- In Round 2, I returned \$3.00 (60%) which seemed fair
- I should be consistent in rewarding similar levels of cooperation
- Returning around 60% rewards their trust while keeping reasonable value

**\*\*Fair return:\*\***

Returning approximately 60% of what I received:

- Return \$3.00
- They'll end with \$6.36 (profit of \$1.36)
- I'll end with \$1.92

'''json

{"return": 3.00}

'''

reasoning:

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$1.64 (33% of their endowment)
- I received \$4.92 after the 3x multiplier
- They currently have \$3.36 remaining
- Total surplus created from their send:  $\$4.92 - \$1.64 = \$3.28$

**\*\*My considerations:\*\***

- They showed modest trust by sending \$1.64 (similar to the \$1.67 I saw in Round 2)
- In Round 2, when I received \$5.01 from a \$1.67 send, I returned \$3.00 (60%)
- I should be consistent in how I respond to similar levels of trust
- Returning about 60% would be \$2.95

**\*\*Fair return calculation:\*\***

- If I return around \$3.00:
  - They end with:  $\$3.36 + \$3.00 = \$6.36$  (profit of \$1.36)
  - I end with:  $\$4.92 - \$3.00 = \$1.92$  (loss of \$0.08 from initial endowment perspective, but positive from game perspective)

Actually, let me think about this differently:

- They sent \$1.64
- This created \$3.28 in surplus ( $\$4.92 - \$1.64$ )
- A fair split of surplus would give each about \$1.64
- So they should end with:  $\$5 + \$1.64 = \$6.64$
- They have \$3.36, so I need to return \$3.28 to get them to \$6.64

But that gives me only \$1.64, which seems low for receiving \$4.92.

Let me be consistent with Round 2 where I returned 60% on a similar send:

- 60% of \$4.92 = \$2.95

...

**usage:**

```
{
 "input_tokens": 2142,
 "output_tokens": 677,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$1.64, it became \$4.92, and you returned \$3.0. Your visible payoff was \$1.92.

Your total visible earnings across all rounds are \$28.119999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My sender history:\*\***

- Round 1: Sent \$3 ? got \$5.56 back (profit \$3.56) ?
- Round 4: Sent \$3.50 ? got \$3.59 back (profit \$0.09) ?
- Round 5: Sent \$3 ? got \$4.06 back (profit \$1.06) ?

**\*\*Observations:\*\***

- \$3 has been my most reliable send amount (2/2 reasonable returns)
- \$3.50 didn't work out well
- Each round is a new co-player, so no personal history
- Total earnings: ~\$28.12 - healthy position

**\*\*Strategy:\*\***

Stick with \$3 as my standard opening send:

- Proven track record
- Shows cooperation and trust
- Creates meaningful value (\$9 for receiver)
- Manages downside risk
- Signals I'm a fair player

'''json

{"send": 3}

'''

**reasoning:**

Looking at my history:

- Round 1 (Sender): Sent \$3, got \$5.56 back - Good!
- Round 2 (Receiver): Received \$5.01, returned \$3.00
- Round 3 (Receiver): Received \$10.98, returned \$5.50
- Round 4 (Sender): Sent \$3.50, got \$3.59 back - Poor return
- Round 5 (Sender): Sent \$3, got \$4.06 back - Okay return
- Round 6 (Receiver): Received \$4.92, returned \$3.00

As SENDER:

- Round 1: Sent \$3 ? returned \$5.56 (profit \$3.56)
- Round 4: Sent \$3.50 ? returned \$3.59 (profit \$0.09)
- Round 5: Sent \$3 ? returned \$4.06 (profit \$1.06)

Total earnings: ~\$28.12

New co-player again.

The pattern shows:

- Sending \$3 has worked reasonably well (2 out of 3 times)
- Sending \$3.50 got poor returns
- Each round is a new partner with unknown behavior

I should continue with a moderate send that balances trust-building with risk management. \$3 seems to be a good baseline that's worked before.

Actually, let me think about the broader strategy. I've been fairly cooperative as a receiver, returning good amounts. As a sender, I've been moderately trusting.

Given that:

1. This is a new co-player
2. \$3 has generally worked well
3. I want to signal cooperation while managing risk
4. Total earnings are healthy at \$28.12

I'll stick with \$3 as my standard opening send.

**usage:**

```
{
 "input_tokens": 2476,
 "output_tokens": 676,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.87 to you. Your visible payoff was \$5.87.

Your total visible earnings across all rounds are \$33.989999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$2.77 (55% of their endowment) - good trust
- I received \$8.31 after the 3x multiplier
- They currently have \$2.23 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 7!
- Last round: I sent them \$3.0, they returned \$3.87 (43% of what they received)
- They showed fair cooperation
- Now they're reciprocating with a solid send of \$2.77

**\*\*My considerations:\*\***

- They treated me fairly last round
- I want to reciprocate and encourage continued cooperation
- Returning around 50% would be fair: ~\$4.15
- This rewards their trust while both of us profit

**\*\*Fair return:\*\***

- Return \$4.00
- They'll end with \$6.23 (profit of \$1.23)
- I'll end with \$4.31 (profit of -\$0.69 relative to endowment, but fair given the exchange)

```
'''json
{"return": 4.00}
'''
```

**reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$2.77 (55% of their endowment) - moderate-high trust
- I received \$8.31 after the 3x multiplier
- They currently have \$2.23 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 7!
- Last round: I sent them \$3.0, they returned \$3.87 to me
- They showed reasonable cooperation last round (43% return rate)

- Now they've sent me \$2.77 (slightly less than my \$3, but still solid)

**\*\*My considerations:\*\***

- This co-player returned \$3.87 on my \$3 send - that was fair/decent
- They're now showing similar trust by sending \$2.77
- I should reciprocate their fairness
- Looking at my past receiver behavior:
  - Round 2: Received \$5.01, returned \$3.00 (60%)
  - Round 3: Received \$10.98, returned \$5.50 (50%)
  - Round 6: Received \$4.92, returned \$3.00 (61%)

**\*\*Fair return:\*\***

They sent me about 55% of their endowment. If I return about 50-55% of what I received:

- Return around \$4.00-\$4.50
- They'll end with \$6.23-\$6.73 (profit of \$1.23-\$1.73)
- I'll end with \$3.81-\$4.31

Given they returned \$3.87 on \$9 (43%), I should probably return a similar rate or slightly better to encourage continued cooperation.

43% of \$8.31 = \$3.57

50% of \$8.31 = \$4.16

I'll return around \$4.00 to be fair and slightly more generous than they were.

**usage:**

```
{
 "input_tokens": 2708,
 "output_tokens": 720,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.77, it became \$8.31, and you returned \$4.0. Your visible payoff was \$4.3100000000000005.

Your total visible earnings across all rounds are \$38.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this round as SENDER:

**\*\*My sender history:\*\***

- Round 1: Sent \$3 ? got \$5.56 back (profit \$3.56) ?
- Round 4: Sent \$3.50 ? got \$3.59 back (profit \$0.09) ?
- Round 5: Sent \$3 ? got \$4.06 back (profit \$1.06) ?
- Round 7: Sent \$3 ? got \$3.87 back (profit \$0.87) ?

**\*\*Observations:\*\***

- \$3 has been very consistent (3/3 good returns)
- Total earnings: \$38.30 - strong position
- This is a NEW co-player

**\*\*Strategy:\*\***

Continue with my proven approach of sending \$3:

- Reliable track record
- Shows cooperative intent
- Creates meaningful value (\$9 for receiver)
- Balanced risk/reward

```
'''json
{"send": 3}
'''
```

**reasoning:**

Looking at this round as SENDER:

**\*\*My sender history:\*\***

- Round 1: Sent \$3 ? got \$5.56 back (profit \$3.56) ?
- Round 4: Sent \$3.50 ? got \$3.59 back (profit \$0.09) ?
- Round 5: Sent \$3 ? got \$4.06 back (profit \$1.06) ?
- Round 7: Sent \$3 ? got \$3.87 back (profit \$0.87) ?

**\*\*Observations:\*\***

- \$3 continues to work well as my standard send (3/3 positive returns)
- Total earnings: \$38.30 - strong position
- New co-player this round (rounds 7-8 was the same co-player)

**\*\*Strategy:\*\***

Continue with \$3 as my reliable opening send:

- Consistent track record
- Shows cooperation
- Creates value (\$9 for receiver)
- Balanced risk/reward
- Establishes trust

I'll send \$3.

**usage:**

```
{
 "input_tokens": 2737,
 "output_tokens": 492,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.0, it became \$9.0, and you saw them return \$3.9 to you. Your visible payoff was \$5.9.

Your total visible earnings across all rounds are \$44.199999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$2.18, so you received \$6.540000000000001. How much do you return? (0-6.540000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$2.18 (44% of their endowment) - moderate trust
- I received \$6.54 after the 3x multiplier
- They currently have \$2.82 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 9!
- Last round: I sent them \$3.0, they returned \$3.90 (43% of what they received)
- They showed fair cooperation
- Now they're reciprocating with \$2.18

**\*\*My considerations:\*\***

- They treated me fairly last round
- I should reciprocate their cooperation
- Returning around 50-55% rewards their trust appropriately
- Want to encourage continued cooperation

**\*\*Fair return:\*\***

- Return \$3.50 (53% of what I received)
- They'll end with \$6.32 (profit of \$1.32)
- I'll end with \$3.04

```
'''json
{"return": 3.50}
'''
```

**Reasoning:**

Looking at this situation:

**\*\*What happened:\*\***

- My co-player sent me \$2.18 (44% of their endowment) - moderate trust
- I received \$6.54 after the 3x multiplier
- They currently have \$2.82 remaining

**\*\*Important context:\*\***

- This is the SAME co-player from Round 9!
- Last round: I sent them \$3.0, they returned \$3.90 (43% of what they received)
- They showed fair cooperation
- Now they're reciprocating with a send of \$2.18

**\*\*My considerations:\*\***

- They treated me fairly last round (returned \$3.90 on \$9 received = 43%)
- I should reciprocate fairly
- They sent \$2.18 which is about 73% of what I sent them (\$3.00)
- Returning around 50-55% would be fair

**\*\*Fair return calculation:\*\***

- 50% of \$6.54 = \$3.27
- 55% of \$6.54 = \$3.60

If I return \$3.50:

- They'll end with \$6.32 (profit of \$1.32)
- I'll end with \$3.04

This seems fair and rewards their trust while maintaining consistency with my previous returns

```
Usage: {
 "input_tokens": 2745,
 "output_tokens": 547,
 "reasoning_tokens": 0
}
```

#### **GAME - Agent\_3 - Sender - dyad\_4**

Timestamp: 2026-08-12T09:29:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_8"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$2.78 to you. Your visible payoff was \$5.279999999999999.

Your total visible earnings across all rounds are \$15.06.

This round, you are the RECEIVER against your current co-player. They sent you \$2.58, so you received \$7.74. How much do you return? (0-7.74)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$2.58 (51.6% of endowment), showing solid trust
- I received \$7.74
- Sender currently has \$2.42

Based on my positive experiences with fair reciprocation, I'll aim for equal final payoffs:

- Total: \$2.42 + \$7.74 = \$10.16
- Equal split: \$5.08 each
- Return needed: \$2.66

{'return': 2.66}

**reasoning:**

I'm the receiver now. Let me analyze:

- Sender sent \$2.58 (51.6% of their endowment)
- I received \$7.74 (after 3x multiplier)
- Sender currently has: \$5 - \$2.58 = \$2.42

My recent experience shows:

- Round 2: I sent \$2.50, got back \$2.43
- Round 3: I sent \$2.50, got back \$2.78

Both returns were generous and aimed at fair splits. I should continue this reciprocal fairness.

For equal final payoffs:

- Total available: \$2.42 (sender's remainder) + \$7.74 (my received) = \$10.16
- Equal split: \$10.16 / 2 = \$5.08 each
- Return needed: \$5.08 - \$2.42 = \$2.66

This would give us both \$5.08, which is fair and reciprocates the sender's trust at 51.6%.

Let me verify:

- Sender final: \$2.42 + \$2.66 = \$5.08
- My final: \$7.74 - \$2.66 = \$5.08

Perfect. I'll return \$2.66.

**usage:**

```
{
 "input_tokens": 1287,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.58, it became \$7.74, and you returned \$2.66. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$20.14.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my track record:

- Sending around 50% has consistently resulted in fair reciprocation
- Other agents are also showing similar moderate trust levels
- This strategy has been mutually beneficial

I'll continue with this successful approach.

{'send': 2.50}

**reasoning:**

Let me review my history:

- Round 1 (receiver): Received from sender who sent \$2.35 (47%), returned \$2.20, payoff \$4.85
- Round 2 (sender): Sent \$2.50 (50%), got back \$2.43, payoff \$4.93
- Round 3 (sender): Sent \$2.50 (50%), got back \$2.78, payoff \$5.28
- Round 4 (receiver): Received from sender who sent \$2.58 (51.6%), returned \$2.66, payoff \$5.08
- Round 5 (sender): ???

My strategy of sending around 50% has been working well - I've been getting fair returns.

The pattern shows:

- When I send ~50% (\$2.50), I get back \$2.43-\$2.78
- Others are sending similar amounts (47-52%)
- Fair reciprocation is the norm

I should continue with this moderate trust approach around 50%.

{'send': 2.50}

**usage:**

```
{
 "input_tokens": 1524,
 "output_tokens": 316,
}
```



```

 "reasoning_tokens": 0
 }
Message 6 role=user
 content:
 Round 6

 In your most recent previous game round, you were the SENDER against another agent. You sent $2.5, it became $7.5, and you saw them return $2.24 to you. Your visible payoff was $4.74.
 Your total visible earnings across all rounds are $24.880000000000003.

 This round, you are the RECEIVER against your current co-player. They sent you $2.72, so you received $8.16. How much do you return? (0-8.16)
 Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
 content:
 Looking at this situation:

 - Sender sent $2.72 (54.4% of endowment), showing strong trust
 - I received $8.16
 - Sender currently has $2.28

 For equal final payoffs:
 - Total: $2.28 + $8.16 = $10.44
 - Equal split: $5.22 each
 - Return needed: $2.94

 {'return': 2.94}
 reasoning:
 Let me analyze this situation:

 - Sender sent $2.72 (54.4% of their $5 endowment) - showing good trust
 - I received $8.16
 - Sender currently has $5 - $2.72 = $2.28

 For equal final payoffs:
 - Total available: $2.28 + $8.16 = $10.44
 - Equal split: $10.44 ÷ 2 = $5.22 each
 - Return needed: $5.22 - $2.28 = $2.94

 This sender showed slightly more trust than average (54.4% vs the ~50% I've been seeing), so I should reciprocate fairly to maintain cooperation.

 Let me return $2.94 to achieve equal payoffs.
 usage:
 {
 "input_tokens": 1725,
 "output_tokens": 298,
 "reasoning_tokens": 0
 }
Message 8 role=user
 content:
 Round 7

 In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $2.72, it became $8.16, and you returned $2.94. Your visible payoff was $5.220000000000001.
 Your total visible earnings across all rounds are $30.1.

 This round, you are the RECEIVER against your current co-player. They sent you $3.36, so you received $10.08. How much do you return? (0-10.08)
 Respond exactly as JSON: {'return': <amount>}
Message 9 role=assistant
 content:
 Looking at this situation:

 - Sender sent $3.36 (67.2% of endowment), showing very high trust - much higher than typical
 - I received $10.08
 - Sender currently has $1.64

```

For equal final payoffs:  
- Total: \$1.64 + \$10.08 = \$11.72  
- Equal split: \$5.86 each  
- Return needed: \$4.22

{'return': 4.22}

**reasoning:**

Looking at this situation:

- Sender sent \$3.36 (67.2% of endowment), showing very high trust
- I received \$10.08
- Sender currently has \$1.64

This is notably higher trust than I've seen in previous rounds (most have been around 50%)  
. I should reciprocate generously.

For equal final payoffs:  
- Total: \$1.64 + \$10.08 = \$11.72  
- Equal split: \$5.86 each  
- Return needed: \$5.86 - \$1.64 = \$4.22

This sender showed significant trust, so I should return a fair amount that rewards that trust.

**usage:**

```
{
 "input_tokens": 1965,
 "output_tokens": 273,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$4.22. Your visible payoff was \$5.86.

Your total visible earnings across all rounds are \$35.96.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my experience:

- My moderate trust strategy (sending ~50%) has been consistently reciprocated fairly
- I've seen varying levels of trust from others, with fair returns in most cases
- Average earnings have been solid with this approach

I'll continue with my tested strategy.

{'send': 2.50}

**reasoning:**

Looking at my history:

- I've been consistently reciprocating fairly when I'm the receiver
- When I'm the sender, I've been sending around \$2.50 (50%)
- The last round showed someone sending \$3.36 (67.2%), which is much higher than average
- I've been getting fair returns overall

My strategy of sending around 50% has been working well. I could consider increasing slightly given the positive reciprocity I've seen, but the moderate approach has been stable and mutually beneficial.

I'll stick with my successful strategy of around 50%.

**usage:**

```
{
 "input_tokens": 1938,
 "output_tokens": 208,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$4.04 to you. Your visible payoff was \$6.54.

Your total visible earnings across all rounds are \$42.5.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

- Sender sent \$3.62 (72.4% of endowment), showing very high trust
- I received \$10.86
- Sender currently has \$1.38

For equal final payoffs:

- Total: \$1.38 + \$10.86 = \$12.24
- Equal split: \$6.12 each
- Return needed: \$4.74

{'return': 4.74}

**reasoning:**

Looking at this situation:

- Sender sent \$3.62 (72.4% of endowment), showing very high trust
- I received \$10.86
- Sender currently has \$1.38

For equal final payoffs:

- Total: \$1.38 + \$10.86 = \$12.24
- Equal split: \$6.12 each
- Return needed: \$4.74

This sender showed exceptional trust (72.4% is quite high). I should reciprocate fairly with equal split approach.

**usage:**

```
{
 "input_tokens": 1936,
 "output_tokens": 234,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.62, it became \$10.86, and you returned \$4.74. Your visible payoff was \$6.119999999999999.

Your total visible earnings across all rounds are \$48.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my performance:

- My moderate trust approach (sending \$2.50) has been effective
- Returns have varied, but I've maintained positive outcomes
- This balanced strategy has proven sustainable across multiple rounds

I'll continue with this tested approach.

{'send': 2.50}

**Reasoning:**

Looking at my history:

- I've been sending \$2.50 consistently as sender

- I've been receiving returns that vary, but generally fair
- As a receiver, I've been returning amounts that aim for equal splits

My experiences:

- Round 4: Received \$7.74, returned \$2.66 (aiming for equal split)
- Round 5: Sent \$2.50, got back \$2.24
- Round 6: Received \$8.16, returned \$2.94 (aiming for equal split)
- Round 7: Received \$10.08, returned \$4.22 (aiming for equal split)
- Round 8: Sent \$2.50, got back \$4.04 (very generous!)
- Round 9: Received \$10.86, returned \$4.74 (aiming for equal split)

My total visible earnings are \$48.62 across 10 rounds, averaging \$4.86 per round, which is below the \$5 endowment.

The \$2.50 sending strategy has been working reasonably well. In round 8, I got an exceptionally good return of \$4.04 on a \$2.50 send. This suggests that continuing with this moderate trust level is reasonable.

I'll stick with my proven strategy of \$2.50.

```
Usage: {
 "input_tokens": 1950,
 "output_tokens": 378,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_8 - Receiver - dyad\_4

Timestamp: 2026-08-12T09:29:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_8"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 10,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.93 to you. Your visible payoff was \$4.93.

Your total visible earnings across all rounds are \$14.96.

This round, you are the RECEIVER against your current co-player. They sent you \$3.22, so you received \$9.66. How much do you return? (0-9.66)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

**\*\*Current Round (Round 4):\*\***

- Co-player sent \$3.22 ? I received \$9.66
- This represents significant trust (sending 64% of their endowment)

**\*\*Fair return analysis:\*\***

- Returning \$4.00 would give:
  - Sender: \$5 - \$3.22 + \$4.00 = \$5.78 (profit of \$0.78)
  - Me: \$9.66 - \$4.00 = \$5.66 (profit of \$0.66)

This creates a fair split where both players benefit from the cooperation.

{'return': 4.00}

reasoning:

Let me analyze this situation:

**\*\*Round 4 - I'm the RECEIVER:\*\***

- My co-player sent me \$3.22
- I received \$9.66 (after 3x multiplier)
- I need to decide how much to return

**\*\*Previous rounds analysis:\*\***

- Round 1 (SENDER): Sent \$3, got back \$2.46 - slightly below fair
- Round 2 (RECEIVER): Received \$9.57, returned \$4.00 - tried to be fair
- Round 3 (SENDER): Sent \$3, got back \$2.93 - again slightly below fair

**\*\*Fair return calculation:\*\***

The sender gave up \$3.22 from their \$5 endowment, creating \$9.66 for me.

- Surplus created:  $\$9.66 - \$3.22 = \$6.44$

A fair split would give us each roughly equal gains:

- If I return \$4.00:

- Sender gets:  $\$5 - \$3.22 + \$4.00 = \$5.78$  (gain of \$0.78)

- I get:  $\$9.66 - \$4.00 = \$5.66$  (gain of \$0.66)

This seems reasonably balanced. The sender sent a significant portion of their endowment, showing trust.

I should return around \$4.00 to be fair and reciprocate their trust.

**usage:**

```
{
 "input_tokens": 1324,
 "output_tokens": 501,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.22, it became \$9.66, and you returned \$4.0. Your visible payoff was \$5.66.

Your total visible earnings across all rounds are \$20.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my experience so far:

**\*\*As SENDER:\*\***

- Round 1: Sent \$3 ? received \$2.46 (modest return)

- Round 3: Sent \$3 ? received \$2.93 (decent return)

**\*\*As RECEIVER:\*\***

- I've consistently returned \$4.00 on ~\$9.60, maintaining fairness

**\*\*Current round:\*\***

- New co-player (randomized pairing)

- Sending \$3 has been my standard opening move

- Creates \$9 of value with moderate risk

I'll continue with the balanced approach that's been working.

{'send': 3}

**reasoning:**

Looking at my history as SENDER:

- Round 1: Sent \$3, received \$2.46 back

- Round 3: Sent \$3, received \$2.93 back

Both times I sent \$3, I received back around \$2.50-\$3.00, which is close to fair but not quite equal splitting of the gains.

As RECEIVER:

- Round 2: Received \$9.57, returned \$4.00

- Round 4: Received \$9.66, returned \$4.00

I've been consistent in returning fairly.

Now Round 5, I'm a SENDER with a new co-player. The pattern suggests that sending around \$3 has been reasonably productive. The returns haven't been amazing (I've profited minimally as sender), but they've been positive.

I should continue with a moderate strategy. Sending \$3 creates value and has worked reasonably well so far.

**usage:**

```
{
 "input_tokens": 1601,
 "output_tokens": 366,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.2 to you. Your visible payoff was \$7.2.

Your total visible earnings across all rounds are \$27.82.

This round, you are the RECEIVER against your current co-player. They sent you \$2.86, so you received \$8.58. How much do you return? (0-8.58)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This is a slightly smaller send than previous rounds

**\*\*Fair return analysis:\*\***

- In previous rounds, I returned ~\$4.00 on receives of ~\$9.60
- This represents about a 42% return rate
- For \$8.58, a proportional return would be around \$3.60

Returning \$3.70 would give:

- Sender: \$5 - \$2.86 + \$3.70 = \$5.84 (modest profit)
- Me: \$8.58 - \$3.70 = \$4.88

This maintains fairness while ensuring both players benefit from cooperation.

{'return': 3.70}

**reasoning:**

Looking at this situation:

**\*\*Current Round (Round 6):\*\***

- Co-player sent \$2.86 ? I received \$8.58
- This shows trust, sending about 57% of their endowment

**\*\*Fair return analysis:\*\***

- Total surplus created: \$8.58 - \$2.86 = \$5.72
- A fair split would give each of us roughly \$2.86 in profit

If I return \$3.50:

- Sender gets: \$5 - \$2.86 + \$3.50 = \$5.64 (profit of \$0.64)
- I get: \$8.58 - \$3.50 = \$5.08 (profit of \$0.08)

That's not quite balanced. Let me try \$4.00:

- Sender gets: \$5 - \$2.86 + \$4.00 = \$6.14 (profit of \$1.14)
- I get: \$8.58 - \$4.00 = \$4.58 (profit of \$-0.42)

Wait, that would give me less than my base \$5. Let me recalculate.

Actually, I don't start with \$5 as the receiver. I start with \$0 and receive the multiplied amount.

So:

- Sender starts with \$5, sends \$2.86, gets back \$X
- Sender final: \$5 - \$2.86 + \$X = \$2.14 + \$X
- I start with \$0, receive \$8.58, return \$X
- My final: \$8.58 - \$X

For a balanced outcome where we both profit similarly:

- Sender profit: \$2.14 + \$X - \$5 = \$X - \$2.86

- My profit:  $\$8.58 - \$X - \$0 = \$8.58 - \$X$
- Setting equal:  $\$X - \$2.86 = \$8.58 - \$X$  gives  $\$X = \$5.72$

That would split the surplus evenly, but feels generous. Instead, I should ensure the sender profits from their trust while keeping a reasonable share for myself. Returning around \$3.80 would give them roughly \$0.94 in profit while I keep \$4.78.

But that's heavily skewed in my favor. Testing \$4.50 or \$5.00 returns gets closer to balance-the sender would profit \$1.64 to \$2.14 while I'd keep \$3.58 to \$4.08. I'm reconsidering what fairness actually means here. In previous exchanges where I received around \$9.50-\$9.66 from sends of \$3.19-\$3.22, returning \$4.00 felt proportional. Using that same ratio (roughly 42% return rate) on this \$8.58 suggests returning around \$3.58-\$3.70. Testing \$3.70 gives the sender \$0.84 profit and me \$4.88, which still favors me but follows the pattern I've established in earlier rounds....

**usage:**

```
{
 "input_tokens": 1883,
 "output_tokens": 1261,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.86, it became \$8.58, and you returned \$3.7. Your visible payoff was \$4.88.

Your total visible earnings across all rounds are \$32.7.

This round, you are the RECEIVER against your current co-player. They sent you \$2.69, so you received \$8.07. How much do you return? (0-8.07)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

**\*\*Current Round (Round 7):\*\***

- Co-player sent \$2.69 ? I received \$8.07
- A moderate send showing trust

**\*\*My return pattern:\*\***

- I've been returning around 42% of what I receive
- This has created mutually beneficial outcomes

For \$8.07 received:

- Returning \$3.50 would give:
  - Sender:  $\$5 - \$2.69 + \$3.50 = \$5.81$  (profit of \$0.81)
  - Me:  $\$8.07 - \$3.50 = \$4.57$

This maintains fairness and ensures both players benefit.

```
{'return': 3.50}
```

**reasoning:**

Looking at the current situation:

**\*\*Current Round (Round 7):\*\***

- Co-player sent \$2.69 ? I received \$8.07
- This is another moderate send

**\*\*My return pattern:\*\***

- Round 2: Received \$9.57, returned \$4.00 (42%)
- Round 4: Received \$9.66, returned \$4.00 (41%)
- Round 6: Received \$8.58, returned \$3.70 (43%)

I've been maintaining around a 42% return rate, which seems fair.

For \$8.07:

- 42% return would be:  $\$8.07 \times 0.42 = \$3.39$

Returning \$3.50 would give:



- Sender:  $\$5 - \$2.69 + \$3.50 = \$5.81$  (profit of  $\$0.81$ )
- Me:  $\$8.07 - \$3.50 = \$4.57$

This maintains my fair approach while ensuring both benefit. Let me go with  $\$3.50$ .

**usage:**

```
{
 "input_tokens": 2202,
 "output_tokens": 424,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send  $\$2.69$ , it became  $\$8.07$ , and you returned  $\$3.5$ . Your visible payoff was  $\$4.57$ .

Your total visible earnings across all rounds are  $\$37.27$ .

This round, you are the SENDER against your current co-player. You have  $\$5$ . How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

**\*\*Previous round (Round 7) with this SAME co-player:\*\***

- They sent me  $\$2.69$  (showing trust)
- I returned  $\$3.50$  (fair return)

**\*\*Current round (Round 8):\*\***

- Now I'm the sender with the same co-player
- They demonstrated moderate cooperation
- I should reciprocate their trust

Since they sent me  $\$2.69$  and I treated them fairly, I'll send  $\$3.00$  to reciprocate their cooperative behavior and potentially continue a mutually beneficial relationship.

```
{'send': 3.00}
```

**reasoning:**

Looking at the situation:

**\*\*Last round (Round 7):\*\***

- I was RECEIVER, my co-player was SENDER
- They sent  $\$2.69$  ? I received  $\$8.07$
- I returned  $\$3.50$
- They got:  $\$5 - \$2.69 + \$3.50 = \$5.81$
- I got:  $\$4.57$

**\*\*This round (Round 8):\*\***

- I am SENDER, they are RECEIVER
- Same co-player!
- They sent me a moderate amount ( $\$2.69$ ) last round
- I returned fairly ( $\$3.50$ )

This is an opportunity to reciprocate their trust. They sent me  $\$2.69$ , so I should send a similar or slightly higher amount to show I'm willing to cooperate fairly with someone who cooperates with me.

Sending  $\$3$  would be reasonable - it's my standard amount and slightly more than what they sent me, showing reciprocal trust.

Actually, let me think about this more carefully. They sent me  $\$2.69$ . I could:

- Send  $\$2.69$  to exactly match their send
- Send  $\$3$  which is my standard amount and slightly more
- Send more to show even stronger reciprocation

I think sending  $\$3$  is appropriate - it matches my typical strategy and shows I'm reciprocating their cooperative behavior.

**usage:**

```
{
 "input_tokens": 2341,
 "output_tokens": 450,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.23 to you. Your visible payoff was \$5.23.

Your total visible earnings across all rounds are \$42.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my pattern:

**\*\*As SENDER:\*\***

- Rounds 1, 3, 5, 8: Consistently sent \$3
- Returns varied from \$2.46 to \$5.20, averaging around \$3.45

**\*\*Current situation (Round 9):\*\***

- New co-player (randomized pairing)
- My strategy of sending \$3 has been reasonably effective
- Creates \$9 of value while maintaining some protection

I'll continue with my established approach.

{ 'send': 3.00 }

**reasoning:**

Let me analyze the situation:

**\*\*Round 9 - I'm the SENDER\*\***

**\*\*My history as SENDER:\*\***

- Round 1: Sent \$3 ? received \$2.46
- Round 3: Sent \$3 ? received \$2.93
- Round 5: Sent \$3 ? received \$5.20 (great return!)
- Round 8: Sent \$3 ? received \$3.23

**\*\*My history as RECEIVER:\*\***

- Round 2: Received \$9.57 ? returned \$4.00
- Round 4: Received \$9.66 ? returned \$4.00
- Round 6: Received \$8.58 ? returned \$3.70
- Round 7: Received \$8.07 ? returned \$3.50

**\*\*Current situation:\*\***

- New co-player (randomized pairing)
- Total earnings so far: \$42.50
- Sending \$3 has been my standard approach
- I've been getting mixed returns, but generally around equal splits

I should continue with my consistent strategy of sending \$3, which:

1. Shows reasonable trust
2. Creates value through the 3x multiplier
3. Has worked fairly well on average
4. Is my established pattern

{ 'send': 3 }

**usage:**

```
{
 "input_tokens": 2298,
 "output_tokens": 425,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.83 to you. Your visible payoff was \$5.83.

Your total visible earnings across all rounds are \$48.33.

This round, you are the RECEIVER against your current co-player. They sent you \$2.79, so you received \$8.370000000000001. How much do you return? (0-8.370000000000001)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

**\*\*Current Round (Round 10):\*\***

- Co-player sent \$2.79 ? I received \$8.37
- A moderate send showing cooperation

**\*\*My return pattern:\*\***

- I've consistently returned around 40-43% of received amounts
- This creates fair, mutually beneficial outcomes

For \$8.37 received:

- Returning \$3.50 would give:
  - Sender: \$5 - \$2.79 + \$3.50 = \$5.71 (profit of \$0.71)
  - Me: \$8.37 - \$3.50 = \$4.87

This maintains my fair approach.

{ 'return': 3.50 }

**Reasoning:**

Looking at this situation:

**\*\*Current Round (Round 10):\*\***

- Co-player sent \$2.79 ? I received \$8.37
- This is a moderate send

**\*\*My return pattern:\*\***

- I've been returning approximately 40-43% of what I receive
- Round 4: Returned \$4.00 on \$9.66 (41.5%)
- Round 6: Returned \$3.70 on \$8.58 (43.1%)
- Round 7: Returned \$3.50 on \$8.07 (43.4%)

For \$8.37 received:

- 42% would be about \$3.52

Returning \$3.50 would give:

- Sender: \$5 - \$2.79 + \$3.50 = \$5.71 (profit of \$0.71)
- Me: \$8.37 - \$3.50 = \$4.87

This maintains consistency with my fair return pattern.

```
Usage: {
 "input_tokens": 2298,
 "output_tokens": 405,
 "reasoning_tokens": 0
}
```